



1.5L and 2.2L Diesel Engine (G22)
Service Manual

*For
Generators*

SAVE THIS MANUAL FOR FUTURE REFERENCE

 WARNING

CANCER AND REPRODUCTIVE HARM

www.P65Warnings.ca.gov.

(000393a)

 WARNING

Breathing diesel engine exhaust exposes you to chemicals known to the State of California to cause cancer and birth defects or other reproductive harm.

- Always start and operate the engine in a well-ventilated area.
- If in an enclosed area, vent the exhaust to the outside.
- Do not modify or tamper with the exhaust system.
- Do not idle the engine except as necessary.

For more information go to

www.P65Warnings.ca.gov/diesel. (000394)

The information in this manual is known to be accurate as of the date of release shown on the cover. However, engineering changes can occur without notice. To obtain the proper parts, be sure to use the serial number of the unit being serviced when ordering from this manual. Always refer to any part notes and changes before ordering. All other parts not shown in the manual can be found on Generac's exploded view drawings at www.generac.com or by contacting your local authorized dealer.

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Section 1: Introduction and Safety

Introduction

Thank you for purchasing a Generac Power Systems Inc. product. This engine has been designed to provide high-performance, efficient operation, and years of use when maintained properly.

Read This Manual Thoroughly



WARNING

Consult Manual. Read and understand manual completely before using product. Failure to completely understand manual and product could result in death or serious injury. (000100a)

If any section of this manual is not understood, contact the nearest Independent Authorized Service Dealer (IASD) or Generac Customer Service at 1-888-436-3722 (1-888-GENERAC), or visit www.generac.com for starting, operating, and servicing procedures. The owner is responsible for proper maintenance and safe use of the unit.

SAVE THESE INSTRUCTIONS for future reference. This manual contains important instructions that must be followed during placement, operation, and maintenance of the unit and its components. Always supply this manual to any individual that will use this unit, and instruct them on how to correctly start, operate, and stop the unit in case of emergency.

Safety Rules

The manufacturer cannot anticipate every possible circumstance that might involve a hazard. The alerts in this manual, and on tags and decals affixed to the unit, are not all inclusive. If using a procedure, work method, or operating technique that the manufacturer does not specifically recommend, verify that it is safe for others and does not render the equipment unsafe.

Throughout this publication, and on tags and decals affixed to the unit, DANGER, WARNING, CAUTION, and NOTE blocks are used to alert personnel to special instructions about a particular operation that may be hazardous if performed incorrectly or carelessly. Observe them carefully. Alert definitions are as follows:

DANGER

Indicates a hazardous situation which, if not avoided, will result in death or serious injury.

(000001)

WARNING

Indicates a hazardous situation which, if not avoided, could result in death or serious injury.

(000002)

CAUTION

Indicates a hazardous situation which, if not avoided, could result in minor or moderate injury.

(000003)

NOTE: Notes contain additional information important to a procedure and will be found within the regular text of this manual.

These safety alerts cannot eliminate the hazards that they indicate. Common sense and strict compliance with the special instructions while performing the action or service are essential to preventing accidents.

IMPORTANT NOTE: Upon disassembly, clearly label components such as pistons, connecting rods, rod caps, main bearing caps, push rods, and lifters.

General Hazards

DANGER

Loss of life. Property damage. Installation must always comply with applicable codes, standards, laws and regulations. Failure to do so will result in death or serious injury. (000190)

DANGER

Automatic start-up. Disconnect utility power and render unit inoperable before working on unit. Failure to do so will result in death or serious injury. (000191)

WARNING

Equipment damage. Only qualified service personnel may install, operate, and maintain this equipment. Failure to follow proper installation requirements could result in death, serious injury, and equipment or property damage. (000182a)



WARNING

Electrocution. Potentially lethal voltages are generated by this equipment. Render the equipment safe before attempting repairs or maintenance. Failure to do so could result in death or serious injury. (000187)

WARNING

Electric shock. Only a trained and licensed electrician should perform wiring and connections to unit. Failure to follow proper installation requirements could result in death, serious injury, and equipment or property damage. (000155a)



WARNING

Moving Parts. Do not wear jewelry when starting or operating this product. Wearing jewelry while starting or operating this product could result in death or serious injury. (000115)



WARNING

Moving Parts. Keep clothing, hair, and appendages away from moving parts. Failure to do so could result in death or serious injury. (000111)



WARNING

Hot Surfaces. When operating machine, do not touch hot surfaces. Keep machine away from combustibles during use. Hot surfaces could result in severe burns or fire. (000108)



WARNING

Risk of Burn. Use protective gloves when handling hot parts. Failure to do so could result in serious injury.

(000386)

WARNING

Risk of injury. Do not operate or service this machine if not fully alert. Fatigue can impair the ability to operate or service this equipment and could result in death or serious injury.

(000215a)

Exhaust Hazards



DANGER

Asphyxiation. Running engines produce carbon monoxide, a colorless, odorless, poisonous gas. Carbon monoxide, if not avoided, will result in death or serious injury.

(000103)



WARNING

Asphyxiation. Always use a battery operated carbon monoxide alarm indoors and installed according to the manufacturer's instructions. Failure to do so could result in death or serious injury.

(000178a)

Fire Hazards



WARNING

Fire and explosion. Installation must comply with all local, state, and national electrical building codes. Noncompliance could result in unsafe operation, equipment damage, death, or serious injury.

(000218)



WARNING

Fire hazard. Use only fully-charged fire extinguishers rated "ABC" by the NFPA. Discharged or improperly rated fire extinguishers will not extinguish electrical fires in automatic standby generators.

(000219)



WARNING

Risk of Fire. Unit must be positioned in a manner that prevents combustible material accumulation underneath. Failure to do so could result in death or serious injury.

(000147)

Explosion Hazards**⚠ DANGER**

Explosion and fire. Fuel and vapors are extremely flammable and explosive. No leakage of fuel is permitted. Keep fire and spark away. Failure to do so will result in death or serious injury. (000192)

⚠ DANGER

Explosion and fire. Connection of fuel source must be completed by a qualified professional technician or contractor. Incorrect installation of this unit will result in death, serious injury, and property and equipment damage. (000151a)

**⚠ DANGER**

Risk of fire. Allow fuel spills to completely dry before starting engine. Failure to do so will result in death or serious injury. (000174)

**⚠ WARNING**

Risk of Fire. Hot surfaces could ignite combustibles, resulting in fire. Fire could result in death or serious injury. (000110)

Lifting Hazards**⚠ WARNING**

Personal injury. Failure to properly connect lifting cables, chains, or straps could result in death, serious injury, or property damage. (000346)

⚠ WARNING

Personal Injury. Do not use lifting eye if there are signs of damage or corrosion. Doing so could result in death, serious injury, or property damage. (000433)

⚠ WARNING

Personal Injury. Do not use lifting eye other than as directed. Doing so could result in death, serious injury, or property damage. (000434)

⚠ WARNING

Personal Injury. Verify all fasteners are properly tightened prior to lifting unit. Failure to do so could result in death, serious injury, or property damage. (000351)

Battery Hazards



⚠ DANGER

Electrocution. Do not wear jewelry while working on this equipment. Doing so will result in death or serious injury.

(000188)



⚠ WARNING

Explosion. Do not dispose of batteries in a fire. Batteries are explosive. Electrolyte solution can cause burns and blindness. If electrolyte contacts skin or eyes, flush with water and seek immediate medical attention.

(000162)



⚠ WARNING

Explosion. Batteries emit explosive gases while charging. Keep fire and spark away. Wear protective gear when working with batteries. Failure to do so could result in death or serious injury.

(000137a)



⚠ WARNING

Electrical shock. Disconnect battery ground terminal before working on battery or battery wires. Failure to do so could result in death or serious injury.

(000164)



⚠ WARNING

Risk of burns. Batteries contain sulfuric acid and can cause severe chemical burns. Wear protective gear when working with batteries. Failure to do so could result in death or serious injury.

(000138a)



⚠ WARNING

Risk of burn. Do not open or mutilate batteries. Batteries contain electrolyte solution which can cause burns and blindness. If electrolyte contacts skin or eyes, flush with water and seek immediate medical attention.

(000163a)

⚠ WARNING

Environmental Hazard. Always recycle batteries at an official recycling center in accordance with all local laws and regulations. Failure to do so could result in environmental damage, death, or serious injury.

(000228)

Always recycle batteries in accordance with local laws and regulations. Contact your local solid waste collection site or recycling facility to obtain information on local recycling processes. For more information on battery recycling, visit the Battery Council International website at: <http://batteryCouncil.org>.

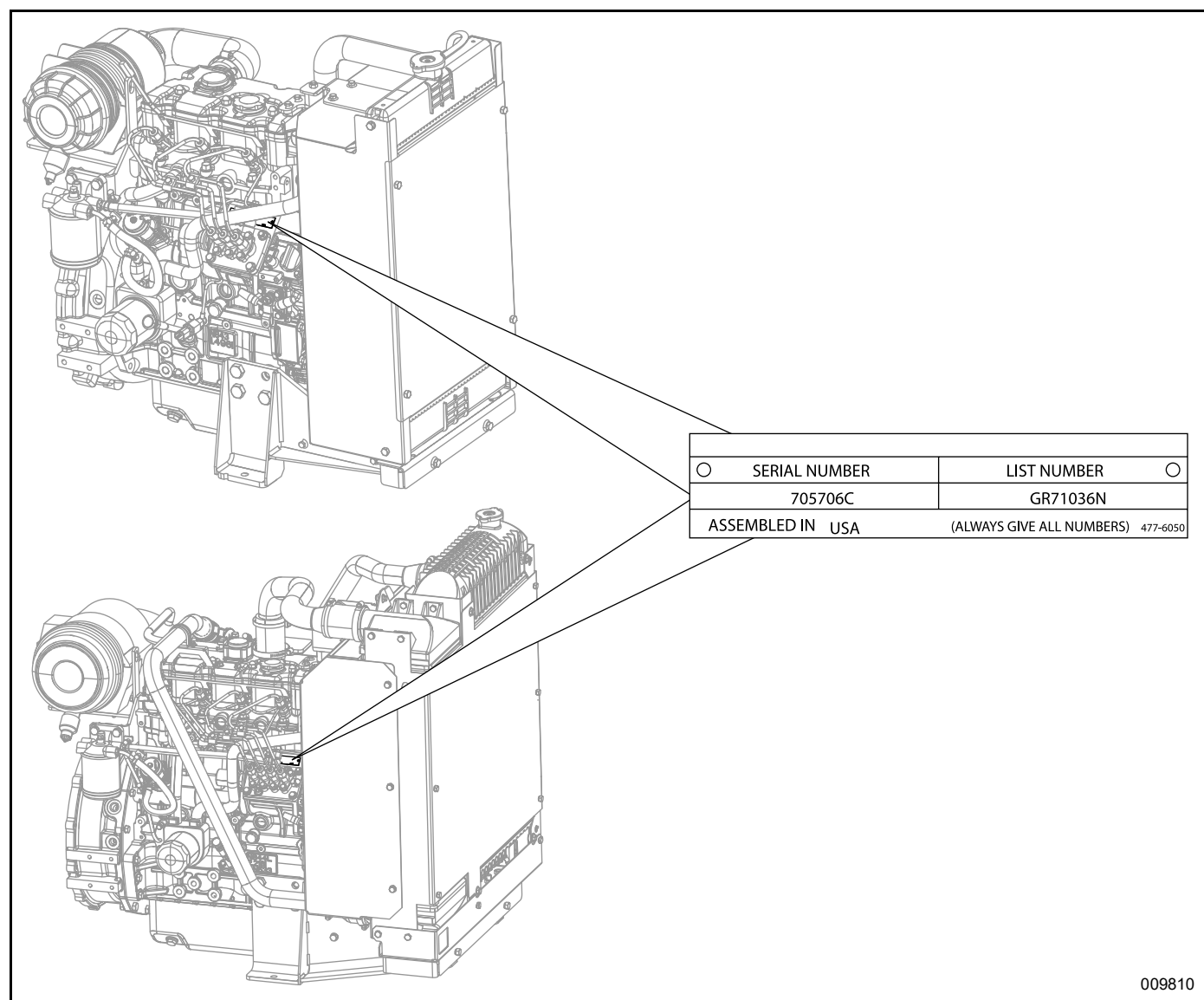
Section 2: General Information

Engine Information

ENGINE SIZE:	1.5 Liter	FUEL:	Diesel	MAKE:	Generac (G22)	ENGINE P/N:	10000038156
ENGINE SIZE:	2.2 Liter	FUEL:	Diesel	MAKE:	Generac (G22)	ENGINE P/N:	10000011378

Engine Identification

The engine identification tag is located on the left side of the engine. The left and right sides are determined by viewing the engine from the flywheel end. The engine identification tag includes information such as the engine list number and engine serial number.



IMPORTANT NOTE: This Engine utilizes various non-traditional thread specifications. Those locations will be called out specifically when available. During servicing, be certain to verify thread pitch and dimension before repairing or replacing any components.

Data Label

Each engine set has a data label containing important information about the engine.

NOTE: The example below is a generic representation only. For actual information on your particular model, refer to the data label(s) affixed to your unit.

G.....Engine family
P.....Type of engine
*****.....The list number of engine
U.....Country of manufacture
O..... The first digit is a production code
00001..... Engine serial number
T..... Year of manufacture

Record for Reference

Information for the following items may be needed to order parts. Locate the information for your engine. Record the information in the appropriate space. Make a copy of this list for your records and keep for future reference.

Engine Model _____
Engine Serial Number _____
Engine Low Idle RPM _____
Engine Full Load RPM _____
Primary Fuel Filter _____
Water Separator Element _____
Secondary Fuel Filter Element _____
Lubrication Oil Filter Element _____
Auxiliary Oil Filter Element _____
Total Lubrication System Capacity _____
Total Cooling System Capacity _____
Air Cleaner Element _____
Fan Drive Belt _____

General Information

Engine Description

NOTE: When ordering new parts, refer to the engine identification number to receive the correct parts. See [Engine Identification](#) for the location of this number.

The engine is a diesel engine controlled with a mechanically actuated fuel injection pump. The engine cylinders are arranged in-line.

The cylinder head assembly has one inlet valve and one exhaust valve for each cylinder. Each cylinder valve has a single valve spring. The pistons have two compression rings and an oil control ring.

It is important to ensure the correct piston height so the piston does not contact the cylinder head. The correct piston height also ensures the efficient combustion of fuel which is necessary to conform to requirements for emissions.

The crankshaft has five main bearing journals. End play for all the engines is controlled by the thrust washers that are on the rear main bearing.

The timing gears are stamped with timing marks to ensure the correct alignment of the gears during assembly. When the No. 1 piston is at top center compression stroke, the teeth stamped on the crankshaft gear and the camshaft gear will be in alignment with the idler gear.

The crankshaft gear turns the idler gear which then turns the camshaft gear.

The fuel injection pump and the fuel priming pump are mounted on the cylinder block. Both pumps are operated by the camshaft lobes.

The fuel injection pump conforms to requirements for emissions. Adjustments to the fuel injection pump timing and high idle should only be made by trained personnel. The fuel injection pumps have mechanical governors that control the engine RPM.

The engine oil pump is a gerotor type pump. The engine oil pump is located in the center of the idler gear. The engine oil pump sends lubricating oil to the main oil gallery through an oil relief valve on the right side of the cylinder block. The rocker arm levers receive pressurized oil through an externally located oil line. The oil line runs from the main oil gallery to the cylinder head.

Coolant from the bottom of the radiator passes through the belt driven centrifugal water pump. The coolant is cooled by the radiator and the temperature is regulated by a water temperature regulator.

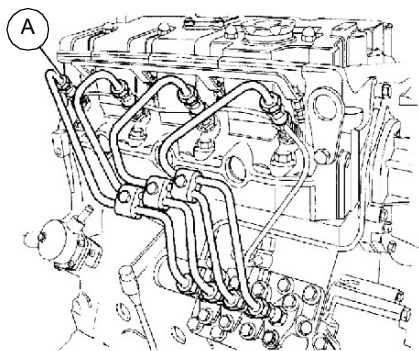
Section 3: Specifications

Engine Specifications

Engine	Item	English	Metric
D2.2L	Maximum Operating Speed	2,800 RPM	
	Cylinders and Arrangement	In-line four cylinder	
	Bore	3.3 in	84 mm
	Stroke	3.9 in	100 mm
	Displacement	135 in ³	2.216 L
	Aspiration	Turbocharged aftercooled	
	Compression Ratio	23.3:1	
	Firing Order	1-3-4-2	
	Rotation (when viewed from flywheel)	Counterclockwise	
	Valve Lash Setting (Inlet)	0.0078 in	0.2 mm
	Valve Lash Setting (Exhaust)	0.0078 in	0.2 mm
	Injection	Indirect	
D1.5L	Maximum Operating Speed	3,000 RPM	
	Cylinders and Arrangement	In-line three cylinder	
	Bore	3.3 in	84 mm
	Stroke	3.5 in	90 mm
	Displacement	91 in ³	1.496 L
	Aspiration	Naturally aspirated	
	Compression Ratio	22.2:1	
	Firing Order	1-2-3	
	Rotation (when viewed from flywheel)	Counterclockwise	
	Valve Lash Setting (Inlet)	0.0078 in	0.2 mm
	Valve Lash Setting (Exhaust)	0.0078 in	0.2 mm
	Injection	Indirect	

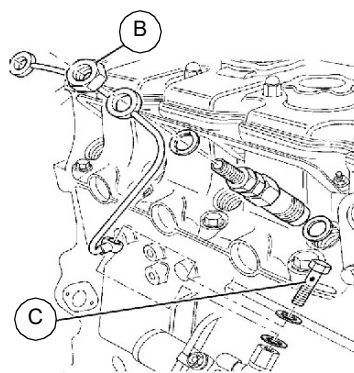
NOTE: Unless otherwise noted, all specifications and procedures throughout this manual apply to the D2.2L and D1.5L engines.

Fuel Injection Lines



006952

Figure 3-1. Fuel Injection Lines



006951

Figure 3-2. Fuel Injection Lines

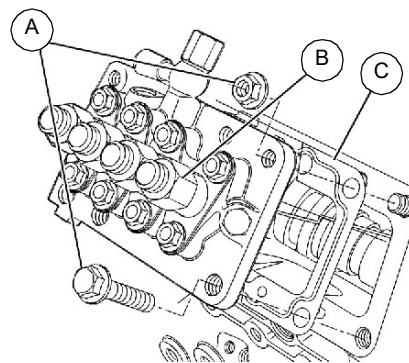
Torque for the union nuts (A) for the fuel injectors:
..... 15 ft-lbs (20 Nm)

Torque for the nut (B)..... 20 ft-lbs (27 Nm)

Torque for the banjo bolt (C)..... **22 in-lbs** (2.5 Nm)

NOTE: All washers must be replaced when the fuel lines are removed.

Fuel Injection Pump



006953

Figure 3-3. Fuel Injection Pump

Type of fuel injection pump..... In-line cassette

Direction of rotation of the camshaft for the fuel injection pump Clockwise from the drive end

Torque for the mounting nuts and setscrews (A):
..... 11 ft-lbs (15 Nm)

Torque for the delivery valve holders (B):
..... 31 ft-lbs (42 Nm)

IMPORTANT NOTE: Changing the injection pump timing is a violation of EPA regulations and will jeopardize compliance of the engine.

The shim (C) adjusts the timing of the fuel injection pump. A thicker shim retards the timing. A thinner shim advances the timing. The timing changes 1 degree for each 0.004 in (0.10 mm) difference in the thickness of the shim. More than one shim can be used. If the fuel injection pump is reinstalled, new shims, which have the same thickness as the original shims, must be installed.

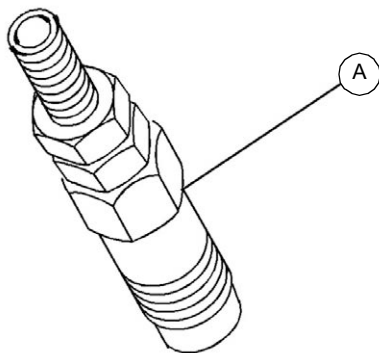
If any of the following new components are installed, new shims which have the same thickness as the original shims must be used.

- Camshaft
- Cylinder block

The following thicknesses of shims are available:

- 0.008 in (0.20 mm)
- 0.010 in (0.25 mm)
- 0.014 in (0.35 mm)
- 0.020 in (0.50 mm)

Fuel Injector



006956

Figure 3-4. Fuel Injector

Torque for the fuel injectors (A) 47 ft-lbs (64 Nm)

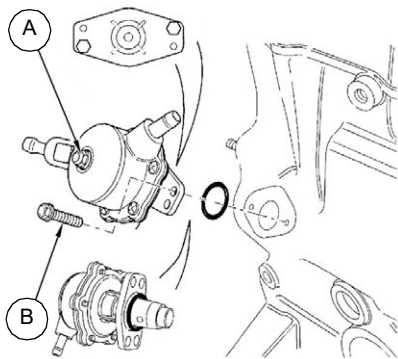
NOTE: Remove the original seat washer from the hole for the fuel injector. Do not reuse the original seat washer.

NOTE: Apply a 0.0787 in (2 mm) bead of Permatex® PermaShield™ or equivalent to the first 0.2362 in (6 mm) of the thread to the fuel injector.

Injector opening pressure 2,132 psi (14,700 kPa)

Leakage in 10 seconds 0 drops

Fuel Transfer Pump



006974

Figure 3-5. Fuel Transfer Pump

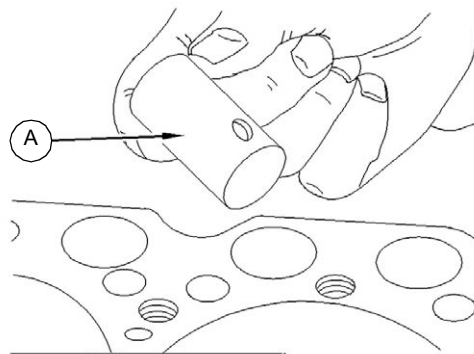
Torque for the adjustment screw (A) for the fuel priming pump 22 in-lbs (2.5 Nm)

NOTE: The fuel inlet for the fuel priming can be rotated 360°. The fuel inlet is adjustable in 15° increments.

NOTE: The flange of the fuel priming pump has two sets of locating holes. This allows the fuel priming pump to be mounted in four positions for the outlet connection.

Tighten the mounting bolts (B) to the following torque:
..... 53 in-lbs (6 Nm)

Lifter Group



007005

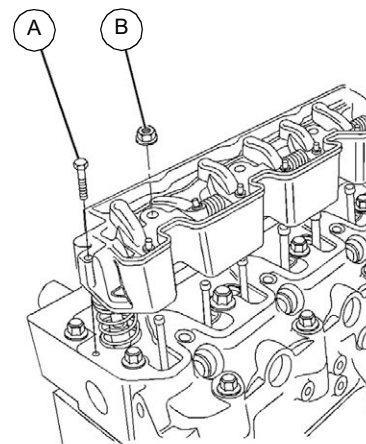
Figure 3-6. Lifter Group

Clearance between lifter and bore in the engine for the lifter:

Standard maximum clearance 0.0023 in (0.058 mm)

Repair limit 0.0031 in (0.080 mm)

Rocker Shaft



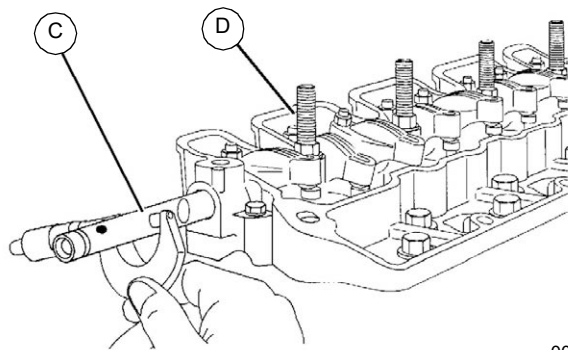
007060

Figure 3-7. Rocker Shaft

NOTE: Before assembly, lubricate the components with clean engine oil.

Torque for setscrews (A) 89 in-lbs (10 Nm)

Torque for nuts (B) 24 ft-lbs (33 Nm)



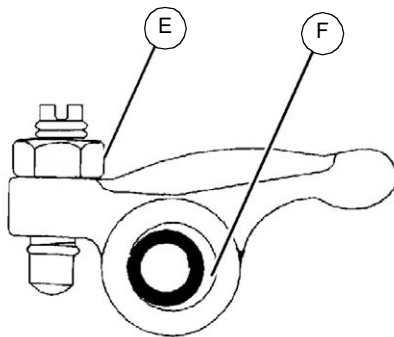
007061

Figure 3-8. Rocker Shaft

Diameter of rocker shaft (C):
 0.5886 to 0.5894 in (14.95 to 14.97 mm)

Service limit 0.5854 in (14.87 mm)

Torque for studs (D) 41 ft-lbs (56 Nm)



007062

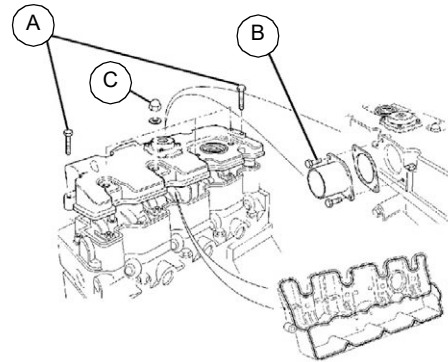
Figure 3-9. Rocker Shaft

Torque for locknut (E) 10 ft-lbs (14 Nm)

Rocker arm minimum permissible clearance (F) on the rocker shaft:
 0.00120 to 0.00366 in (0.030 to 0.093 mm)

Maximum permissible clearance: 0.008 in (0.2 mm)

Valve Mechanism Cover



007095

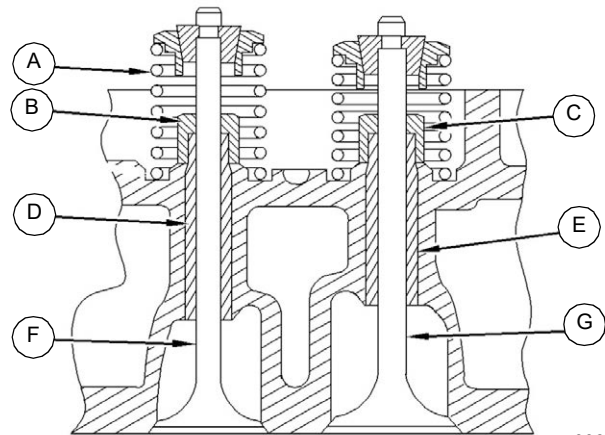
Figure 3-10. Valve Mechanism Cover

Torque for setscrews (A) for valve mechanism cover:
 89 in-lbs (10 Nm)

Torque for setscrews (B) for connector on inlet manifold:
 10 ft-lbs (14 Nm)

Torque for cap nuts (C) for valve mechanism cover:
 10 ft-lbs (14 Nm)

Cylinder Head Valves



006902

Figure 3-11. Cylinder Head Valves

Valve spring (A):
 Standard free length 1.378 in (35.0 mm)

Service limit for the standard free length:
 1.319 in (33.5 mm)

Standard test force: 17.8 lbs (79 N)

Service limit for the standard test force:
 15.4 lbs (68.6 N)

Length under test force 1.197 in (30.4 mm)

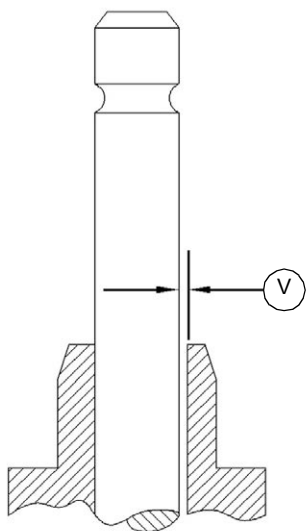
Valve guide seal (B) for exhaust valve:

Identification Black garter spring

Label on the black garter spring "EX"

Valve guide seal (C) for the inlet valve:

Identification Silver garter spring



006903

Figure 3-12. Cylinder Head Valves

Exhaust valve guide (D):

Clearance (V) between exhaust valve and valve guide:
..... 0.0020 to 0.0030 in (0.050 to 0.075 mm)

Service limit 0.010 in (0.25 mm)

Inlet valve guide (E):

Clearance (V) between inlet valve and valve guide:
..... 0.0012 to 0.0024 in (0.030 to 0.060 mm)

Service limit 0.008 in (0.2 mm)

Exhaust valve stem (F):

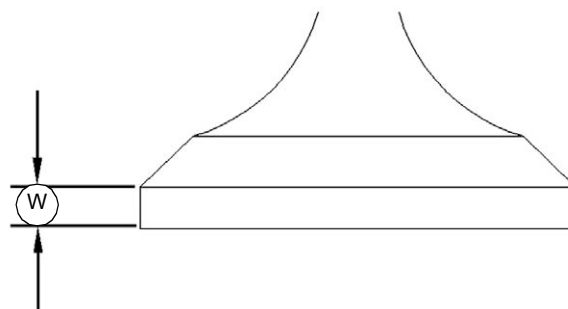
Diameter of exhaust valve stem:
..... 0.2732 to 0.2738 in (6.940 to 6.955 mm)

Service limit 0.269 in (6.84 mm)

Inlet valve stem (G):

Diameter of inlet valve stem:
..... 0.27382 to 0.27441 in (6.955 to 6.970 mm)

Service limit 0.271 in (6.89 mm)



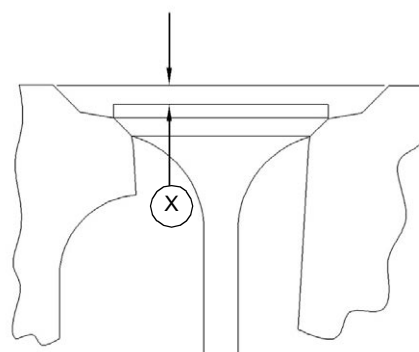
007093

Figure 3-13. Valve Head Thickness

Thickness (W) of valve head:

..... 0.03642 to 0.04232 in (0.925 to 1.075 mm)

Service limit 0.020 in (0.5 mm)



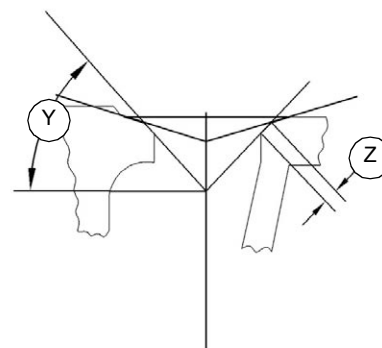
007091

Figure 3-14. Valve Depth Below Cylinder Head Face

Valve depth (X) below cylinder head face:

Inlet and exhaust valves:
..... 0.0256 to 0.0374 in (0.65 to 0.95 mm)

Service limit 0.071 in (1.8 mm)



007094

Figure 3-15. Valve Seat Angle and Contact Face

Valve seat angle (Y) 45°

Contact face (E):

Exhaust valve (D2.2L engines)
..... 0.0764 to 0.0850 in (1.94 to 2.16 mm)

Exhaust valve (D1.5L engines)
..... 0.0653 to 0.0681 in (1.66 to 1.73 mm)

Exhaust Valve Service limit 0.098 in (2.5 mm)

Inlet valve (D2.2L engines)
..... 0.0591 to 0.0790 in (1.50 to 2.00 mm)

Inlet valve (D1.5L engines)
..... 0.0653 to 0.0736 in (1.66 to 1.87 mm)

Inlet Valve Service limit..... 0.098 in (2.5 mm)

Cylinder Head

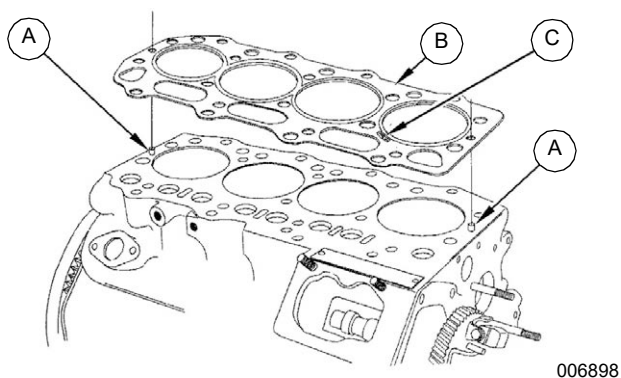


Figure 3-16. Cylinder Head

The dowel pins (A) in the cylinder block hold the cylinder head gasket in the correct position when the cylinder head is installed.

Cylinder head gasket (B):

Table 3-1. Selection of Head Gasket (D2.2L Engines)

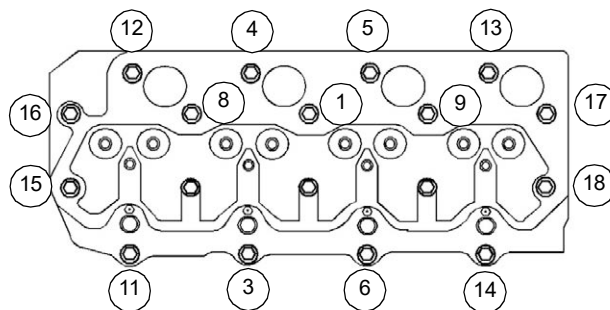
Piston Height Above Top Face of Cylinder Block	Gasket Thickness
0.0138 to 0.020 in (0.35 to 0.5 mm)	0.047 in (1.2 mm)
0.020 to 0.024 in (0.5 to 0.6 mm)	0.051 in (1.3 mm)

Table 3-2. Selection of Head Gasket (D1.5L Engines)

Piston Height Above Top Face of Cylinder Block	Gasket Thickness
0.0236 to 0.0272 in (0.6 to 0.69 mm)	0.051 in (1.3 mm)
0.0276 to 0.0311 in (0.7 to 0.79 mm)	0.055 in (1.4 mm)

The stamped marking on the cylinder head gasket (C) must face upward. This ensures the cylinder head gasket is installed correctly.

Tightening Procedure for Cylinder Head (D2.2L Engines)



006901

Figure 3-17. Tightening Procedure for Cylinder Head Bolts

Use the following procedure to tighten the bolts for the cylinder head.

- Put clean engine oil on the threads of the bolts. The bolts are tightened in the numerical sequence shown in [Figure 3-17](#).

Torque for bolts.....75 ft-lbs (101 Nm)

- Repeat the procedure in Step 1 to tighten all bolts to the correct torque.

Tightening Procedure for Cylinder Head (D1.5L Engines)

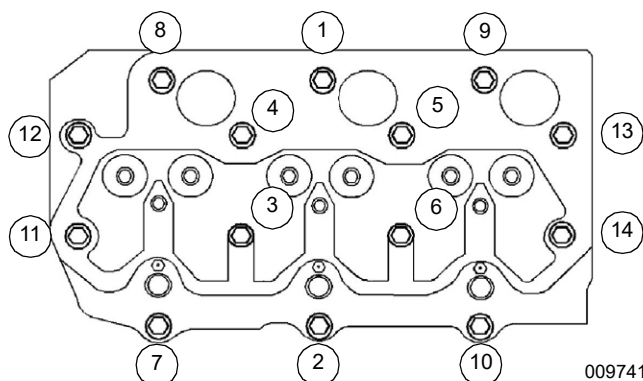


Figure 3-18. Tightening Procedure for Cylinder Head Bolts

Use the following procedure to tighten the bolts for the cylinder head.

1. Put clean engine oil on the threads of the bolts. The bolts are tightened in the numerical sequence shown in [Figure 3-18](#).

Torque for bolts 75 ft-lbs (101 Nm)

2. Repeat the Step 1 to tighten all bolts to the correct torque.

Measuring Cylinder Head Distortion

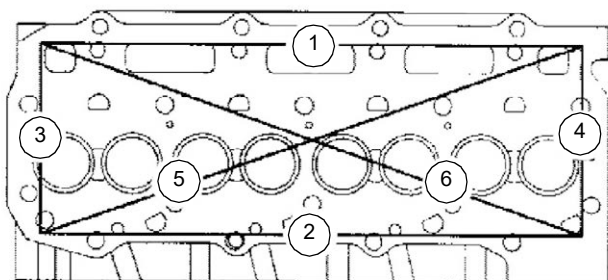


Figure 3-19. Measure Cylinder Head Distortion

Distortion of cylinder head:
..... 0.000 to 0.002 in (0.00 to 0.05 mm)

Maximum service limit..... 0.005 in (0.12 mm)

Maximum limit for regrinding the cylinder head:
..... 0.006 in (0.15 mm)

NOTE: Use a straight edge and a feeler gauge to check the six positions shown in [Figure 3-19](#) for distortion.

See [Inspect Cylinder Head](#) for the procedures for measuring the cylinder head.

NOTE: If the cylinder head has been machined, check the valve depth below the cylinder head face. See [Cylinder Head Valves](#) for valve depth.

Turbocharger

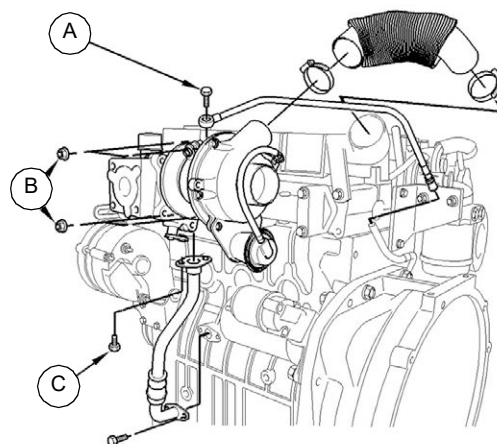


Figure 3-20. Turbocharger

Torque for banjo bolt (A) for oil supply line:
..... 13.3 ft-lbs (18 Nm)

Torque for nuts (B) that secure the turbocharger to the exhaust manifold..... 18 ft-lbs (25 Nm)

Torque for setscrews (C) for oil drain tube:
..... 7 ft-lbs (10 Nm)

Exhaust Manifold

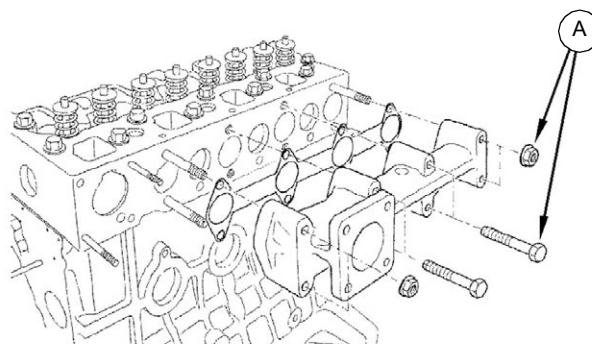


Figure 3-21. Exhaust Manifold

Torque for nuts and setscrews (A) for exhaust manifold:
..... 18.4 ft-lbs (25 Nm)

Camshaft

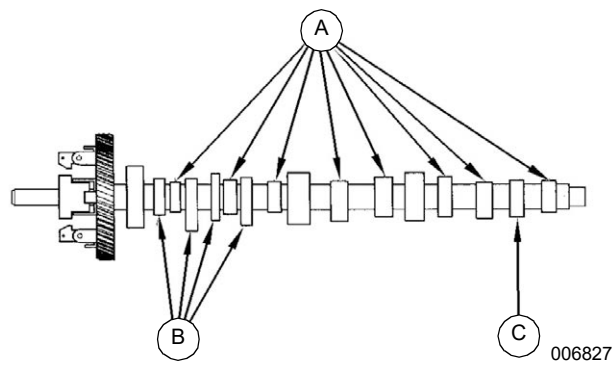


Figure 3-22. Camshaft

- Height (A) of camshaft lobe for inlet and exhaust valves:
..... 1.3564 to 1.3585 in (34.453 to 34.507 mm)
- Service limit 1.3270 in (33.7 mm)
- Height (B) of camshaft lobe for fuel injection pump:
..... 1.6512 to 1.6559 in (41.940 to 42.060 mm)
- Service limit 1.6450 in (41.8 mm)
- Height (C) of camshaft lobe for fuel priming pump:
..... 1.2559 to 1.2598 in (31.900 to 32.000 mm)
- Service limit 1.1810 in (30.0 mm)

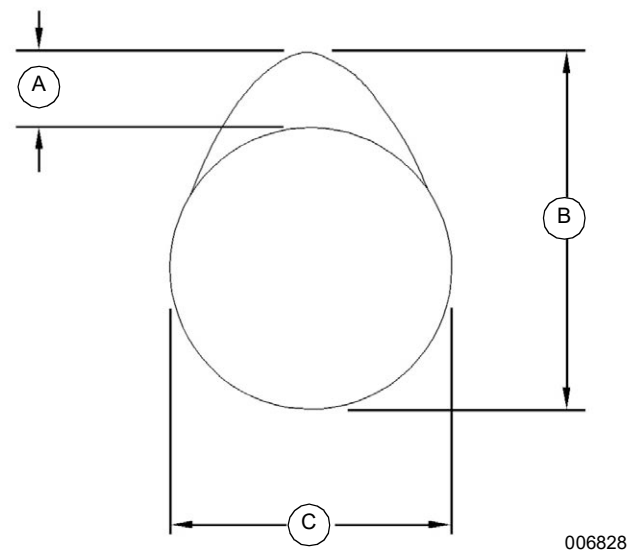
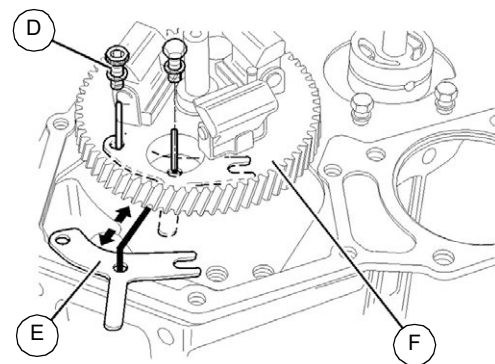


Figure 3-23. Camshaft

- To determine lobe lift, use the following procedure:
1. Measure height of camshaft lobe (B).
 2. Measure base circle (C).

3. Subtract base circle found in Step 2 from height of camshaft lobe found in Step 1. The difference is the actual camshaft lobe lift (A).



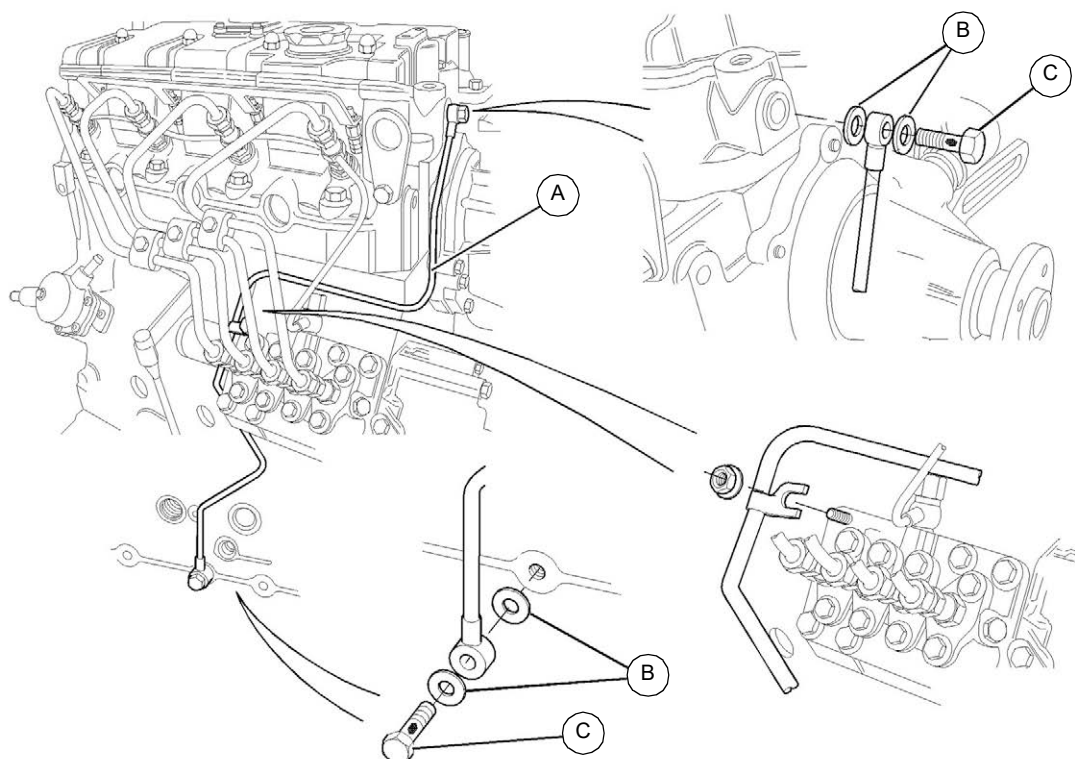
006829

Item	Description
A	Setscrews for retainer plate
B	Retainer plate for camshaft
C	Camshaft gear

Figure 3-24. Camshaft

- Tighten the setscrews (D) for the retainer plate for the camshaft to the following torque:
- 8.0 ft-lbs (11 Nm)

Engine Oil Lines



006921

Item	Description
A	Engine oil line
B	Washers
C	Banjo bolts

Figure 3-25. Engine Oil Lines

NOTE: The washers must be replaced when the engine oil line is removed.

Torque for banjo bolts (C)9 ft-lbs (12 Nm)

NOTE: When the engine oil relief valve is installed, verify all components are clean. Lightly lubricate all components with clean engine oil.

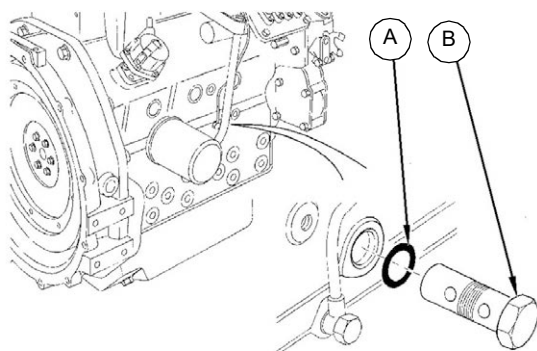
A new O-ring (A) should be used when the engine oil relief valve is installed.

Tighten engine oil relief valve (B) to the following torque:
47 ft-lbs (64 Nm)

Engine oil pressure at the engine oil relief valve (D2.2L engines)
.....51 to 65 psi (352 to 448 kPa)

Engine oil pressure at the engine oil relief valve (D1.5L engines)
.....38 to 52 psi (262 to 359 kPa)

IMPORTANT NOTE: Always remove the engine oil relief valve before removing or installing the crankshaft.



006932

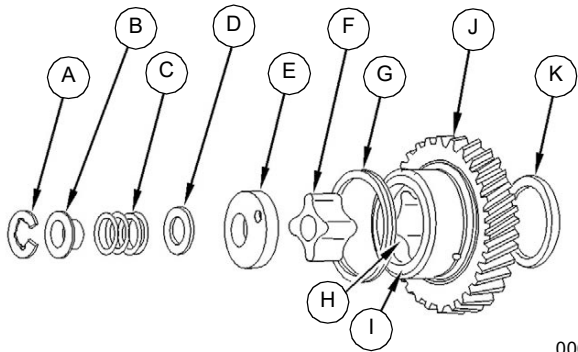
Figure 3-26. Engine Oil Relief Valve

Engine Oil Pump



Equipment Damage. Do not turn the crankshaft if the front housing is not installed. Doing so could result in engine damage.

(000567)



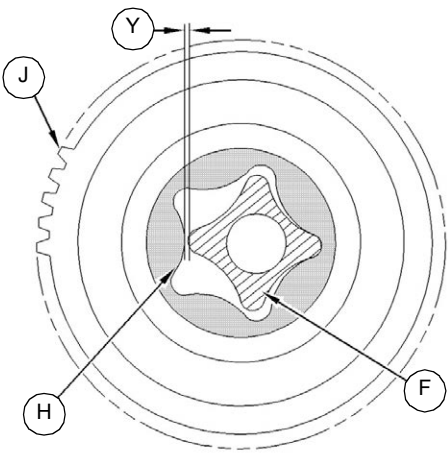
006928

Item	Description
A	Circlip
B	Collar
C	Spring
D	Shim
E	Oil pump cover
F	Inner rotor
G	Spring
H	Outer rotor
I	Bushing
J	Idler gear
K	Thrust washer

Figure 3-27. Engine Oil Pump

Inner rotor (F):
Number of lobes 4

Outer rotor (H):
Number of lobes 5

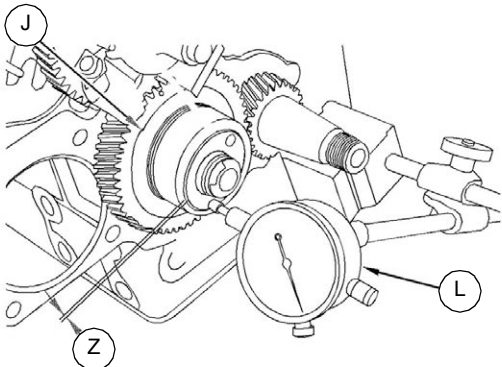


006929

Figure 3-28. Engine Oil Pump

Clearance (Y) between inner rotor and outer rotor:
..... 0.0004 to 0.0006 in (0.01 to 0.15 mm)

Service Limit..... 0.0098 in (0.25 mm)



006930

Figure 3-29. Engine Oil Pump

When the components of the oil pump are installed on the front housing, measurement (Z) between circlip (A) and collar (B) must not exceed the following distance:
..... 0.004 to 0.006 in (0.10 to 0.15 mm)

Service limit..... 0.008 in (0.20 mm)

The distance between the faces is adjusted with shims (D). The following sizes of shims are available:

- 0.004 in (0.10 mm)
- 0.006 in (0.15 mm)
- 0.008 in (0.20 mm)
- 0.02 in (0.50 mm)

Engine Oil Pan

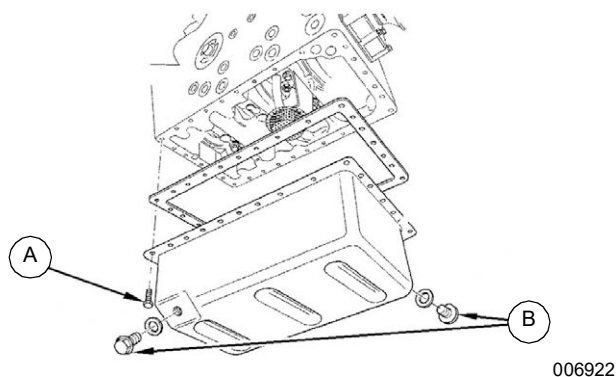


Figure 3-30. Engine Oil Pan

Tighten bolts (A) for engine oil pan to the following torque:
..... 8 ft-lbs (11 Nm)

Tighten drain plugs (B) of engine oil pan to the following torque:
..... 26 ft-lbs (35 Nm)

NOTE: Install a new gasket when the engine oil pan is removed or replaced.

Oil Suction Tube and Oil Strainer

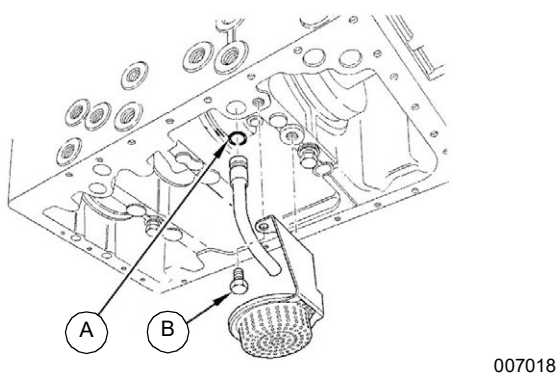


Figure 3-31. Oil Suction Tube and Oil Strainer

Tighten bolts (B) for oil strainer to the following torque:
..... 8 ft-lbs (11 Nm)

NOTE: Install a new O-ring (A) in the hole of the cylinder block when the oil suction tube is removed or replaced.

Water Temperature Regulator

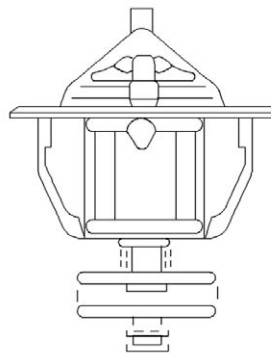


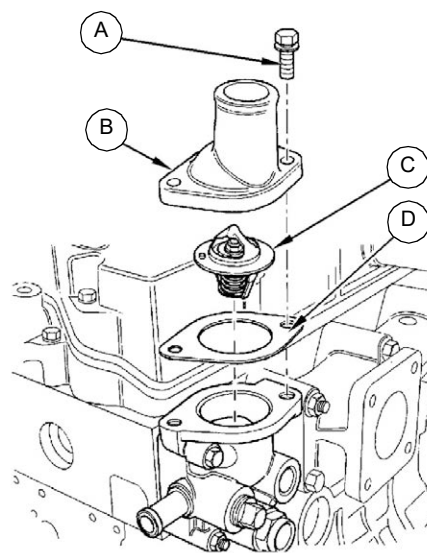
Figure 3-32. Water Temperature Regulator

Opening temperature of the water temperature regulator (D2.2L engines): 180 °F (82 °C)

Opening temperature of the water temperature regulator (D1.5L engines): 176 to 183 °F (80 to 84 °C)

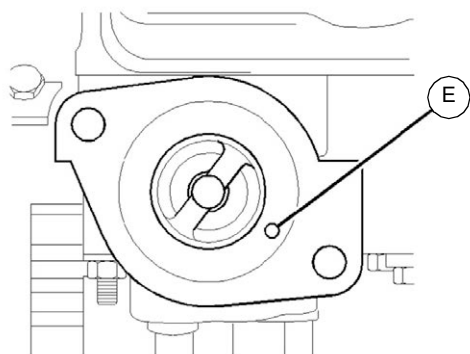
Fully open temperature of the water temperature regulator: 203 °F (95 °C)

Water Temperature Regulator Housing



Item	Description
A	Setscrews
B	Cover
C	Water temperature regulator
D	Gasket

Figure 3-33. Water Temperature Regulator Housing



007036

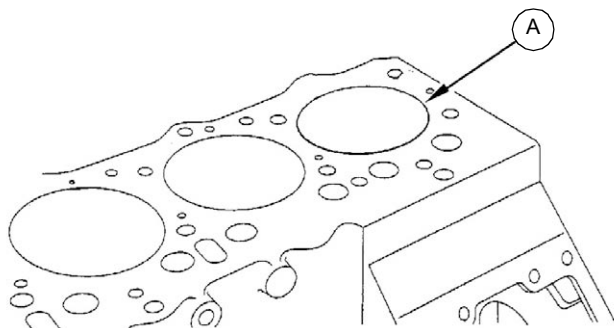
Figure 3-34. Water Temperature Regulator

NOTE: Verify the water temperature regulator is seated correctly in the housing.

NOTE: Verify the jiggle pin (E) on the water temperature regulator is correctly positioned.

Torque for the two setscrews (A) for the water temperature regulator housing 10 ft-lbs (14 Nm)

Cylinder Block

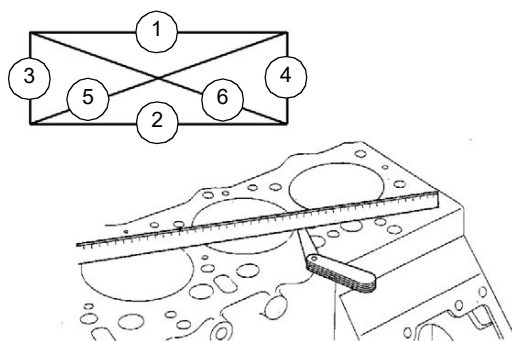


006872

Figure 3-35. Cylinder Block

Diameter of the bore in the cylinder block:
..... 3.3071 to 3.3078 in (84.000 to 84.019 mm)

Service limit 3.3150 in (84.200 mm)



006871

Figure 3-36. Cylinder Block

Flatness of top of cylinder block:

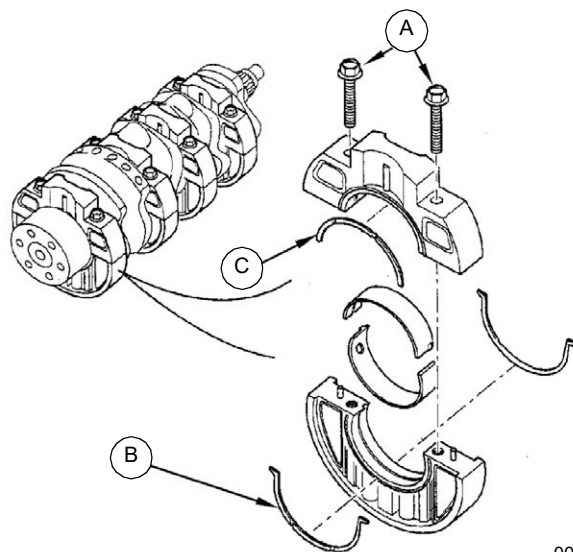
..... Less than 0.002 in (0.05 mm)

Service limit 0.005 in (0.12 mm)

NOTE: Use a straight edge and a feeler gauge to check the six positions in **Figure 3-36** for flatness.

NOTE: The front bush for the crankshaft must be installed with the chamfer toward the cylinder block. Verify the oil hole in the front bush for the crankshaft is aligned with the oil hole in the cylinder block.

Crankshaft



006855

Figure 3-37. Crankshaft (Typical)

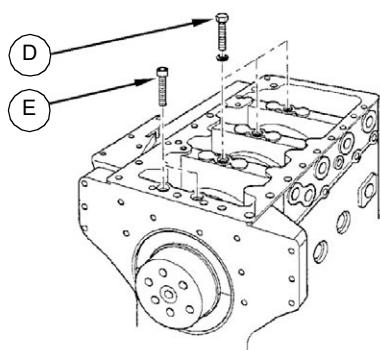
Tighten bolts (A) of holder for the main bearing to the following torque 38.3 ft-lbs (52 Nm)

Thrust washers (B):

..... 0.1161 to 0.1181 in (2.95 to 3.00 mm)

Service limit 0.1102 in (2.80 mm)

NOTE: Verify the oil grooves of all thrust washers (C) are toward the crankshaft.



006856

Figure 3-38. Crankshaft (Typical)

Tighten retaining bolts (D) for crankshaft to the following torque 38.3 ft-lbs (52 Nm)

Tighten Allen head screws (E) to the following torque:
..... 19.9 ft-lbs (27 Nm)

Crankshaft end play:
..... 0.0040 to 0.0118 in (0.10 to 0.30 mm)

Service limit..... 0.0197 in (0.50 mm)

NOTE: If crankshaft end play exceeds the service limit, check thrust washers for wear.

See [Connecting Rod Bearing Journal](#) for information on the connecting rod bearing journals of the crankshaft.

See [Main Bearing Journal](#) for information on the main bearing journals of the crankshaft.

Connecting Rod Bearing Journal

Table 3-3. Diameter of the Connecting Rod Bearing Journals

Journals	Diameter	Service Limit
Standard	2.04582 to 2.04626 in (51.964 to 51.975 mm)	2.0433 in (51.90 mm)
Undersize (0.010 in (0.25 mm))	2.03598 to 2.03641 in (51.714 to 51.725 mm)	2.0335 in (51.65 mm)
Undersize (0.020 in (0.50 mm))	2.02614 to 2.02660 in (51.464 to 51.475 mm)	2.0236 in (51.40 mm) ¹

¹ If the diameter of the connecting rod bearing journal is less than the maximum undersize service limit, the crankshaft must be replaced.

Clearance between connecting rod bearing and connecting rod bearing journal:

Standard clearance:

..... 0.00138 to 0.00335 in (0.035 to 0.085 mm)

Service limit..... 0.0079 in (0.20 mm)

Main Bearing Journal

Table 3-4. Diameter of Main Bearing Journals

Journals	Diameter	Service Limit
Standard	2.67550 to 2.67597 in (67.957 to 67.970 mm)	2.6732 in (67.90 mm)

Clearance between main bearing and main bearing journal:

Standard clearance:

..... 0.0017 to 0.0040 in (0.044 to 0.102 mm)

Service limit..... 0.0079 in (0.20 mm)

Connecting Rod

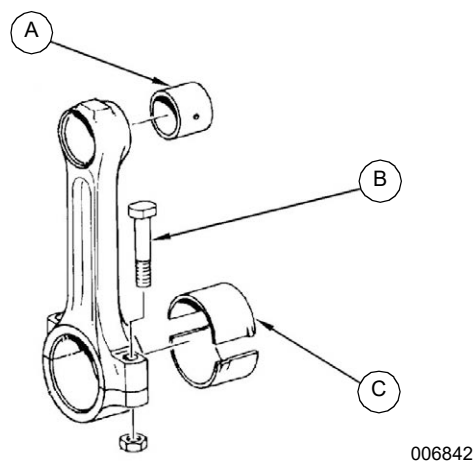


Figure 3-39. Connecting Rod

- Clearance between piston pin and piston pin bearing (A):
..... 0.00040 to 0.00099 in (0.010 to 0.025 mm)
- Service limit 0.004 in (0.10 mm)
- Torque for nut and bolt (B)..... 38 ft-lbs (52 Nm)
- Clearance between connecting rod bore and connecting rod bearing (C):
..... 0.004 to 0.012 in (0.10 to 0.30 mm)
- Service limit 0.0276 in (0.70 mm)

Markings on the Connecting Rod

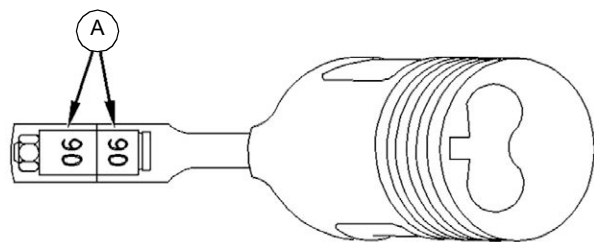


Figure 3-40. Markings on Connecting Rod

The pistons and connecting rods are matched to each cylinder. Note the position of each connecting rod and piston for correct assembly.

Identification marks (A) on the connecting rod and on the connecting rod cap must be matched and aligned. When the connecting rod is installed correctly, the marks should face the right side of the engine.

See [Inspect Connecting Rod](#) for the procedure to measure distortion and parallelism of the connecting rod.

Piston and Rings

Markings on the Piston

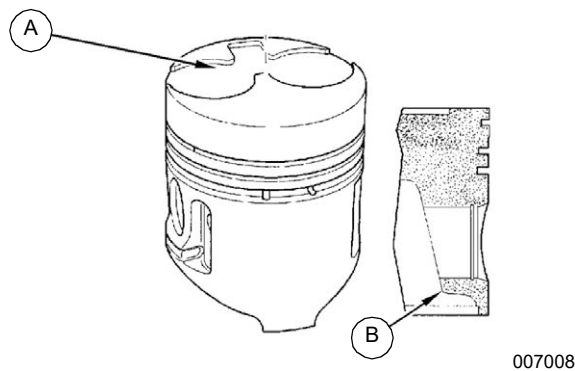


Figure 3-41. Markings on Piston

The chamber (A) on top of piston must face the right side of the engine.

The name (B) on the inside of piston must align with the stamped number on the connecting rod.

Piston and Piston Rings

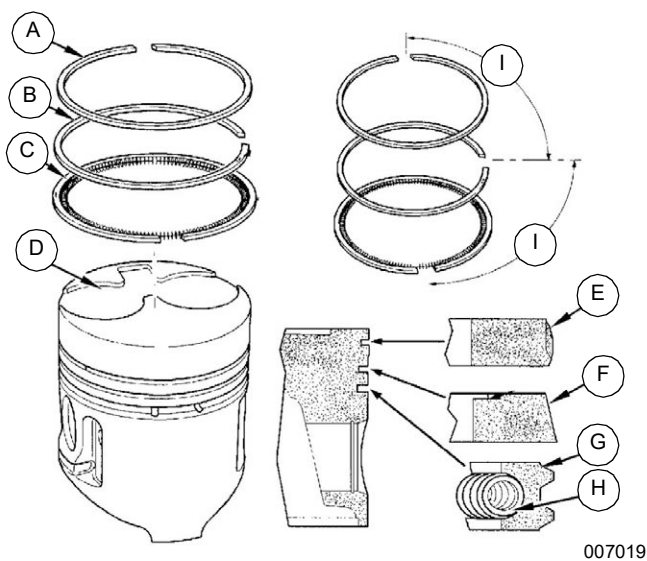


Figure 3-42. Piston and Piston Rings

Use a feeler gauge to measure clearance between piston ring groove and piston ring. If the clearance is greater than the service limit, use a new piston ring and check the clearance.

If the clearance is within the service limit, renew the piston rings. If the clearance is outside of the service limit, replace the piston.

- Shape of top piston ring (E):
- Naturally aspirated Barrel face
- Turbocharged Half keystone

Clearance between piston ring groove and top piston ring (D1.5L engines):

..... 0.0028 to 0.0043 in (0.07 to 0.11 mm)

NOTE: It is difficult to measure the wear of the top piston ring. When either the intermediate ring (B) or the oil control ring (C) is outside the service limit, renew all the rings.

NOTE: Install the letters "T" or "RN" toward the top of the piston. New top piston rings have a red identification mark which must be on the left of the ring gap when the top piston ring is installed on an upright piston.

NOTE: New top piston rings for D2.2L engines have a yellow identification mark which must be on the left of the ring gap when the top piston ring is installed on an upright piston.

NOTE: The top surface of the piston ring should be equally distant from the top face of the cylinder block at all points before the piston ring end gap is measured with a feeler gauge.

Shape of intermediate ring (F) Taper

Clearance between piston ring groove and intermediate ring (F) 0.0016 to 0.0032 in (0.04 to 0.08 mm)

Service limit for clearance of intermediate ring (F):
..... 0.0098 in (0.25 mm)

Gap of intermediate ring (F) (D2.2L engines):
..... 0.020 to 0.023 in (0.5 to 0.7 mm)

Gap of intermediate ring (F) (D1.5L engines):
..... 0.0079 to 0.0157 in (0.2 to 0.4 mm)

Service limit for gap of intermediate ring (F):
..... 0.047 in (1.2 mm)

NOTE: Install the word "Top" toward the top of the piston. New intermediate rings have a green identification mark which must be on the left of the ring gap when the intermediate ring is installed on an upright piston.

NOTE: The top surface of the piston ring should be equally distant from the top face of the cylinder block at all points before the piston ring end gap is measured with a feeler gauge.

Clearance between piston ring groove and oil control ring (C) 0.0008 to 0.0024 in (0.02 to 0.06 mm)

Service limit for clearance of oil control ring (C):
..... 0.0059 in (0.15 mm)

Gap of oil control ring (C):
..... 0.0098 to 0.0197 in (0.25 to 0.50 mm)

Service limit for gap of oil control ring (C):
..... 0.039 in (1.0 mm)

The oil control ring has two components. The two components of the oil control ring are installed in the following order:

1. Spring (H)
2. Oil control ring (G)

NOTE: A latch pin is used to hold both ends of the spring of the oil control ring in position. The ends of the spring of the oil control ring must be 180° opposite the end gap of the oil control ring.

NOTE: The top surface of the piston ring should be equally distant from the top face of the cylinder block at all points before the piston ring end gap is measured with a feeler gauge.

Diameter of piston skirt (D):
..... 3.3050 to 3.3056 in (83.948 to 83.963 mm)

Service limit 3.295 in (83.7 mm)

Clearance of piston skirt (D) to cylinder wall:
..... 0.00150 to 0.00283 in (0.0380 to 0.0720 mm)

Service limit 0.010 in (0.25 mm)

Diameter of piston pin:
..... 1.1021 to 1.1024 in (27.994 to 28.000 mm)

Service limit 1.1016 in (27.98 mm)

Clearance between the hole for the piston pin and the piston pin:
..... -0.0004 to +0.0004 in (-0.001 to +0.011 mm)

Service limit 0.0008 in (0.02 mm)

NOTE: See [Cylinder Head](#) for the piston height for a given gasket thickness.

Housing (Front)

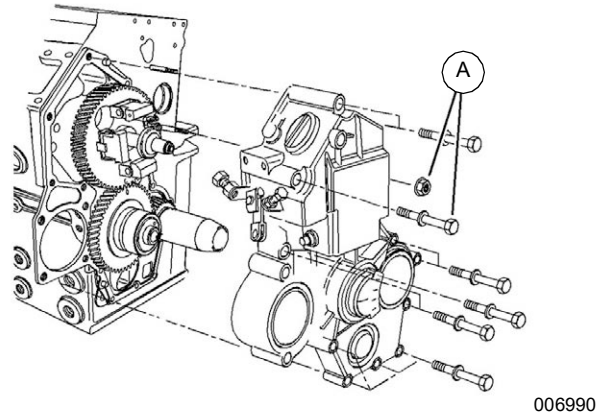


Figure 3-43. Housing (Front)

Tighten setscrews and nuts (A) to the following torque:
..... 7 ft-lbs (10 Nm)

NOTE: Note the positions of the setscrews when the bolts are removed. The setscrews have different lengths.

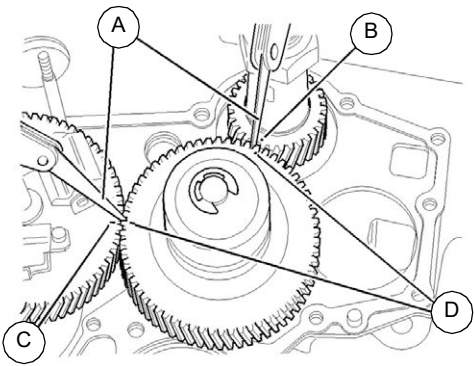
Gear Group (Front)

Table 3-5. Ignition Timing (D2.2L Engine)

Maximum Rated Engine Speed (RPM)	Ignition Timing (BTDC)
2,800	23 ± 1°

Table 3-6. Ignition Timing (D1.5L Engine)

Maximum Rated Engine Speed (RPM)	Ignition Timing (BTDC)
3,000	22 ± 1°



Item	Description
A	Feeler gauge
B	Timing mark on crankshaft gear
C	Timing mark on camshaft gear
D	Timing mark on idler gear

Figure 3-44. Gear Group (Front)

Minimum backlash for all gears:
..... 0.003 in (0.08 mm)

Maximum backlash for all gears:
..... 0.010 in (0.25 mm)

NOTE: If the backlash is greater than the maximum backlash, replace the camshaft gear, the idler gear and the crankshaft gear.

When the idler gear is installed on the shaft of the oil pump, align a timing mark on idler gear (D) with the timing mark on crankshaft gear (B). Also, align the other timing mark on idler gear (D) with the timing mark on camshaft gear (C).

Flywheel

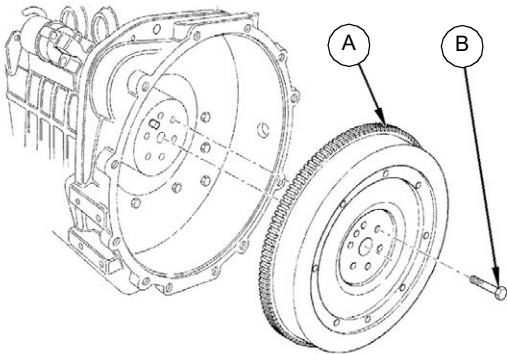


Figure 3-45. Flywheel

Heat flywheel ring gear (A) to the following temperature:

.....248 to 302 °F (120 to 150 °C)

NOTE: Replace ring gear if excessively worn. If excessive wear is not present, remove the ring gear and install the ring gear at 90 degrees from the original position. Heat ring gear evenly.

Tighten setscrews (B) on flywheel to the following torque:
.....55 ft-lbs (74 Nm)

Maximum flywheel runout 0.008 in (0.20 mm)

Flywheel Housing

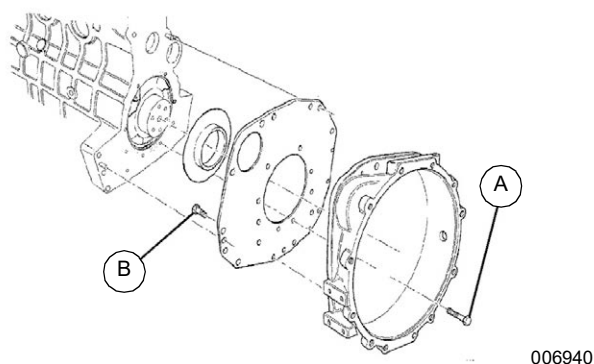


Figure 3-46. Flywheel Housing

Torque for setscrews (A) for back plate:
..... 18.8 ft-lbs (25 Nm)

Torque for setscrews (B) for flywheel housing:
..... 18.8 ft-lbs (25 Nm)

Crankshaft Pulley

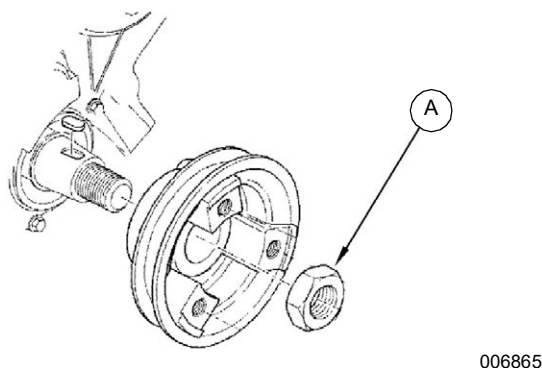


Figure 3-47. Flywheel Housing

Torque for crankshaft pulley nut (A):
.....224 ft-lbs (304 Nm)

Belt Tension Chart

Table 3-7. Belt Tension Chart

Initial Belt Tension ¹	Used Belt Tension ²
90 to 110 lb (400 to 489 N)	60 to 80 lb (267 to 356 N)

¹ Initial Belt Tension refers to a new belt.

² Used Belt Tension refers to a belt that has been in operation for 30 minutes or more at the rated speed.

Fan Drive

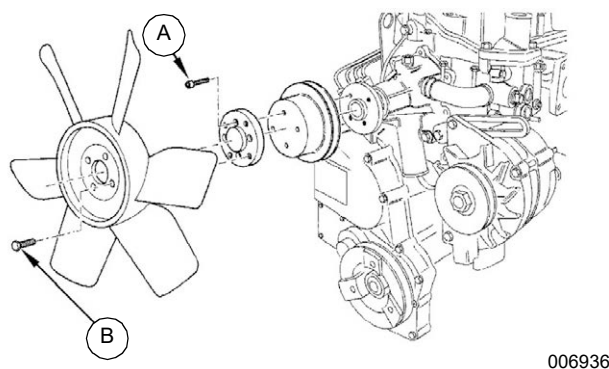


Figure 3-48. Fan Drive

Tighten Allen head screws (A) for the adapter to the following torque..... 8 ft-lbs (11 Nm)

Tighten setscrews (B) for the fan to the following torque:
..... 8 ft-lbs (11 Nm)

Engine Lifting Bracket

The engine is equipped with two engine lifting brackets.
Tighten setscrew on each engine lifting bracket to the following torque 20 ft-lbs (26 Nm)

Alternator and Regulator

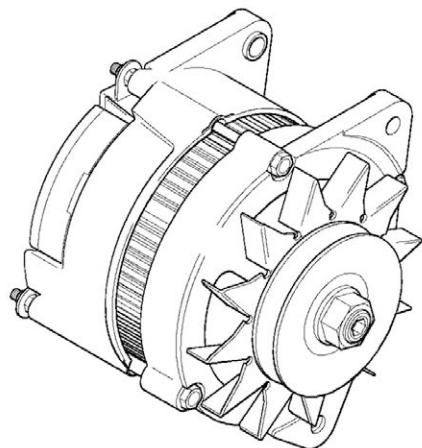


Figure 3-49. Alternator and Regulator

The rotation of the alternator is clockwise when the alternator is viewed from the pulley.
The regulator of the alternator is sealed. The regulator is a non-serviceable part.

- Polarity..... Negative ground to the case
- Rotation Either direction
- Output voltage 14.0 ± 0.5 volts
- Rated voltage 12 volts
- Rated current output..... 55, 65 or 85 amperes

Electric Starting Motor

Starting Motor

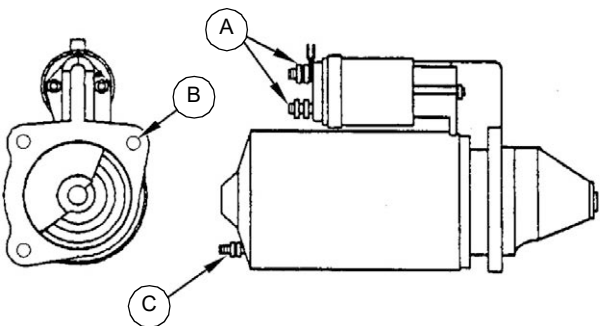


Figure 3-50. Starting Motor and Solenoid

- No load conditions at 77 °F (25 °C):
- RPM with no load 4,000 to 6,000
- Maximum current..... 540 A
- Current draw with no load 130 A
- Voltage 11.5 V
- Rated voltage 12 V
- Power rating 2.0 kW
- Minimum average cranking RPM 130
- Starting motor solenoid:
- Pull-in current 54.5 A
- Hold-in current..... 10.5 A
- Tighten the battery terminal nut (A) to the following torque: 11 ft-lbs (15 Nm)
- Tighten the two mounting bolts (B) to the following torque: 37 ± 4 ft-lbs (50 ± 6 Nm)
- Tighten the nut (C) for the switch terminal to the following torque 9 to 12 **in-lbs** (1.0 to 1.3 Nm)
- Maximum resistance of the starter cable at 68 °F (20 °C) and at 12 V 0.04 ohms

Start Relays

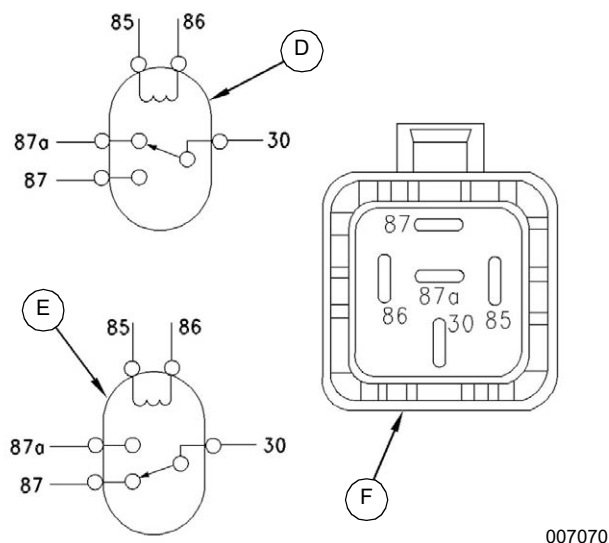


Figure 3-51. Start Relays Schematic

When the ignition switch (D) is in the OFF or RUN position, or the relay switch is closed across the contact "30" and the contact "87a".

When the ignition switch (E) is in the START position, the relay switch is closed across the contact "30" and the contact "87". Then, the starting motor is engaged.

Maximum voltage drop in closed position:

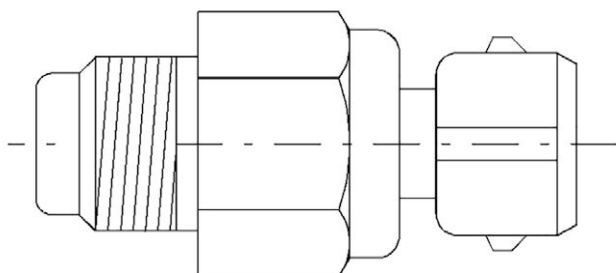
..... 0.150 to 0.250 V

Connector pins on start relay (F)

Resistance between contact "85" and contact "86":

..... 7.2 ± 0.7 ohms

Coolant Temperature Switch

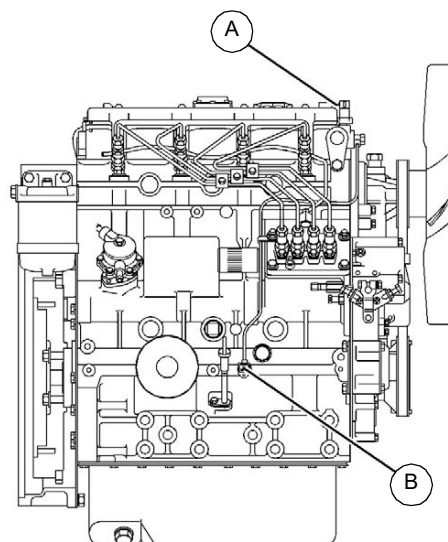


006847

Figure 3-52. Coolant Temperature Switch (Typical)

Tighten coolant temperature switch to the following torque..... 20 ft-lbs (27 Nm)

Engine Oil Pressure Switch



006923

Figure 3-53. Engine Oil Pressure Switch

NOTE: The engine oil pressure switch can be found on the valve mechanism cover or on the oil pressure switch.

Torque for engine oil pressure switch on valve mechanism cover (A):

..... 97 in-lbs (11 Nm)

(Amp) Brown connector

Pressure rating..... 4.26 psi (29.4 kPa)

Green sticker with a red dot..... (Amp) Black connector

Pressure rating..... 4.26 psi (29.4 kPa)

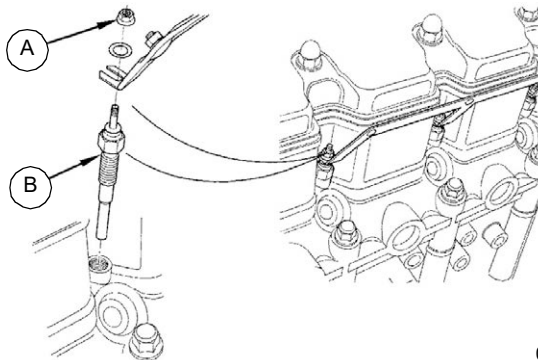
Torque for engine oil pressure switch on cylinder block (B):

..... 17 ft-lbs psi (23 Nm)

(Amp) black connector

Pressure rating..... 10 psi (68.9 kPa)

Glow Plugs



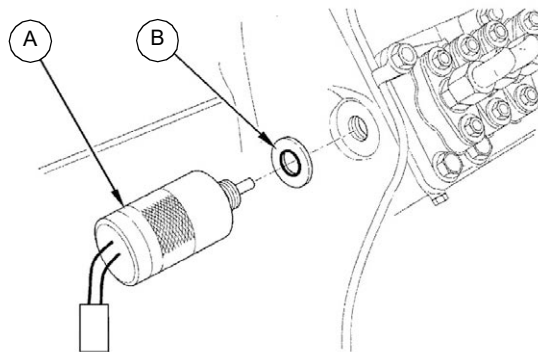
006981

Figure 3-54. Glow Plugs

Torque for nut (A) on bus bar:..... 11 **in-lbs** (1.2 Nm)

Torque for glow plug (B) 13 **ft-lbs** (18 Nm)

Fuel Shutoff Solenoid



006961

Figure 3-55. Fuel Shutoff Solenoid

Tighten fuel shutoff solenoid (A) to the following torque:
..... 13 **ft-lbs** (17 Nm)

NOTE: Verify washer (B) is replaced during installation of the fuel shutoff solenoid (A).

Section 4: Disassembly and Assembly

Fuel Filter Base

CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

IMPORTANT NOTE: Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids. Dispose of all fluids according to local regulations and mandates.

NOTE: Place identification marks on all hoses for installation purposes. Plug all hoses and all the ports in the fuel filter base. This helps prevent fluid loss and contaminants from entering the system.

Removal Procedure

1. Turn the fuel supply OFF.
2. See [Figure 4-1](#). Loosen hose clamps (D) and disconnect hoses (E).
3. If necessary, remove fuel filter element (F) from fuel filter base (A). See [Replace Fuel System Filter](#).
4. Remove fasteners (B) and remove fuel filter base (A) from the mounting bracket.
5. If necessary, remove plugs (F) and washers (not shown) from fuel filter base (A). Remove tube assemblies (C) and rubber olives (not shown) from fuel filter base (A).

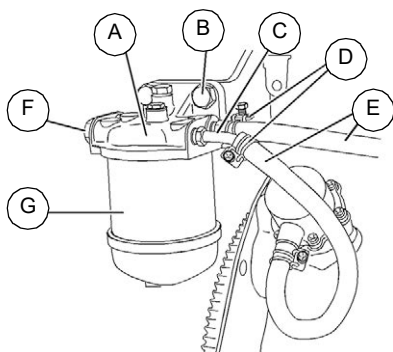


Figure 4-1. Remove Fuel Filter Base

006950

Installation Procedure

CAUTION

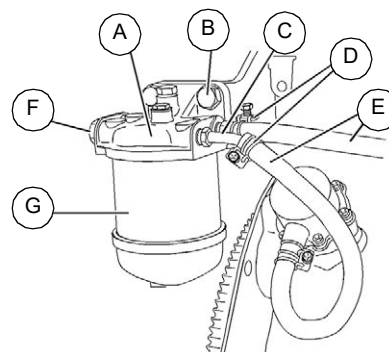
Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

NOTE: If the engine is equipped with a hand priming pump, the hand priming pump is mounted on the fuel filter base. The assembly of the fuel filter base and the hand priming pump is not serviceable.

1. Verify the fuel filter base is clean and free from damage. If necessary, replace the fuel filter base.
2. See [Figure 4-2](#). If necessary, install new rubber olives (not shown) onto tube assemblies (C). Install tube assemblies (C) to fuel filter base (A). Ensure correct orientation of the tube assemblies. Tighten the nuts to 80 in-lbs (9 Nm).
3. Install washers (not shown) onto plugs (F). Install plugs (F) to fuel filter base (A). Tighten the plugs to 17 ft-lbs (23 Nm).
4. Align fuel filter base (A) with the mounting bracket. Install fasteners (B). Tighten the fasteners to 37 ft-lbs (50 Nm).
5. If necessary, install a new fuel filter element (G) to fuel filter base (A). See [Replace Fuel System Filter](#).
6. Connect hoses (E) and tighten hose clamps (D).

NOTE: Verify the hoses do not contact any other engine components.



006950

Figure 4-2. Install Fuel Filter Base

7. Turn fuel supply ON.
8. Remove the air from the fuel system. See [Prime Fuel System](#).

Fuel Transfer Pump (Electrical Fuel Transfer Pump)

Removal Procedure

⚠ CAUTION

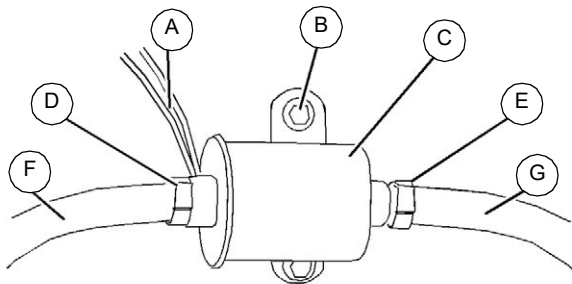
Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

IMPORTANT NOTE: Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids. Dispose of all fluids according to local regulations and mandates.

NOTE: Place identification marks on all hoses for installation purposes. Plug all hoses and all the ports in the fuel filter base. This helps prevent fluid loss and helps keep contaminants from entering the system.

1. Turn fuel supply OFF.
2. Turn battery disconnect switch OFF.
3. See [Figure 4-3](#). Disconnect harness assembly (A).
4. Loosen hose clamps (D) and (E). Disconnect hoses (F) and (G).
5. Remove bolts (B) and remove electric transfer pump (C).



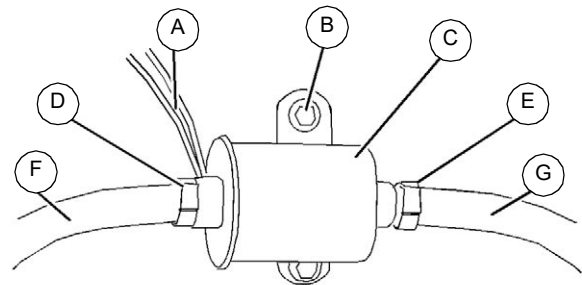
006975

Figure 4-3. Remove Fuel Transfer Pump

Install Procedure

1. Verify the electric transfer pump is clean and free from damage. If necessary, replace the electric transfer pump.
2. See [Figure 4-4](#). Position electric transfer pump (C) on the mounting and install bolts (B).
3. Tighten bolts (B) to 79 in-lbs (9 Nm).

4. Connect hoses (F) and (G). Tighten hose clamps (D) and (E).
5. Connect harness assembly (A).



006975

Figure 4-4. Install Fuel Transfer Pump

6. Turn fuel supply ON.
7. Turn battery disconnect switch ON.
8. Remove the air from the fuel system. See [Prime Fuel System](#).

Fuel Transfer Pump (Mechanical Fuel Transfer Pump)

Removal Procedure

⚠ CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

IMPORTANT NOTE: Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids. Dispose of all fluids according to local regulations and mandates.

NOTE: Place identification marks on all hoses for installation purposes. Plug all hoses and all the ports in the fuel filter base. This helps prevent fluid loss and helps prevent contaminants from entering the system.

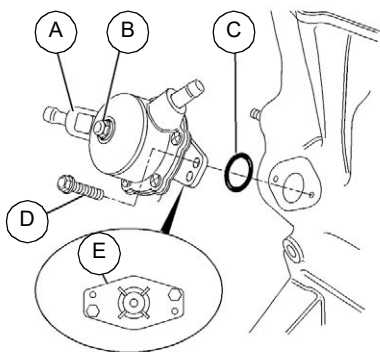
1. Turn fuel supply OFF.

NOTE: See [Figure 4-5](#). The fuel transfer pump can be oriented in two positions. Before removing the fuel transfer pump from the cylinder block, note the orientation of flange (E) on fuel transfer pump (A) for assembly.

2. Loosen the hose clamps and disconnect the hoses (not shown) from fuel transfer pump (A).
3. Evenly loosen bolts (D) and remove fuel transfer pump (A) from the cylinder block.

NOTE: To remove the fuel transfer pump, it may be necessary to rotate the crankshaft until the operating plunger of the fuel transfer pump is not under pressure.

4. Remove O-ring seal (C) from fuel transfer pump (A).



006973

Figure 4-5. Remove Fuel Transfer Pump

Installation Procedure



Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

IMPORTANT NOTE: Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids. Dispose of all fluids according to local regulations and mandates.

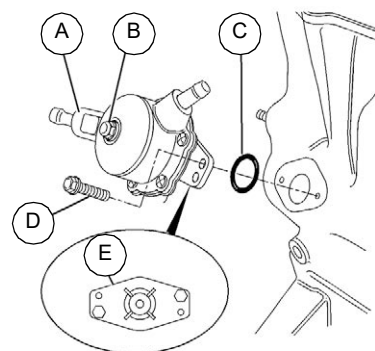
NOTE: Place identification marks on all hoses for installation purposes. Plug all hoses and all the ports in the fuel filter base. This helps prevent fluid loss and helps keep contaminants from entering the system.

1. See [Figure 4-5](#). Clean the mating surfaces of the cylinder block and flange (E) on the fuel transfer pump.

NOTE: Verify the camshaft lobe for the fuel transfer pump is at minimum lift before the fuel transfer pump is installed. The fuel transfer pump can be oriented in two positions. Verify the fuel transfer pump is oriented in the correct position.

2. See [Figure 4-6](#). Install a new O-ring seal (C) to fuel transfer pump (A).
3. Lubricate the operating plunger of fuel transfer pump (A) with clean engine oil.
4. Position fuel transfer pump (A) on the cylinder block. Verify the operating plunger is positioned correctly on the camshaft lobe. Install and tighten bolts (D) to 53 **in-lbs** (6 Nm).
5. Connect the hoses (not shown) to fuel transfer pump (A) and tighten.

NOTE: The inlet for the fuel transfer pump can be rotated 360 degrees by loosening bolt (B). The fuel inlet is adjustable in 15 degree increments. If adjustment is made to the position of the fuel inlet, tighten bolt (B) to 22 **in-lbs** (2.5 Nm).



006973

Figure 4-6. Install Fuel Transfer Pump

6. Turn fuel supply ON.
7. Prime the fuel system. See [Prime Fuel System](#).

Fuel Injection Lines

Removal Procedure



Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

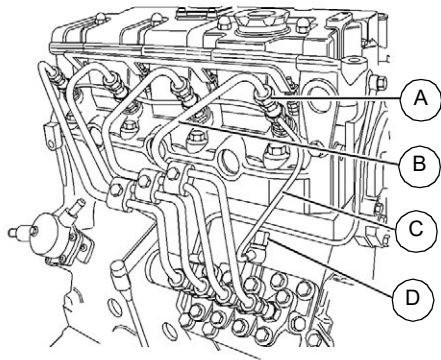
IMPORTANT NOTE: Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids. Dispose of all fluids according to local regulations and mandates.

IMPORTANT NOTE: Do not let the tops of fuel injectors turn when the fuel line nuts are loosened or tightened.

IMPORTANT NOTE: Verify fuel injector is not defective prior to install.

NOTE: Place identification marks on all hoses for installation purposes. Plug all hoses and ports in the fuel filter base. This helps prevent fluid loss, and helps keep contaminants from entering the system.

1. Turn fuel supply OFF.
2. See [Figure 4-7](#) and [Figure 4-8](#). Remove nuts (A) for fuel injection lines (B) from the fuel injectors.
3. Remove nuts (A) for fuel injection lines (B) from the fuel injection pump.
4. Remove fuel injection lines (B) from the engine as a unit.
5. Use suitable caps to plug the open ports of the fuel injection pump immediately.
6. Remove banjo bolt (D) from fuel return line (C). Remove washers (H).



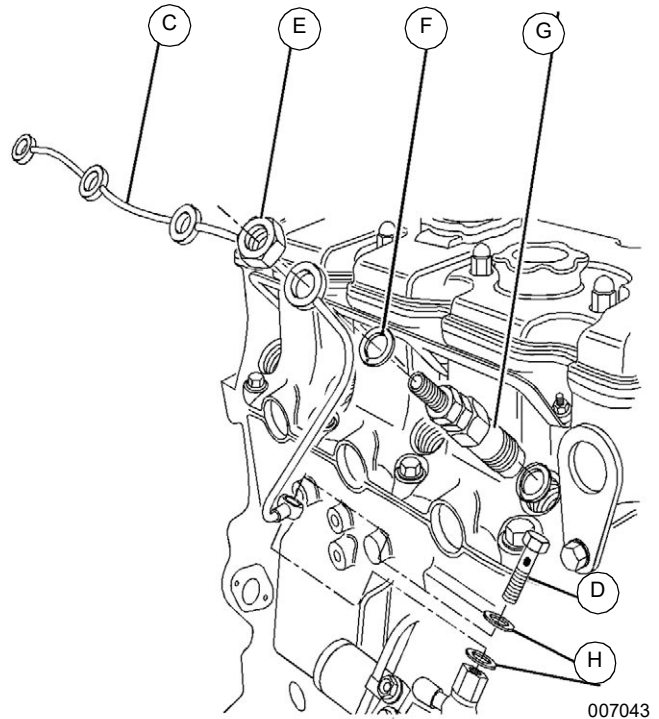
007042

Figure 4-7. Remove Hardware for Fuel Injection Lines

7. Remove nuts (E) from fuel injectors (G).

NOTE: Verify the fuel return line is not distorted when the nuts are loosened.

8. Remove fuel return line (C) and washers (F) from fuel injectors (G).
9. Use suitable cap to plug the fuel injectors immediately.



007043

Figure 4-8. Remove Hardware for Fuel Injection Lines

Installation Procedure



Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

IMPORTANT NOTE: Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids. Dispose of all fluids according to local regulations and mandates.

NOTE: Place identification marks on all hoses for installation purposes. Plug all hoses and all the ports in the fuel filter base. This helps prevent fluid loss and helps keep contaminants from entering the system.

IMPORTANT NOTE: Do not let the tops of fuel injectors turn when the fuel line nuts are loosened or tightened.

IMPORTANT NOTE: Verify fuel injector is not defective prior to install.

1. See [Figure 4-9](#) and [Figure 4-10](#). Remove the caps from fuel injectors (G). Install new washers (F) and fuel return line (C) to fuel injectors (G).

NOTE: The washers (F) have two small holes.

2. Install nuts (E) to fuel injectors (G).
3. Install new washers (H) to fuel return line (C) and install banjo bolt (D) to the fuel injection pump. Tighten banjo bolt (D) to 62 **in-lbs** (7 Nm).
4. Tighten nuts (E) to 20 ft-lbs (27 Nm).

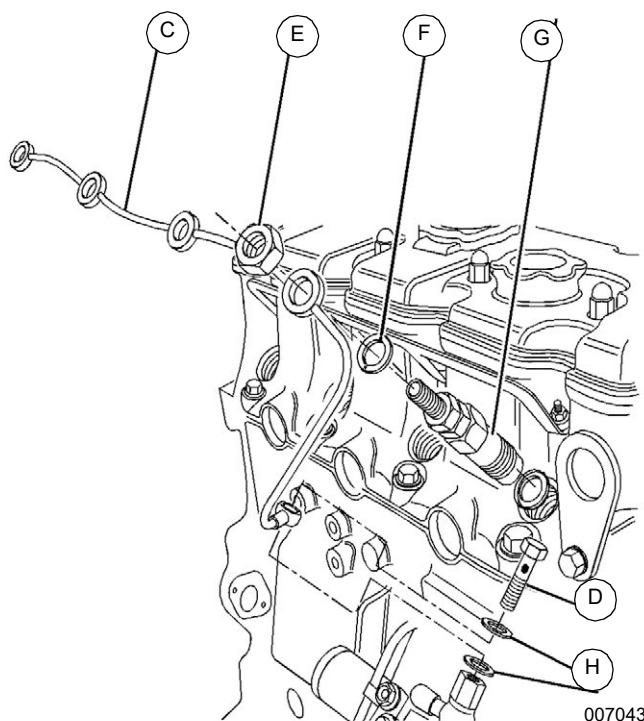
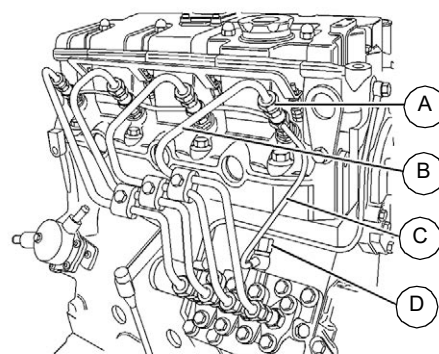


Figure 4-9. Install Hardware for Fuel Injection Lines

5. Remove the caps from the outlet connections of the fuel injection pump. Install the fuel injection lines to the engine as a unit.
6. Connect fuel injection lines (B) to fuel injectors (G). Tighten the union nuts (A) finger tight.
7. Use an injector pipe nut tool to tighten union nuts (A) at the fuel injection pump to 17 ft-lbs (23 Nm), starting with the center union nut.
8. Use an injector pipe nut tool to tighten union nuts (A) at the fuel injectors.
9. Tighten union nuts (A) to 17 ft-lbs (23 Nm).



007042

Figure 4-10. Install Hardware for Fuel Injection Lines

10. Turn fuel supply ON.
11. Prime the fuel system. See [Prime Fuel System](#).

Fuel Shutoff Solenoid

Removal Procedure

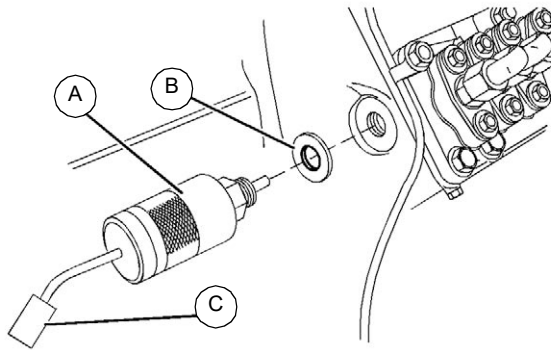


Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

IMPORTANT NOTE: Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids. Dispose of all fluids according to local regulations and mandates.

1. Turn battery disconnect switch OFF.
2. See [Figure 4-11](#). Disconnect electrical connection (C) from the harness assembly (not shown). Mark all connections for installation.
3. Remove fuel shutoff solenoid (A) from the fuel injection pump housing by rotating the fuel shutoff solenoid in a counter-clockwise direction.
4. Remove sealing washer (B) from fuel shutoff solenoid (A).



007046

Figure 4-11. Remove Fuel Shutoff Solenoid

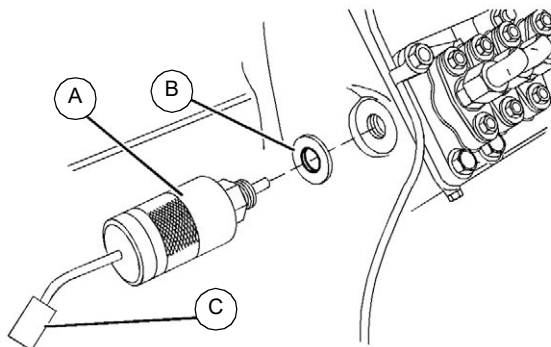
Installation Procedure



Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. See **Figure 4-12**. Install sealing washer (B) to fuel shutoff solenoid (A).
2. Install fuel shutoff solenoid (A) into the fuel injection pump housing by rotating the fuel shutoff solenoid clockwise. Tighten the fuel shutoff solenoid to 12 ft-lbs (17 Nm).
3. Connect electrical connection (C) to the harness assembly (not shown).



007046

Figure 4-12. Install Fuel Shutoff Solenoid

4. Turn battery disconnect switch ON.

Fuel Injection Pump

Removal Procedure

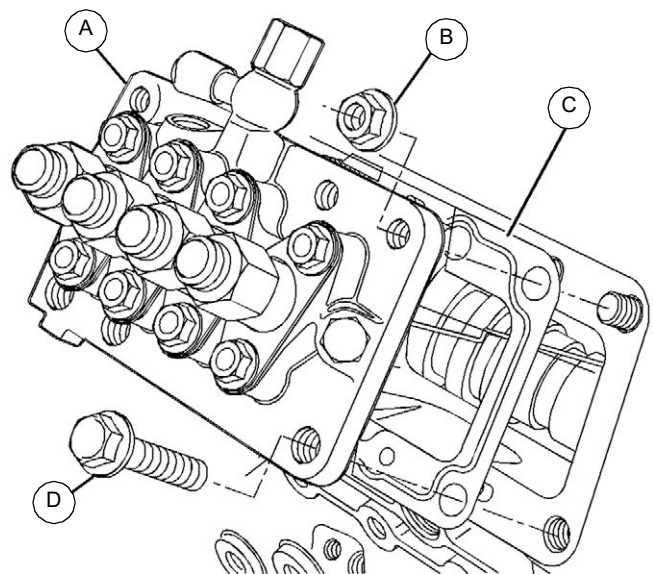


Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

IMPORTANT NOTE: Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids. Dispose of all fluids according to local regulations and mandates.

1. Remove **Fuel Injection Lines**.
2. See **Figure 4-13** and **Figure 4-14**. Gradually loosen bolts (D) and nuts (B) that fasten the fuel injection pump to the cylinder block.
3. Carefully raise fuel injection pump (A) from the cylinder block and remove clip (G) that connects link (F) to fuel rack control (E).
4. Remove fuel injection pump (A) from the cylinder block.
5. Remove shims (C) from the mounting face of the cylinder block.



006998

Figure 4-13. Remove Fuel Injection Pump

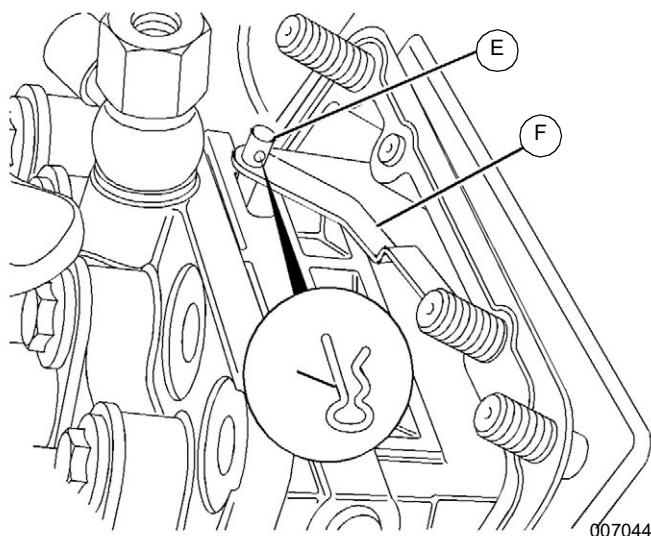


Figure 4-14. Remove Fuel Injection Pump

IMPORTANT NOTE: Changing the injection pump timing is a violation of EPA regulations and will jeopardize compliance of the engine.

NOTE: Record the thickness of each shim and the number of shims for reassembly. The fuel injection timing is determined by the thickness of the shim pack between the fuel injection pump and the mounting face on the cylinder block. See [Fuel Injection Pump](#) for more information.

Installation Procedure

CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. Clean the mating surfaces of the cylinder block and the fuel injection pump.
2. See [Figure 4-15](#) and [Figure 4-16](#). New shims (C) must be used during assembly. Install the correct thickness and the correct number of shims on the mounting face of the cylinder block. See [Fuel Injection Pump](#) for more information.
3. Position fuel injection pump (A) close to the mounting face of the cylinder block, and connect link (F) and fuel rack control (E) with clip (G).
4. Align fuel injection pump (A) with the studs on the cylinder block. Install the fuel injection pump to the cylinder block.
5. Install bolts (D) and nuts (B). Verify the tube clip for the engine oil line is secured by the appropriate fastener.

6. Evenly tighten bolts (D) and nuts (B) to 11 ft-lbs (15 Nm).

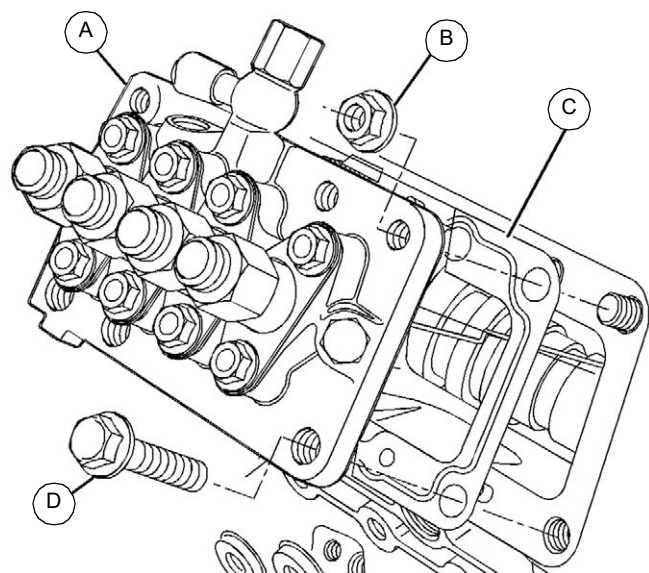


Figure 4-15. Install Fuel Injection Pump

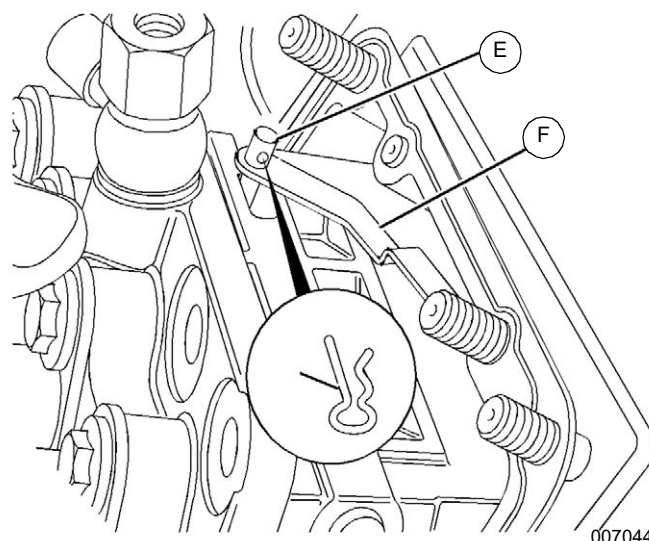


Figure 4-16. Install Fuel Injection Pump

7. Install the fuel injection lines. See [Fuel Injection Lines](#).
8. Install the fuel shutoff solenoid. See [Fuel Shutoff Solenoid](#).

Fuel Injector

Removal Procedure

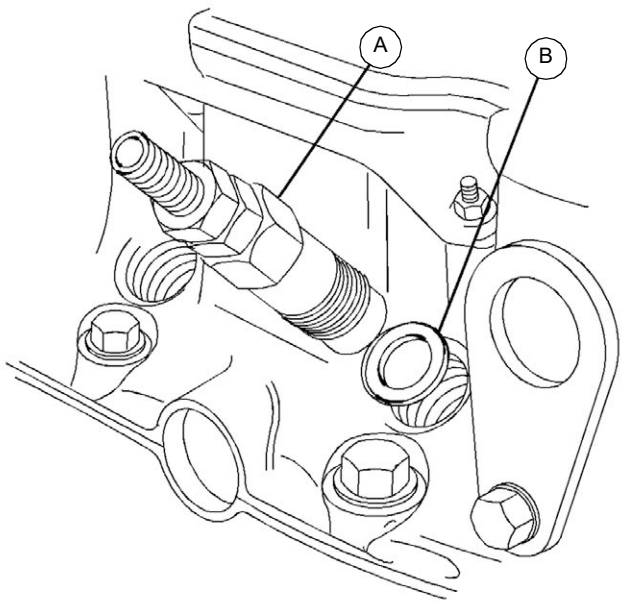
⚠ CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

IMPORTANT NOTE: Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids. Dispose of all fluids according to local regulations and mandates.

1. Remove fuel injection lines. See [Fuel Injection Lines](#).
2. See [Figure 4-17](#). Use a deep socket to remove fuel injector (A) from the cylinder head.
3. Remove seat washer (B) from the cylinder head.
4. Cap all openings or plug all openings immediately.



007045

Figure 4-17. Remove Fuel Injector

Installation Procedure

⚠ CAUTION

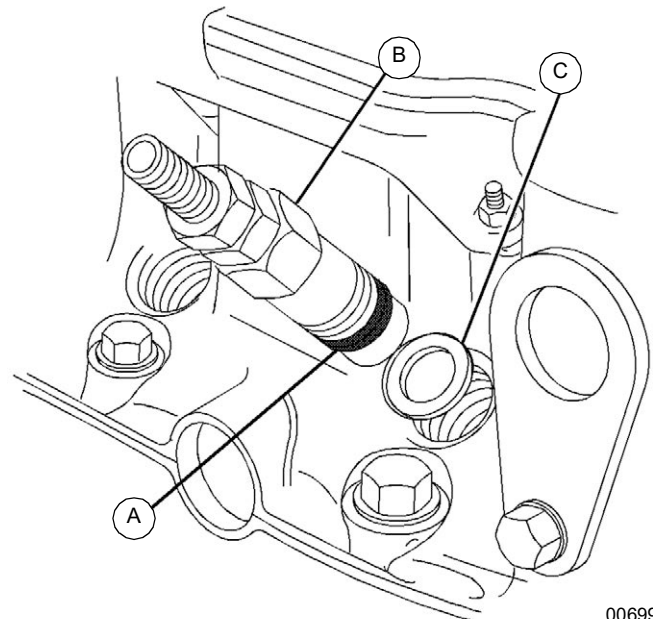
Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. Clean the bore for the fuel injector in the cylinder head. Verify no debris enters the cylinder. Clean the threads on the body of the fuel injector.
2. See [Figure 4-18](#). Install new seat washer (C) into the bore for the fuel injector in the cylinder head.
3. Apply a bead of Permatex® PermaShield™ or equivalent (A) to the first two threads of the fuel injector that engage into the cylinder head. The bead should have a diameter of 0.08 in (2 mm) and a length of 0.25 in (6 mm).

NOTE: Verify Permatex® PermaShield™ or equivalent (A) does not cover the body of the fuel injector below the threads.

4. Install fuel injector (B) into the cylinder head. Use a deep socket to tighten the fuel injector to 47 ft-lbs (64 Nm).
5. Install fuel injection lines. See [Fuel Injection Lines](#).



006999

Figure 4-18. Install Fuel Injector

Turbocharger

Removal Procedure

CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

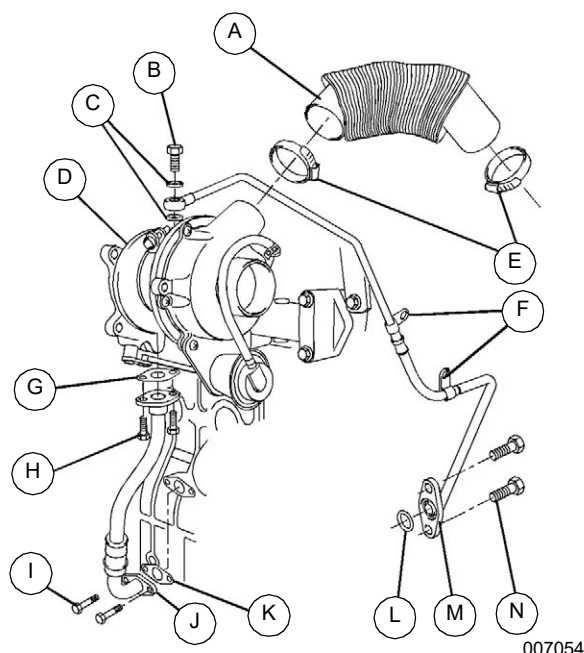
IMPORTANT NOTE: Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids. Dispose of all fluids according to local regulations and mandates.

NOTE: Plug and cap all open ports and tube assemblies.

1. See [Figure 4-19](#) and [Figure 4-20](#). Loosen hose clamps (E) and remove air inlet hose (A).
2. Remove Allen head screws (U), exhaust elbow (S) and gasket (R) from the turbocharger.
3. Remove banjo bolt (B) and washers (C). Remove the fasteners and the spacers (not shown) for tube clips (F). Remove bolts (N) and remove tube assembly (M) from the cylinder block. Remove O-ring seal (L).
4. Remove bolts (H) and disconnect tube assembly (J) from the turbocharger. Remove joint (G).

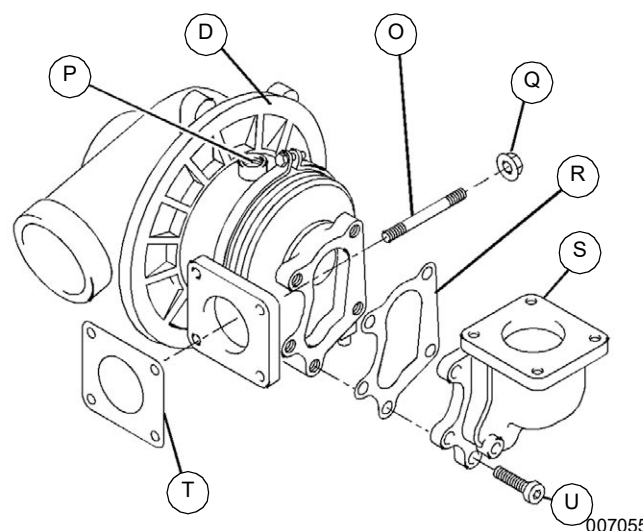
If necessary, remove bolts (I) and tube assembly (J) from the cylinder block. Remove joint (K).

5. Remove nuts (Q), turbocharger (D) and gasket (T) from the exhaust manifold. If necessary, remove studs (O) from the exhaust manifold.



007054

Figure 4-19. Remove Turbocharger



007055

Figure 4-20. Remove Turbocharger

NOTE: Do not use the actuator rod of the wastegate to lift the turbocharger.

Installation Procedure



Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. Verify the turbocharger is clean and free from damage. Inspect the turbocharger for wear. If the turbocharger is worn, the complete turbocharger must be replaced.
2. Test the actuator for correct operation. See [Test Wastegate](#) for more information. If the actuator is damaged or the actuator does not operate within the specified limits, the complete turbocharger must be replaced.
3. See [Figure 4-21](#) and [Figure 4-22](#). Clean the mating surfaces of the exhaust manifold. If necessary, install studs (O) to the exhaust manifold. Tighten the studs to 13 ft-lbs (18 Nm). Install a new gasket (T) over the studs.
4. Position turbocharger (D) onto the exhaust manifold. Install nuts (Q) and tighten to 18 ft-lbs (25 Nm).

NOTE: Do not use the actuator rod of the wastegate to lift the turbocharger.

5. Verify tube assemblies (J) and (M) are clean and free from damage. If necessary, replace the tube assemblies.
6. If necessary, position a new joint (K) and tube assembly (J) onto the cylinder block. Install bolts (I) and finger tighten.
7. Position a new joint (G) on tube assembly (J). Align tube assembly (J) to the bottom of the turbocharger. Install bolts (H). Tighten the bolts finger tight.
8. Tighten bolts (H) and (I) to 89 in-lbs (10 Nm).
9. Lubricate the bearings of turbocharger (D) with clean engine oil through oil inlet port (P). Rotate the shaft of the turbocharger to distribute the lubricant.
10. Install a new O-ring seal (L) to tube assembly (M). Position tube assembly (M) against the cylinder block. Install bolts (N). Tighten the bolts to 89 in-lbs (10 Nm).
11. Install new washers (C) and banjo bolt (B) to tube assembly (M). Position tube assembly (M) onto turbocharger (D). Tighten the banjo bolt finger tight.
12. If necessary, install the spacer and install the fasteners (not shown) to tube clips (F). Tighten fasteners to 89 in-lbs (10 Nm).
13. Tighten banjo bolt (B) to 13 ft-lbs (18 Nm).

NOTE: Verify the tube assembly does not come into contact with any other components.

14. Clean the mating surfaces of exhaust elbow (S). Position a new gasket (R) and exhaust elbow (S) onto turbocharger (D). Install Allen head screws (U). Tighten the bolts to 24 ft-lbs (32 Nm).
15. Verify inlet hose (A) is clean and free from defects. Loosely install hose clamps (E) to air inlet hose (A). Install the air inlet hose to the connection of the inlet manifold (not shown) and to the turbocharger. Tighten the hose clamps.

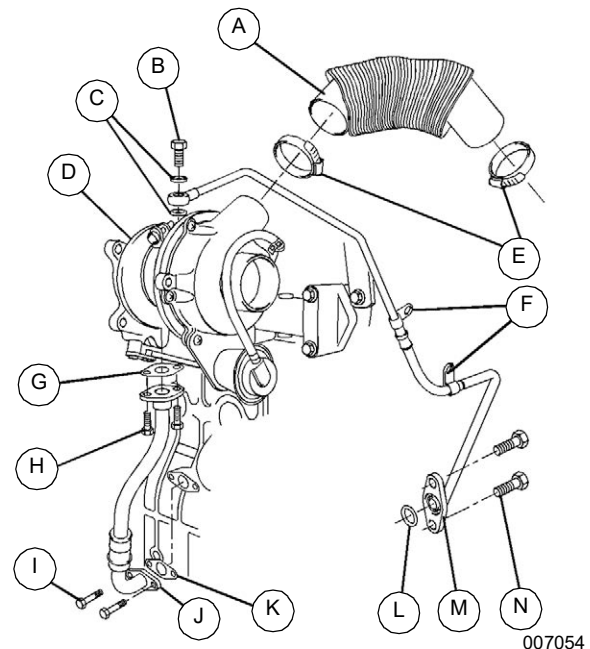


Figure 4-21. Install Turbocharger

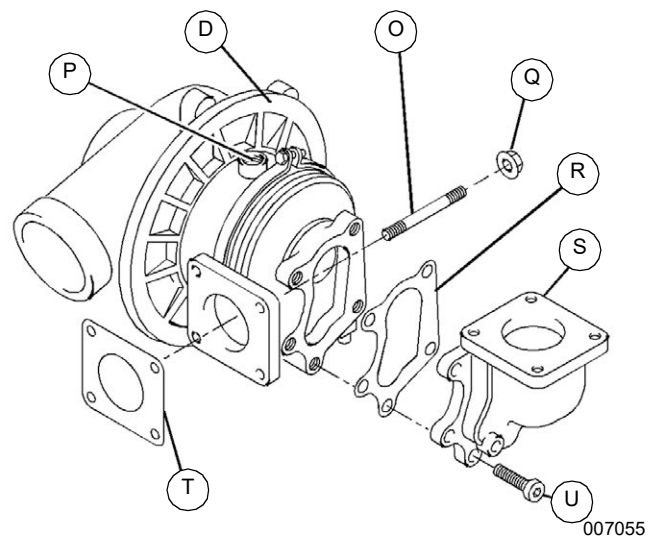


Figure 4-22. Install Turbocharger

Exhaust Manifold

Removal Procedure



Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

NOTE: Remove turbocharger, if equipped. See [Turbocharger](#).

1. See [Figure 4-23](#). Loosen nuts (E) and bolts (F).

NOTE: To prevent distortion of the exhaust manifold, loosen the outer fasteners first.

2. Remove nuts (E) and bolts (F).

NOTE: Identify bolts of different lengths so they can be installed in the correct positions.

3. Remove exhaust manifold (D) from cylinder head (A). Note the orientation of the exhaust manifold for installation.
4. Remove gasket (C) from cylinder head (A).
5. If necessary, remove exhaust manifold studs (B) from cylinder head (A).

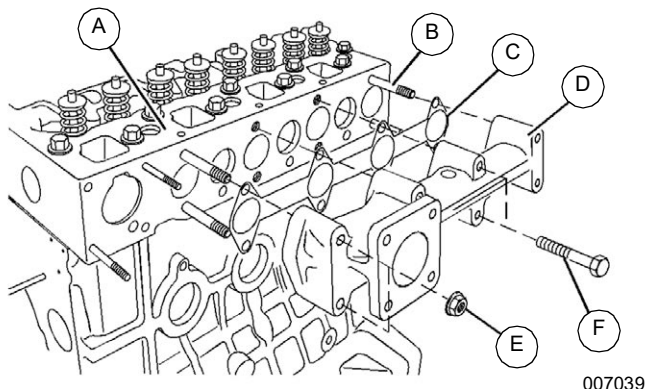


Figure 4-23. Remove Exhaust Manifold

Installation Procedure



Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. Verify the mating surfaces of the cylinder head and the exhaust manifold are clean and free from damage.

2. See [Figure 4-24](#). If necessary, install exhaust manifold studs (B) to cylinder head (A).
3. Install a new exhaust manifold gasket (C) to cylinder head (A).
4. Align exhaust manifold (D) with studs (B) and install the exhaust manifold to cylinder head (A).

NOTE: Verify the exhaust manifold is installed in the correct orientation.

5. Install nuts (E) and bolts (F) finger tight.

NOTE: Verify bolts of different lengths are installed in the correct positions.

6. For D2.2L engines, tighten nuts (E) and bolts (F) to 18 ft-lbs (25 Nm), tightening the inner bolts first.
7. For D1.5L engines, tighten nuts (E) and bolts (F) to 89 in-lbs (10 Nm), tightening the inner bolts first.

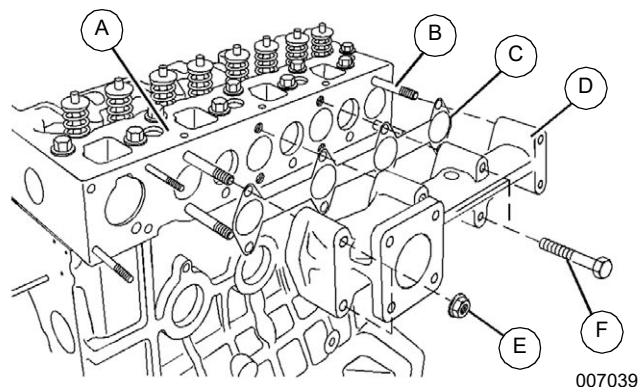


Figure 4-24. Remove Exhaust Manifold

NOTE: Install the turbocharger, if equipped. See [Turbocharger](#).

Inlet and Exhaust Valve Springs

Removal Procedure



⚠ WARNING

Vision Loss. Eye protection is required for the following procedure. Failure to wear proper eye protection could result in serious injury to the eyes. (000248)

⚠ WARNING

Personal Injury. Properly compress valve springs using a valve spring compressor. Failure to do so could cause the spring to eject, and could result in serious injury. (000193a)

⚠ CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage. (000357)

IMPORTANT NOTE: Install suitable plugs to the inlet ports of the cylinder head to prevent the entry of loose parts into the engine.

IMPORTANT NOTE: Plug the apertures for the push rods in the cylinder head to prevent the entry of loose parts into the engine.

NOTE: The following procedure should be used to remove the valve springs when the cylinder head is installed to the engine. See [Inlet and Exhaust Valves](#) for the procedure to remove the valve springs from a cylinder head that has been removed from the engine.

NOTE: Verify the appropriate piston is at the top center position before the valve spring is removed. Failure to do so could allow the valve to drop into the cylinder bore.

1. Remove the rocker shaft assembly. See [Rocker Shaft and Pushrod](#).
2. See [Figure 4-25](#). Install valve spring compressor (A) and adapter (B) in position on the cylinder head to compress a valve spring for the appropriate piston.
3. Use valve spring compressor to compress valve spring (E) and open the valve slightly.

NOTE: Do not compress the spring so the valve spring retainer (D) touches the valve stem seal.

4. Carefully rotate the crankshaft until the piston touches the valve.

NOTE: Do not use excessive force to turn the crankshaft. The use of force can result in bent valve stems.

5. Continue to rotate the crankshaft and gradually release the pressure on the valve spring compressor until the piston is at top center. The valve is now held in a position that allows the valve spring to be safely removed.

IMPORTANT NOTE: Verify the valve spring is compressed squarely.

6. Use valve spring compressor to compress valve spring (E). Remove valve keepers (C).

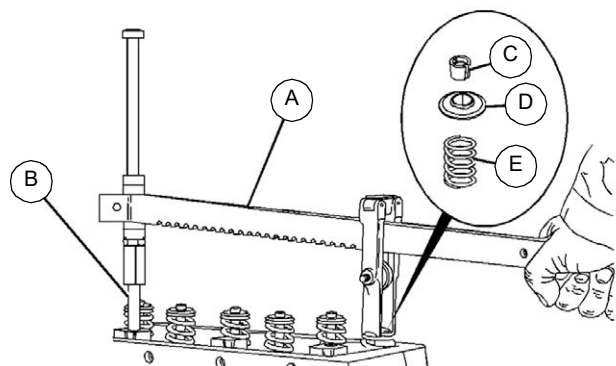
NOTE: If all valve springs require replacement the procedure can be carried out on two cylinders at the same time. The procedure can be carried out on the following pairs of cylinders: 1 with 4 and 2 with 3. Verify all the valve springs are installed before changing from one pair of cylinders to another pair of cylinders.

IMPORTANT NOTE: Do not turn the crankshaft while the valve springs are removed.

7. Apply sufficient pressure to valve spring compressor to allow removal of the valve keepers (C).

NOTE: Do not compress the spring so the valve spring retainer (D) touches the valve stem seal.

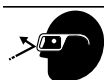
8. Remove valve keepers (C).
9. Slowly release the pressure on valve spring compressor.
10. Remove valve spring retainer (D) and remove valve spring (E).
11. Remove valve spring compressor and adapter.



006994

Figure 4-25. Remove Inlet and Exhaust Valve Springs

Installation Procedure



⚠ WARNING

Vision Loss. Eye protection is required for the following procedure. Failure to wear proper eye protection could result in serious injury to the eyes.

(000248)

⚠ WARNING

Personal Injury. Properly compress valve springs using a valve spring compressor. Failure to do so could cause the spring to eject, and could result in serious injury.

(000193a)

⚠ CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

IMPORTANT NOTE: Install suitable plugs to the inlet ports of the cylinder head to prevent the entry of loose parts into the engine.

IMPORTANT NOTE: Plug the apertures for the push rods in the cylinder head to prevent the entry of loose parts into the engine.

IMPORTANT NOTE: Do not turn the crankshaft while the valve springs are removed.

1. Inspect the valve springs for the correct length. See [Cylinder Head Valves](#).
2. See [Figure 4-26](#). Install valve spring (E) onto the cylinder head. Position valve spring retainer (D) onto valve spring (E).
3. Install valve spring compressor (A) and adapter (B) in the appropriate position on the cylinder head to compress the valve spring.
4. Apply sufficient pressure to valve spring compressor to install valve keepers (C).

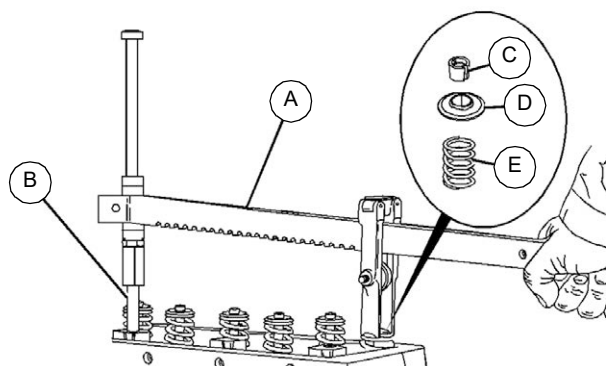
NOTE: Do not compress the spring so valve spring retainer (D) touches the valve stem seal.

5. Install the valve spring keepers.
6. Carefully release the pressure on valve spring compressor.

NOTE: Verify the valve keepers are correctly seated.

7. Remove valve spring compressor. Verify all the valves are secured in place by a valve spring and valve keepers. Rotate the crankshaft through about 45 degrees to clear the piston from the valve. Lightly strike the top of the valve with a soft hammer to verify the valve keepers are properly installed.

8. Install the rocker shaft assembly. See [Rocker Shaft and Pushrod](#).



006994

Figure 4-26. Install Inlet and Exhaust Valve Springs

Inlet and Exhaust Valves

Removal Procedure



⚠ WARNING

Vision Loss. Eye protection is required for the following procedure. Failure to wear proper eye protection could result in serious injury to the eyes.

(000248)

⚠ WARNING

Personal Injury. Properly compress valve springs using a valve spring compressor. Failure to do so could cause the spring to eject, and could result in serious injury.

(000193a)

⚠ CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. Clean the bottom face of the cylinder head. Check the depth of the valves below the face of the cylinder head before the valve springs are removed. See [Cylinder Head Valves](#) for the correct dimensions.
2. Place a temporary identification mark on the heads of the valves to identify the correct position.

NOTE: Do not stamp the heads of the valves. Stamping or punching the heads of the valves could cause the valves to fracture.

IMPORTANT NOTE: Verify the valve spring is compressed squarely.

- See [Figure 4-27](#) and [Figure 4-28](#). Use valve spring compressor (A) to compress the appropriate valve spring (D). Remove valve keepers (B).

NOTE: Do not compress the valve spring so the valve spring retainer (C) touches valve stem seal (E).

- Remove valve spring compressor (A).
- Remove valve spring retainer (C). Remove valve spring (D).
- Remove valve (F).
- Remove valve stem seal (E).
- Repeat Steps 3 to 7 for the remaining valves.

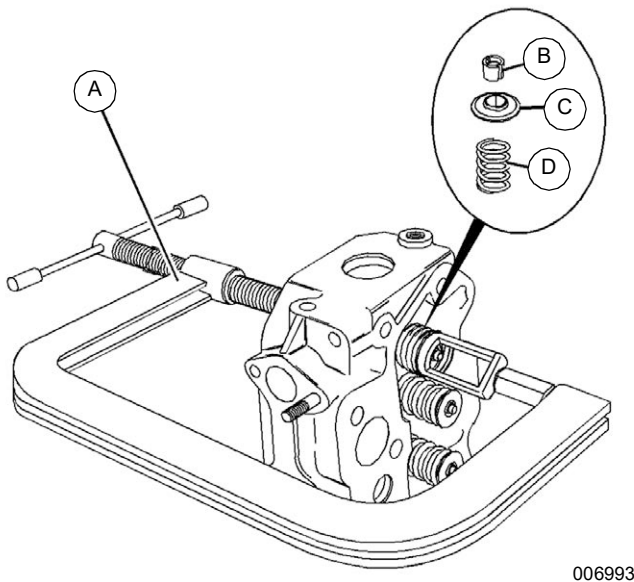


Figure 4-27. Valve Spring Compressor

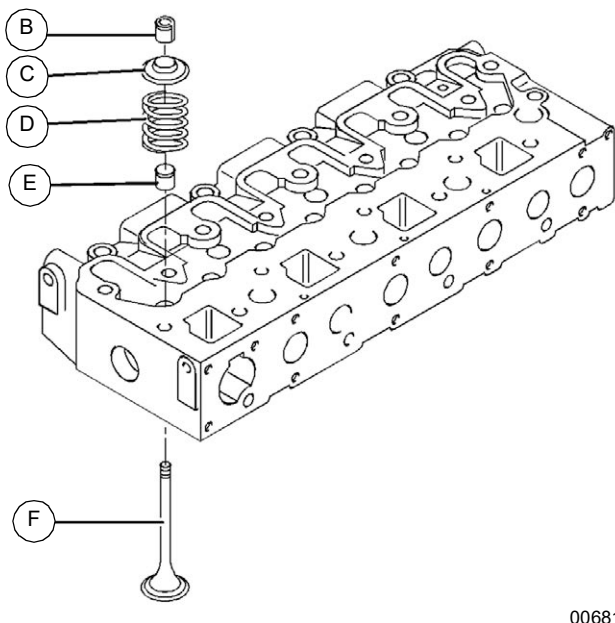


Figure 4-28. Remove Inlet and Exhaust Valves

Installation Procedure



⚠ WARNING

Vision Loss. Eye protection is required for the following procedure. Failure to wear proper eye protection could result in serious injury to the eyes.

(000248)

⚠ WARNING

Personal Injury. Properly compress valve springs using a valve spring compressor. Failure to do so could cause the spring to eject, and could result in serious injury.

(000193a)

⚠ CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

- Clean all components of the cylinder head assembly. Verify all ports, coolant passages and lubrication passages in the cylinder head are free from debris. Follow Steps 2 through 6 to inspect the components of the cylinder head assembly. Replace any components that are worn or damaged.
- Inspect the cylinder head for wear and damage. See [Inspect Cylinder Head](#).
- Inspect the valve seats for wear and damage. See [Cylinder Head Valves](#) for more information.
- Inspect the valve guides for wear and damage. See [Cylinder Head Valves](#) and [Inspect Valve Guide](#).
- Inspect the valves for wear and damage. See [Cylinder Head Valves](#).
- Inspect the valve springs for the correct length. See [Cylinder Head Valves](#).
- See [Figure 4-29](#), [Figure 4-30](#) and [Figure 4-31](#). Use valve stem seal replacer (G) to install new valve stem seals (E) onto each of the valve guides.

NOTE: The outer face of the valve guides must be clean and dry before installing the valve stem seals.

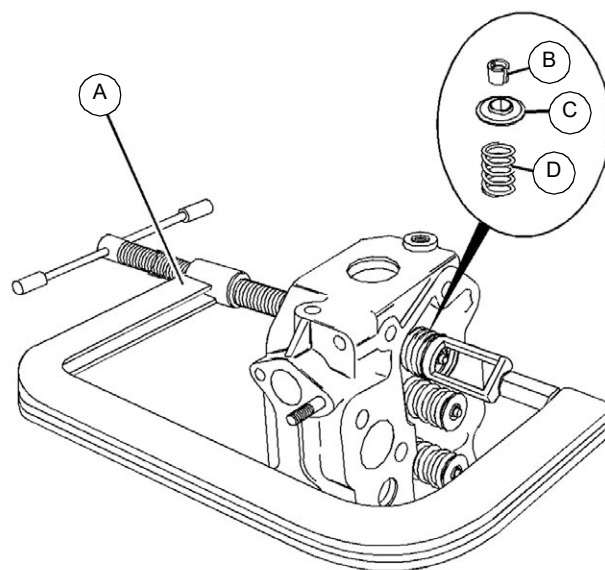
- Lubricate the stem of valve (F) with clean engine oil. Install valve (F) in the appropriate position in the cylinder head. Check the depth of the valve below the face of the cylinder head. See [Inspect Valve Depth](#).
- Install valve spring (D) to the cylinder head. Position valve spring retainer (C) onto valve spring (D).

IMPORTANT NOTE: Verify the valve spring is compressed squarely.

10. Use valve spring compressor (A) to compress valve spring (D). Install valve keepers (B).

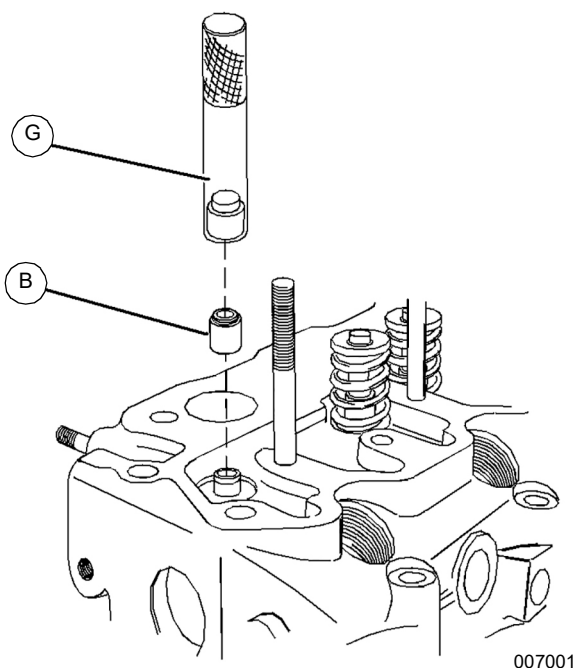
NOTE: Do not compress the spring so valve spring retainer (C) touches valve stem seal (E).

11. Remove valve spring compressor (A).
12. Repeat Steps 9 through 11 for the remaining valves.
13. Place the cylinder head on a suitable support.
14. Verify the heads of the valves are not obstructed. Gently strike the top of the valves with a soft hammer to ensure valve keepers (B) are properly installed.
15. Install the cylinder head. See [Cylinder Head](#).



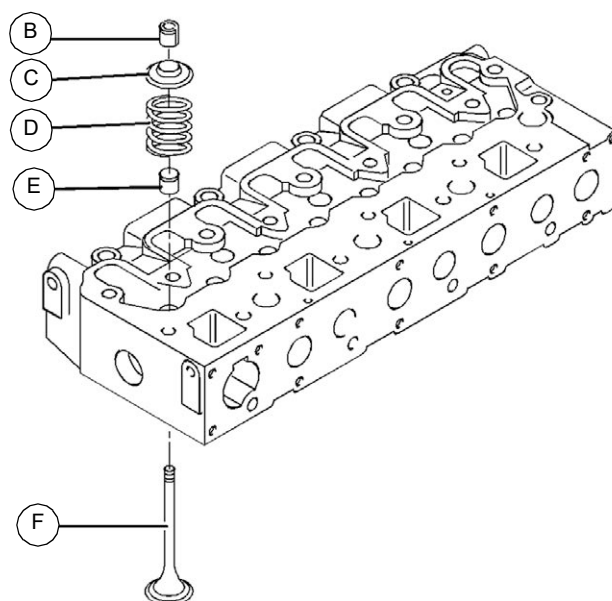
006993

Figure 4-30. Valve Spring Compressor



007001

Figure 4-29. Install New Valve Stem Seals



006815

Figure 4-31. Remove Inlet and Exhaust Valves

Engine Oil Line

Removal Procedure

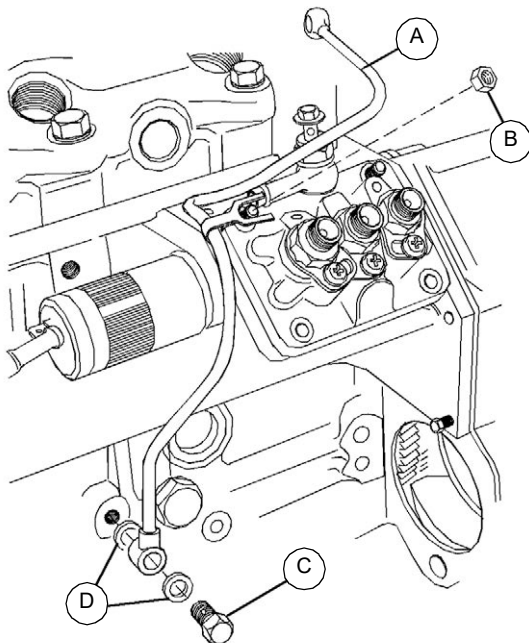
⚠ CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

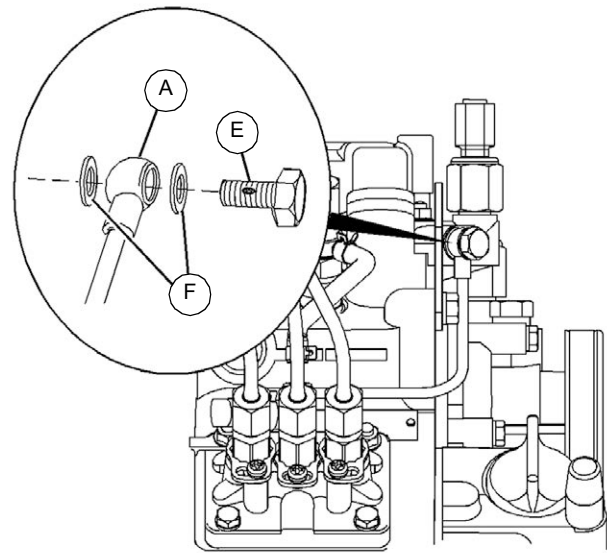
IMPORTANT NOTE: Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids. Dispose of all fluids according to local regulations and mandates.

1. See [Figure 4-32](#) and [Figure 4-33](#). Loosen nut (B) that attaches the clip on oil line (A) to the fuel injection pump.
2. Remove banjo bolt (C) and washers (D) from the cylinder block.
3. Remove banjo bolt (E) and washers (F) from the cylinder head.
4. Remove oil line (A) from the engine.



006920

Figure 4-32. Remove Engine Oil Line



006919

Figure 4-33. Remove Engine Oil Line

Installation Procedure

⚠ CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. See [Figure 4-34](#) and [Figure 4-35](#). Place oil line (A) on the engine. Verify the clip on the oil line is located below nut (B).
2. Position banjo bolt (C) and new washers (D) onto oil line (A). Install the banjo bolt and oil line to the cylinder block finger tight.
3. Position banjo bolt (E) and new washers (F) onto oil line (A). Install the banjo bolt and the oil line to the cylinder head finger tight.
4. Tighten nut (B) that attaches the clip on oil line (A) to the fuel injection pump to 133 **in-lbs** (15 Nm).
5. Tighten banjo bolts (C) and (E) to 106 **in-lbs** (12 Nm).

Engine Air Cooler

Removal Procedure

CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

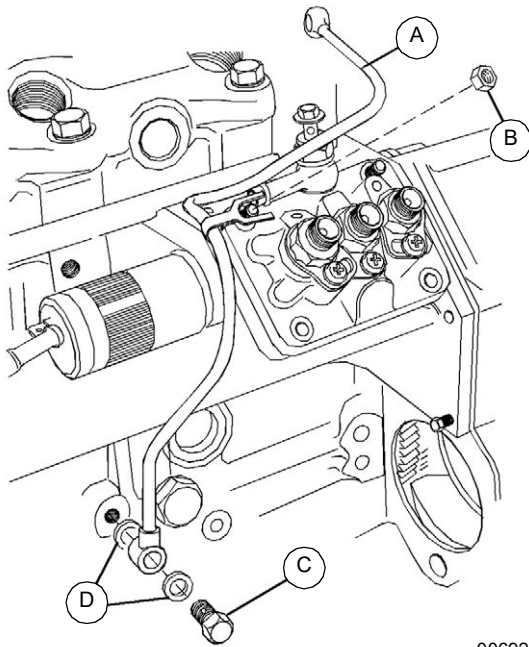
(000357)

IMPORTANT NOTE: Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids. Dispose of all fluids according to local regulations and mandates.

1. Drain the coolant into a suitable container. See [Change Coolant](#) or [Check Cooling System Coolant Level](#) for the correct procedure.
2. Drain the engine lubricating oil into a suitable container. See [Drain Engine Oil](#).
3. See [Figure 4-36](#). Use a strap wrench to remove oil filter element (E).
4. Loosen hose clamps (B) and disconnect hoses (A). Note the positions of the coolant inlet and the coolant outlet for installation.
5. Remove adapter (D) and remove oil cooler (C) from the cylinder block.

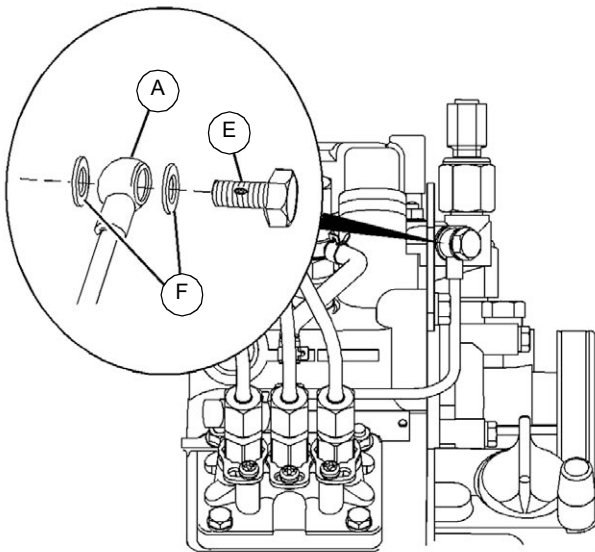
NOTE: Make a temporary mark to show the orientation of the oil cooler for installation.

6. Remove O-ring seal (F) from oil cooler (C).



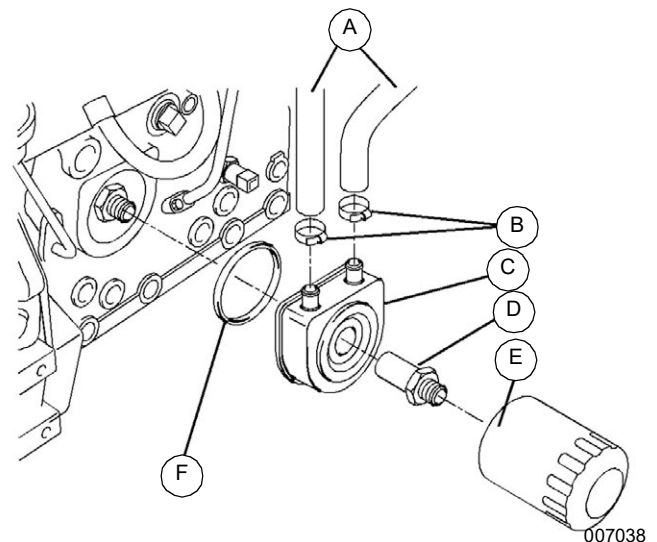
006920

Figure 4-34. Install Engine Oil Line



006919

Figure 4-35. Install Engine Oil Line



007038

Figure 4-36. Remove Engine Oil Cooler

Installation Procedure

⚠ CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. Verify oil cooler is clean and free from damage. Clean the mating surfaces of the cylinder block.
2. See [Figure 4-37](#). Install a new O-ring seal (F) to oil cooler (C). Position oil cooler (C) on the cylinder block and install adapter (D). Tighten adapter (D) to 25 ft-lbs (34 Nm).

NOTE: Verify oil cooler is correctly oriented.

3. Connect hoses (A) to the coolant inlet and the coolant outlet on oil cooler (C). Tighten hose clamps (B).
4. Install a new oil filter element (E).
5. Fill the cooling system to the correct level. See [Change Coolant](#) or [Check Cooling System Coolant Level](#) for the correct procedure.

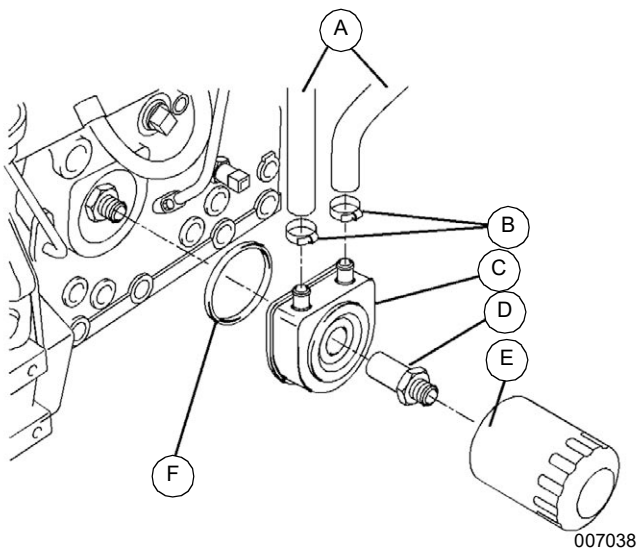


Figure 4-37. Install Engine Oil Cooler

6. Fill the engine oil pan to the correct level. See [Replace Oil Filter](#) for the correct procedure.

Engine Oil Relief Valve

Removal Procedure

⚠ CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

IMPORTANT NOTE: Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids. Dispose of all fluids according to local regulations and mandates.

1. See [Figure 4-38](#). Remove engine oil relief valve (B) from the cylinder block.
2. Remove O-ring seal (A) from engine oil relief valve (B).

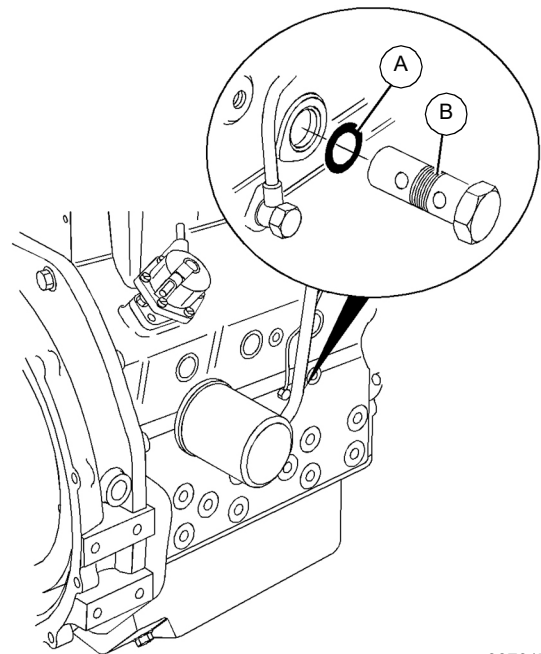


Figure 4-38. Remove Engine Oil Relief Valve

Installation Procedure

CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. See **Figure 4-39**. Install a new O-ring seal (A) to engine oil relief valve (B).
2. Lubricate engine oil relief valve (B) with clean engine oil.
3. Install engine oil relief valve (B) into the cylinder block. Tighten the engine oil relief valve to 47 ft-lbs (64 Nm).

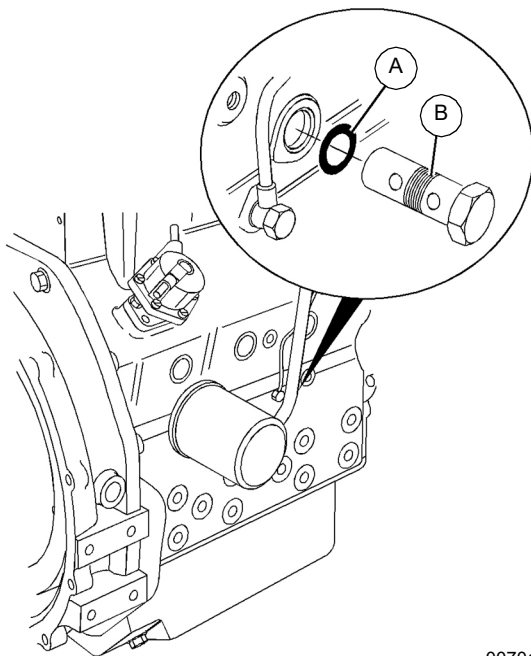


Figure 4-39. Install Engine Oil Relief Valve

Engine Oil Pump

Removal Procedure



WARNING

Vision Loss. Eye protection is required for the following procedure. Failure to wear proper eye protection could result in serious injury to the eyes.

(000248)

WARNING

Personal Injury. Always wear all necessary protective equipment and use recommended tools while performing the following procedure. Failure to do so could result in serious injury.

(000556)

CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

IMPORTANT NOTE: Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids. Dispose of all fluids according to local regulations and mandates.

IMPORTANT NOTE: Verify the front housing is installed before turning the crankshaft.

1. See **Figure 4-40**, **Figure 4-41** and **Figure 4-42**. Remove C-clip (I) that retains idler gear (C) onto idler hub (A).
2. Remove the following components from idler hub: (A) collar (H), spring (G), shim (F), and oil pump cover (E).
3. Remove idler gear (C) from idler hub (A).
4. Remove inner rotor (D) from idler hub (A).
5. Remove thrust washer (B) from idler hub (A).
6. Use hub removal tool (A0001151334 for D1.5L engines or 10000034090 for 2.2L engines) and slide hammer puller (10000034146) to remove idler hub (A).

NOTE: If necessary, follow Step 7 through Step 10 to remove oil strainer (M) and suction pipe (K).

7. Remove the engine oil pan. See **Engine Oil Pan** for the correct procedure.
8. Remove bolts (L) and oil strainer (M) from the cylinder block.

9. Remove suction pipe (K) from the cylinder block.
10. Remove O-ring seal (J) from the suction pipe.

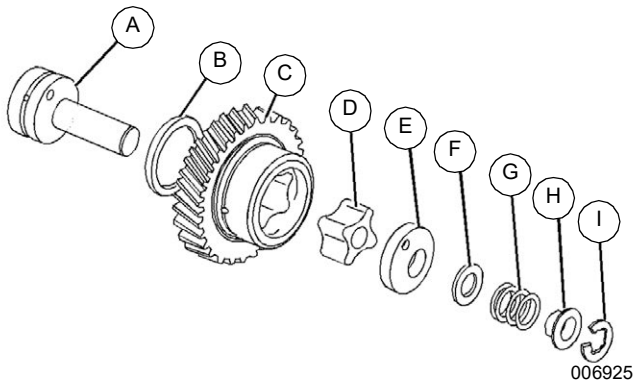


Figure 4-40. Remove Engine Oil Pump

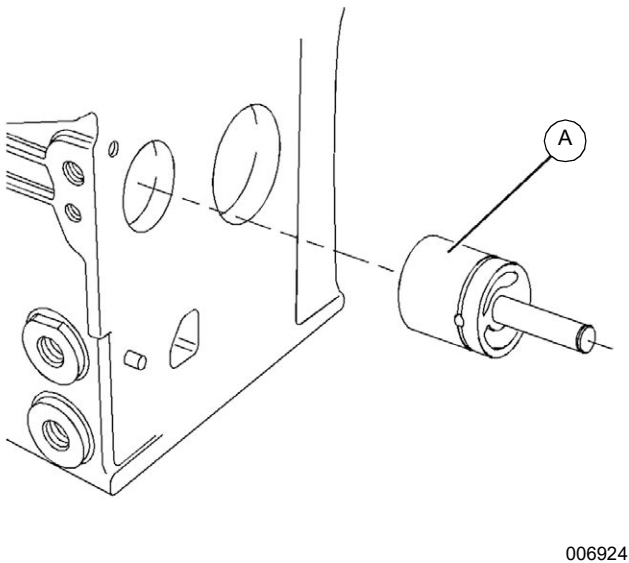


Figure 4-41. Remove Engine Oil Pump

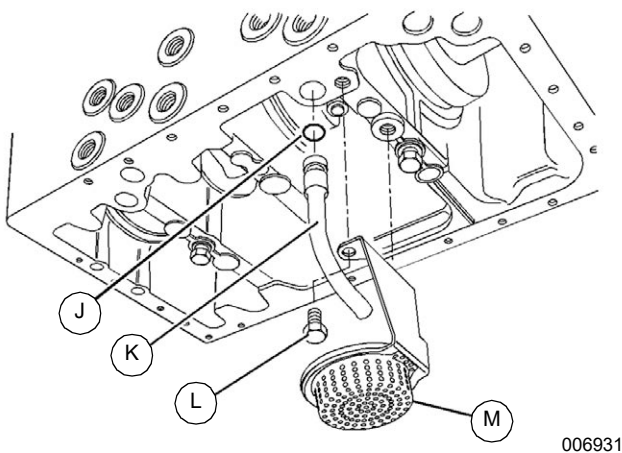


Figure 4-42. Remove Engine Oil Pump

Installation Procedure

CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

IMPORTANT NOTE: Verify the front housing is installed before turning the crankshaft.

Idler Hub

1. See [Figure 4-43](#) and [Figure 4-44](#). Install the pin of alignment tool (A) to the cylinder block.
2. Install idler hub (B) into the guide plate of alignment tool (A). Align the guide plate of alignment tool (A) with pin of alignment tool (A).
3. Use a hammer to strike the guide plate of alignment tool (A). Drive idler hub (B) into the cylinder block until the guide plate contacts the front of the cylinder block.
4. Remove alignment tool (A).
5. Install crankshaft (R). See [Crankshaft](#).
6. Position a new joint (O) on the cylinder block. Align plate (N) with the dowels in the cylinder block and install the plate. Install bolts (Q) and tighten to 89 in-lbs (10 Nm).
7. Install camshaft (R). See [Camshaft](#).

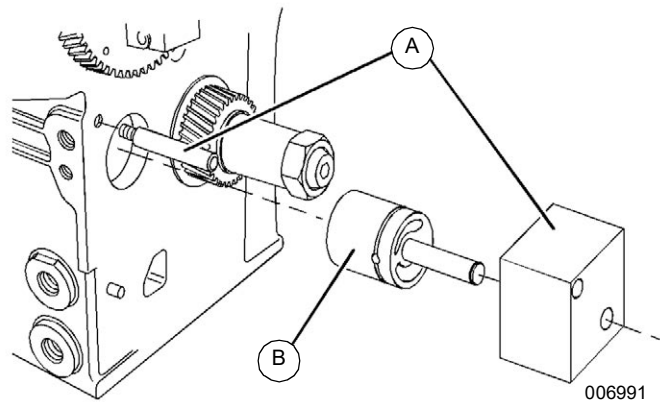
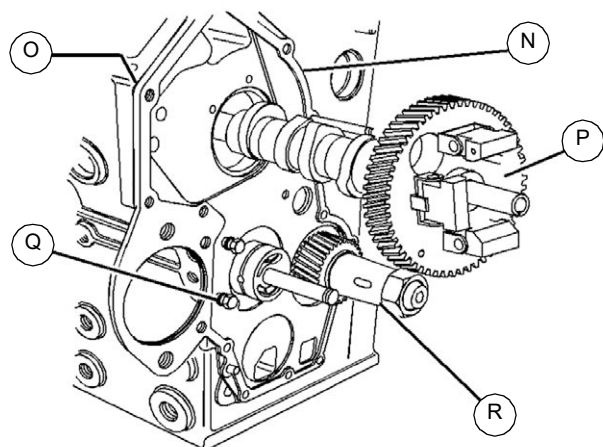


Figure 4-43. Install Idler Hub

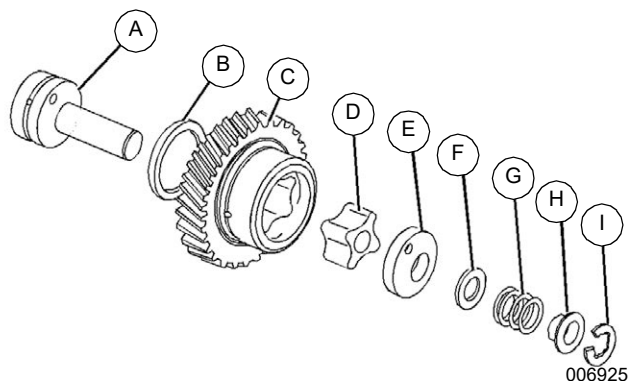


006992

Figure 4-44. Install Idler Hub

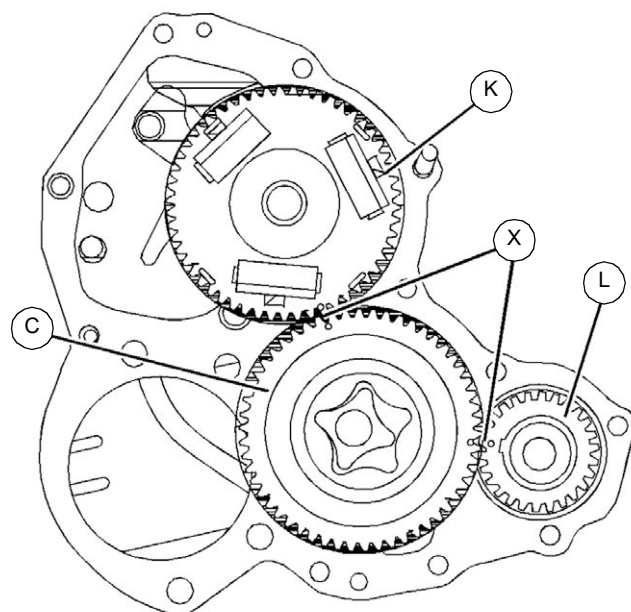
Engine Oil Pump

1. See [Figure 4-45](#), [Figure 4-46](#) and [Figure 4-47](#). Install thrust washer (B). Lubricate the thrust washer with clean engine oil.
2. Apply multipurpose grease to the faces of inner rotor (D) and to the vanes of idler gear (C).
3. Align timing marks (X) on idler gear (C) with the respective timing marks on gears (K) and (L). Install idler gear (C) onto idler hub (A).
4. Install inner rotor (D) to idler gear (C).
5. Install the following items to idler hub (A):
 - Oil pump cover (E)
 - Shim (F)
 - Spring (G)
 - Collar (H)
6. Install retaining ring (I) onto idler hub (A).
7. Use a feeler gauge (J) to measure the end play of the engine oil pump. See [Engine Oil Pump](#).



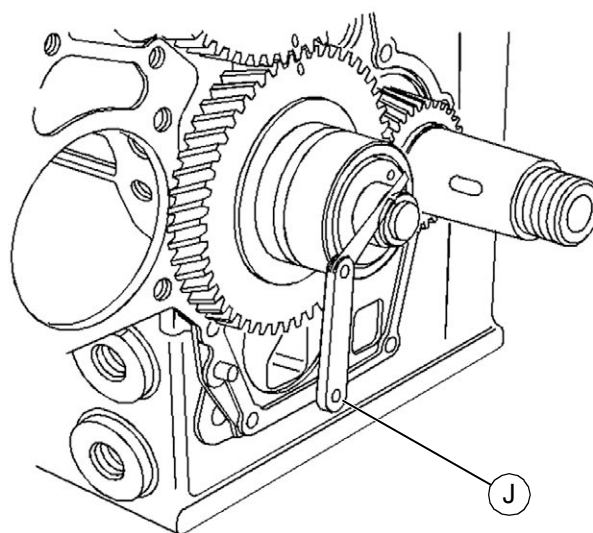
006925

Figure 4-45. Install Engine Oil Pump



006926

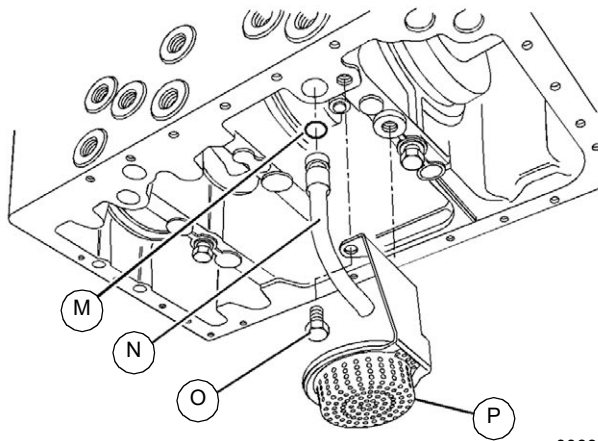
Figure 4-46. Install Engine Oil Pump



007279

Figure 4-47. Install Engine Oil Pump

8. If the oil strainer was removed, follow Steps 9 through 11 to install the oil strainer.
9. See [Figure 4-48](#). Install a new O-ring seal (M) on oil tube assembly (N).
10. Install oil suction pipe (N) in the cylinder block.
11. Install oil strainer (P) to the cylinder block and tighten bolts (O) to 97 in-lbs (11 Nm).



006931

Figure 4-48. Install Engine Oil Pump

12. Install the front housing. See [Housing \(Front\)](#).
13. Install the engine oil pan. See [Engine Oil Pan](#).

Water Pump

⚠ CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

Removal Procedure

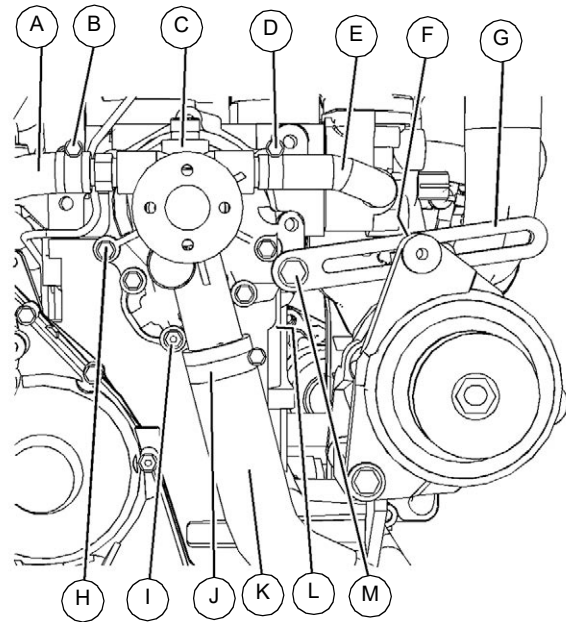
1. Remove the fan. See [Fan](#).
2. If necessary, remove the alternator. See [Alternator](#).
3. Drain the coolant into a suitable container for storage or disposal. See [Change Coolant](#) or [Check Cooling System Coolant Level](#).
4. See [Figure 4-49](#). Loosen hose clamp (J) and disconnect hose (K) from water pump (C).
5. Loosen hose clamp (B) and disconnect hose (A) from water pump (C).
6. Loosen hose clamps (D) and disconnect hose (E) from water pump (C) and the thermostat housing.
7. Remove bolt (M) for alternator bracket (G). Position the alternator bracket away from water pump (C).

NOTE: The alternator bracket on some engines is secured by two bolts.

8. Remove bolts (H) and nut (I) from water pump (A). Remove the water pump from the cylinder block.

NOTE: If necessary, gently tap the water pump with a soft faced hammer to loosen the water pump.

9. Remove gasket (L) (not shown).



007101

Figure 4-49. Remove Water Pump

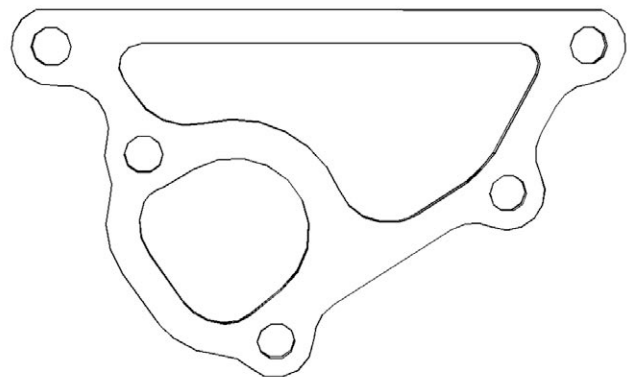
Installation Procedure

⚠ CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. Verify the gasket surfaces of the water pump and the cylinder block are clean and free from damage.
2. See [Figure 4-50](#), [Figure 4-51](#) and [Figure 4-52](#). Verify the bolt holes in position (N) and stud (O) are clean and free from oil and coolant.
3. Install a new gasket (L) (not shown) over stud (O).
4. Verify the gasket is correctly oriented.



007002

Figure 4-50. Orientation of Water Pump to Block Gasket

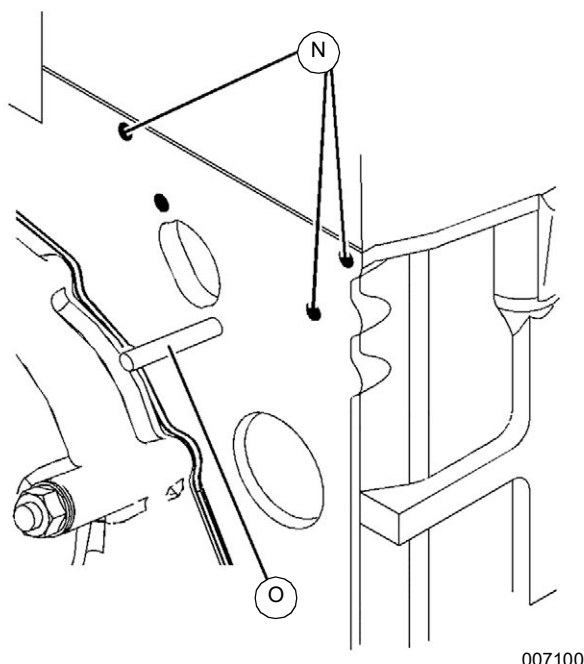


Figure 4-51. Install Water Pump

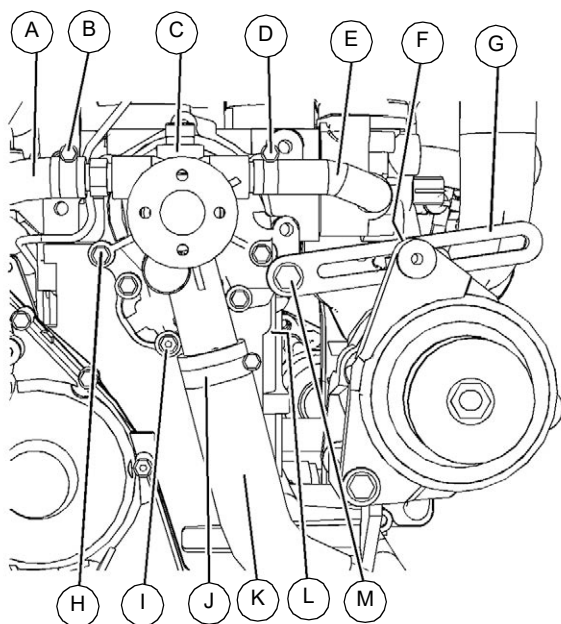


Figure 4-52. Install Water Pump

5. See [Figure 4-52](#) and [Figure 4-53](#). Apply Loctite™ 242 to the first two threads of bolts (H) and nut (I).
6. Install water pump (C).
7. Install bolts (H) and nut (I) hand tight. Verify gasket (L) (not shown) is correctly positioned.
8. Tighten bolts (H) and nut (I) to 44 **in-lbs** (5 Nm) in the sequence shown in [Figure 4-53](#).

9. Tighten bolts (H) and nut (I) to 89 **in-lbs** (10 Nm) in the sequence shown in [Figure 4-53](#).
10. Connect hose (A) to water pump (C). Securely tighten hose clamp (B).
11. Connect hose (E) to water pump (C). Securely tighten hose clamp (D).
12. Connect hose (K) to water pump (C). Securely tighten hose clamp (J).
13. Position alternator bracket (G) onto water pump (C). Loosely install bolt (M) to the alternator bracket.

NOTE: The alternator bracket on some engines is secured by two bolts.

14. Fill the cooling system with coolant. See [Change Coolant](#) or [Check Cooling System Coolant Level](#) for the correct procedure.

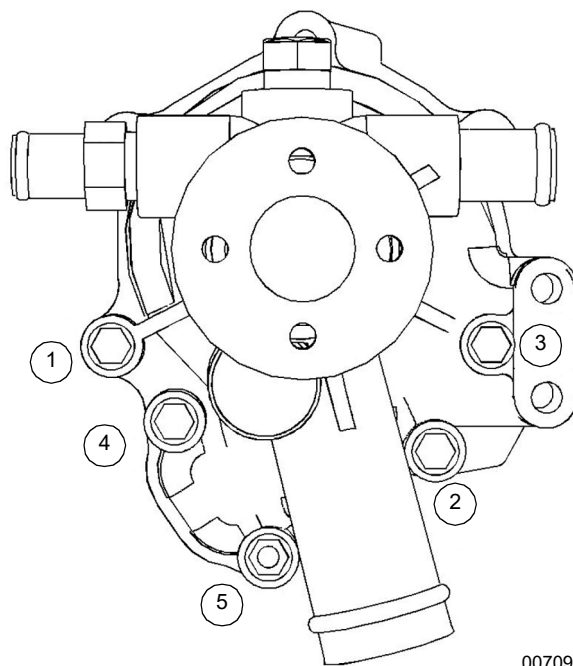


Figure 4-53. Install Water Pump

15. If necessary, install the alternator. See [Alternator](#) for the correct procedure.
16. Install the fan. See [Fan](#) for the correct procedure.

Type Two Water Pump

Removal Procedure



Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

IMPORTANT NOTE: Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids. Dispose of all fluids according to local regulations and mandates.

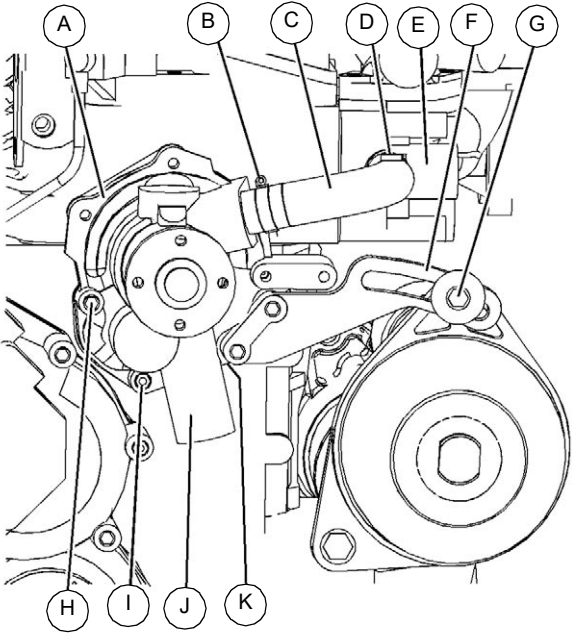
1. Remove the fan. See [Fan](#) for the correct procedure.
2. Drain the coolant from the cooling system into a suitable container for storage or disposal. See [Change Coolant](#) or [Check Cooling System Coolant Level](#).
3. Loosen the hose clamp and disconnect the coolant hose assembly from inlet connection (J) on water pump (A).
4. Loosen hose clamp (B) and hose clamp (D) for hose assembly (C). Disconnect the hose assembly from water pump (A) and water temperature regulator (E).
5. Remove bolt and washer (G) from adjusting bracket (F).
6. Remove bolts (H) and nut (I) from water pump (A). Support the water pump as the hardware is removed.

NOTE: Note position of different length bolts for installation purposes.

7. Remove water pump (A) and adjusting bracket (F) from the cylinder block.

NOTE: If necessary, gently tap the water pump with a soft faced hammer to loosen the water pump.

8. Remove gasket (K) (not shown).



007056

Figure 4-54. Remove Water Pump

Installation Procedure

Part Description	Qty.
Loctite™ 242	1

1. Verify the gasket surfaces of the water pump and the cylinder block are clean and free from damage.
2. Verify all components of the water pump are free from wear or damage. Replace any components that are worn or damaged.
3. See [Figure 4-55](#). Verify the bolt holes in position (N) and stud (M) are clean and free from oil and coolant.

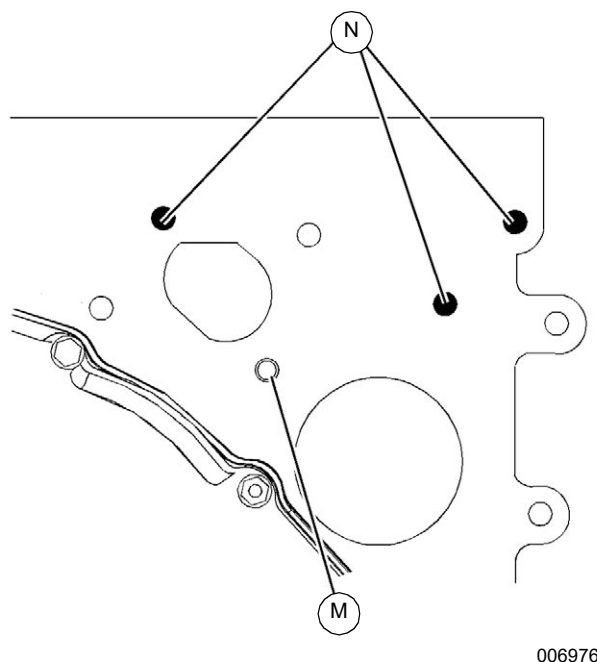


Figure 4-55. Water Pump Gasket

4. Install a new gasket (not shown) over stud. Verify the gasket is correctly oriented, as shown in [Figure 4-56](#).

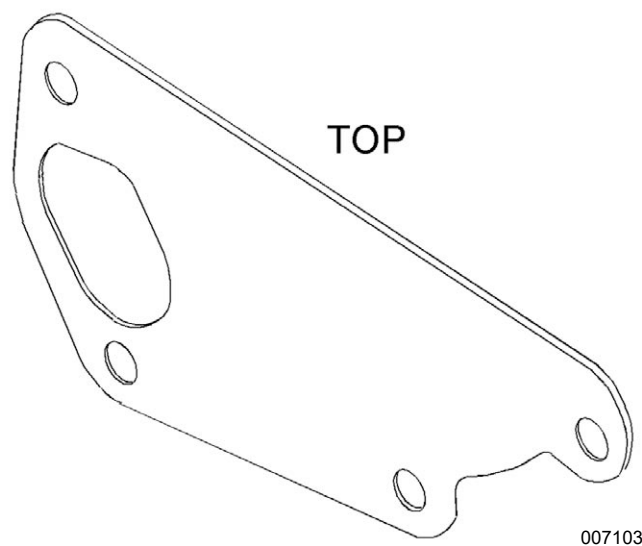


Figure 4-56. Water Pump Gasket Orientation

5. See [Figure 4-57](#) and [Figure 4-58](#). Apply Loctite™ 242 to the first two threads of bolts (H) and nut (I).
6. Install water pump (A) and adjusting bracket (F).
7. Install bolts (H) and nut (I) hand tight. Verify the gaskets are correctly positioned and the different length bolts are installed into the correct position.
8. Tighten bolts (H) and nut (I) to 44 **in-lbs** (5 Nm) in the sequence shown in [Figure 4-58](#).

9. Tighten bolts (H) and nut (I) to 89 **in-lbs** (10 Nm) in the sequence shown in [Figure 4-58](#).
10. Connect the coolant hose assembly to inlet connection (J) on water pump (A). Securely tighten the hose clamp.
11. Connect hose assembly (C) to water pump (A) and water temperature regulator (E). Securely tighten hose clamp (B) and hose clamp (D).
12. Install bolt and washer (G) to adjusting bracket (F).

NOTE: The bolt will be tightened after the alternator belt has been installed.

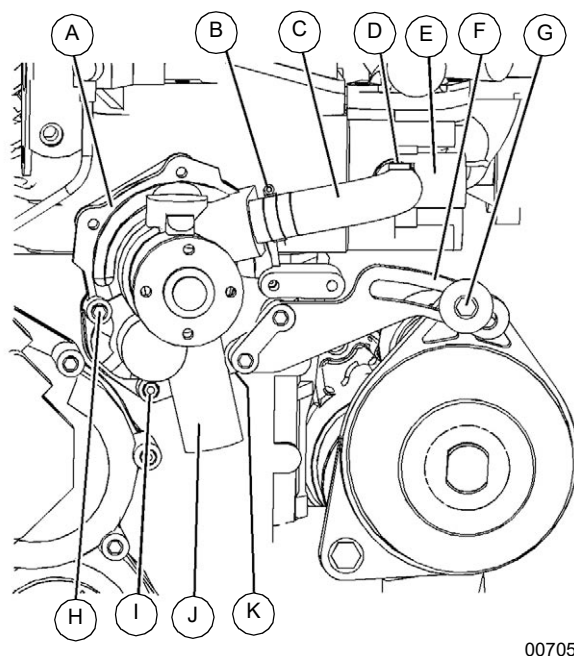


Figure 4-57. Install Water Pump

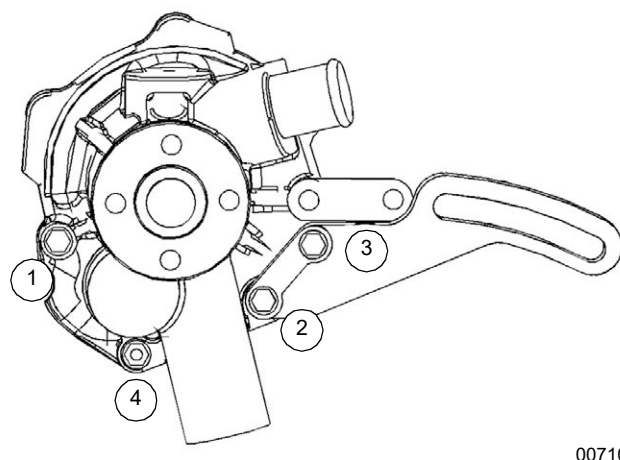


Figure 4-58. Install Water Pump

13. Install the fan. See [Fan](#) for the correct procedure.

14. Fill the cooling system with coolant. See [Change Coolant](#) or [Check Cooling System Coolant Level](#).

Water Temperature Regulator Housing

Removal Procedure

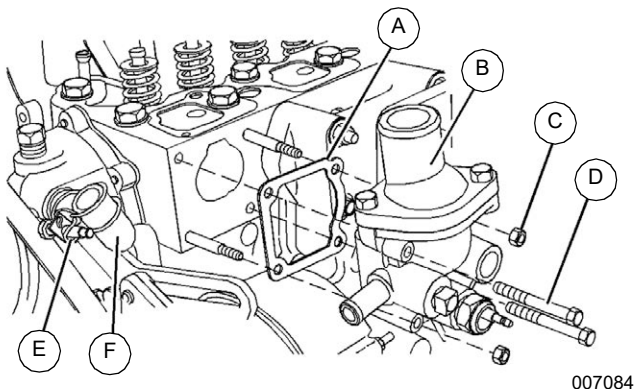


Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

IMPORTANT NOTE: Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids. Dispose of all fluids according to local regulations and mandates.

1. Drain the coolant from the cooling system into a suitable container for storage or disposal. See [Change Coolant](#) or [Check Cooling System Coolant Level](#).
2. See [Figure 4-59](#). Loosen hose clamps (E) and disconnect hose (F) from water temperature regulator housing (B).
3. Remove bolts (D) and nuts (C).
4. Remove water temperature regulator housing (B) from the cylinder head.
5. Remove joint (A) from the cylinder head.
6. If necessary, remove the water temperature regulator. See [Water Temperature Regulator](#).



007084

Figure 4-59. Remove Water Temperature Regulator Housing

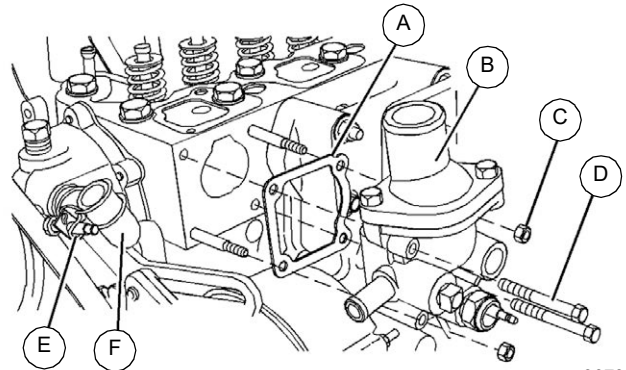
Installation Procedure



Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. Verify the mating surfaces of the cylinder head and the water temperature regulator housing are clean and free from damage.
2. See [Figure 4-60](#). Install a new joint (A) to the cylinder head.
3. Install water temperature regulator housing (B) to the cylinder head.
4. Install bolts (D) and nuts (C). Tighten the fasteners to 89 in-lbs (10 Nm).
5. Connect hose (F) to water temperature regulator housing (B). Tighten hose clamps (E).
6. If necessary, install the water temperature regulator. See [Water Temperature Regulator](#).



007084

Figure 4-60. Install Water Temperature Regulator Housing

7. Fill the cooling system with coolant. See [Change Coolant](#) or [Check Cooling System Coolant Level](#).

Water Temperature Regulator

Removal Procedure



Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

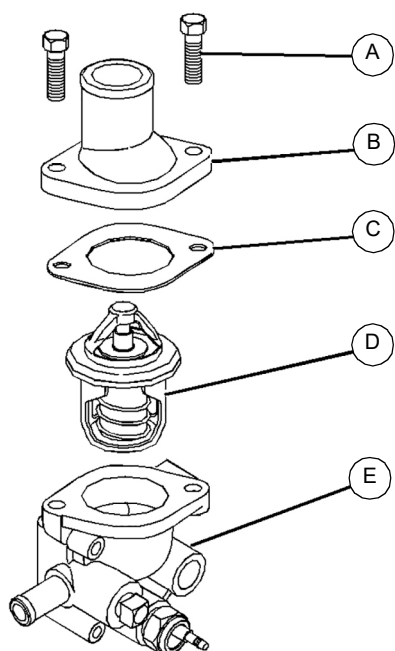
IMPORTANT NOTE: Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid

with suitable containers before opening any compartment or disassembling any component containing fluids. Dispose of all fluids according to local regulations and mandates.

1. Drain the coolant from the cooling system into a suitable container for storage or disposal. See [Change Coolant](#) or [Check Cooling System Coolant Level](#).
2. See [Figure 4-61](#). Remove fasteners (A).
3. Remove outlet connection (B).

NOTE: Identify the orientation of the outlet connection for installation.

4. Remove joint (C).
5. Remove water temperature regulator (D) from water temperature regulator housing (E).



007106

Figure 4-61. Remove Water Temperature Regulator

Installation Procedure



Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

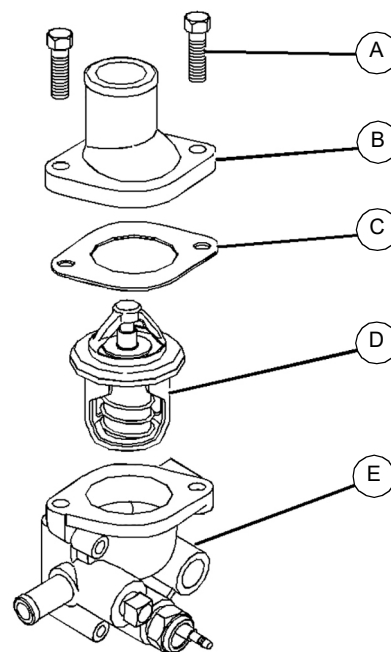
(000357)

1. Inspect the water temperature regulator for wear, damage, and correct operation. See [Test Water Temperature Regulator](#). If necessary, replace the water temperature regulator.
2. Verify the mating surfaces of the outlet connection and the water temperature regulator housing are clean and free from damage.

3. See [Figure 4-62](#). Install water temperature regulator (D) to water temperature regulator housing (E). See [Water Temperature Regulator Housing](#) for the correct orientation of the water temperature regulator.
4. Install a new joint (C).
5. Install outlet connection (B) to water temperature regulator housing (E).

NOTE: Verify the correct orientation of the outlet connection.

6. Install fasteners (A) and tighten to 124 in-lbs (14 Nm).



007106

Figure 4-62. Remove Water Temperature Regulator

7. Fill the cooling system with coolant to the correct level. See [Change Coolant](#) or [Check Cooling System Coolant Level](#).

Flywheel

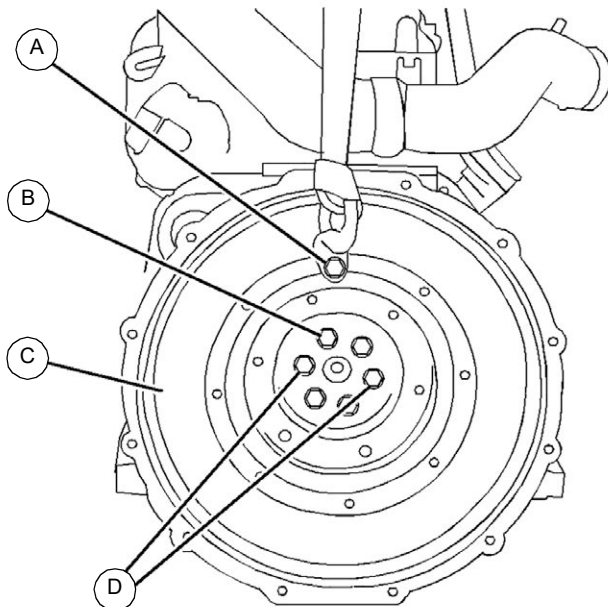
Removal Procedure

⚠ CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

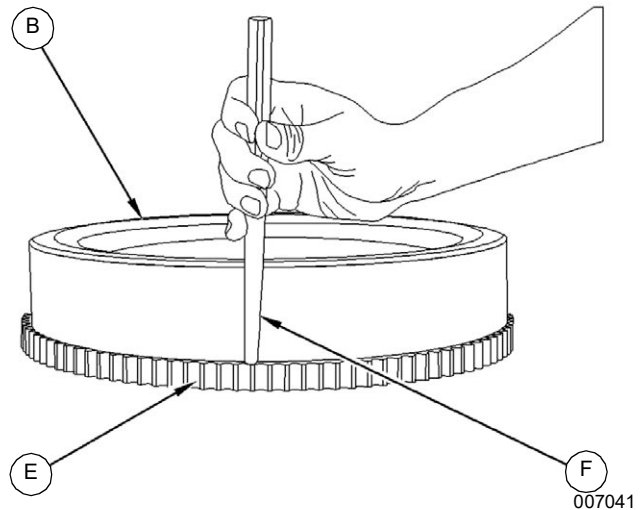
(000357)

1. Remove the electric starting motor. See [Electric Starting Motor](#).
2. Remove the GEN-END/AC alternator.
3. See [Figure 4-63](#) and [Figure 4-64](#). Install lifting bracket (A) and a suitable lifting device to flywheel (C). The weight of the flywheel is approximately 93 lbs (42 kg).
4. Remove two bolts (B) from positions (C) on the flywheel.
5. Install guide stud (B) in positions (D) on the flywheel.
6. Remove the remaining bolts (B) and remove flywheel (C).
7. Inspect ring gear (E) and flywheel (B) for wear or damage. If the ring gear or the flywheel is worn or damaged, use new parts for replacement.
8. To remove ring gear (E) from flywheel (B), place the flywheel on a suitable support. Use a hammer (not shown) and a punch (F) to remove the ring gear from the flywheel.



007040

Figure 4-63. Remove Flywheel



007041

Figure 4-64. Remove Flywheel

Installation Procedure



⚠ WARNING

Risk of Burn. Use protective gloves when handling hot parts. Failure to do so could result in serious injury.

(000386)

⚠ CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. See [Figure 4-65](#) and [Figure 4-66](#). If the ring gear was removed, follow Steps 2 through 4 to install ring gear (E) to flywheel (C).
2. Identify the orientation of the new ring gear to install the ring gear correctly onto the flywheel.

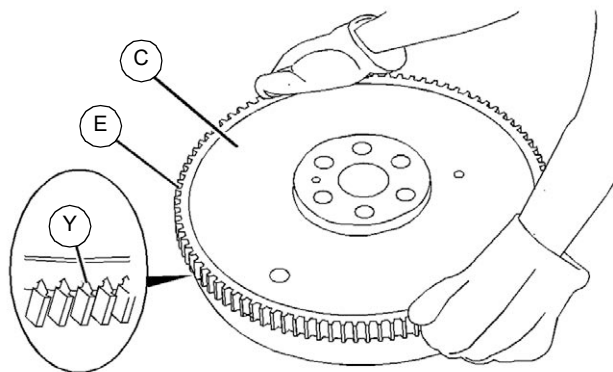
NOTE: The chamfered side of gear teeth (Y) must face toward the starting motor when the flywheel is installed. This will ensure correct engagement of the starting motor.

3. Heat ring gear (E) in an oven to a maximum temperature of 302 °F (150 °C) prior to installation.

NOTE: Do not use a torch to heat the ring gear.

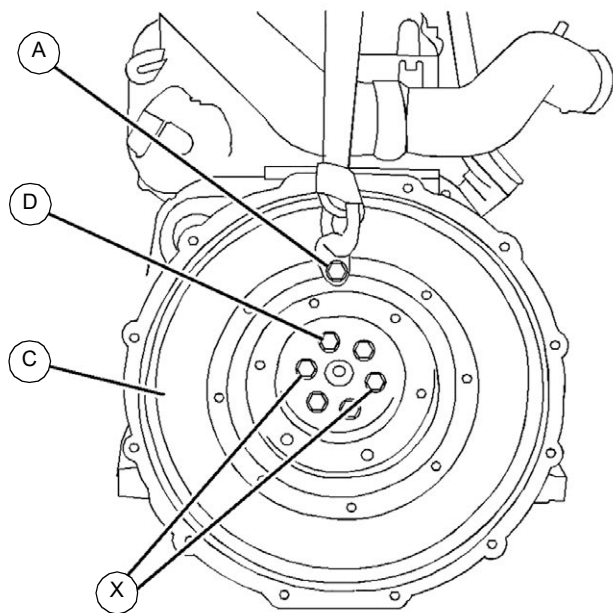
4. Verify the orientation of ring gear (E) is correct and quickly install the ring gear onto flywheel (C).
5. Thoroughly clean the flywheel housing. Inspect the crankshaft rear seal for leaks. If there are any oil leaks replace the crankshaft rear seal. [Crankshaft Rear Seal](#).
6. Install guide stud (B) to positions (X) in the crankshaft.

7. Install guide stud (A) and a suitable lifting device to flywheel (C). The weight of the flywheel is approximately 93 lbs (42 kg).
8. Position flywheel (C) onto guide stud (B). Align the hole in the flywheel with the roll pin in the crankshaft.
9. Install bolts (D) finger tight.
10. Remove guide stud (B) and install the two remaining bolts (D).
11. Use a suitable tool to prevent flywheel (C) from rotating. Tighten bolts (D) to 54 ft-lbs (74 Nm).



006997

Figure 4-65. Install Flywheel



007040

Figure 4-66. Install Flywheel

12. Remove lifting bracket (A) and the lifting device from the flywheel.
13. Install the electric starting motor. See [Electric Starting Motor](#).
14. Install the alternator.

Crankshaft Rear Seal

Removal Procedure

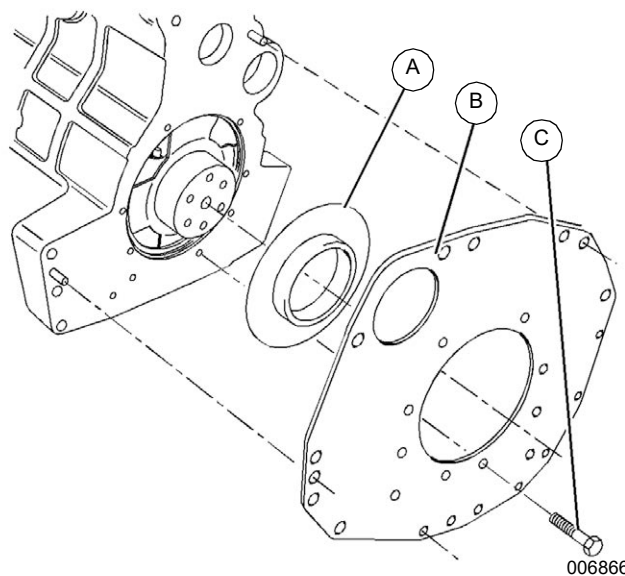


Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

IMPORTANT NOTE: Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids. Dispose of all fluids according to local regulations and mandates.

1. Remove the flywheel. See [Flywheel](#).
2. See [Figure 4-67](#). Remove bolts (C) and remove the flywheel housing or back plate (B) from the cylinder block. See [Flywheel Housing](#) for more information.
3. Remove crankshaft rear seal (A) from the cylinder block. Discard the crankshaft oil seal.



006866

Figure 4-67. Remove Crankshaft Rear Seal

Installation Procedure



Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. See [Figure 4-68](#). Verify the mating surface of cylinder block (E) and the flywheel housing or back plate (B) are clean and free from damage. Inspect dowels (D). If the dowels are damaged, use new parts for replacement.
2. Verify crankshaft flange (F) is clean and free from damage. It is possible to reclaim a crankshaft flange that has a worn seal surface, or a damaged seal surface by installing a wear sleeve. See [Crankshaft Wear Sleeve \(Rear\)](#) for more information.
3. Apply clean engine oil to the flange of crankshaft (F) around the running surface of crankshaft rear seal.
4. Align a new crankshaft rear seal (A) with the flange of crankshaft (F). Carefully install crankshaft rear seal (A) onto the crankshaft flange.
5. Apply a continuous bead of silicone sealant (A) to the rear face of the cylinder block around the edge of crankshaft rear seal (A). Apply a continuous bead of silicone sealant around plug (X).
6. Align the flywheel housing or back plate (B) to dowels (D). Install the flywheel housing or back plate (B) and install bolts (C) to cylinder block (E). See [Flywheel Housing](#).
7. Tighten bolts (C) to 19 ft-lbs (25 Nm).

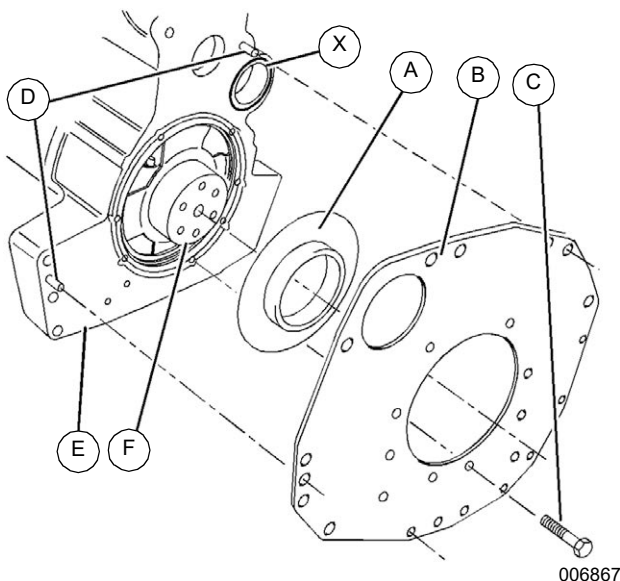


Figure 4-68. Install Crankshaft Rear Seal

8. Install the flywheel. See [Flywheel](#).

Crankshaft Wear Sleeve (Rear)

Removal Procedure



Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

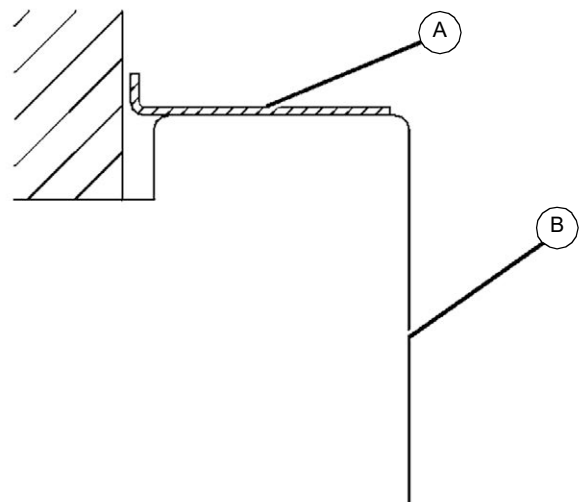
1. Remove the crankshaft rear seal. See [Crankshaft Rear Seal](#).

NOTE: Wear sleeves are used to reclaim worn seal surfaces or damaged seal surfaces. Wear sleeves are not original equipment.

2. Use a sharp tool to score a deep line across crankshaft wear sleeve (A).

NOTE: Take care to avoid damaging the crankshaft.

3. Insert a thin blade between crankshaft wear sleeve (A) and crankshaft (B) below the scored line. The crankshaft wear sleeve will separate along the line.
4. Remove crankshaft wear sleeve (A) from crankshaft (B). Discard the crankshaft wear sleeve.



006868

Figure 4-69. Remove Crankshaft Wear Sleeve

Installation Procedure

CAUTION

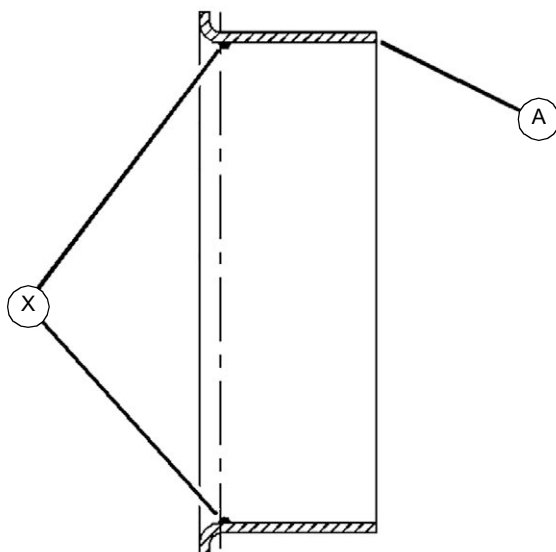
Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. Verify the crankshaft is thoroughly clean and dry. Remove any areas of raised damage.
2. See [Figure 4-70](#) and [Figure 4-71](#). Apply a small continuous bead of liquid gasket to the inner surface of crankshaft wear sleeve (A) at position (X). Apply the bead of liquid gasket 0.2 in (5.0 mm) from the flange end of the crankshaft wear sleeve.
3. Align crankshaft wear sleeve (A) with crankshaft (B). Position installation tool (C) that is provided with the crankshaft wear sleeve over the crankshaft. Use a hammer to drive the crankshaft wear sleeve onto the crankshaft. Verify dimension (Y) is 0.098 ± 0.004 in (2.5 ± 0.1 mm).

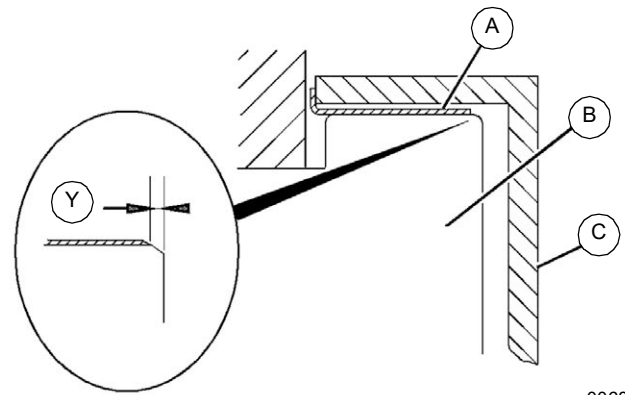
NOTE: Dimension (Y) is the distance from the edge of the crankshaft wear sleeve from the rear face of the crankshaft.

4. Remove installation tool (C).
5. Verify the crankshaft wear sleeve has no rough edges.



006869

Figure 4-70. Install Crankshaft Wear Sleeve



006870

Figure 4-71. Install Crankshaft Wear Sleeve

6. Install a new crankshaft rear seal. See [Crankshaft Rear Seal](#).

Flywheel Housing (Engines with Flywheel Housing and Back Plate)

Removal Procedure

CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. Remove the flywheel. See [Flywheel](#).
2. Support the engine.
3. Install a suitable lifting device to the flywheel housing. Support the weight of the housing. The weight of the flywheel housing is approximately 66 lbs (30 kg).
4. See [Figure 4-72](#) and [Figure 4-73](#). Remove bolts (C) and (D) that fasten flywheel housing (H) to the cylinder block.
5. Remove bolts (F) on each side of the cylinder block. These bolts fasten flywheel housing (H) to the cylinder block.
6. Remove bolts (A) that fasten back plate (B) to flywheel housing (H).
7. Remove nuts (E) and the bolts (not shown) that fasten back plate (B) to flywheel housing (H).
8. Carefully remove flywheel housing (H) from back plate (B) and dowels (G). Use a suitable lifting device to remove the flywheel housing.

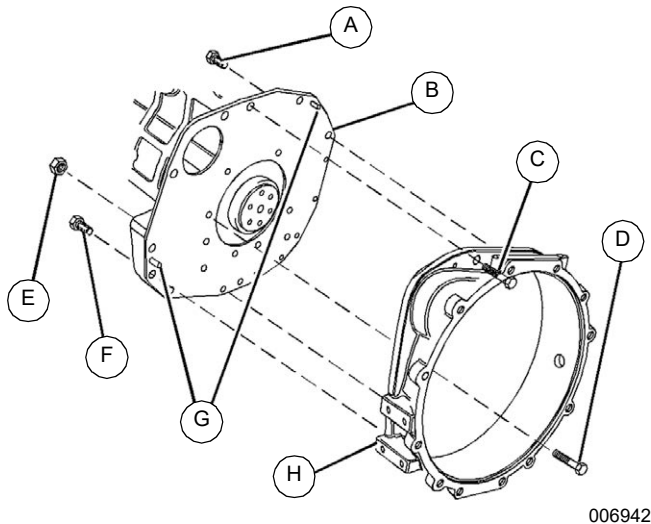


Figure 4-72. Remove Flywheel Housing

- dowels are damaged, use new parts for replacement.
2. Use a suitable lifting device to align flywheel housing (H) to dowels (G). The weight of the flywheel housing is approximately 66 lbs (30 kg). Install the flywheel housing to the back plate.
3. Install nuts (E) and bolts (not shown) that fasten back plate (B) to flywheel housing (H).
4. Install bolts (A) that fasten back plate (B) to flywheel housing (H).
5. Install bolts (F) on each side of the cylinder block. These bolts fasten flywheel housing (H) to the cylinder block.
6. Remove the lifting device from the flywheel housing.
7. Install bolts (C) and (D) that fasten flywheel housing (H) to the cylinder block.
8. Tighten the fasteners to 19 ft-lbs (25 Nm).

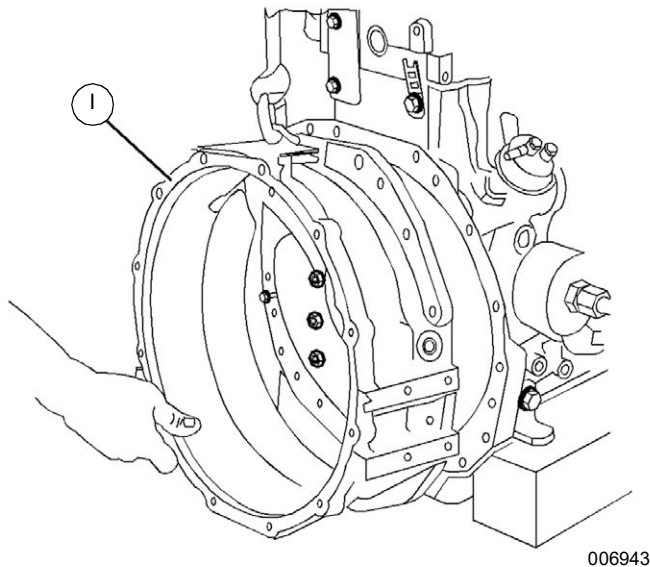


Figure 4-73. Remove Flywheel Housing

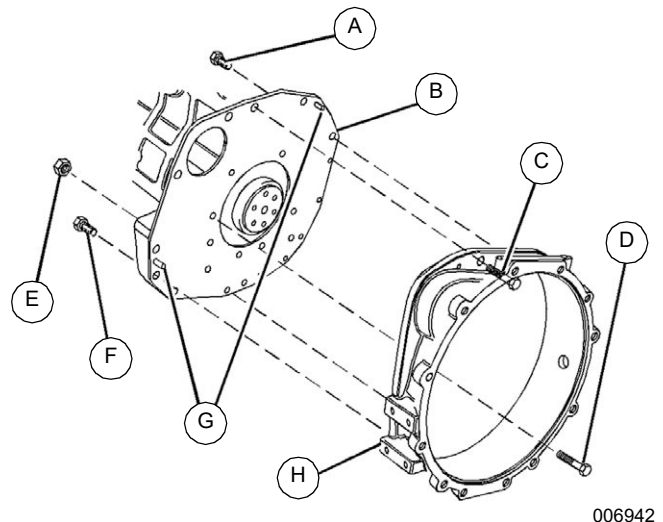


Figure 4-74. Install Flywheel Housing

Installation Procedure

CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. See [Figure 4-74](#) and [Figure 4-75](#). Verify the mating surfaces of back plate (B) and flywheel housing (H) are clean and free from damage. Inspect the crankshaft rear seal (not shown) for leaks. If necessary, replace the crankshaft rear seal. See [Crankshaft Rear Seal](#) for the correct procedure. Inspect dowels (G) for damage. If the

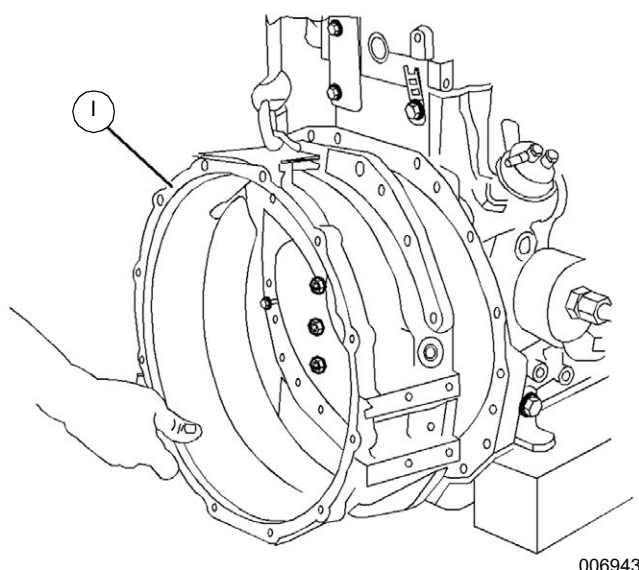


Figure 4-75. Install Flywheel Housing

9. Install the flywheel. See [Flywheel](#).

Flywheel Housing

Removal Procedure



Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. Support the engine.
2. Install a suitable lifting device to the flywheel housing. Support the weight of the housing. The weight of the flywheel housing is approximately 66 lbs (30 kg).
3. See [Figure 4-76](#). Remove bolts (A) that fasten flywheel housing (B) to the cylinder block.
4. Use the lifting device to remove flywheel housing (B) from the engine.

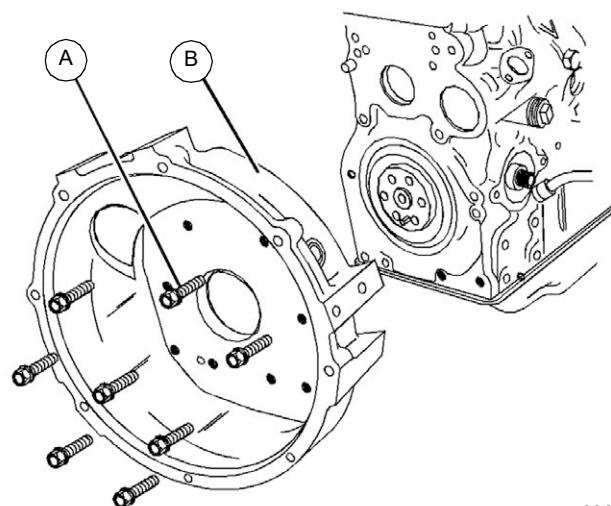


Figure 4-76. Install Flywheel Housing

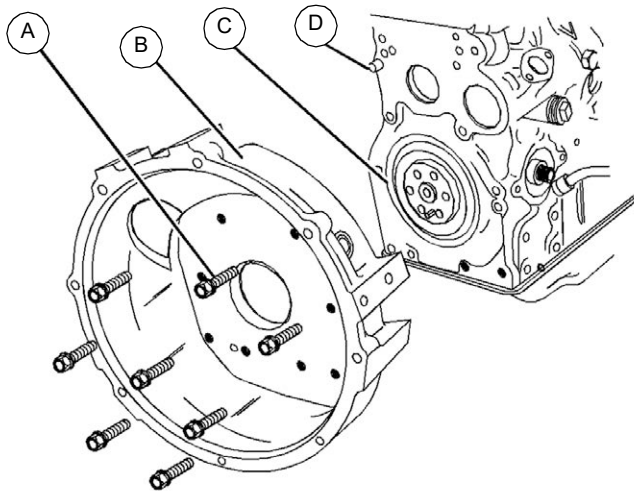
Installation Procedure



Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. See [Figure 4-77](#). Verify the mating surfaces of flywheel housing (B) are clean and free from damage. Inspect the crankshaft rear seal (C) for leaks. If necessary, replace the crankshaft rear seal. See [Crankshaft Rear Seal](#) for the correct procedure. Inspect dowels (D) for damage. If the dowels are damaged, use new parts for replacement.
2. Use a suitable lifting device to align flywheel housing (B) to dowels (D) and install the flywheel housing. The weight of the flywheel housing is approximately 66 lbs (30 kg).
3. Install bolts (A) and tighten to 19 ft-lbs (25 Nm).
4. Remove the lifting device from the flywheel housing.



006945

Figure 4-77. Install Flywheel Housing

5. Install the flywheel. See [Flywheel](#).

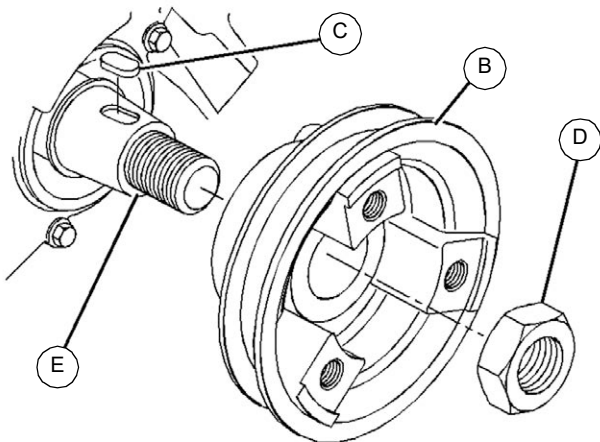
Crankshaft Pulley

Removal Procedure

1. Remove the V-belt. See [V-Belts](#) for the correct procedure.
2. See [Figure 4-78](#) and [Figure 4-79](#). Loosen nut (D).

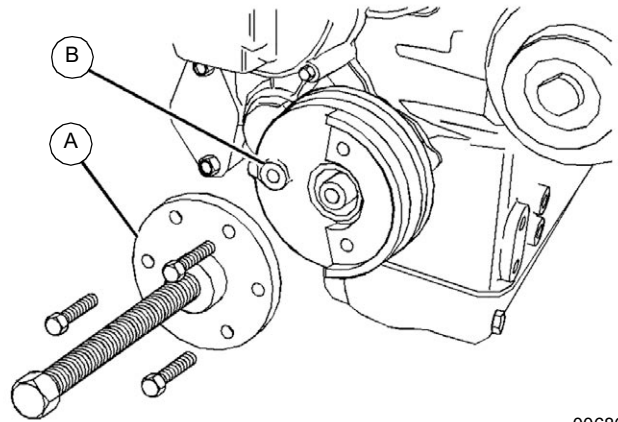
NOTE: Do not remove the nut at this time.

3. Install puller and bolts (A) to crankshaft pulley (B). Use puller and bolts (A) to remove crankshaft pulley (B) from crankshaft (E). Remove puller and bolts (A).
4. Remove nut (D).
5. Remove pulley (B) from crankshaft (E).
6. Remove woodruff key (C) from crankshaft (E).



006864

Figure 4-78. Remove Crankshaft Pulley

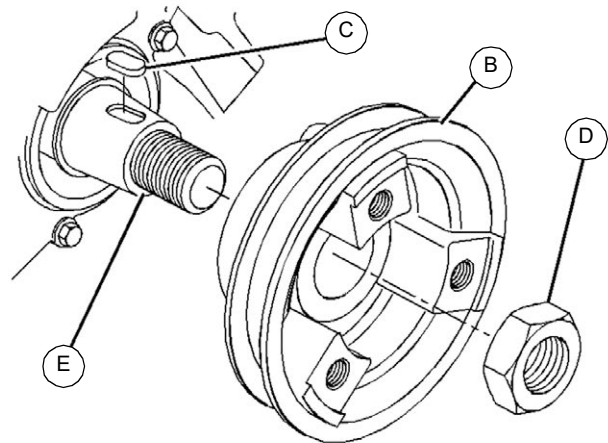


006863

Figure 4-79. Remove Crankshaft Pulley

Installation Procedure

1. Verify the following components are clean and free from damage: the taper of the crankshaft, the woodruff key and the bore of the crankshaft pulley. Replace any damaged components. Verify the taper of the crankshaft and the bore of the crankshaft pulley are clean and dry before assembly.
2. See [Figure 4-80](#). Install woodruff key (C) to crankshaft (E).
3. Install crankshaft pulley (B) to crankshaft (E).
4. Lubricate nut (D) with clean engine oil and install the nut. Tighten nut (D) to 224 ft-lbs (304 Nm).



006864

Figure 4-80. Install Crankshaft Pulley

5. Install the V-belt to the engine. See [V-Belts](#) for the correct procedure.

Crankshaft Front Seal

Removal Procedure

⚠ CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

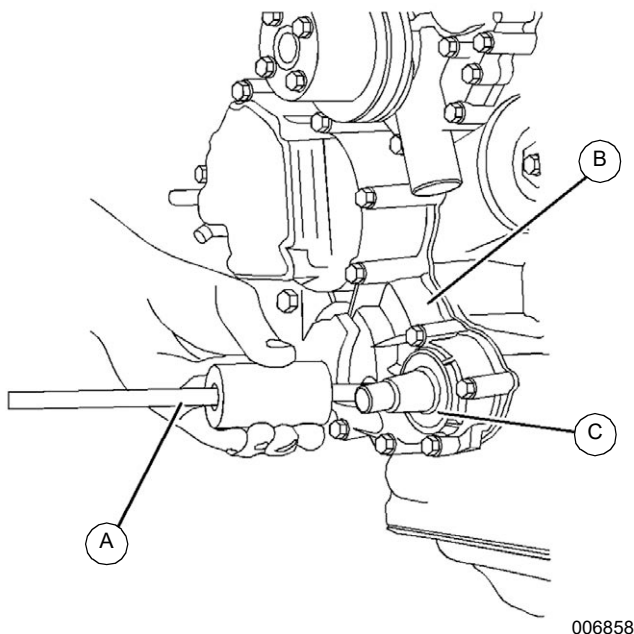
(000357)

1. Remove the crankshaft pulley. See [Crankshaft Pulley](#).
2. See [Figure 4-81](#). Use a 0.158 in (4 mm) drill to make three holes in crankshaft front seal (10000034146) (C).

IMPORTANT NOTE: Verify the main lip is used to remove the crankshaft front seal. Do not damage the edge of the housing for the crankshaft front seal.

NOTE: Do not damage the crankshaft during the removal process of the crankshaft front seal.

3. Use slide hammer puller (A) to carefully remove crankshaft front seal (C). Alternate the position of slide hammer puller (A) from one hole to another hole. This will allow even removal of the crankshaft front seal from front housing (B).



006858

Figure 4-81. Remove Crankshaft Front Seal

Alternative Removal Procedure

⚠ CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. Remove the crankshaft pulley. See [Crankshaft Pulley](#) for the correct procedure.
2. Remove the front housing. See [Housing \(Front\)](#) for the correct procedure.

NOTE: This is an alternative procedure to remove the crankshaft front oil seal. This procedure may be used if the front housing has been removed from the engine.

3. Use a suitable mandrel and a press to remove the oil seal from the front cover. Verify the front cover is supported to prevent damage to the cover while the oil seal is being removed.

Installation Procedure

⚠ CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. See [Figure 4-82](#). Verify the bore in front housing (A) and the nose of the crankshaft are clean and free from damage.
2. Lubricate the lip of a new crankshaft front seal (B) with clean engine oil.
3. Position crankshaft front seal (B) in the bore of front housing (A).

IMPORTANT NOTE: Verify the lip of the crankshaft front seal that is spring loaded is facing toward the inside of the front housing and that it is square with the bore of the housing for the crankshaft front seal.

4. Use a suitable tool to install crankshaft front seal (B) to front housing (A).

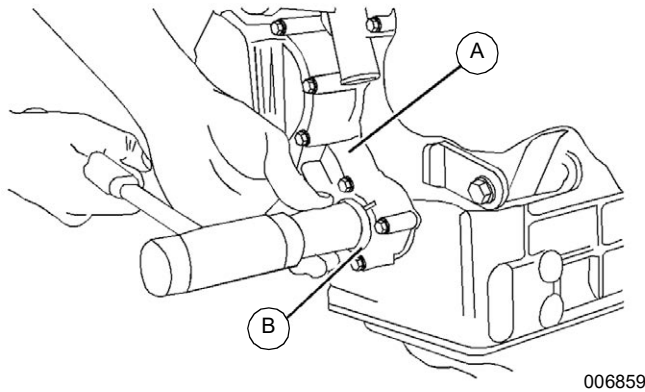


Figure 4-82. Install Crankshaft Front Seal

5. Install the crankshaft pulley. See [Crankshaft Pulley](#) for the correct procedure.

Housing (Front)

Removal Procedure

⚠ CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

IMPORTANT NOTE: Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids. Dispose of all fluids according to local regulations and mandates.

1. Remove the fuel injection pump. See [Fuel Injection Pump](#).
2. Remove the crankshaft pulley. See [Crankshaft Pulley](#).
3. See [Figure 4-83](#). If the engine is equipped with a turbocharger, disconnect the hose from connection (A).
4. Remove nuts (D) and bolts (I).
5. Remove cover (H) and O-ring seal (G).
6. Remove fasteners (C). Identify the position of the different fasteners for installation.
7. Carefully remove front housing (F) from plate (E). Verify pin (B) remains in the front housing.

NOTE: The front housing is aligned to the plate and to the cylinder block with dowels.

8. Remove the joint (not shown).

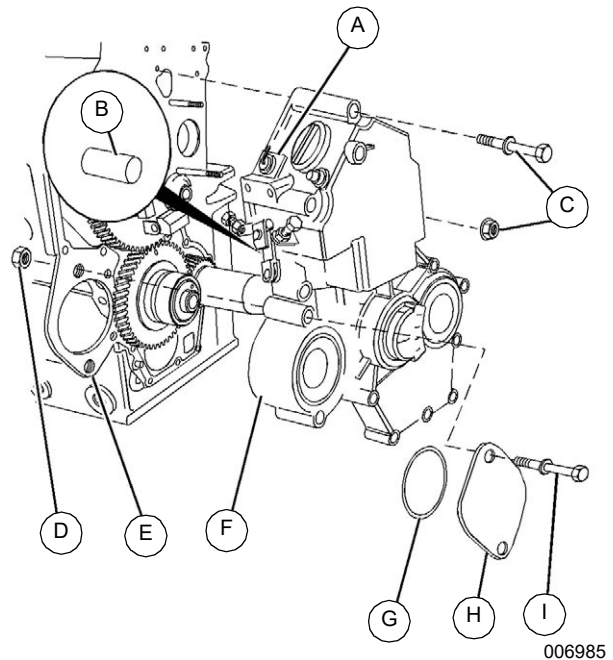


Figure 4-83. Remove Crankshaft Front Seal

Disassembly Procedure



⚠ WARNING

Vision Loss. Eye protection is required for the following procedure. Failure to wear proper eye protection could result in serious injury to the eyes.

(000248)

⚠ WARNING

Personal Injury. Always wear all necessary protective equipment and use recommended tools while performing the following procedure. Failure to do so could result in serious injury.

(000556)

⚠ CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. Record the governor settings. See [Adjust Governor](#).
2. Remove the engine front housing. See [Housing \(Front\)](#).
3. See [Figure 4-84](#). Remove spring (D) from arm (B) on the assembly of throttle control (A).

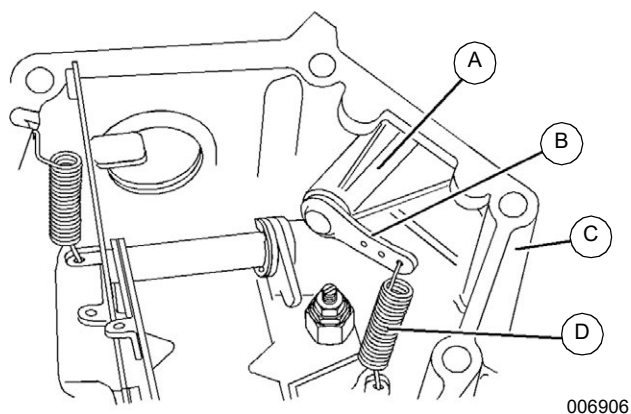


Figure 4-84. Disassemble Housing (Front)

4. See [Figure 4-85](#). Remove bolts (E) and remove the assembly of throttle control (A) from front housing (C).
5. Remove joint (F).

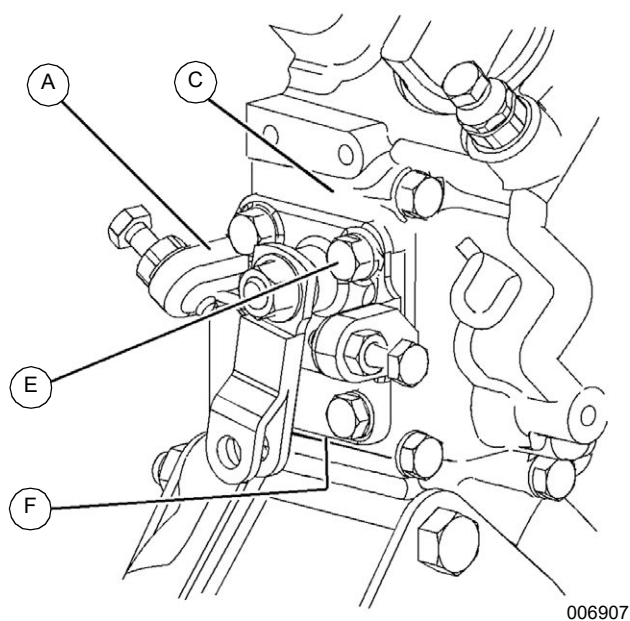


Figure 4-85. Disassemble Housing (Front)

6. See [Figure 4-86](#). Remove shaft (I) from front housing (C).
7. Remove lever assembly (G) from front housing (C).
8. Remove sealing washer (H) from shaft (I).

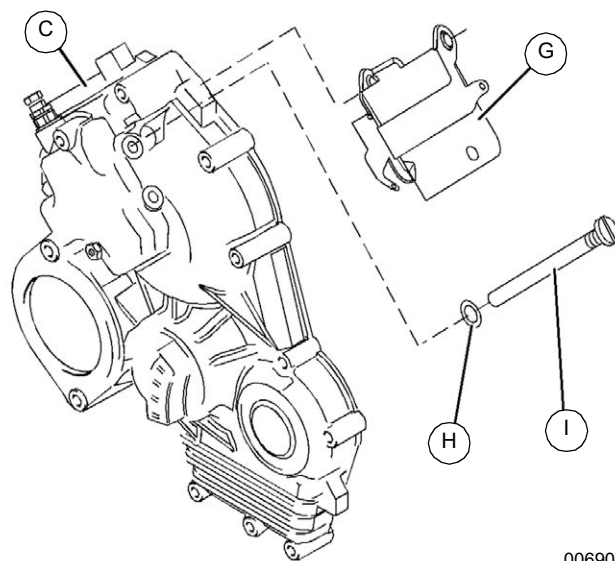


Figure 4-86. Disassemble Housing (Front)

9. See [Figure 4-87](#). If the engine is equipped with an angleich unit, remove angleich unit (J) from lever assembly (G).

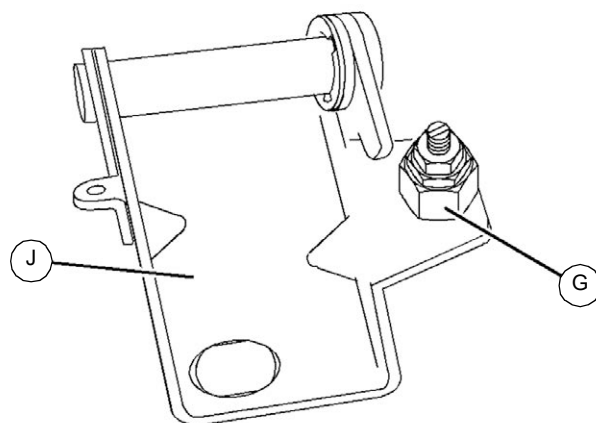
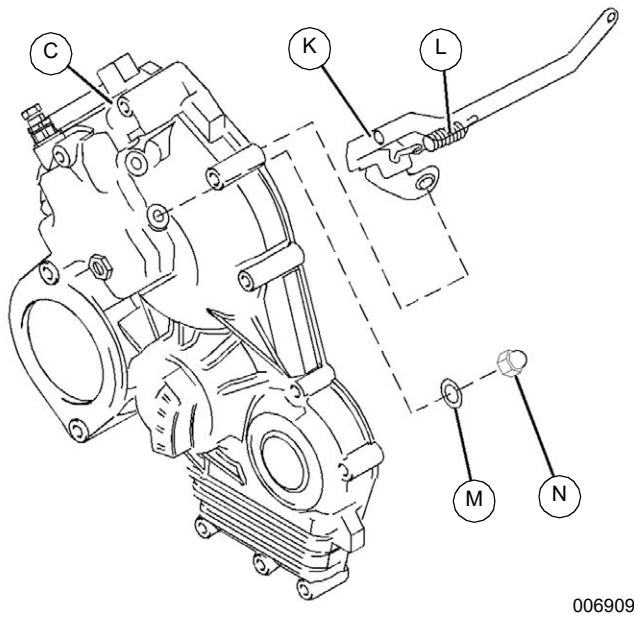


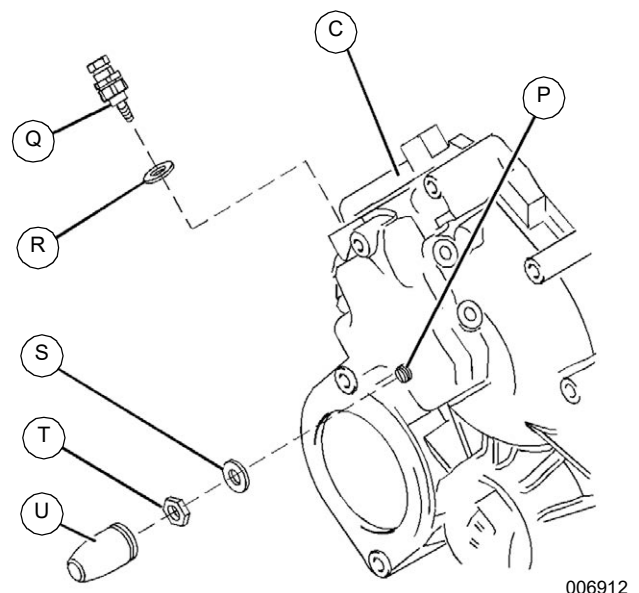
Figure 4-87. Remove Angleich Unit

10. See [Figure 4-88](#) and [Figure 4-89](#). Disconnect spring (L) from front housing (C).
11. Remove nut (N) and remove sealing washer (M) from front housing (C).
12. Remove shaft (O) and remove lever assembly (K) from front housing (C).



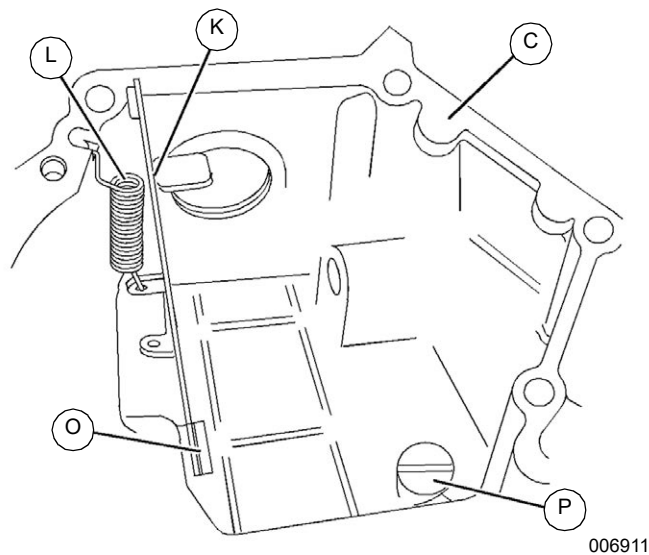
006909

Figure 4-88. Disassemble Housing (Front)



006912

Figure 4-90. Disassemble Housing (Front)



006911

Figure 4-89. Disassemble Housing (Front)

13. See [Figure 4-90](#). Remove spring assembly (Q) and remove sealing washer (R) from front housing (C).

NOTE: The position of fuel screw (P) controls the emissions of the engine. Do not remove the fuel screw unless a different front housing will be installed. Do not adjust the fuel screw. See [Adjust Governor](#).

14. If necessary, remove cap (U), locknut (T) and sealing washer (S) from front housing (C). Remove fuel screw (P).

Assembly Procedure



WARNING

Vision Loss. Eye protection is required for the following procedure. Failure to wear proper eye protection could result in serious injury to the eyes.

(000248)

WARNING

Personal Injury. Properly compress valve springs using a valve spring compressor. Failure to do so could cause the spring to eject, and could result in serious injury.

(000193a)

CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

NOTE: See [Figure 4-91](#) and [Figure 4-92](#). The position of stop (P) controls the emissions of the engine. Verify correct adjustment of the stop.

1. If the original front housing has been replaced by a different item, follow Steps 2 through 5.
2. Install stop (P) to front housing (C).
3. Install a new sealing washer (S) to stop (P).
4. Loosely install locknut (T) to stop (P). The procedure used to set the position of the stop is described in [Record Governor Settings](#).
5. Install a new emissions label to front housing (C).

6. Install a new sealing washer (R) to spring assembly (Q). Loosely install spring assembly (Q) to front housing (C).

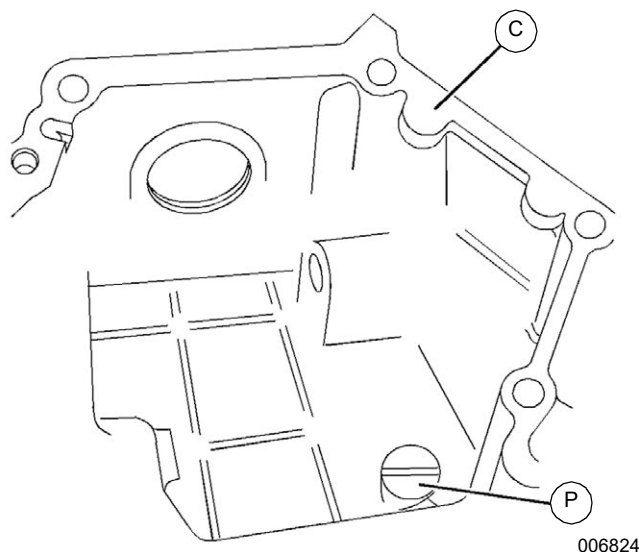


Figure 4-91. Assemble Housing (Front)

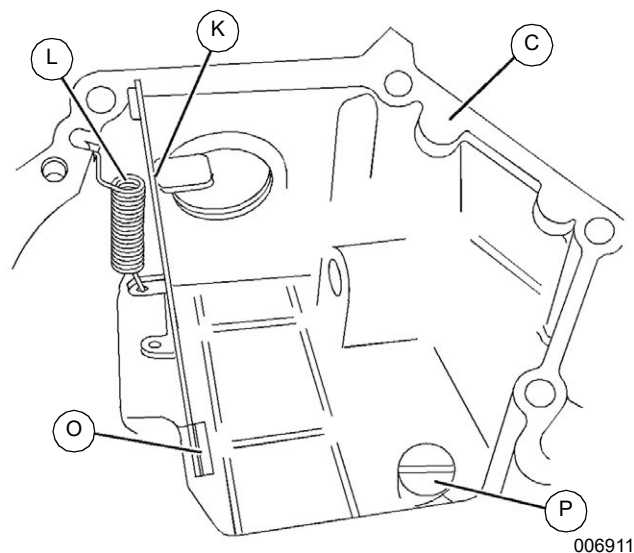


Figure 4-93. Assemble Housing (Front)

8. See [Figure 4-94](#). Install a new sealing washer (M) and nut (N) to front housing (C). Tighten the nut to 53 in-lbs (6 Nm).
9. Connect spring (L) to front housing (C).

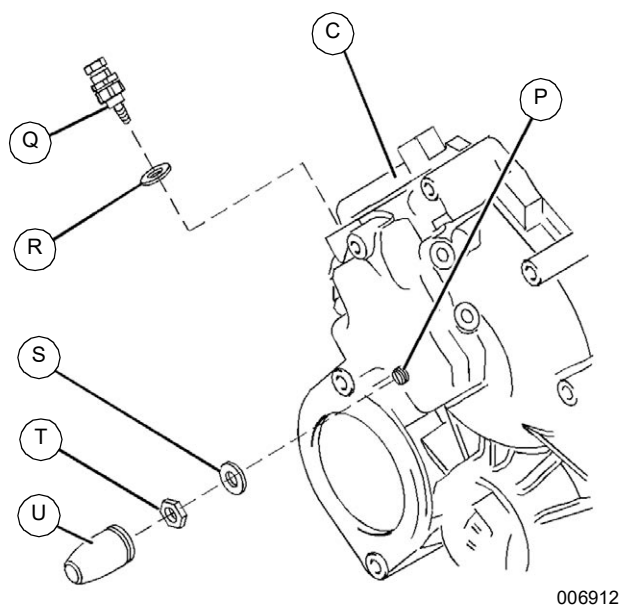


Figure 4-92. Assemble Housing (Front)

7. See [Figure 4-93](#). Place lever assembly (K) in position in front housing (C) and install shaft (O).

NOTE: Verify the correct orientation of the lever assembly.

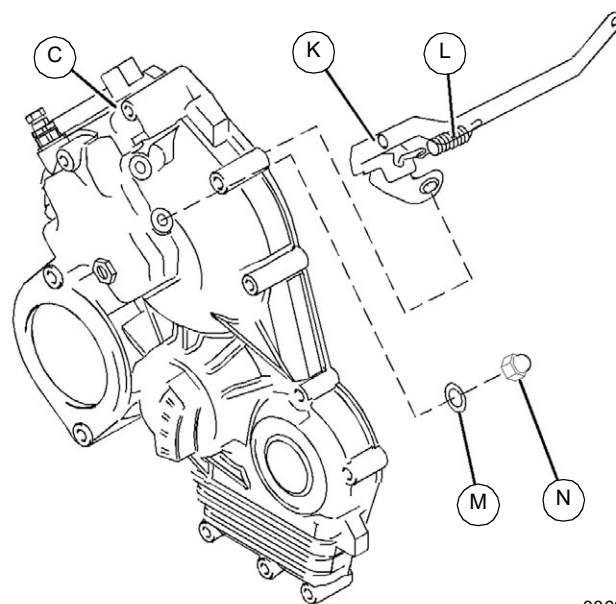


Figure 4-94. Assemble Housing (Front)

10. If the engine is equipped with an angleich unit, follow Steps 11 and 12 to install the angleich unit.
11. See [Figure 4-95](#) and [Figure 4-96](#). Apply Loctite™ 275 Thread Lock to the first two threads of angleich unit (J).
12. Install angleich unit (J) to lever assembly (G). Tighten the angleich unit to 53 in-lbs (6 Nm).

NOTE: Verify Loctite™ 243 Thread Lock is not allowed to contaminate the plunger assembly of the angleich unit. Contamination of the plunger assembly will render the angleich unit inoperative.

13. Install a new sealing washer (H) to shaft (I).
14. Apply Loctite™ 243 Thread Lock to the first two threads of shaft (I).
15. Place lever assembly (G) in position in front housing (C) and install shaft (I). Tighten the shaft to 142 in-lbs (16 Nm).

NOTE: Verify the correct orientation of the lever assembly.

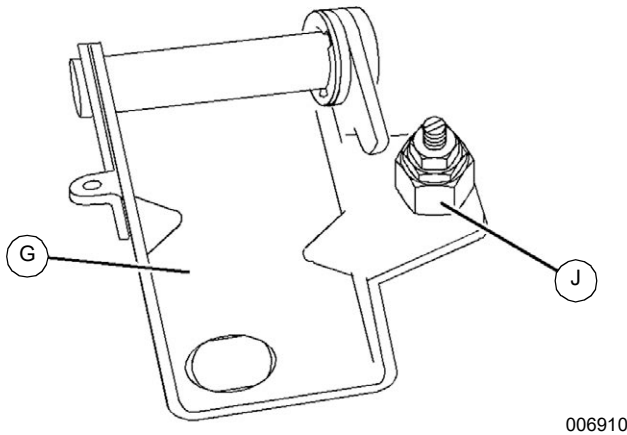


Figure 4-95. Assemble Housing (Front)

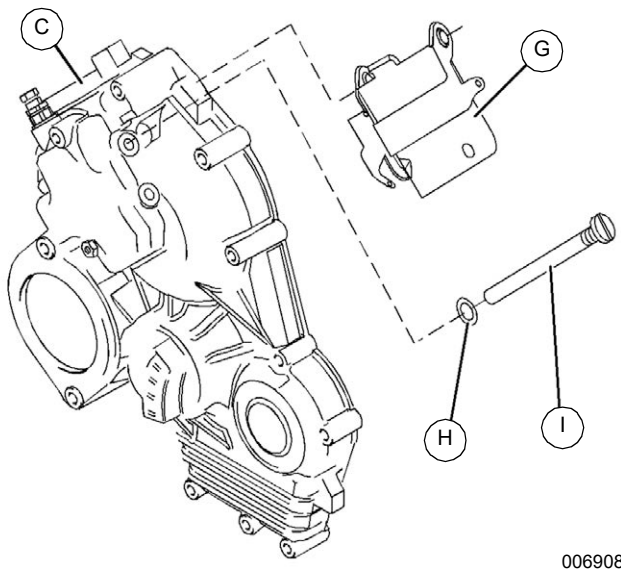


Figure 4-96. Assemble Housing (Front)

16. See [Figure 4-97](#). Position a new joint (F) and position the assembly of throttle control (A) onto front housing (C).
17. Install and tighten bolts (E) to 97 in-lbs (11 Nm).

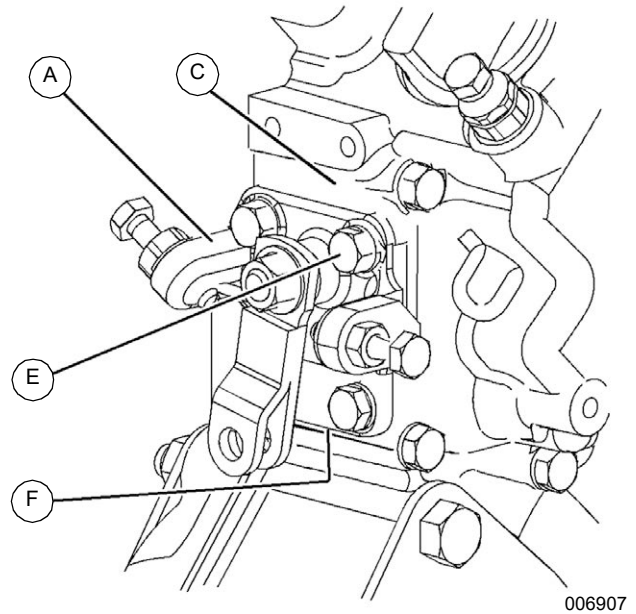


Figure 4-97. Assemble Housing (Front)

18. See [Figure 4-98](#). Install spring (D).
19. If the position of stop (P) has been altered, the stop must be reset. See [Record Governor Settings](#).

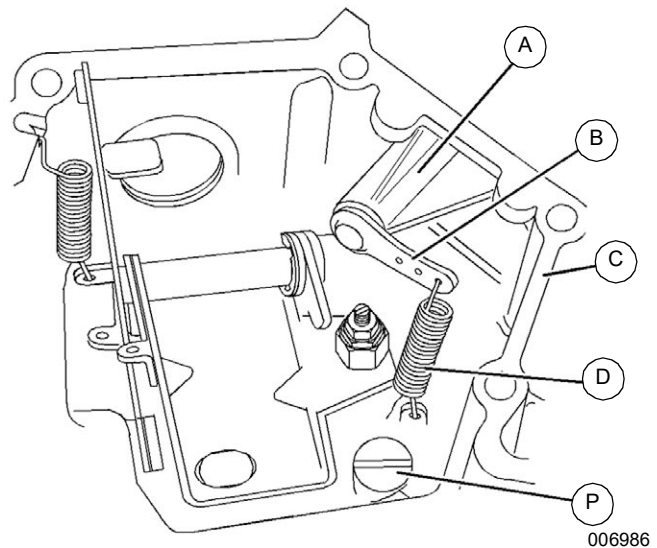


Figure 4-98. Assemble Housing (Front)

20. Install the front housing. See [Housing \(Front\)](#) for the correct procedure.
21. See [Figure 4-92](#). Set the position of the spring assembly (Q). Tighten the spring assembly to 25 ft-lbs (34 Nm).
22. See [Figure 4-92](#). Install a new cap (U) to nut (T).

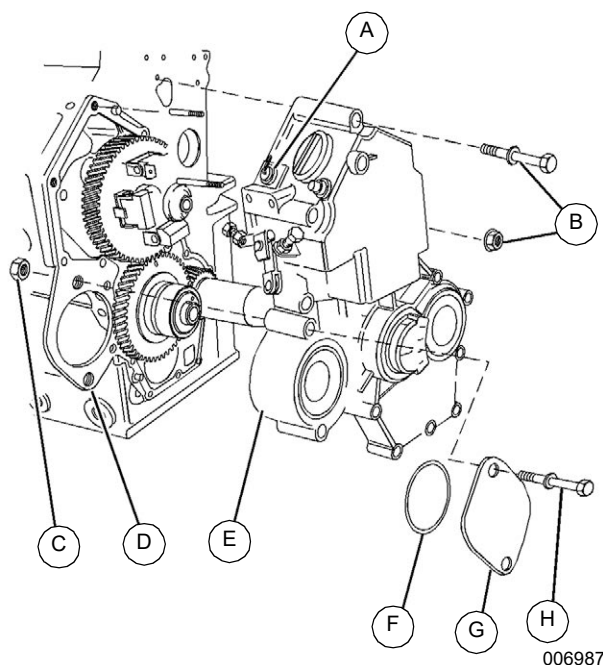
Installation Procedure

CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

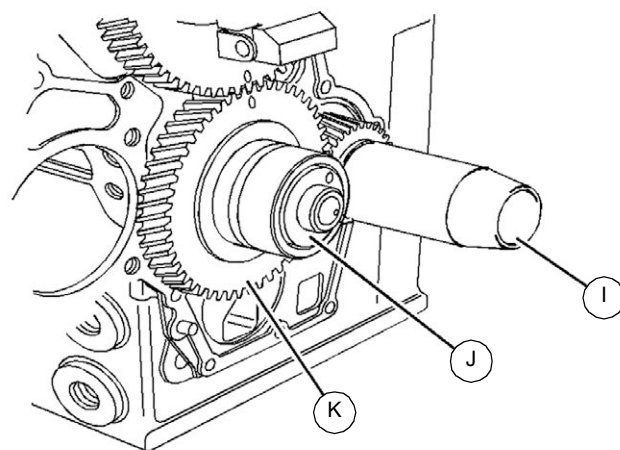
(000357)

1. See [Figure 4-99](#), [Figure 4-100](#) and [Figure 4-101](#). Verify the assembly of front housing (E) and the nose of the crankshaft are clean and free from damage. Clean the mating surface of plate (D).
2. Install seal protector (I) on the nose of the crankshaft.
3. Align pin (L) with hole (M) in oil pump cover (J). Verify the oil pump cover is concentric to idler gear assembly (K).
4. Install a new joint to plate (D).
5. Guide the linkage for the fuel injection pump into position so the link is free to move inside the aperture on the cylinder block for the fuel injection pump. Align front housing (E) to the dowels in plate (D). Install front housing (E) to plate (D).
6. Install fasteners (B) and tighten to 89 in-lbs (10 Nm).
7. Remove seal protector (I) from the nose of the crankshaft.
8. Install a new O-ring seal (F) to front housing (E). Position cover (G) on front housing (E) and install nuts (C) and bolts (H).
9. Tighten nuts (C) to 37 ft-lbs (50 Nm).
10. If the engine is equipped with a turbocharger, connect the hose to connection (A).



006987

Figure 4-99. Install Housing (Front)



006988

Figure 4-100. Install Housing (Front)

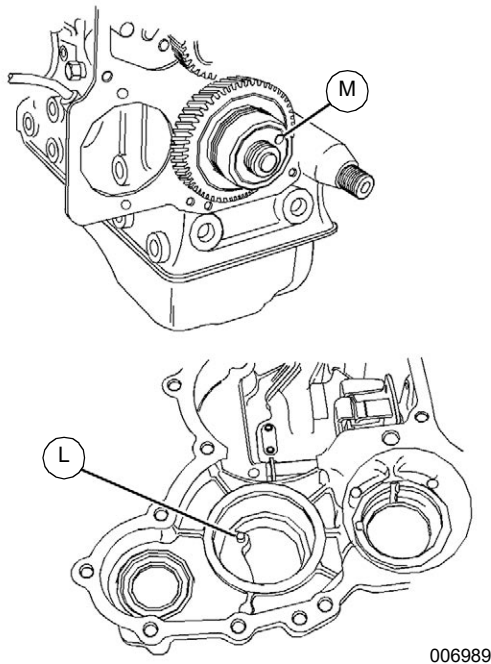


Figure 4-101. Install Housing (Front)

11. Install the crankshaft pulley. See [Crankshaft Pulley](#).
12. Install the fuel injection pump. See [Fuel Injection Pump](#).

Crankcase Breather

Removal Procedure



⚠ WARNING

Vision Loss. Eye protection is required for the following procedure. Failure to wear proper eye protection could result in serious injury to the eyes.

(000248)

⚠ WARNING

Personal Injury. Always wear all necessary protective equipment and use recommended tools while performing the following procedure. Failure to do so could result in serious injury.

(000556)

⚠ CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

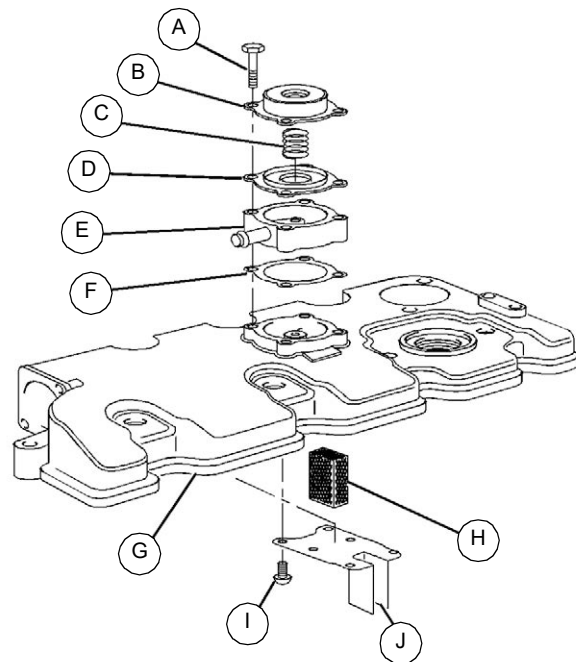
IMPORTANT NOTE: Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and

repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids. Dispose of all fluids according to local regulations and mandates.

1. See [Figure 4-102](#). Remove bolts (A) and remove the assembly of cover (B), spring (C) and diaphragm (D). Note the orientation of cover (B).
2. Remove spring (C) and diaphragm (D) from cover (B).
3. Remove adapter (E) and joint (F) from valve mechanism cover (G).

NOTE: Make a temporary mark to show the adapter orientation for installation.

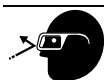
4. If necessary, follow Steps 5 through 7 to remove the gauze for the breather.
5. Remove valve mechanism cover (G). See [Valve Mechanism Cover](#).
6. Remove screws (I) and carefully remove plate (J).
7. Remove gauze (H) from valve mechanism cover (G).



006852

Figure 4-102. Remove Crankcase Breather

Installation Procedure



⚠ WARNING

Vision Loss. Eye protection is required for the following procedure. Failure to wear proper eye protection could result in serious injury to the eyes. (000248)

⚠ WARNING

Personal Injury. Always wear all necessary protective equipment and use recommended tools while performing the following procedure. Failure to do so could result in serious injury. (000556)

⚠ CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage. (000357)

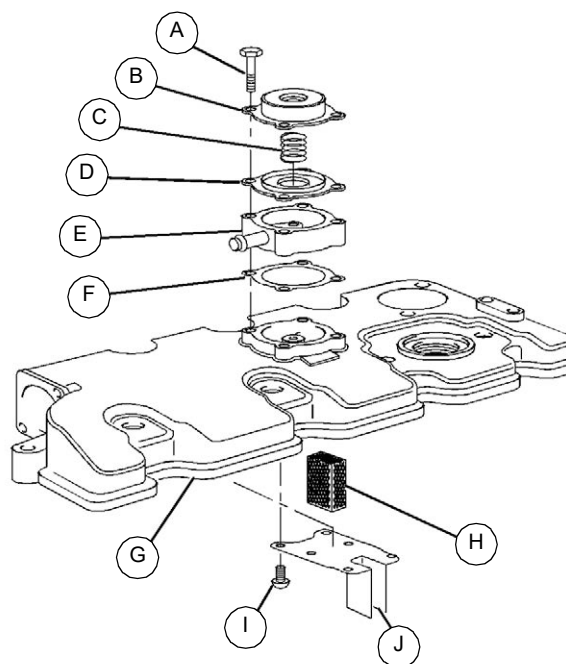
1. See **Figure 4-103**. Clean and inspect all parts. Replace any parts that are worn or damaged. Verify the cavity for the breather in the valve mechanism cover is clean. Verify vent hole in adapter (E) and the vent hole in cover (B) are free from restriction.
2. If necessary, follow Steps 3 through 6 to install the gauze for the breather.
3. Install gauze (H) to valve mechanism cover (G).
4. Position plate (J) on valve mechanism cover (G) and install screws (I).
5. Tighten screws (I) to 13 **in-lbs** (1.5 Nm).
6. Install valve mechanism cover (G). See **Valve Mechanism Cover**.
7. Position a new joint (F) on the valve mechanism cover and install adapter (E).

NOTE: Verify the correct orientation of the adapter. The vent hole should face upward.

8. Install diaphragm (D) and spring (C) to cover (B).
9. Position the assembly of cover (B), spring (C) and diaphragm (D) onto valve mechanism cover (G).

NOTE: Verify the correct orientation of the cover.

10. Install bolts (A). Use torque wrench (A) to tighten bolts to 27 **in-lbs** (3 Nm).

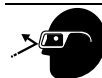


006852

Figure 4-103. Remove Crankcase Breather

Crankcase Breather (Naturally Aspirated Engines)

Removal Procedure



⚠ WARNING

Vision Loss. Eye protection is required for the following procedure. Failure to wear proper eye protection could result in serious injury to the eyes. (000248)

⚠ WARNING

Personal Injury. Always wear all necessary protective equipment and use recommended tools while performing the following procedure. Failure to do so could result in serious injury. (000556)

⚠ CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage. (000357)

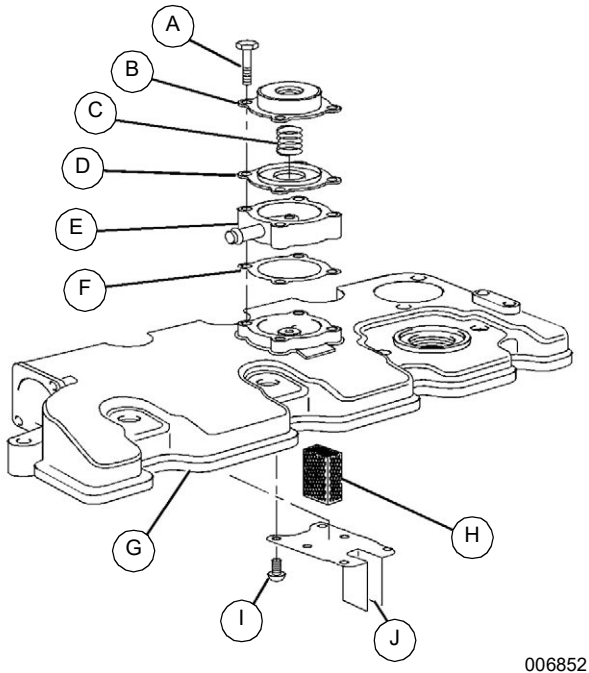
IMPORTANT NOTE: Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component

containing fluids. Dispose of all fluids according to local regulations and mandates.

1. See [Figure 4-104](#). Remove bolts (A) and remove the assembly of cover (B), spring (C) and diaphragm (D). Note the orientation of cover (B).
2. Remove spring (C) and diaphragm (D) from cover (B).
3. Remove adapter (E) and joint (F) from valve mechanism cover (G).

NOTE: Make a temporary mark to show the orientation of the adapter for installation.

4. If necessary, follow Steps 5 through 7 to remove the gauze for the breather.
5. Remove valve mechanism cover (G). See [Valve Mechanism Cover](#).
6. Remove screws (I) and carefully remove plate (J).
7. Remove gauze (H) from valve mechanism cover (G).



006852

Figure 4-104. Remove Crankcase Breather

Installation Procedure



WARNING

Vision Loss. Eye protection is required for the following procedure. Failure to wear proper eye protection could result in serious injury to the eyes. (000248)

WARNING

Personal Injury. Always wear all necessary protective equipment and use recommended tools while performing the following procedure. Failure to do so could result in serious injury. (000556)

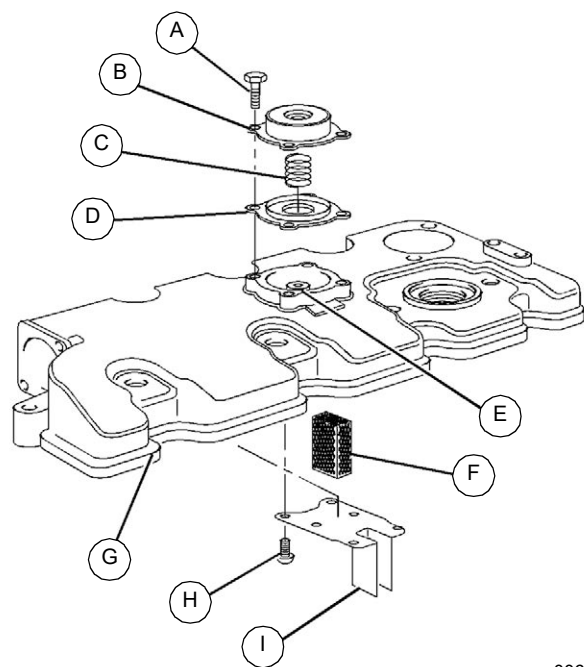
CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage. (000357)

1. See [Figure 4-104](#). Clean and inspect all parts. Replace any parts that are worn or damaged. Verify the cavity for the breather in the valve mechanism cover is clean. Verify vent hole (E) and the vent hole in cover (B) are free from restriction.
2. If necessary, follow Steps 3 through 6 to install the gauze for the breather.
3. Install gauze (F) to valve mechanism cover (G).
4. Position plate (I) on valve mechanism cover (G) and install screws (H).
5. Tighten screws (H) to 13 **in-lbs** (1.5 Nm).
6. Install valve mechanism cover (G). See [Valve Mechanism Cover](#).
7. Install diaphragm (D) and spring (C) to cover (B).
8. Position the assembly of cover (B), spring (C) and diaphragm (D) onto valve mechanism cover (G).

NOTE: Verify the correct orientation of the cover.

9. Install bolts (A). Use torque wrench to tighten bolts to 27 **in-lbs** (3 Nm).



006850

Figure 4-105. Install Crankcase Breather

Valve Mechanism Cover

Removal Procedure

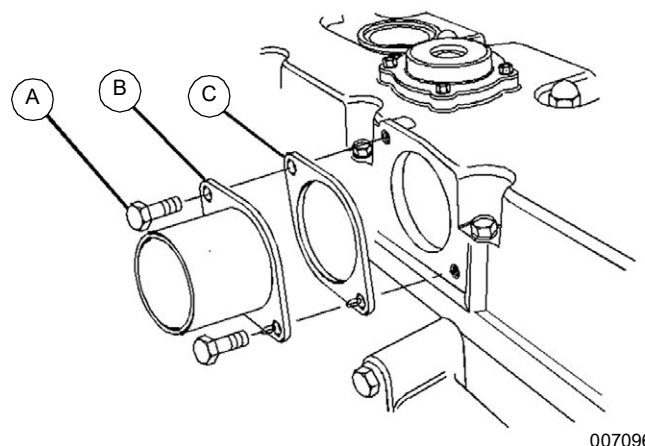


Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

IMPORTANT NOTE: Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids. Dispose of all fluids according to local regulations and mandates.

1. See [Figure 4-106](#). Loosen the hose clamps and disconnect the hose (not shown) from connection (B).



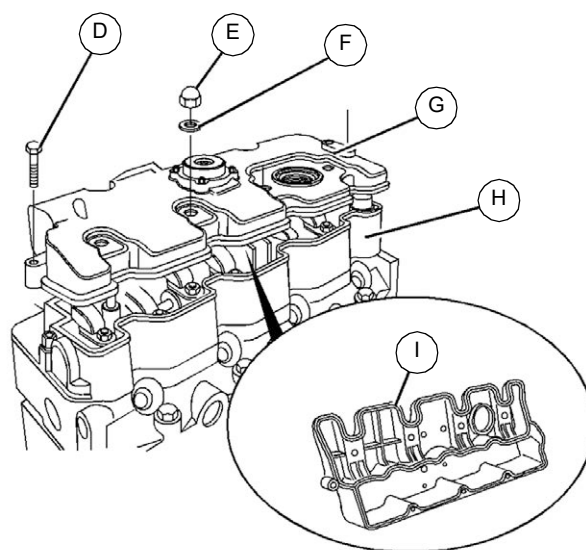
007096

Figure 4-106. Remove Valve Mechanism Cover

2. See [Figure 4-107](#). Remove nuts (E) and washers (F) from valve mechanism cover (G).
3. Remove bolts (D) from valve mechanism cover (G).

NOTE: Loosen the bolts evenly to avoid distortion of the valve mechanism cover.

4. Remove valve mechanism cover (G) from base (H). Remove joint (I) from valve mechanism cover (G).
5. If necessary, remove bolts (A) and remove connection (B) from valve mechanism cover (G). Remove gasket (B).
6. If necessary, remove the crankcase breather. See [Crankcase Breather](#) or [Crankcase Breather \(Naturally Aspirated Engines\)](#).



007097

Figure 4-107. Remove Valve Mechanism Cover

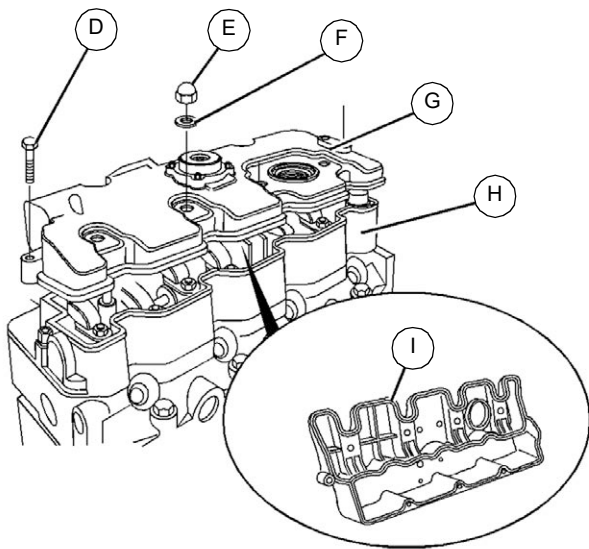
Installation Procedure

⚠ CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

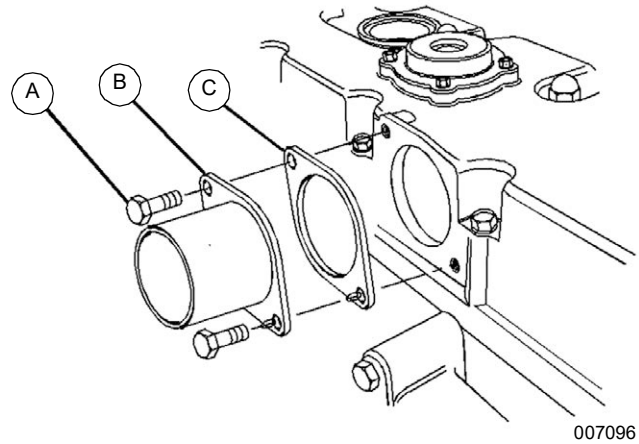
(000357)

1. See [Figure 4-108](#) and [Figure 4-109](#). Verify the valve mechanism cover is clean and free from damage. Clean the mating surface of base (H).
2. If necessary, install the crankcase breather. See [Crankcase Breather](#) or [Crankcase Breather \(Naturally Aspirated Engines\)](#).
3. Install a new joint (I) to valve mechanism cover (G).
4. Position valve mechanism cover (G) onto housing (H) and install bolts (D) finger tight.
5. Install new washers (F) and nuts (E) finger tight.
6. Evenly tighten bolts (D) to 89 **in-lbs** (10 Nm). Tighten nuts (E) to 124 **in-lbs** (14 Nm).
7. If necessary, position a new gasket (C) onto valve mechanism cover (G) and install connection (B).
8. Install bolts (A) and tighten to 124 **in-lbs** (14 Nm).
9. Connect hose (not shown) to connection (B) and tighten hose clamps.



007097

Figure 4-108. Install Valve Mechanism Cover



007096

Figure 4-109. Install Valve Mechanism Cover

Rocker Shaft and Pushrod

Removal Procedure

⚠ CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. Remove the engine oil line. See [Engine Oil Line](#).
2. Remove the valve mechanism cover. See [Valve Mechanism Cover](#).
3. See [Figure 4-110](#). Loosen the nuts and adjustment screws (D) on all rocker arms.
4. Remove bolts (A) and nuts (B) from rocker shaft assembly (C).
5. Remove rocker shaft assembly (C) from the cylinder head.
6. Place temporary identification marks on caps (E) and pushrods (F).

NOTE: Identification will ensure the pushrods and the caps can be reinstalled in the original positions. Do not interchange the positions of used pushrods or caps.

7. Remove pushrods (F) from the cylinder head. Remove caps (E) from the valve stems.

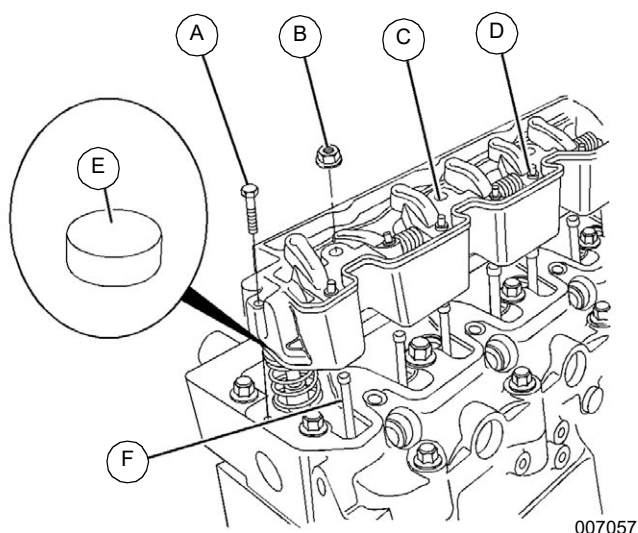


Figure 4-110. Remove Rocker Shaft and Pushrod

8. See [Figure 4-111](#). Remove joint (G) from the base of rocker shaft assembly (C).

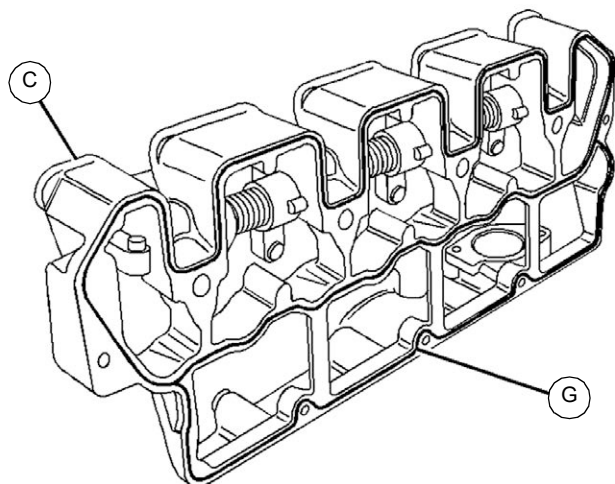
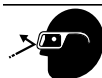


Figure 4-111. Remove Rocker Shaft and Pushrod

Disassembly Procedure



⚠ WARNING

Vision Loss. Eye protection is required for the following procedure. Failure to wear proper eye protection could result in serious injury to the eyes. (000248)

⚠ WARNING

Personal Injury. Always wear all necessary protective equipment and use recommended tools while performing the following procedure. Failure to do so could result in serious injury. (000556)

⚠ CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage. (000357)

1. Remove the rocker shaft and the pushrods. See [Rocker Shaft and Pushrod](#).
2. Make a temporary identification mark on each rocker arm assembly to show the location.

NOTE: Used components must be reinstalled in the original location. Do not interchange components.

3. See [Figure 4-112](#). To remove rocker shaft (F), install a suitable bolt into the end of the rocker shaft. Use the bolt to pull the rocker shaft from base (H).
4. Remove rocker arms (C), washers (E), and springs (D).
5. Remove O-ring seal (G) from rocker shaft (F).
6. Remove adjustment screws (A) and nuts (B) from rocker arms (C).

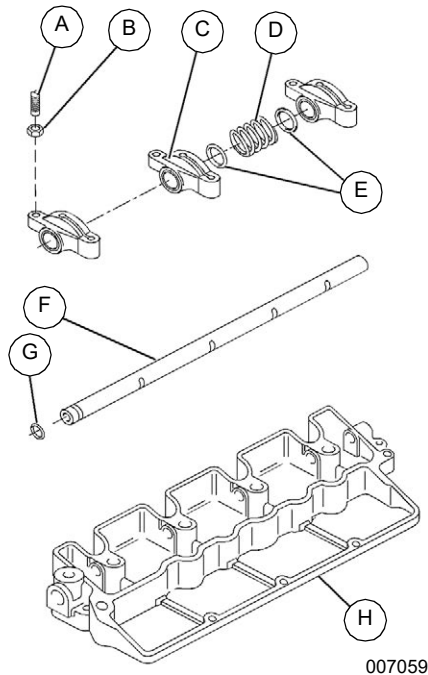


Figure 4-112. Disassemble Rocker Shaft and Pushrod

Assembly Procedure



WARNING

Vision Loss. Eye protection is required for the following procedure. Failure to wear proper eye protection could result in serious injury to the eyes. (000248)

WARNING

Personal Injury. Always wear all necessary protective equipment and use recommended tools while performing the following procedure. Failure to do so could result in serious injury. (000556)

CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage. (000357)

1. Verify all components are clean and free from wear or damage. See [Rocker Shaft](#) for more information. If necessary, replace any components that are worn or damaged.
2. See [Figure 4-113](#). Lubricate adjustment screw (A) with clean engine oil. Install adjustment screw (A) and nut (B) to rocker arm (C). Repeat the process for the remaining rocker arms.
3. Install a new O-ring seal (G) onto rocker shaft (F). Lubricate the rocker shaft with clean engine oil.

4. Position the end of rocker shaft (F) into the bore in base (H). Align recesses (X) on the rocker shaft with holes (Y) in the base.
5. Gradually insert rocker shaft (F) into the base (H). Install rocker arm (C).
6. Continue to insert rocker shaft (F) and install the following components to the rocker shaft:
 - Rocker arm (C)
 - Washer (E)
 - Spring (D)
 - Washer (E)
 - Rocker arm (C)
7. Repeat Steps 5 and 6 to install the remaining components.
8. Verify rocker shaft (F) is fully engaged into base (H). Verify recesses (X) on the rocker shaft are aligned with holes (Y) in the base.

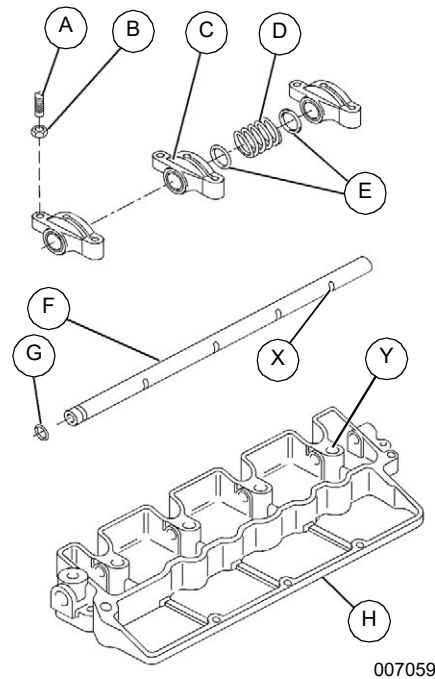


Figure 4-113. Disassemble Rocker Shaft and Pushrod

9. Install the rocker shaft and the pushrods. See [Rocker Shaft and Pushrod](#).

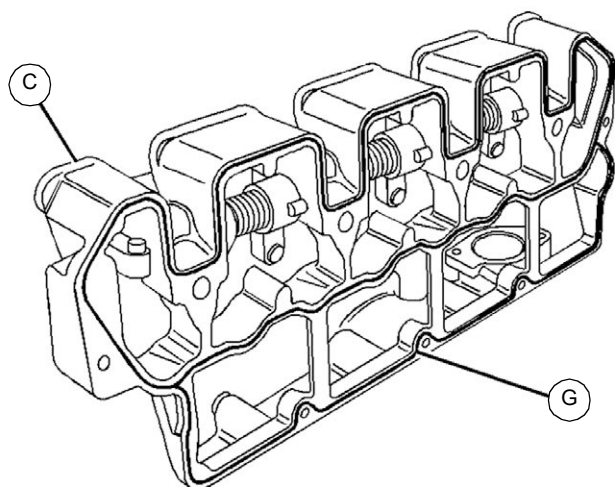
Installation Procedure

⚠ CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. Verify all components are clean and free from wear or damage. See [Rocker Shaft](#) for more information. Replace any components that are worn or damaged.
2. See [Figure 4-114](#). Install a new joint (G) to the base of rocker shaft assembly (C).



007058

Figure 4-114. Install Rocker Shaft and Pushrod

3. See [Figure 4-115](#). Apply clean engine oil to both ends of pushrods (F). Install the pushrods to the engine with the cup upward. Verify the pushrods are installed correctly in the socket of the valve lifters.

NOTE: If the pushrods have been used, verify they are installed in original positions.

4. Lubricate the top of the valve stems with clean engine oil. Install caps (E) to the valve stems.

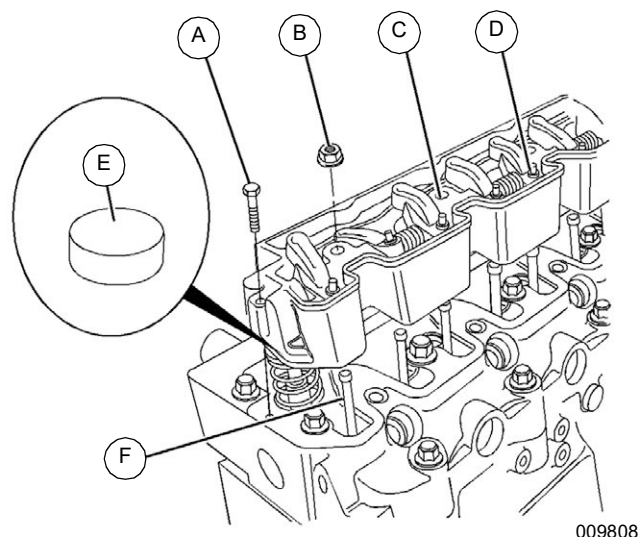
NOTE: If the caps have been used, verify they are installed in original positions.

5. Install rocker shaft assembly (C) to the cylinder head.

NOTE: Verify the ends of adjustment screws (D) are correctly seated in ends of pushrods (F).

6. Install nuts (B). Use a deep socket to evenly tighten the nuts. Begin at the center of the rocker shaft and work outward. Tighten nuts (B) to 24 ft-lbs (33 Nm).
7. Install bolts (A) and tighten to 88 in-lbs (10 Nm).

8. Adjust the valve lash. See [Inspect and Adjust Engine Valve Lash](#) for the correct procedure.



009808

Figure 4-115. Install Rocker Shaft and Pushrod

Cylinder Head

Removal Procedure

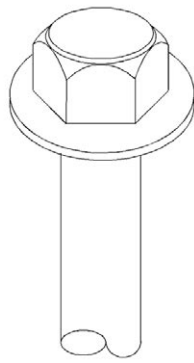
⚠ CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

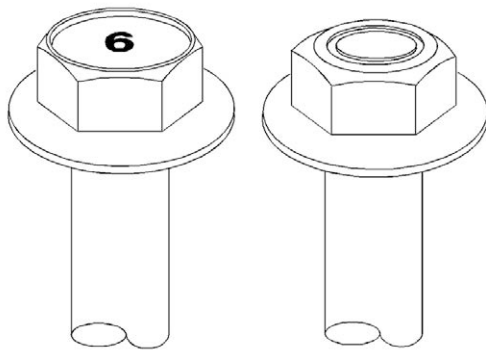
IMPORTANT NOTE: Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids. Dispose of all fluids according to local regulations and mandates.

NOTE: One of two types of cylinder head bolts can be installed. Type One bolts can be reused if the bolts are serviceable. See [Figure 4-116](#). Type Two bolts must be discarded and new bolts installed. See [Figure 4-117](#).



007089

Figure 4-116. Type One Cylinder Head Bolts



007088

Figure 4-117. Type Two Cylinder Head Bolts

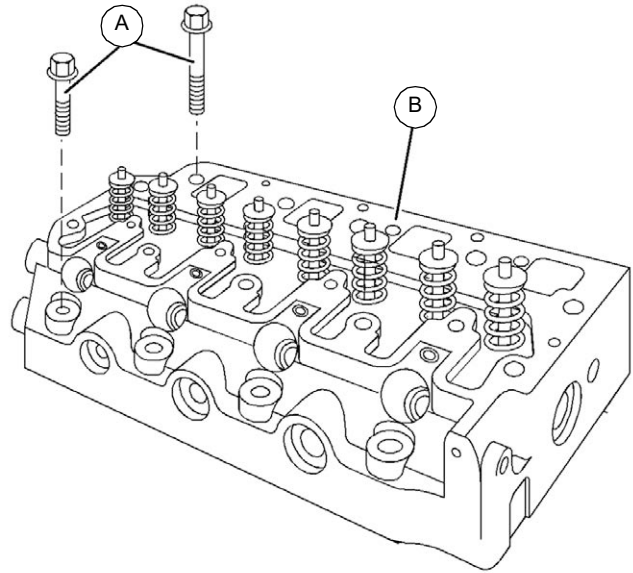
1. Remove the exhaust manifold. See [Exhaust Manifold](#) for the correct procedure.
2. Remove the fuel filter base. See [Fuel Filter Base](#) for the correct procedure.
3. Remove the fuel injectors. See [Fuel Injector](#) for the correct procedure.
4. Remove the glow plugs. See [Glow Plugs](#) for the correct procedure.
5. Remove the rocker shaft and the pushrods. See [Rocker Shaft and Pushrod](#).
6. Remove the water pump. See [Water Pump](#) for the correct procedure.
7. Drain the coolant from the cooling system into a suitable container for storage or for disposal. See [Change Coolant](#) or [Check Cooling System Coolant Level](#) for the correct draining procedure.
8. See [Figure 4-118](#). Gradually loosen bolts (A) in reverse numerical order to prevent distortion of the cylinder head.
9. Loosen the cylinder head bolts in the reverse order to the sequence shown in [Figure 4-119](#) and [Figure 4-120](#).
10. See [Figure 4-118](#). Remove bolts (A) from cylinder head (B).

NOTE: Type Two bolts must be discarded.

11. Attach a suitable lifting device to cylinder head (B). The weight of the cylinder head is approximately 66 lbs (30 kg). Carefully lift the cylinder head off the cylinder block.

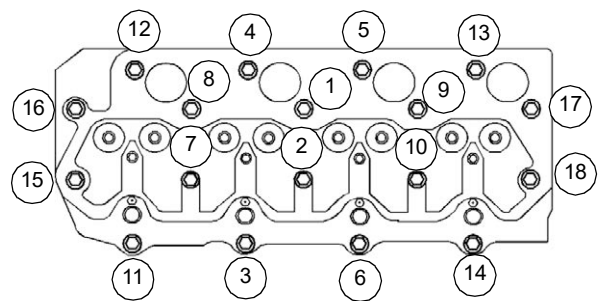
NOTE: Do not use a lever to separate the cylinder head from the cylinder block. Do not damage machined surfaces of the cylinder head during the removal procedure. Avoid contamination of the cylinder bores with coolant or with debris.

NOTE: Place the cylinder head on a surface that will not scratch the cylinder head face.



006874

Figure 4-118. Remove Bolts from Cylinder Head



006873

Figure 4-119. Cylinder Head Bolt Torque Sequence for D2.2L Engines

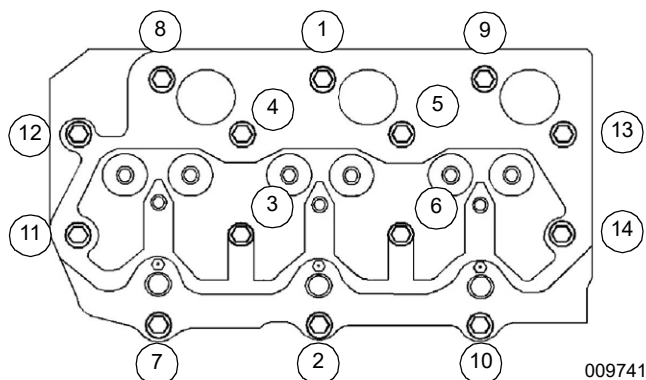


Figure 4-120. Cylinder Head Bolt Torque Sequence for D1.5L Engines

12. See [Figure 4-121](#). Note the position of dowels (C) in the cylinder block. Do not remove the dowels unless damaged.
13. If necessary, remove the water temperature regulator from the cylinder head. See [Water Temperature Regulator](#).

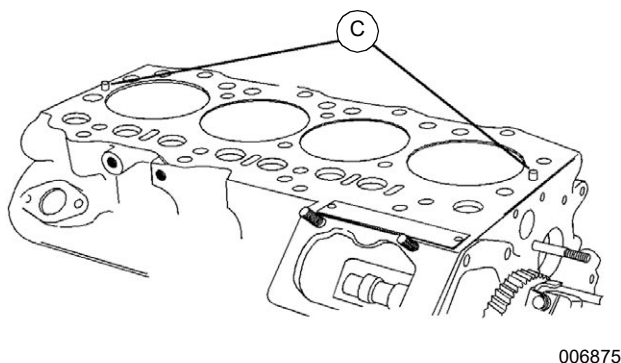


Figure 4-121. Remove Cylinder Head

Installation Procedure for Type One Cylinder Head Bolts

CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. Thoroughly clean the gasket surfaces of the cylinder head and the cylinder block. Do not damage the gasket surfaces of the cylinder head or the cylinder block. Verify no debris enters the cylinder bores, the coolant passages, or the lubricant passages.
2. Inspect the gasket surface of the cylinder head for distortion. See [Cylinder Head](#) for more information. If the mating surface of the cylinder head is distorted beyond maximum permitted limits, replace the cylinder head.

3. See [Figure 4-122](#), [Figure 4-123](#) and [Figure 4-124](#). Inspect dowels (C) for damage. If necessary, replace the dowels in the cylinder block.
4. Install guide bolt (A) to the cylinder block.
5. Verify the new cylinder head gasket (D) is the same thickness as the cylinder head gasket that was previously removed. The thickness of the cylinder head gasket is indicated by the number that is shown at position (X). See [Cylinder Head](#) for more information.
6. Align cylinder head gasket (D) with guide bolt (A) and with dowels (C). Install the cylinder head gasket onto the cylinder block.
7. Use a suitable lifting device to lift the cylinder head. The weight of the cylinder head is approximately 66 lbs (30 kg).
8. Use guide bolt (A) to align cylinder head (F) with the cylinder block. Install the cylinder head to the cylinder block.

NOTE: Verify the cylinder head is correctly positioned onto dowels (C).

9. Remove guide bolt (A).

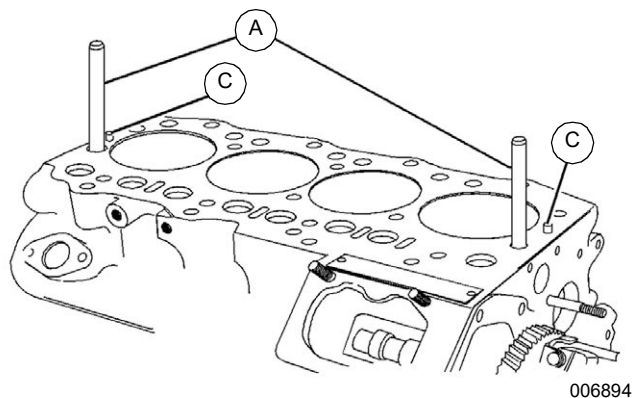


Figure 4-122. Install Cylinder Head

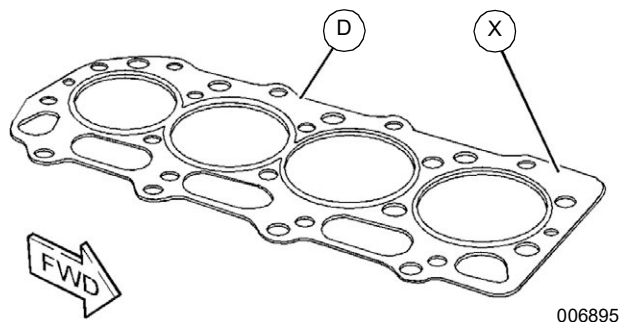
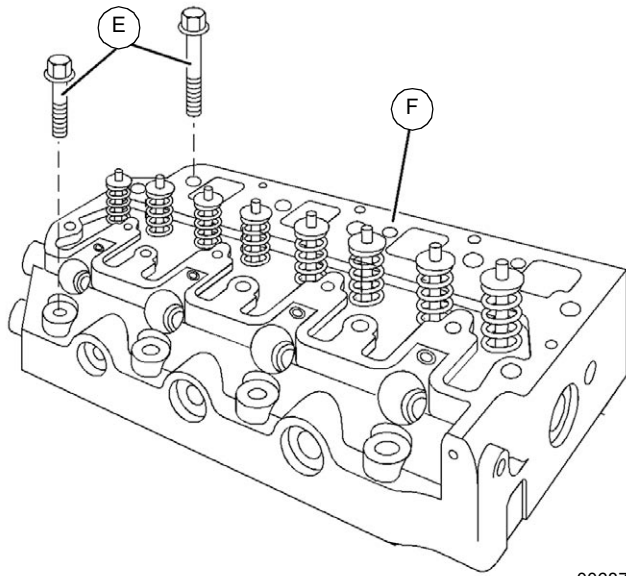


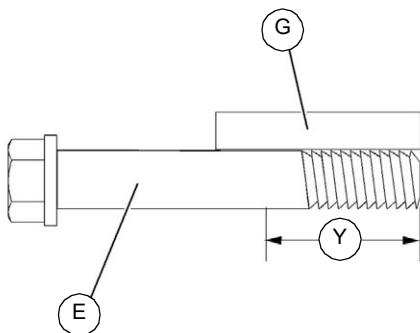
Figure 4-123. Install Cylinder Head



006874

Figure 4-124. Install Cylinder Head

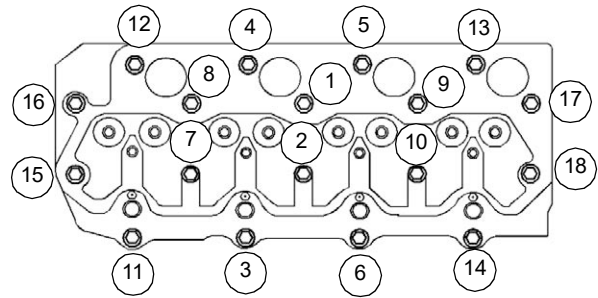
10. See [Figure 4-124](#) and [Figure 4-125](#). Clean Type One bolts (E). Follow Step 11 through Step 12 to inspect the Type One bolts.
11. Check the length of Type One bolts (E).
12. Use straight edge (G) to check the threads of bolts (E).
13. If any Type One bolts show visual reduction in the diameter of the thread over length (Y), replace the bolts with a new set of Type Two bolts. If Type One bolts are replaced with a new set of Type Two bolts, see [Installation Procedure for Type Two Cylinder Head Bolts](#).
14. Install bolts (G) to cylinder head (F).



006896

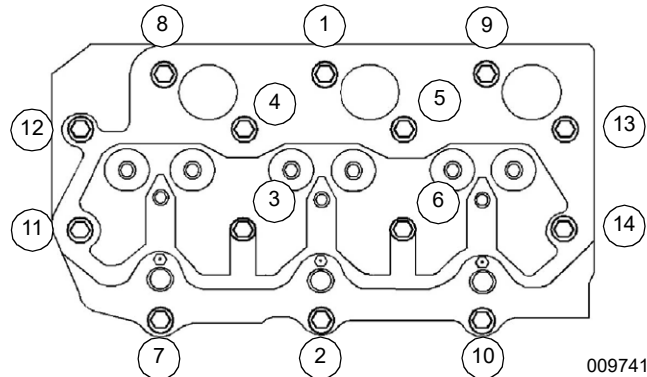
Figure 4-125. Install Cylinder Head

15. Tighten the bolts to 75 ft-lbs (101 Nm). Use the numerical sequence shown in [Figure 4-126](#) and [Figure 4-127](#).



006873

Figure 4-126. Cylinder Head Bolt Torque Sequence for D2.2L Engines



009741

Figure 4-127. Cylinder Head Bolt Torque Sequence for D1.5L Engines

16. If necessary, install the water temperature regulator to the cylinder head. See [Water Temperature Regulator](#).
17. Install the water pump. See [Water Pump](#).
18. Install the rocker shaft and the pushrods. See [Rocker Shaft and Pushrod](#).
19. Install the glow plugs. See [Glow Plugs](#).
20. Install the fuel injectors. See [Fuel Injector](#).
21. Install the fuel filter base. See [Fuel Filter Base](#).
22. Install the exhaust manifold. See [Exhaust Manifold](#).

Installation Procedure for Type Two Cylinder Head Bolts

CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. Thoroughly clean the gasket surfaces of the cylinder head and the cylinder block. Do not damage the gasket surfaces of the cylinder head or the cylinder block. Verify no debris enters the

cylinder bores, the coolant passages, or the lubricant passages.

2. Inspect the gasket surface of the cylinder head for distortion. See [Cylinder Head](#) for more information. If the mating surface of the cylinder head is distorted beyond maximum permitted limits, replace the cylinder head.
3. See [Figure 4-128](#), [Figure 4-129](#) and [Figure 4-130](#). Inspect dowels (C) for damage. If necessary, replace the dowels in the cylinder block.
4. Install guide bolt (A) to the cylinder block.
5. Verify the new cylinder head gasket (D) is the same thickness as the cylinder head gasket that was previously removed. The thickness of the cylinder head gasket is indicated by the number shown at Position (X). See [Cylinder Head](#) for more information.
6. Align cylinder head gasket (D) with guide bolt (A) and with dowels (C). Install the cylinder head gasket onto the cylinder block.
7. Use a suitable lifting device to lift the cylinder head. The weight of the cylinder head is approximately 66 lbs (30 kg).
8. Use guide bolt (A) to align cylinder head (C) with the cylinder block. Install the cylinder head to the cylinder block.

NOTE: Verify the cylinder head is correctly positioned onto dowels (C).

9. Remove guide bolt (A).
10. Install new bolts (E) to cylinder head (C).

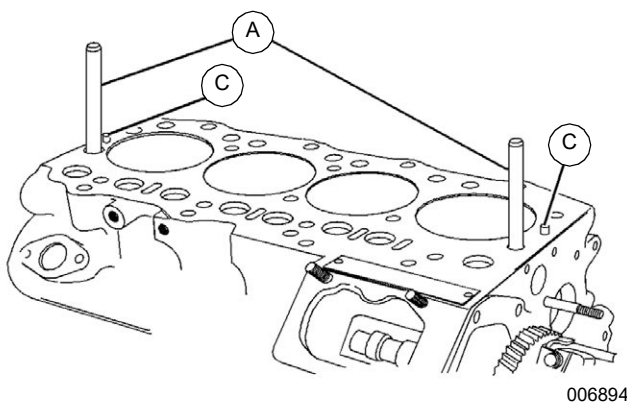


Figure 4-128. Install Cylinder Head

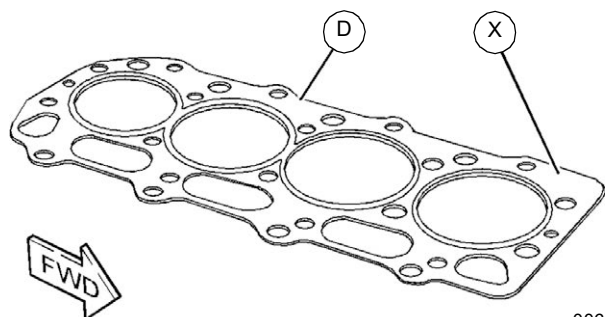


Figure 4-129. Install Cylinder Head

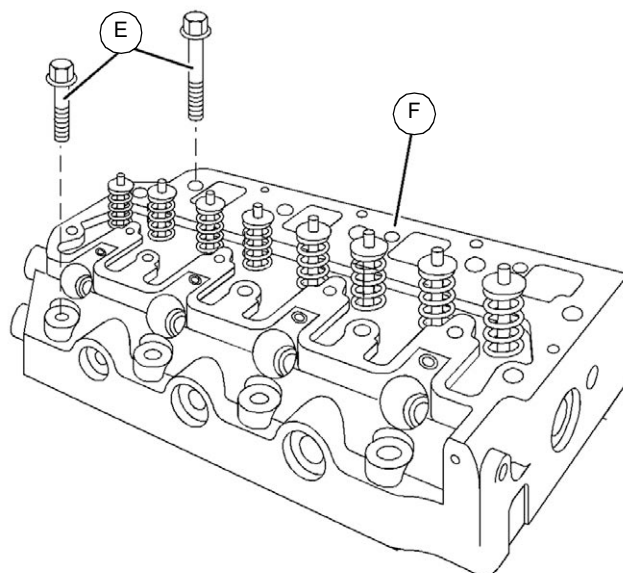
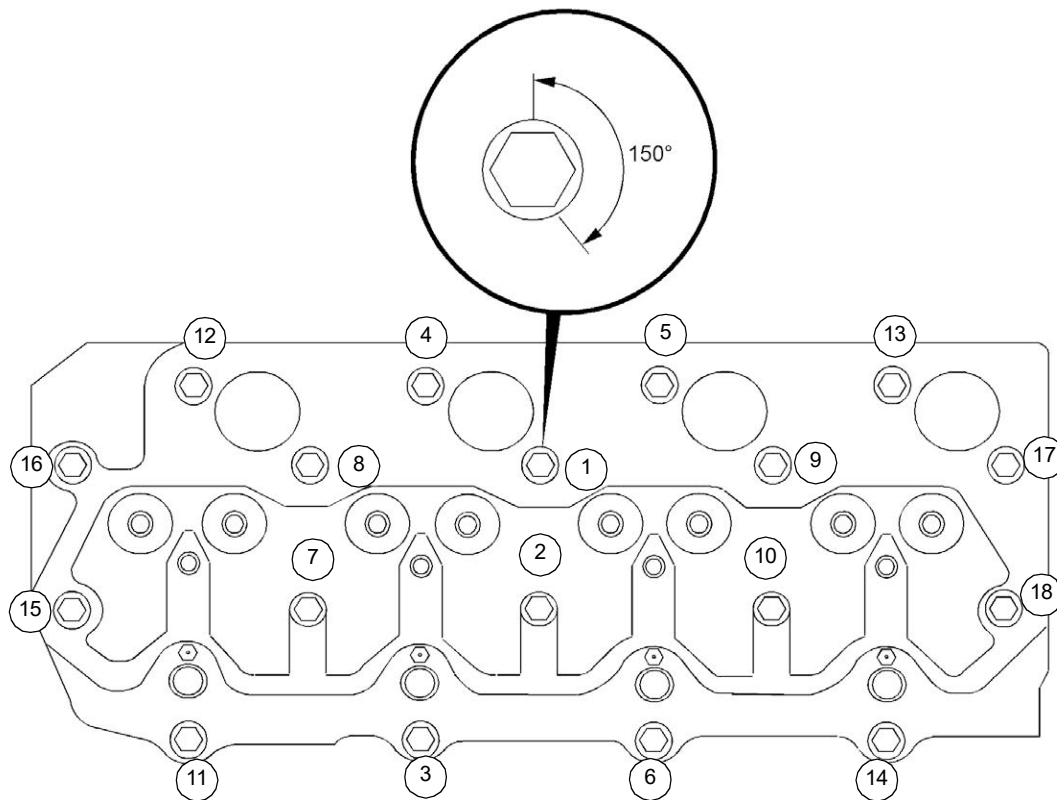


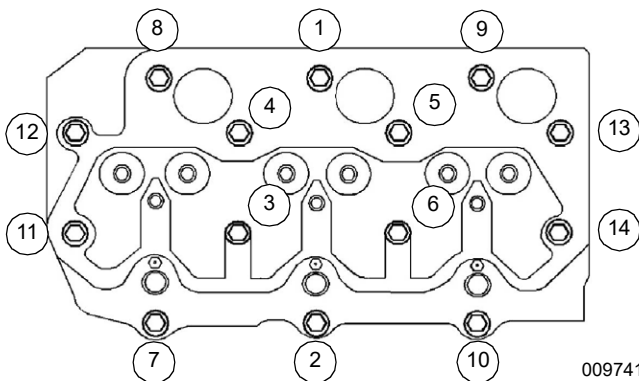
Figure 4-130. Install Cylinder Head

11. See [Figure 4-131](#) and [Figure 4-132](#). Tighten the bolts to 37 ft-lbs (50 Nm) using the numerical sequence shown. Use angle gauge (10000034194) to turn the bolts a further 150 degrees in the numerical sequence.
12. If necessary, install the water temperature regulator to the cylinder head. See [Water Temperature Regulator](#).



006897

Figure 4-131. Bolt Tightening Sequence for D2.2L Engines



009741

Figure 4-132. Bolt Tightening Sequence for D1.5L Engines

13. Install the water pump. See [Water Pump](#).
14. Install the rocker shaft and the pushrods. See [Rocker Shaft and Pushrod](#).
15. Install the glow plugs. See [Glow Plugs](#).
16. Install the fuel injectors. See [Fuel Injector](#).
17. Install the fuel filter base. See [Fuel Filter Base](#).
18. Install the exhaust manifold. See [Exhaust Manifold](#).

Lifter Group

Removal Procedure

⚠ CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

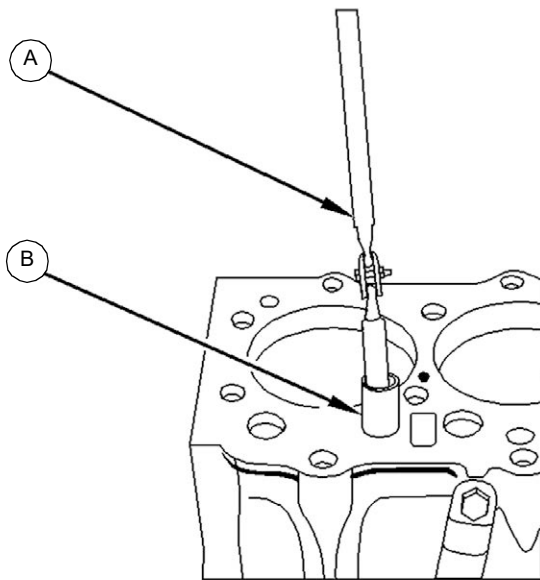
(000357)

IMPORTANT NOTE: Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids. Dispose of all fluids according to local regulations and mandates.

1. Remove the cylinder head. See [Cylinder Head](#).
2. See [Figure 4-133](#). Use telescopic magnet (A) to remove lifter (B) from the cylinder block.

NOTE: Make a temporary identification mark on each lifter to identify the correct location.

3. Repeat Step 2 for the remaining lifters.



007006

Figure 4-133. Remove Lifters

Installation Procedure

⚠ CAUTION

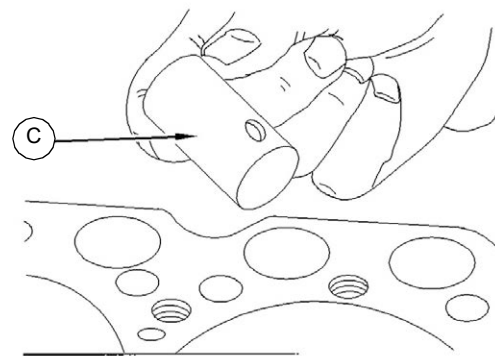
Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

NOTE: It is strongly recommended that all lifters be replaced when a new camshaft is installed.

1. See [Figure 4-134](#). Clean lifters (C). Follow Steps 2 through 4 to inspect the lifters. Replace lifters that are worn or damaged.
2. Inspect the seat of the pushrod in the lifter for wear or damage. Verify oil holes in the lifter are not restricted.
3. Inspect the shank of the lifter for wear or damage. See [Lifter Group](#) for more information.
4. Inspect the face of the lifter that runs on the camshaft for visual wear or damage.
5. Lubricate lifters (C) with clean engine oil.
6. Install lifters (C) into the cylinder block.

NOTE: The lifters should be free to rotate. If the lifters have been used, verify the lifters are installed in the original positions.



007005

Figure 4-134. Install Lifters

7. Install the cylinder head. See [Cylinder Head](#).

Camshaft

Removal Procedure

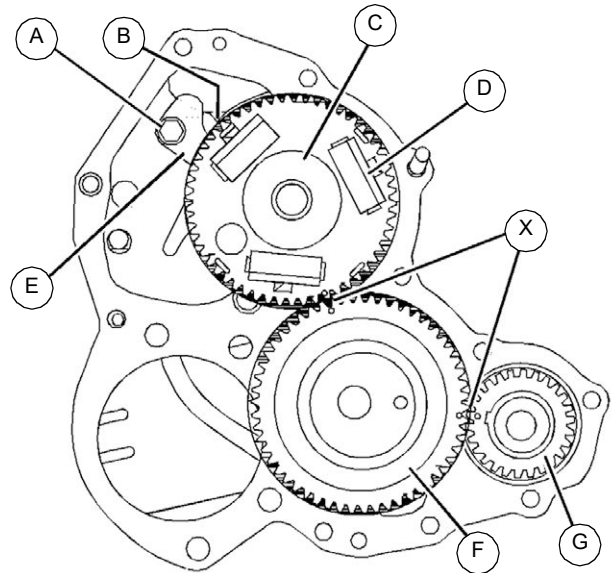


Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

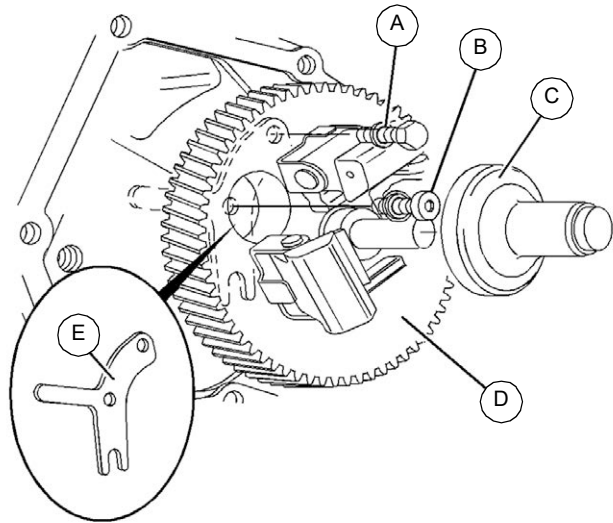
IMPORTANT NOTE: Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids. Dispose of all fluids according to local regulations and mandates.

1. Remove the lifters. See [Lifter Group](#).
2. Remove the front housing. See [Housing \(Front\)](#).
3. If the engine is equipped with a mechanical fuel transfer pump, remove the fuel transfer pump. See [Fuel Transfer Pump \(Mechanical Fuel Transfer Pump\)](#).
4. See [Figure 4-135](#), [Figure 4-136](#) and [Figure 4-137](#). Remove slider (C) from camshaft gear (D).
5. Before any rotation of gears is performed to align timing marks (X), remove the following components from the engine oil pump:
 - C-clip (H)
 - Collar (I)
 - Spring (J)
 - Shims (K)
 - Oil pump cover (L)
6. Remove the engine oil pump components to prevent damage to the oil pump idler hub. See [Engine Oil Pump](#) for the correct procedure.
7. Rotate camshaft gear (D) to align the access hole in the camshaft gear with fastener (B).
8. Remove bolt (B). Rotate camshaft gear (D) to align the access hole in the camshaft gear with bolt (A). Remove bolt (A).
9. Remove camshaft retainer (E).
10. Rotate the crankshaft until timing marks (X) are aligned on the following gears:
 - Crankshaft gear (G)
 - Camshaft gear (D)
 - Idler gear (F)



006830

Figure 4-135. Remove Slider from Camshaft Gear



006831

Figure 4-136. Remove Slider from Camshaft Gear

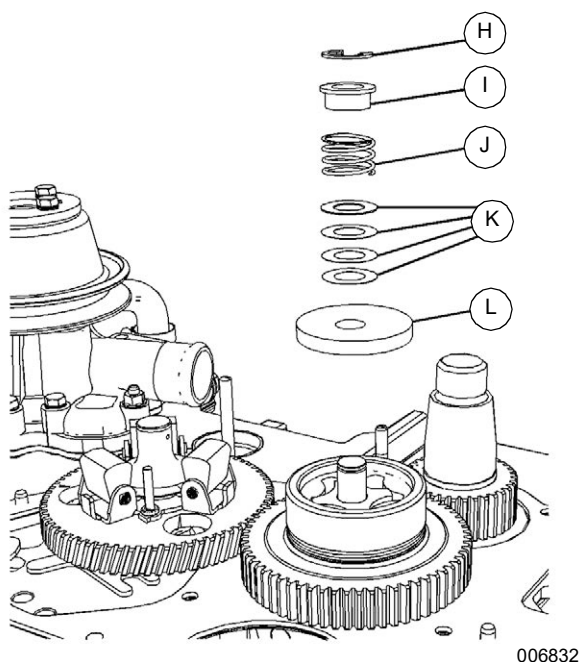


Figure 4-137. Remove Slider from Camshaft Gear

11. See **Figure 4-138**. Carefully remove camshaft assembly (M) from the cylinder block.

NOTE: Verify the lobes of the camshaft and the camshaft bearings are not damaged.

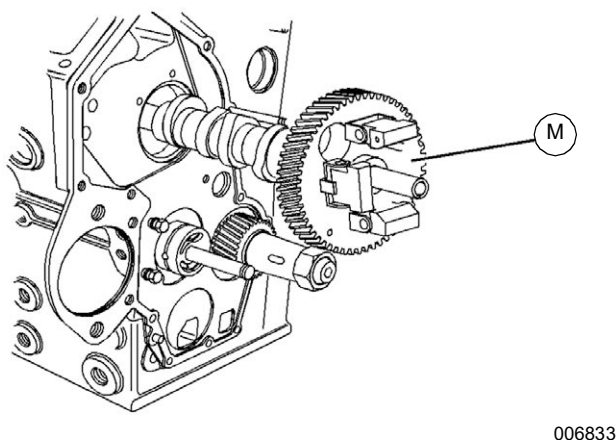


Figure 4-138. Remove Camshaft Assembly

Disassembly Procedure



Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. Remove the camshaft assembly. See **Camshaft**.

2. See **Figure 4-139**. Use combination puller to remove gear (E) from camshaft (A).

NOTE: The gear should be positioned on a suitable support to prevent damage to the governor flyweights during disassembly.

3. Remove spacer (D) and woodruff key (B) from camshaft (A).
4. Use combination puller to remove bearing (C) from camshaft (A).

NOTE: Identify the orientation of the bearing for installation.

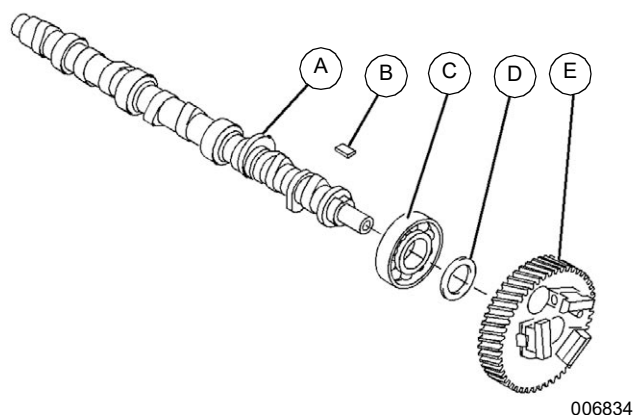


Figure 4-139. Remove Gear from Camshaft

Assembly Procedure



Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. Verify all components of the camshaft assembly are clean and free from damage.
2. Lubricate the nose of camshaft (A) with clean engine oil. Use a suitable press to install bearing (C) to the camshaft.

NOTE: Verify the bearing is installed in the correct orientation. The camshaft bearing should be pressed squarely onto the camshaft or damage to the bearing may occur. Do not press on the outer face of the bearing.

3. Install spacer (D) and woodruff key (B) to camshaft (A).
4. Align gear (E) with woodruff key (B). Use a suitable press to install the gear to the nose of camshaft (A).

NOTE: The gear should be positioned on a suitable support to prevent damage to the governor flyweights during installation.

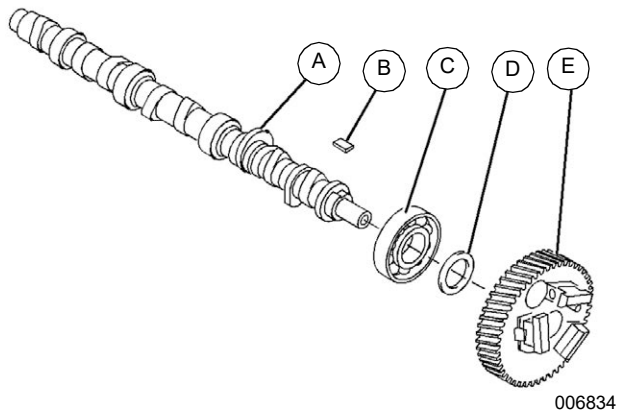


Figure 4-140. Assemble Gear to Camshaft

5. Install the camshaft assembly. See [Camshaft](#).

Installation Procedure



Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. Verify the assembly of the camshaft is clean and free from damage.
2. Lubricate the bearings of the camshaft with clean engine oil.
3. See [Figure 4-141](#), [Figure 4-142](#), [Figure 4-143](#) and [Figure 4-144](#). Carefully install camshaft assembly (M) into the cylinder block. Verify timing marks (X) are aligned on the following gears:
 - Crankshaft gear (G)
 - Camshaft gear (D)
 - Idler gear (F)

NOTE: Do not damage the lobes of the camshaft or the camshaft bearings.

4. Place camshaft retainer (E) in position. Align the holes in the retainer with the holes in the cylinder block.
5. Rotate camshaft gear (D) to align the access hole in the camshaft gear with the hole for fastener (B).
6. Install bolt (B) and tighten to 89 **in-lbs** (10 Nm). Rotate camshaft gear (D) to align the access hole in the camshaft gear with the hole for bolt (A). Install bolt (A) and tighten to 89 **in-lbs** (10 Nm).
7. After the rotation of gears is performed and timing marks (X) are aligned, install the following components to the engine oil pump:
 - Spring (J)
 - Shims (K)
 - Oil pump cover (L)

8. See [Engine Oil Pump](#).
9. Install slider (C) to camshaft gear (D).
10. If the engine is equipped with a mechanical fuel transfer pump, install the fuel transfer pump. See [Fuel Transfer Pump \(Mechanical Fuel Transfer Pump\)](#).

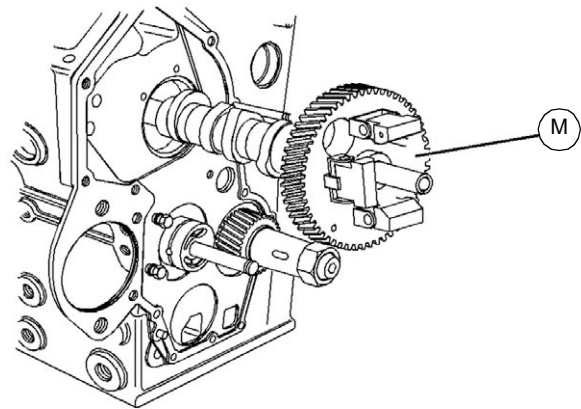


Figure 4-141. Install Camshaft Assembly

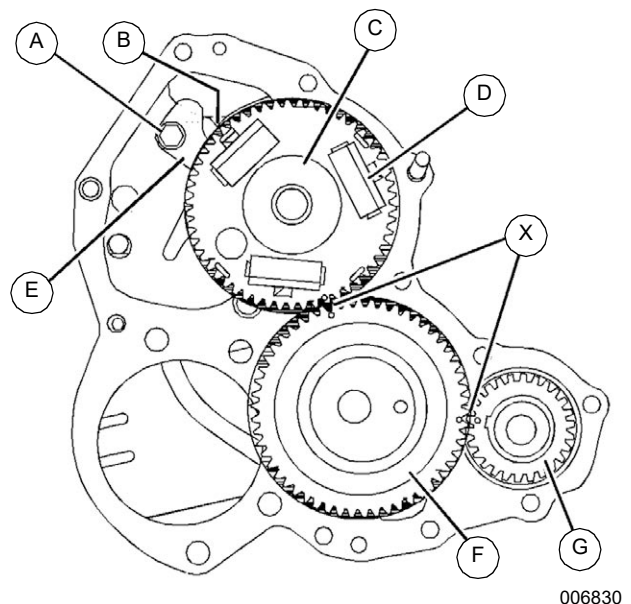
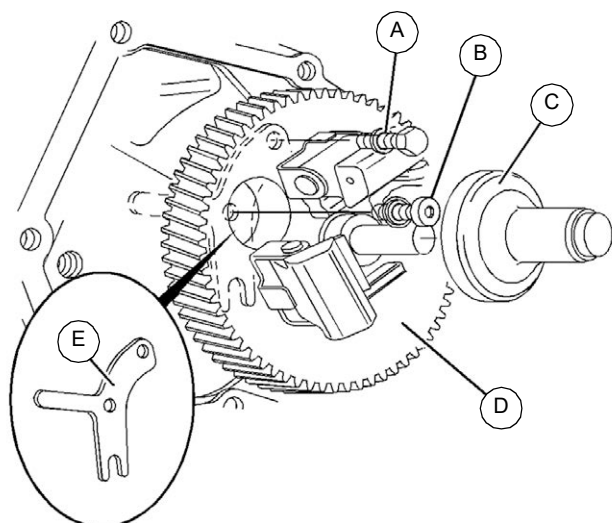
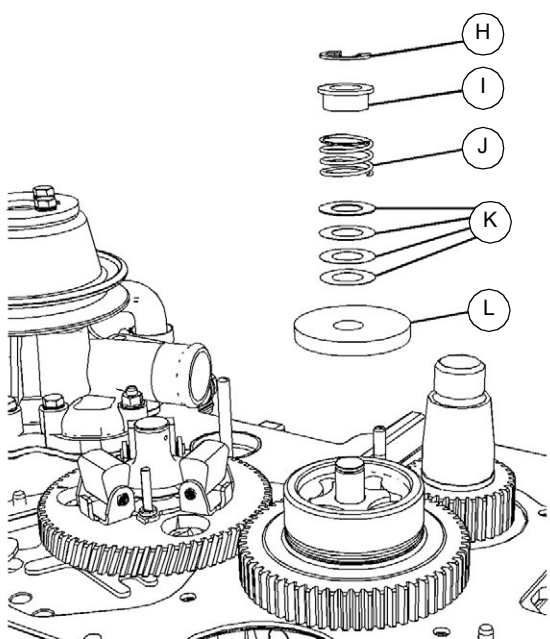


Figure 4-142. Install Camshaft Assembly



006831

Figure 4-143. Install Camshaft Assembly



006832

Figure 4-144. Install Slider to Camshaft Gear

11. Install the lifters. See [Lifter Group](#).
12. Install the front housing. See [Housing \(Front\)](#).

Engine Oil Pan

Removal Procedure

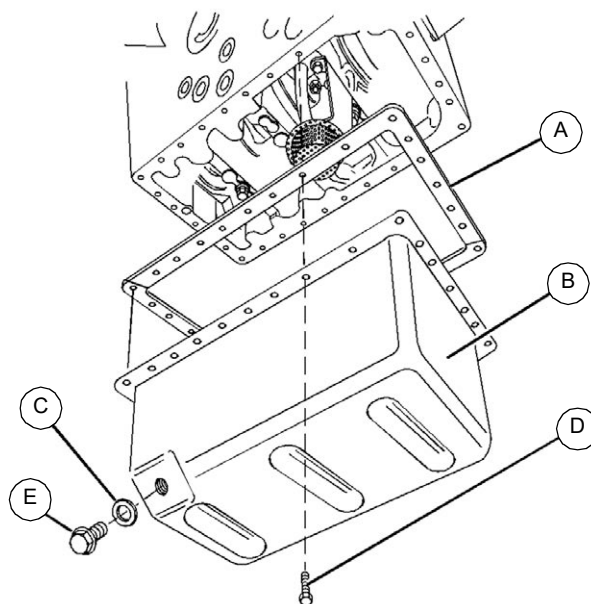


Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

IMPORTANT NOTE: Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids. Dispose of all fluids according to local regulations and mandates.

1. See [Figure 4-145](#). Remove drain plugs (E) from engine oil pan (B). Drain the engine oil into a suitable container for storage or disposal. [Change Engine Oil and Filter](#).
2. Remove sealing washers (C) from drain plugs (E).
3. Remove bolts (D) from engine oil pan (B).
4. Remove engine oil pan (B) from the engine.
5. Remove joint (A).



007012

Figure 4-145. Remove Engine Oil Pan

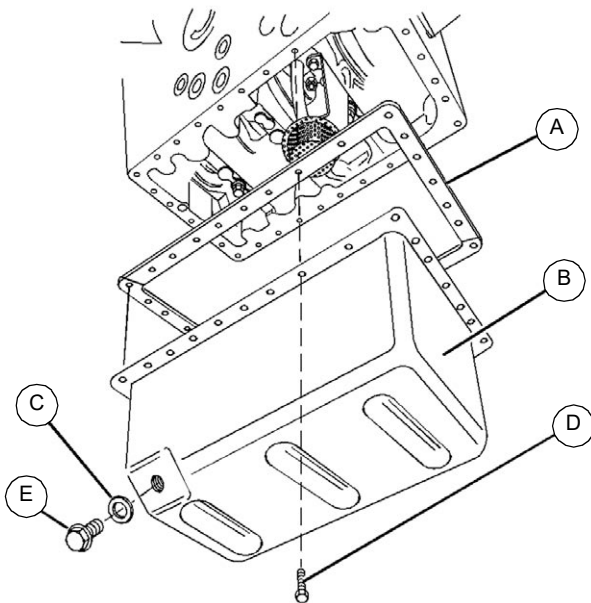
Installation Procedure



Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. Verify the engine oil pan is clean and free from damage. Clean the mating surface of the cylinder block.
2. See [Figure 4-146](#). Install a new joint (A) and engine oil pan (B) to the engine.
3. Install bolts (D) and tighten to 97 in-lbs (11 Nm).
4. Install drain plugs (E) and sealing washers (C) to oil pan (B). Tighten the drain plugs to 25 ft-lbs (34 Nm).
5. Fill the engine oil pan to the correct level. See [Change Engine Oil and Filter](#) for the correct procedure.



007012

Figure 4-146. Install Engine Oil Pan

Pistons and Connecting Rods

Removal Procedure



Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. Remove the cylinder head. See [Cylinder Head](#).

2. Remove the suction pipe. See [Engine Oil Pump](#).
3. Position the piston being removed in the bottom center position.
4. Use ridge reamer to remove the carbon from the inner surface of the cylinder bore.
5. See [Figure 4-147](#). The connecting rod and the connecting rod cap should have an etched number (X) on the side. The number on the connecting rod and the connecting rod cap must match. Verify connecting rod (D) and connecting rod cap (B) are marked for the correct location. If necessary, make a temporary mark on the connecting rod and the connecting rod cap to identify the cylinder number.

NOTE: Do not stamp the connecting rod assembly.

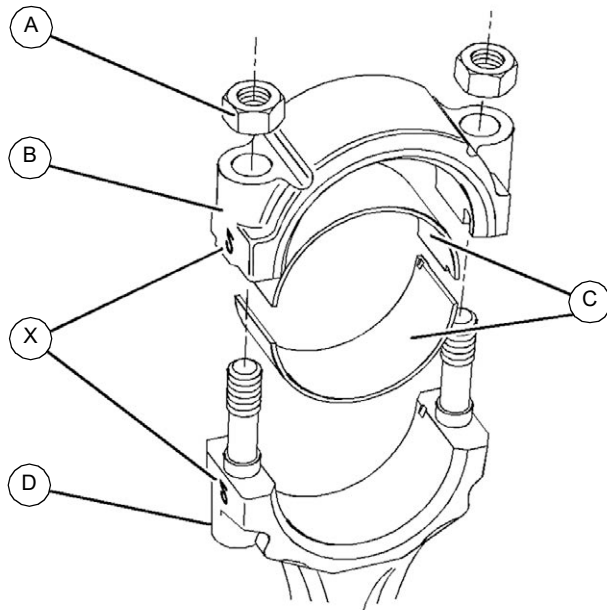
6. Remove nuts (A) and remove connecting rod cap (B) from connecting rod (D).

NOTE: Use tape or rubber tubing on the threads of the connecting rod bolts to protect the crankshaft journals. The sharp edges of the connecting rod bolts could damage the crankshaft journals.

7. See [Figure 4-148](#). Carefully push piston (E) and the connecting rod assembly out of the cylinder bore. Lift the piston out of the top of the cylinder block.

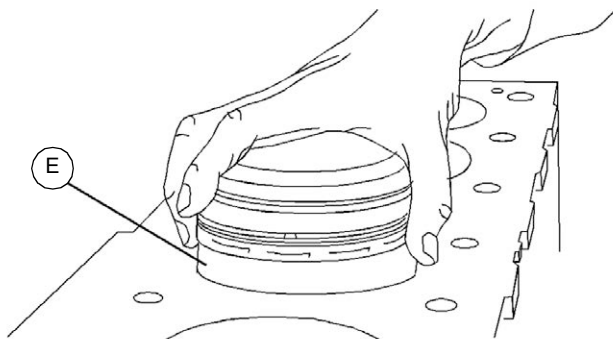
NOTE: Make an identification mark underneath the piston on the pin boss to identify the position on the piston for installation. Always mark the front pin boss.

8. See [Figure 4-147](#). Keep connecting rod bearings (C) with the respective connecting rod (D) and cap (B).
9. Repeat Step 3 through Step 8 to remove the remaining pistons and connecting rods.



007029

Figure 4-147. Remove Pistons and Connecting Rods



007027

Figure 4-148. Remove Pistons and Connecting Rods

Disassembly Procedure

CAUTION

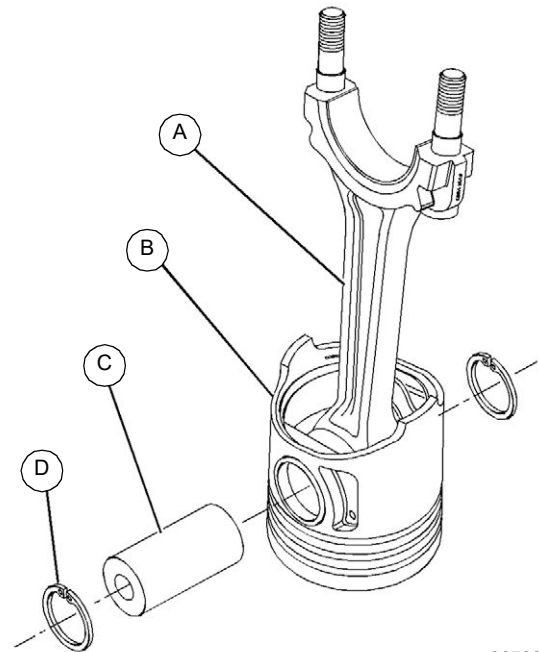
Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. Remove the pistons and connecting rods. See [Pistons and Connecting Rods](#).
2. Make a temporary mark on the components of the piston and connecting rod assembly. This will ensure the components of each piston and connecting rod assembly can be reinstalled in the original cylinder. Mark the front of the piston and the front of the connecting rod. Do not interchange components.

3. See [Figure 4-149](#). Place the piston and connecting rod assembly on a suitable surface with the connecting rod upward. Use retaining ring pliers to remove retaining rings (D).
4. Remove piston pin (C) and connecting rod (A) from piston (B).

NOTE: If the piston pin cannot be removed by hand, heat the piston to a temperature of $113 \pm 9^\circ\text{F}$ ($45 \pm 5^\circ\text{C}$). Do not use a torch to heat the piston.

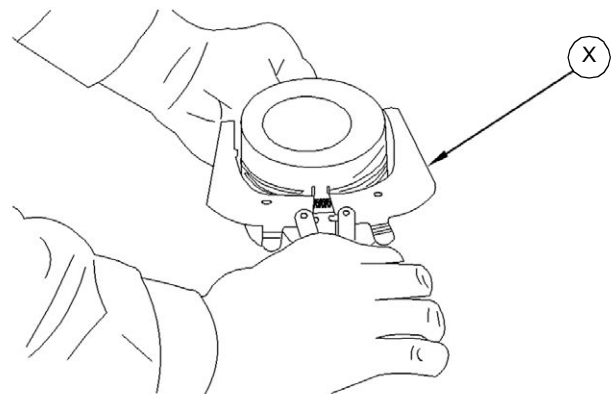


007028

Figure 4-149. Disassemble Pistons and Connecting Rods

5. See [Figure 4-150](#) and [Figure 4-151](#). Place the piston on a suitable surface with the crown upward. Use ring expander (X) to remove compression rings (E) and (F), and oil control ring (G) from piston (B).

NOTE: Identify the orientation of the piston rings for installation.



007021

Figure 4-150. Disassemble Pistons and Connecting Rods

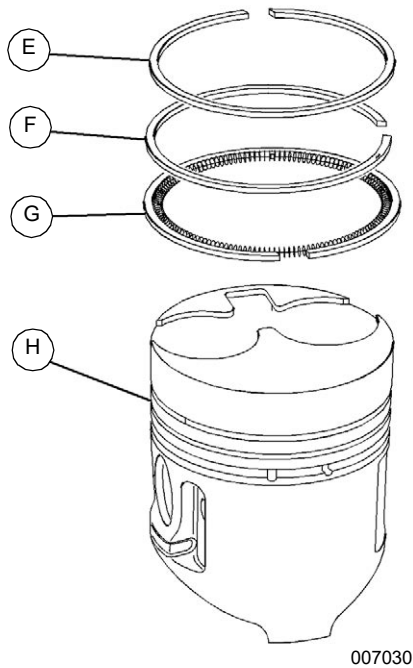


Figure 4-151. Disassemble Pistons and Connecting Rods

6. See [Figure 4-152](#). Remove nuts (K) and connecting rod cap (J) from connecting rod (A).
7. Remove the lower half of connecting rod bearing (I) from connecting rod cap (J). Remove the upper half of connecting rod bearing (H) from connecting rod (A). Keep the bearing shells together.

IMPORTANT NOTE: Removal of the piston pin bushing in the connecting rod must be carried out by personnel with the correct training. Special machinery is also required. For more information refer to your authorized Caterpillar dealer.

8. Inspect the connecting rod for wear or damage. If necessary, replace the connecting rod or replace the bushing for the piston pin.

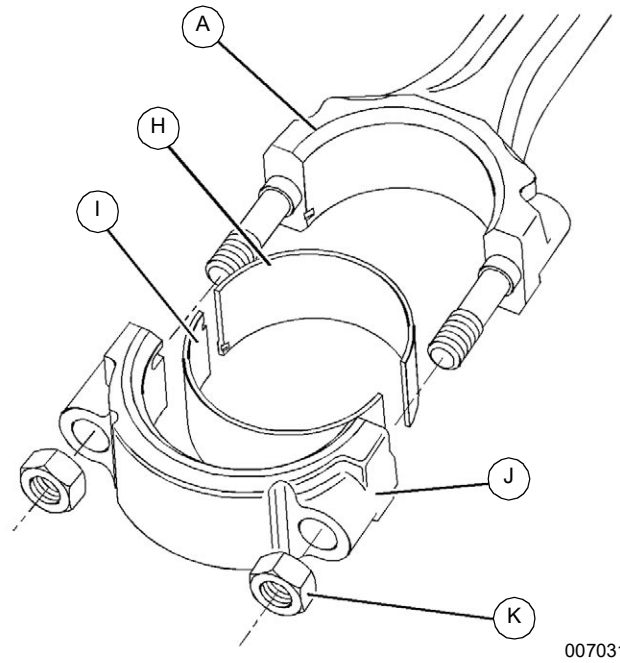


Figure 4-152. Disassemble Pistons and Connecting Rods

Assembly Procedure

CAUTION

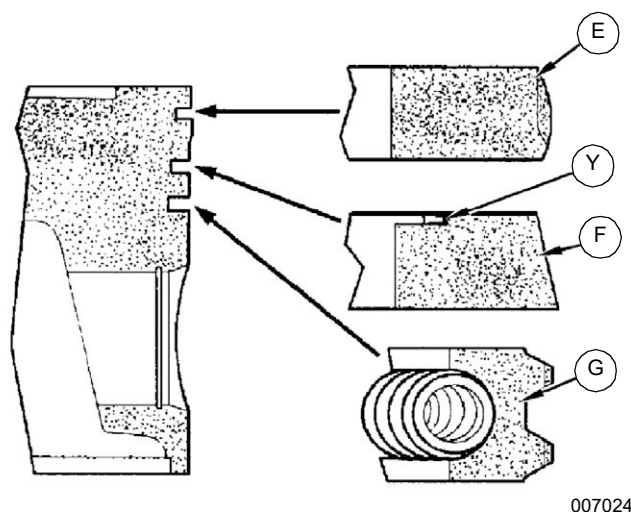
Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. Verify all components are clean and free from wear or damage. If necessary, replace any components that are worn or damaged.
2. Follow Step 3 through Step 7 to install the piston rings to the piston.
3. See [Figure 4-153](#) and [Figure 4-154](#). Position the spring for oil control ring (G) into the oil ring groove in the piston. The central wire must be located inside the end of the spring.
4. Use ring expander (X) to install oil control ring (G) over the spring.

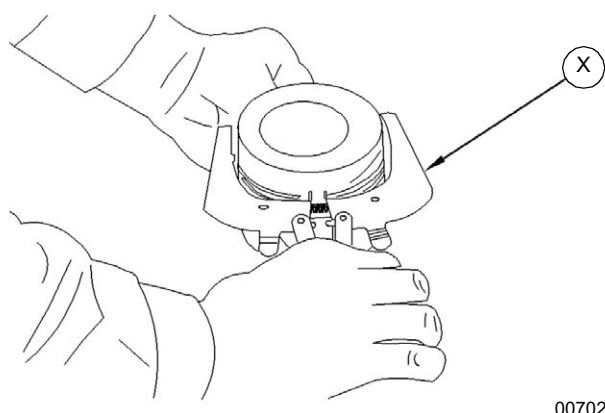
NOTE: Verify the central wire is 180 degrees from the ring gap.

5. Use ring expander (X) to install intermediate compression ring (F) into the second groove in the piston. The letter "T" and the chamfer (Y) on the inner face of the ring must be upward.
6. Use ring expander (X) to install top compression ring (E) into the top groove in the piston. The letter "T" must be upward.
7. Position the piston ring gaps at 120 degrees away from each other.



007024

Figure 4-153. Assemble Pistons and Connecting Rods



007021

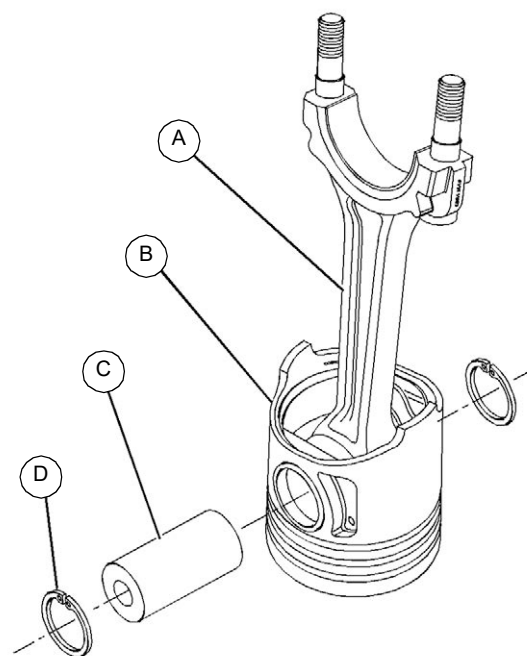
Figure 4-154. Assemble Pistons and Connecting Rods

8. See [Figure 4-155](#). Lubricate the bushing in connecting rod (A) and lubricate the bore for the piston pin (C) in piston (B) with clean engine oil.
9. Place the piston on a suitable surface with the crown downward. Install connecting rod (A) and piston pin (C) to piston (B). The name Shibaura inside the piston must align with the stamped number on the connecting rod. Verify correct orientation of the connecting rod in the piston.

NOTE: If the piston pin cannot be installed by hand, heat the piston to a temperature of $113 \pm 9^\circ\text{F}$ ($45 \pm 5^\circ\text{C}$). Do not use a torch to heat the piston.

10. Use retaining ring pliers to install retaining rings (D) to the piston pin bore in piston (B).

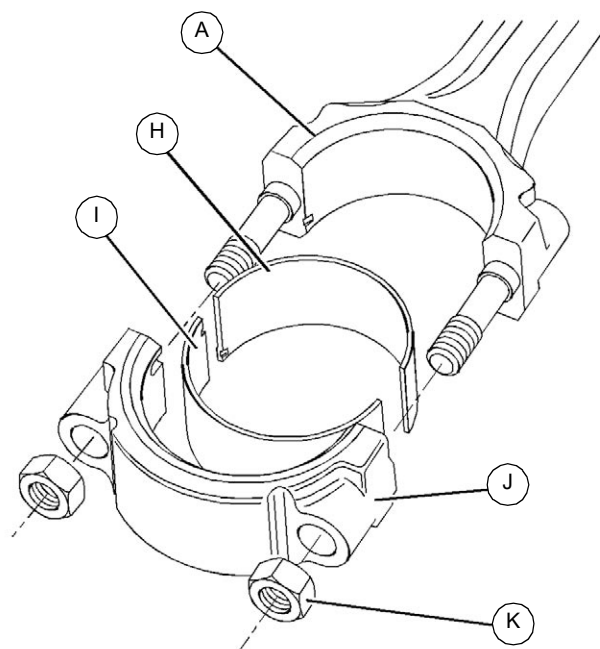
NOTE: Verify the retaining rings are seated in the grooves in the piston.



007028

Figure 4-155. Assemble Pistons and Connecting Rods

11. See [Figure 4-156](#). Install the upper half of connecting rod bearing (H) to connecting rod (A).
12. Install the lower half of connecting rod bearing (I) to connecting rod cap (J).



007031

Figure 4-156. Assemble Pistons and Connecting Rods

13. Install the pistons and the connecting rods. See [Pistons and Connecting Rods](#).

Installation Procedure

⚠ CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. If the connecting rod caps were temporarily installed, remove the connecting rod caps. If necessary, thoroughly clean all the components.
2. Apply clean engine oil to the cylinder bore, to the piston rings, to the outer surface of the piston and to the connecting rod bearings.

NOTE: Install the connecting rod bearings dry when clearance checks are performed. See [Bearing Clearance](#). Apply clean engine oil to the connecting rod bearings during final assembly.

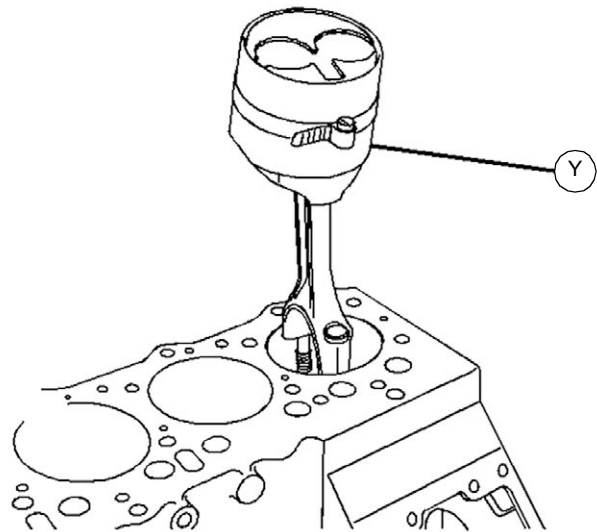
3. Rotate the crankshaft until the crankshaft pin is at the bottom center position. Lubricate the crankshaft pin with clean engine oil.
4. See [Figure 4-157](#) and [Figure 4-158](#). Verify the gaps for the piston rings are at 120 degrees away from each other. Install piston ring compressor (Y) onto piston (E). Use tape or rubber tubing on the connecting rod bolts to protect the crankshaft journals.

NOTE: Verify piston ring compressor (Y) is installed correctly and the piston can easily slide from the tool. Verify the piston and the connecting rod assembly are installed in the correct cylinder. Align number (X) on the side of the connecting rod to the right side of the cylinder block. The right-hand side is determined from the flywheel end of the engine.

5. Carefully push the piston and the connecting rod assembly into the cylinder bore and onto the crankshaft pin.

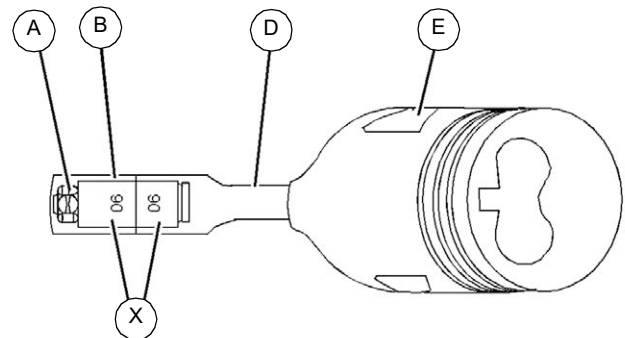
NOTE: Do not damage the finished surface of the crankshaft pin.

6. Install connecting rod cap (B) onto connecting rod (D). Verify number (X) on the connecting rod cap matches number (X) on the connecting rod. Install nuts (A).
7. Tighten nuts (A) to 38 ft-lbs (52 Nm).
8. Repeat Steps 1 through 7 for the remaining pistons and connecting rods.
9. Verify the installed connecting rod assembly has tactile side play. Carefully rotate the crankshaft to ensure there is no binding.



007025

Figure 4-157. Install Pistons and Connecting Rods



007026

Figure 4-158. Install Pistons and Connecting Rods

10. Install the suction pipe. See [Engine Oil Pump](#).
11. Install the cylinder head. See [Cylinder Head](#).

Connecting Rod Bearings

Removal Procedure

⚠ CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

NOTE: If all connecting rod bearings require replacement, the procedure can be carried out on two cylinders at the same time. The procedure can be carried out on the following pairs of cylinders. 1 with 4 and 2 with 3. Verify both pairs of the connecting rod bearings are installed before changing from one pair of cylinders to

another pair of cylinders. See [Connecting Rod Bearings](#).

1. Remove the suction pipe. See [Engine Oil Pump](#).

NOTE: See [Figure 4-159](#). The connecting rod and the connecting rod cap should have matching numbers at position (X). If necessary, make a temporary mark on connecting rod (A) and on connecting rod cap (E).

2. Rotate the crankshaft until the piston is at the bottom center position. Remove nuts (F) and remove connecting rod cap (E) from connecting rod (A).
3. Remove lower connecting rod bearing (D) from connecting rod cap (E).
4. Carefully push connecting rod (A) into the cylinder bore. Remove upper connecting rod bearing (C) from the connecting rod.

NOTE: Use tape or rubber tubing on connecting rod bolts (B) to protect the crankshaft journals. The sharp edges of the connecting rod bolts could damage the crankshaft journals.

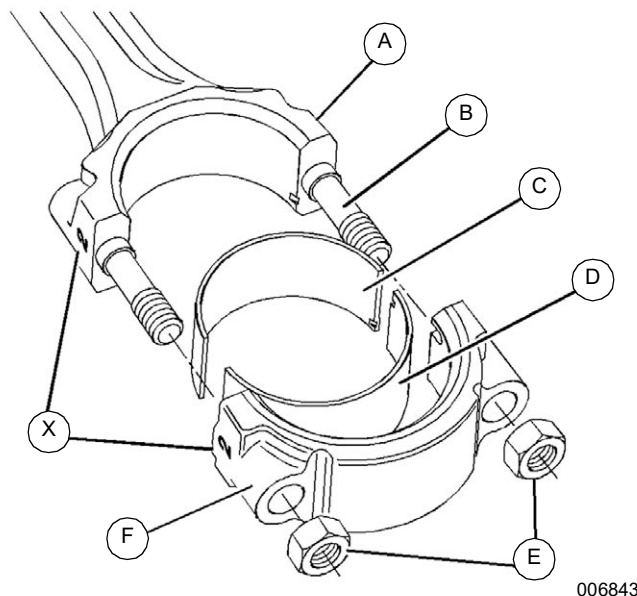


Figure 4-159. Remove Connecting Rod Bearings

connecting rod bearings are clean and free from wear or damage. If necessary, replace the connecting rod bearings.

2. See [Figure 4-160](#). Clean the bearing surface of connecting rod (A) and connecting rod cap (E). Verify number (X) on connecting rod cap (E) aligns with number (X) on connecting rod (A).
3. Install upper connecting rod bearing (C) to connecting rod (A). Lubricate the bearing surface of the connecting rod bearing with clean engine oil.
4. Carefully pull connecting rod (A) against the crankshaft pin.

NOTE: Use tape or rubber tubing on connecting rod bolts (B) to protect the crankshaft journals. The sharp edges of the connecting rod bolts could damage the crankshaft journals.

5. Clean the connecting rod cap. Install lower connecting rod bearing (D) to connecting rod cap (E).
6. Lubricate the pin of the crankshaft and lubricate lower connecting rod bearing (C) with clean engine oil.

NOTE: When the connecting rod caps are installed, verify the identification marks are aligned.

7. Install connecting rod cap (E).
8. Install nuts (D).
9. Tighten nuts (D) to 38 ft-lbs (52 Nm).
10. Verify the installed connecting rod assembly has tactile side play. Carefully rotate the crankshaft to verify there is no binding.

NOTE: If all connecting rod bearings require replacement, the procedure can be carried out on two cylinders at the same time. The procedure can be carried out on the following pairs of cylinders. 1 with 4 and 2 with 3. Verify both pairs of the connecting rod bearings are installed before changing from one pair of cylinders to another pair of cylinders. See [Connecting Rod Bearings](#).

Installation Procedure



Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. Inspect the pins of the crankshaft for damage. If the crankshaft is damaged, replace or recondition the crankshaft. See [Crankshaft](#). Verify the

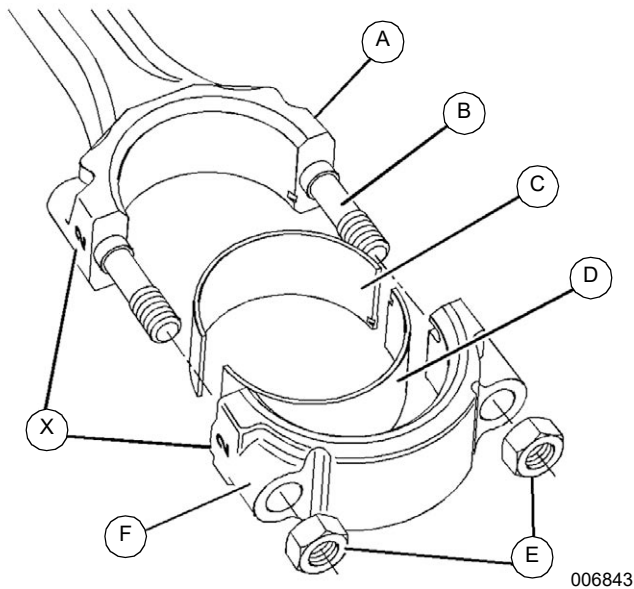


Figure 4-160. Install Connecting Rod Bearings

11. Install the suction pipe. See [Engine Oil Pump](#).

Crankshaft Main Bearings

Removal Procedure



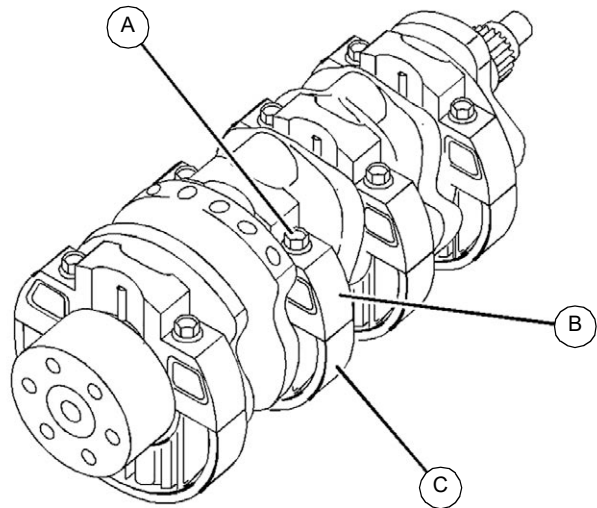
Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. Remove the crankshaft assembly. See [Crankshaft](#).

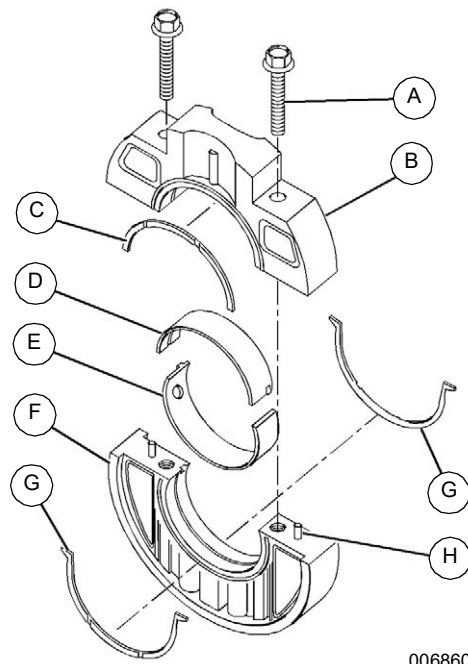
Crankshaft Main Bearings

1. Verify the bearing caps are marked for orientation and the correct position.
2. See [Figure 4-161](#) and [Figure 4-162](#). Remove bolts (A) and upper main bearing cap (B) from lower main bearing cap (F).
3. For D1.5L engines, remove thrust washers (G).
4. For D2.2L engines, remove thrust washers (C) and (G).
5. Remove main bearings (D) and (E) from the main bearing caps. Keep the main bearings with the respective bearing caps.
6. Do not remove dowels (H) from the main bearing caps.



006862

Figure 4-161. Remove Crankshaft Main Bearings

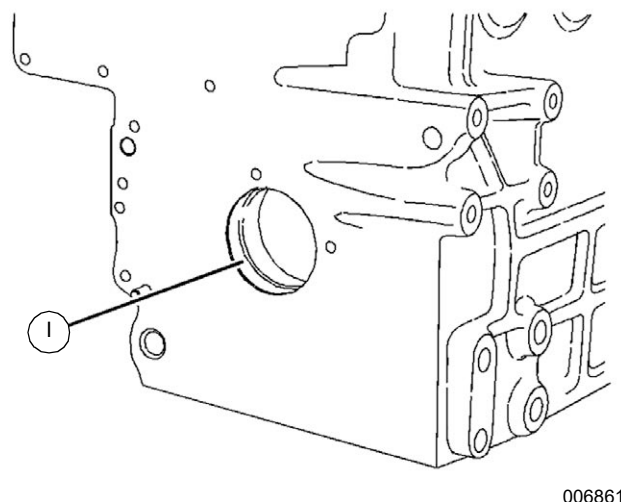


006860

Figure 4-162. Remove Crankshaft Main Bearings

Crankshaft Bearing (Front)

1. See [Figure 4-163](#). Use driver to remove crankshaft bearing (I) from the cylinder block.



006861

Figure 4-163. Remove Crankshaft Bearing (Front)

Installation Procedure

Crankshaft Main Bearings

1. Clean the crankshaft and inspect the crankshaft for wear or damage. See [Crankshaft](#) for more information.
2. Clean the main bearings, main bearing caps and thrust washers. Inspect all components for wear or damage.
3. See [Figure 4-164](#) and [Figure 4-165](#). Install upper main bearing (D) to upper main bearing cap (B). Verify the locating tab for the main bearing is seated in the slot in the main bearing cap.

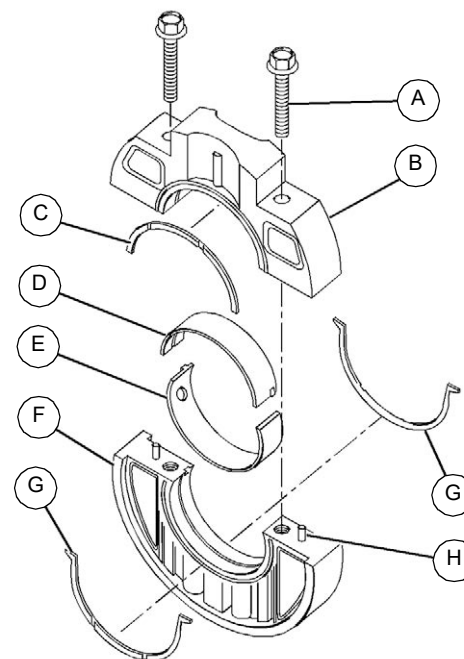
NOTE: The upper bearing has a groove.

4. Install lower main bearing (E) to lower main bearing cap (F). Verify the locating tab for the main bearing is seated in the slot in the main bearing cap.

NOTE: The lower bearing is a plain bearing that has oil holes.

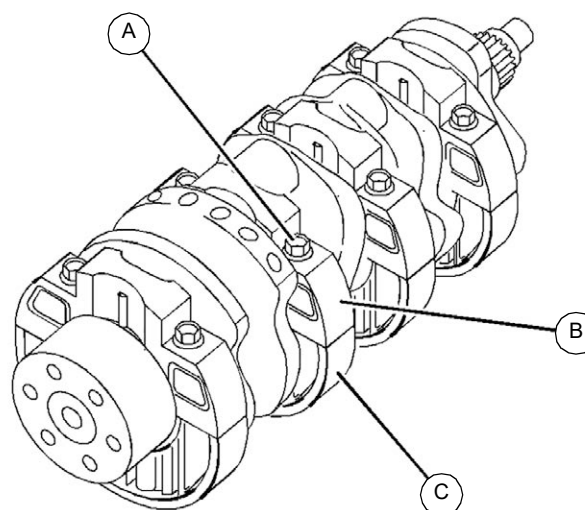
5. Lubricate main bearings (D) and (E) with clean engine oil.
6. Position the upper half of main bearing cap (B) in position on the crankshaft. Position the lower half of main bearing cap (F) in position on the crankshaft. Verify the correct location and orientation of the bearing caps. The locating tabs for the upper and the lower bearings should be on the same side of the engine. Verify dowels (H) are in the correct position. This will ensure the two halves of the main bearing caps are aligned.
7. Install thrust washers (C) and (G).

NOTE: Verify the thrust washers are aligned correctly and the oil grooves are facing the crankshaft.



006860

Figure 4-164. Install Crankshaft Main Bearings



006862

Figure 4-165. Install Crankshaft Main Bearings

Crankshaft Bearing (Front)

1. See [Figure 4-166](#). Use driver (A) to install crankshaft bearing (I) to the cylinder block. Follow Steps 2 through 4 to install the crankshaft bearing.
2. Verify oil supply hole (L) is aligned with the oil gallery in the cylinder block.
3. Verify joint (J) is upward.
4. Verify chamfer (K) faces toward the cylinder block.

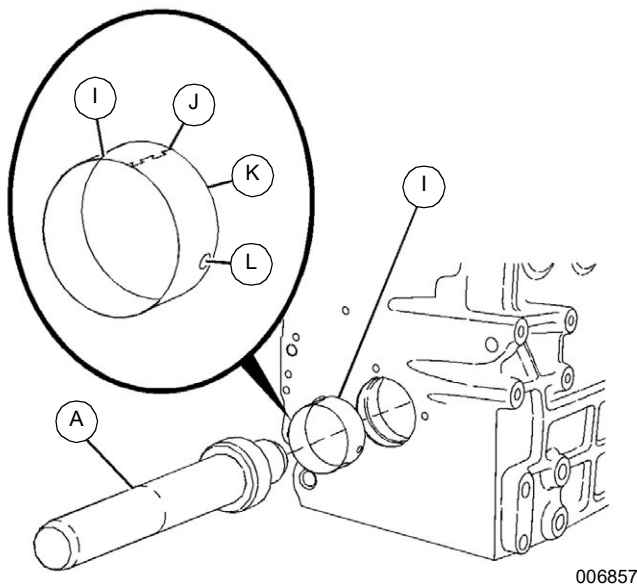


Figure 4-166. Install Crankshaft Bearing (Front)

5. Install the crankshaft assembly. See [Crankshaft](#).

Crankshaft

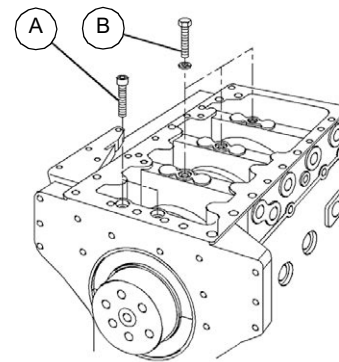
Removal Procedure

⚠ CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. Remove the engine oil relief valve. See [Engine Oil Relief Valve](#).
2. Remove the crankshaft rear seal. See [Crankshaft Rear Seal](#).
3. Remove the engine oil pump and the suction pipe. See [Engine Oil Pump](#).
4. Remove the pistons and the connecting rods. See [Pistons and Connecting Rods](#).
5. The engine should be mounted in a suitable stand with the rear end upward.
6. See [Figure 4-167](#). Remove bolts (B) from the cylinder block.
7. Remove Allen head screws (A) from the cylinder block.

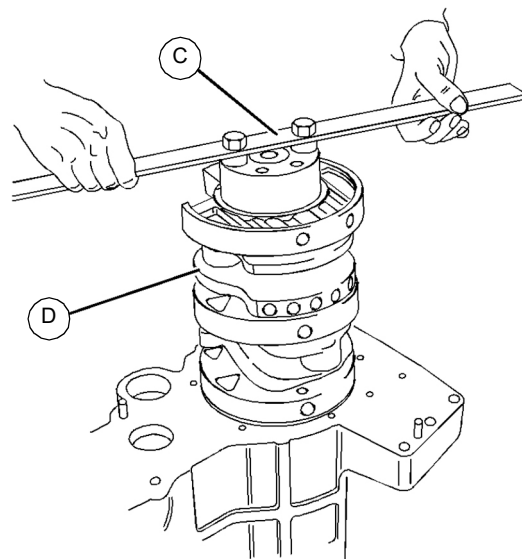


006854

Figure 4-167. Remove Crankshaft

8. See [Figure 4-168](#). Attach a suitable lifting device (C) to crankshaft assembly (D).
9. Carefully lift crankshaft assembly (D) from the cylinder block. If necessary, gently tap the nose of the crankshaft with a soft faced hammer.

NOTE: Install the nut for the crankshaft pulley to protect the nose of the crankshaft. Do not scratch any of the finished surfaces on the crankshaft.



006853

Figure 4-168. Remove Crankshaft

Installation Procedure

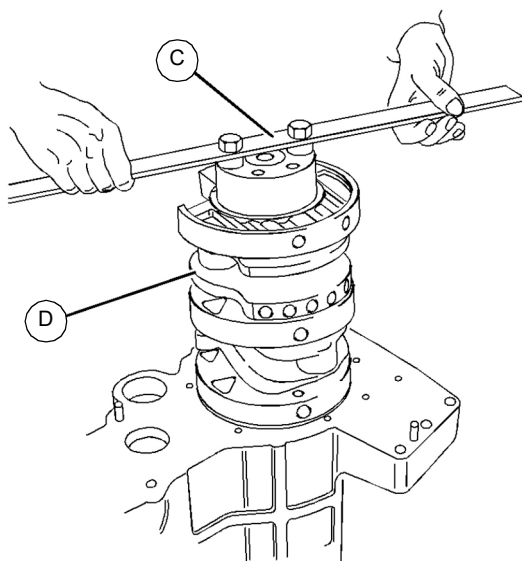
⚠ CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

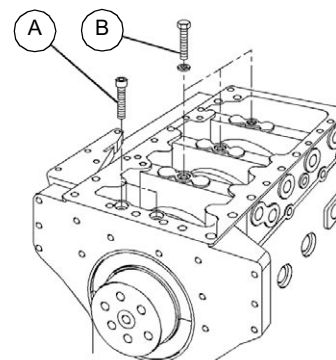
(000357)

1. The engine should be mounted in a suitable stand with the rear end upward.

2. Verify the crankshaft assembly is clean and free from damage.
3. See [Figure 4-169](#) and [Figure 4-170](#). Attach a suitable lifting device (C) to crankshaft assembly (D). Lubricate the front journal of the crankshaft with clean engine oil.
4. Align the holes in the crankshaft assembly for bolts (B) with the holes in the cylinder block. Align the holes in the crankshaft assembly for Allen head screws (A) with the holes in the cylinder block. Carefully lower the crankshaft assembly (D) into the cylinder block. If necessary, gently tap the rear of the crankshaft with a soft faced hammer.
5. Remove lifting device (C) from crankshaft assembly (D).
6. Install Allen head screws (A) finger tight.
7. Install bolts (B) and tighten to 38 ft-lbs (52 Nm).
8. Tighten Allen head screws (A) to 20 ft-lbs (27 Nm).
9. Rotate the crankshaft so there is no binding.
10. Check the crankshaft end play. Push the crankshaft toward the front of the engine. Install dial indicator group to the cylinder block and the rear face of the crankshaft. Push the crankshaft toward the rear of the engine. Use dial indicator group to measure the crankshaft end play. See [Crankshaft](#) for the maximum permissible crankshaft end play.



006853

Figure 4-169. Install Crankshaft

006854

Figure 4-170. Install Crankshaft

11. Install the pistons and the connecting rods. See [Pistons and Connecting Rods](#).
12. Install the engine oil pump and the suction pipe. See [Engine Oil Pump](#).
13. Install the crankshaft rear seal. See [Crankshaft Rear Seal](#).
14. Install the engine oil relief valve. See [Engine Oil Relief Valve](#).

Bearing Clearance

Measurement Procedure

CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

NOTE: Generac does not recommend the checking of the actual clearances of the bearing shells particularly on small engines. Checking of the actual clearances of the bearing shells carries the possibility of obtaining inaccurate results and of damaging the bearing shell or the journal surfaces. Each bearing shell is quality checked for specific wall thickness.

NOTE: The measurements should be within specifications and the correct bearings should be used. If the crankshaft journals and the bores for the block and the rods were measured during disassembly, no further checks are necessary. However, if the technician still wants to measure the bearing clearances, Plastigauge (A) is an acceptable method. Plastigauge (A) is less accurate on journals with small diameters if clearances are less than 0.004 in (0.10 mm).

IMPORTANT NOTE: Lead wire, shim stock or a dial bore gauge can damage the bearing surfaces.

The technician must use Plastigauge (A) correctly. The following points must be remembered:

- Verify the backs of the bearings and the bores are clean and dry.
- Verify the bearing locking tabs are properly seated in the tab grooves.
- The crankshaft must be free of oil at the contact points of Plastigauge (A).

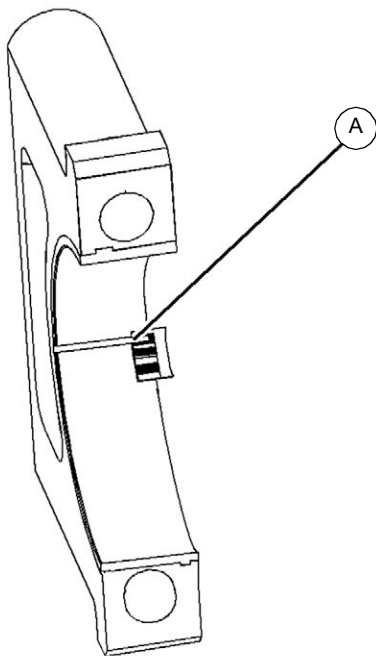
1. See [Figure 4-171](#). Put a piece of Plastigauge (A) on the crown of the bearing in the cap.

NOTE: Do not allow Plastigauge (A) to extend over the edge of the bearing.

2. Use the correct torque-turn specifications to install the bearing cap. Do not use an impact wrench. Be careful not to dislodge the bearing when the cap is installed.

NOTE: Do not turn the crankshaft when Plastigauge (A) is installed.

3. Carefully remove the cap, but do not remove Plastigauge (A). Measure the width of Plastigauge (A) while Plastigauge (A) is in the bearing cap or on the crankshaft journal.
4. Remove all of Plastigauge (A) before installing the bearing cap.



006825

Figure 4-171. Bearing Clearance

NOTE: When Plastigauge (A) is used, the readings can sometimes be unclear. For example, all parts of Plastigauge (A) are not the same width. Measure the major width to verify the parts are within the specification range. See [Connecting Rod Bearing Journal](#) and [Main Bearing Journal](#) for the correct clearances.

Coolant Temperature Switch

Removal Procedure

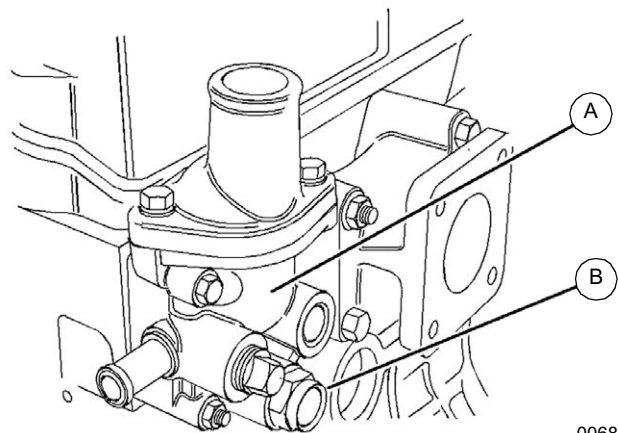


Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

IMPORTANT NOTE: Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids. Dispose of all fluids according to local regulations and mandates.

1. Drain the coolant from the cooling system into a suitable container for storage or disposal. See [Change Coolant](#) or [Check Cooling System Coolant Level](#).
2. See [Figure 4-172](#). Disconnect the harness assembly (not shown) from coolant temperature switch (B).
3. Remove coolant temperature switch (B) from water temperature regulator housing (A).



006846

Figure 4-172. Remove Coolant Temperature Switch

Installation Procedure



Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. Inspect the coolant temperature switch for damage and correct operation. See [Test Coolant](#)

Temperature Switch. If necessary, replace the coolant temperature switch.

2. See **Figure 4-173**. Install coolant temperature switch (B) to water temperature regulator housing (A). Use a deep socket to tighten the coolant temperature switch to 20 ft-lbs (27 Nm).

NOTE: If a used coolant temperature switch is installed, apply a thin layer of pipe sealant to the threads of the coolant temperature switch (A).

3. Connect the harness assembly (not shown) to coolant temperature switch (B).

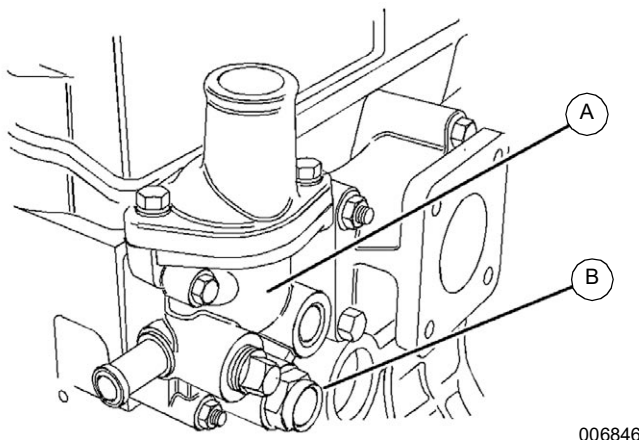


Figure 4-173. Install Coolant Temperature Switch

4. Fill the cooling system with coolant to the correct level. See **Fill Cooling System**.

Engine Oil Pressure Switch

Removal Procedure



Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

IMPORTANT NOTE: Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids. Dispose of all fluids according to local regulations and mandates.

NOTE: The engine oil pressure switch may be located in the valve mechanism cover or in the cylinder block. See **Engine Oil Pressure Switch** for the correct location.

1. See **Figure 4-174**. Disconnect the harness assembly (not shown) from engine oil pressure switch (A).
2. If engine oil pressure switch (A) is located in the valve mechanism cover, remove the engine oil pressure switch from the valve mechanism cover.

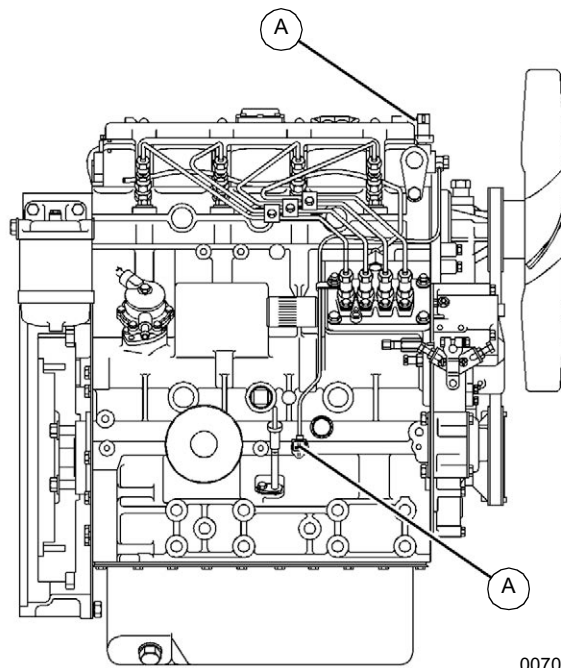


Figure 4-174. Remove Engine Oil Pressure Switch

3. See **Figure 4-175**. If the engine oil pressure switch (A) is located in the cylinder block, follow Step 4 through Step 6 to remove the engine oil pressure switch.
4. Remove engine oil pressure switch (A) and adapter (D) as an assembly from oil line (B).
5. Remove washers (C).
6. Remove engine oil pressure switch (A) from adapter (D).

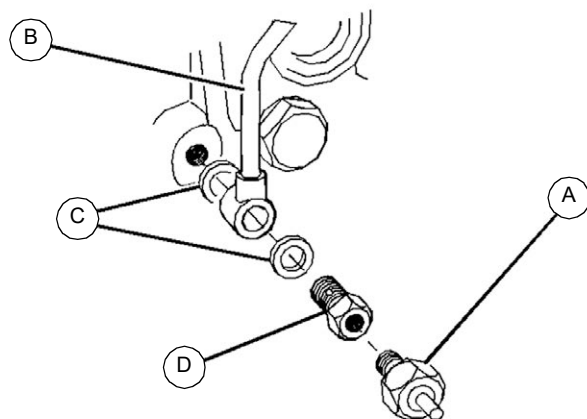


Figure 4-175. Remove Engine Oil Pressure Switch

Installation Procedure

CAUTION

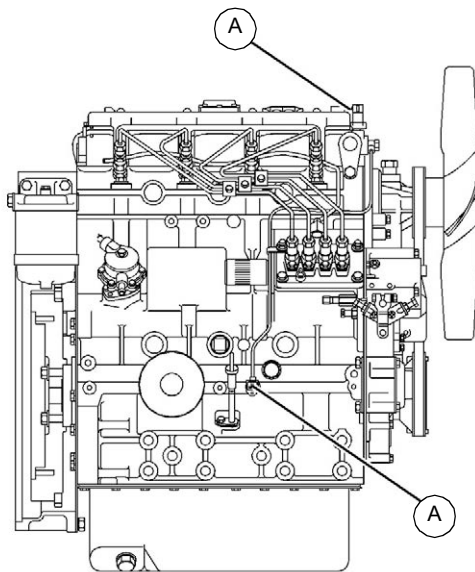
Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

NOTE: The engine oil pressure switch may be located in the valve mechanism cover or in the cylinder block. See [Engine Oil Pressure Switch](#) for the correct location.

NOTE: If a used engine oil pressure switch is installed, apply a thin layer of pipe sealant to the threads of the engine oil pressure switch.

1. See [Figure 4-176](#). If the engine oil pressure switch (A) is located in the valve mechanism cover, install the engine oil pressure switch and tighten to 97 in-lbs (11 Nm).



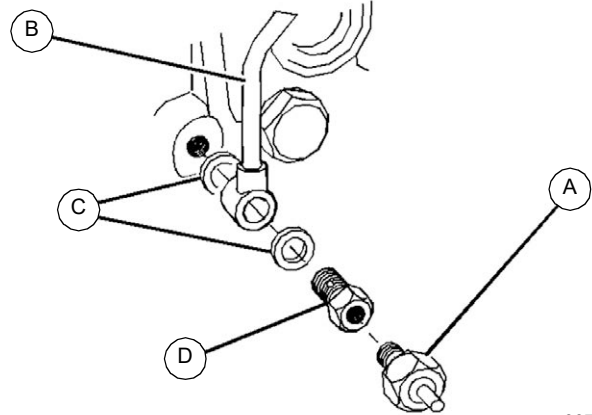
006923

Figure 4-176. Install Engine Oil Pressure Switch

2. See [Figure 4-177](#). If engine oil pressure switch (A) is located in the cylinder block follow Step 3 through Step 5 to install the engine oil pressure switch.
3. Install engine oil pressure switch (A) to adapter (D). Tighten the engine oil pressure switch to 17 ft-lbs (23 Nm).

NOTE: If a used engine oil pressure switch is installed, apply a thin layer of pipe sealant to the threads of the engine oil pressure switch.

4. Position new washers (C) onto oil line (B) and install the assembly of engine oil pressure switch (A) and adapter (D) to the cylinder block.
5. Tighten adapter (D) to 97 in-lbs (11 Nm).
6. Connect the harness assembly (not shown) to engine oil pressure switch (A).



007013

Figure 4-177. Install Engine Oil Pressure Switch

Glow Plugs

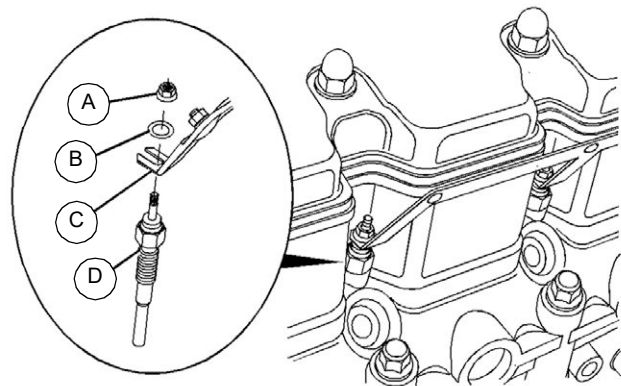
Removal Procedure

CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. Turn the battery disconnect switch OFF.
2. See [Figure 4-178](#). Remove nuts (A) and washers (B) from bus bar (C).
3. Remove bus bar (C) from glow plugs (D).
4. Remove glow plugs (D) from the cylinder head.



006980

Figure 4-178. Remove Glow Plugs

Installation Procedure

CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. Verify the threads of the glow plugs are clean and free from damage. Replace any damaged glow plugs.
2. See [Figure 4-179](#). Install glow plugs (D) into the cylinder head and tighten to 159 **in-lbs** (18 Nm).
3. Position bus bar (C) onto glow plugs (D).
4. Install washers (B) and nuts (A) to glow plugs (D). Use torque wrench to tighten the nuts to 10.6 **in-lbs** (1.2 Nm).

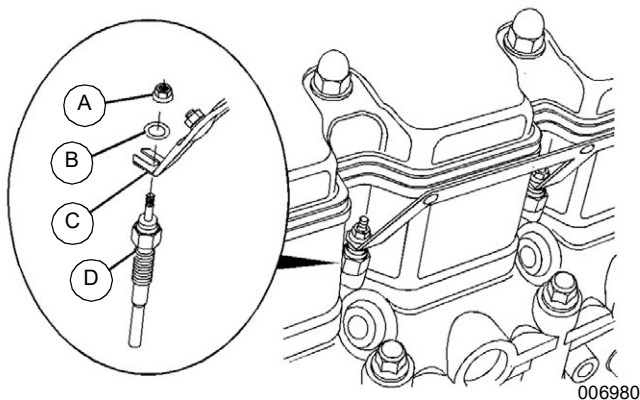


Figure 4-179. Install Glow Plugs

5. Turn the battery disconnect switch ON.

V-Belts

Removal Procedure

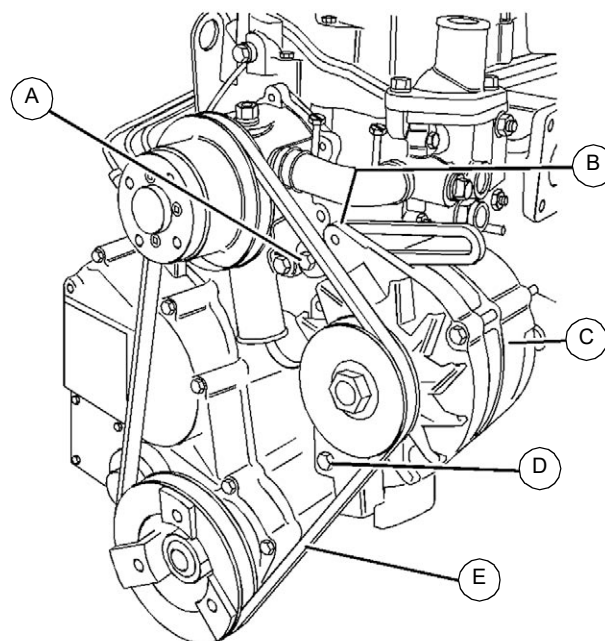
CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. If the engine is equipped with fan guards, remove the fan guards.
2. See [Figure 4-180](#). Loosen bolts (A), (B) and (D).
3. Push alternator (C) toward the engine and remove V-belt (E).

NOTE: Mark the direction of rotation if the V-belt will be reused.



007098

Figure 4-180. Remove V-Belts

Installation Procedure

CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. Check the condition of the V-belt and replace if worn or damaged.
2. See [Figure 4-181](#). Install V-belt (E) in position on the engine.

NOTE: A used V-belt should be installed in the original direction of rotation.

3. Slide alternator (C) away from the engine. See [Table 3-7](#) for information on the correct belt tension. Tighten bolts (A), (B) and (D) to 18 ft-lbs (25 Nm).
4. If the engine is equipped with fan guards, install the fan guards.

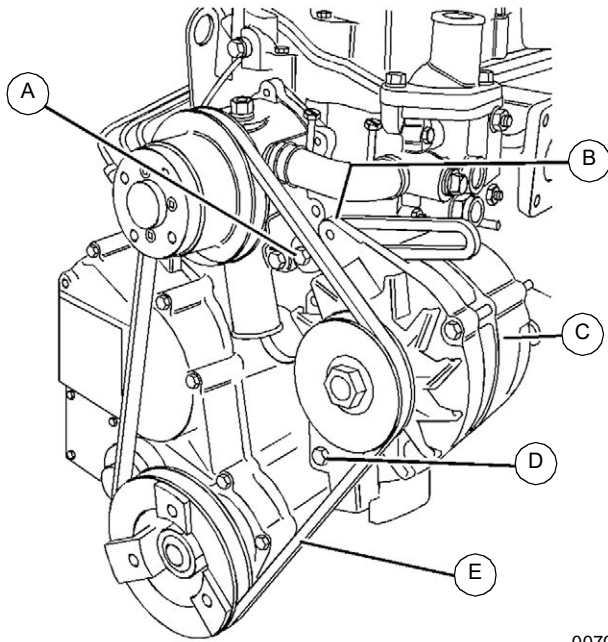


Figure 4-181. Install V-Belts

007098

Fan

Removal Procedure

⚠ CAUTION

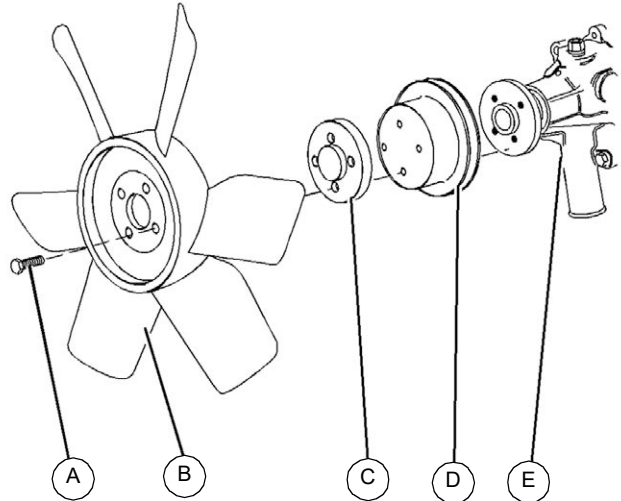
Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. Remove the V-Belt. See [V-Belts](#).
2. See [Figure 4-182](#). Remove bolts (A) and fan (B).

NOTE: Mark the orientation of the fan for installation.

3. Remove spacer (C) and pulley (D) from water pump (E).



006937

Figure 4-182. Remove Fan

Installation Procedure

⚠ CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. Verify all components are clean and free from damage. Replace any components that are worn or damaged.
2. See [Figure 4-183](#). Position pulley (D) and spacer (C) on water pump (E). Verify the bolt holes are aligned.
3. Install fan (B).

NOTE: Verify the correct orientation of the fan.

4. Install bolts (A) and tighten to 97 **in-lbs** (11 Nm).

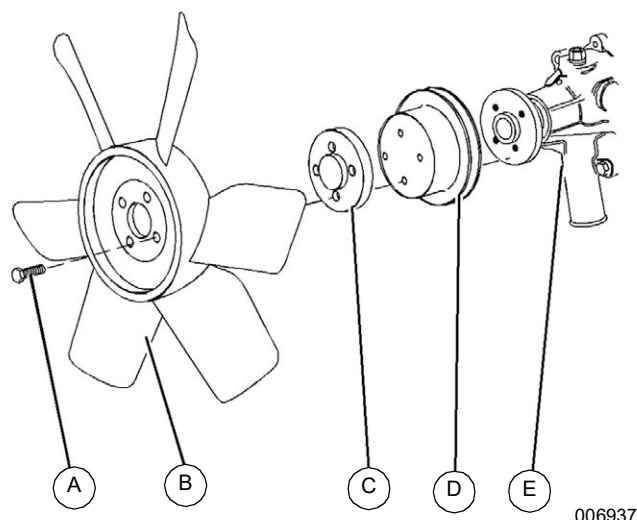


Figure 4-183. Install Fan

5. Install the V-Belt. See [V-Belts](#).

Alternator

Removal Procedure

CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. Remove the V-Belt. See [V-Belts](#).
2. Turn the battery disconnect switch OFF.
3. Make temporary identification marks on the connections of the harness assembly.
4. See [Figure 4-184](#). Disconnect the harness assembly (not shown) from alternator (E).
5. Remove bolt (D) and washer (C) from alternator (E).
6. Remove washer (G) and nut (F). Remove bolt (H) and alternator (E) from the mounting bracket.
7. If necessary, remove bolt (A) and remove adjusting link (B).

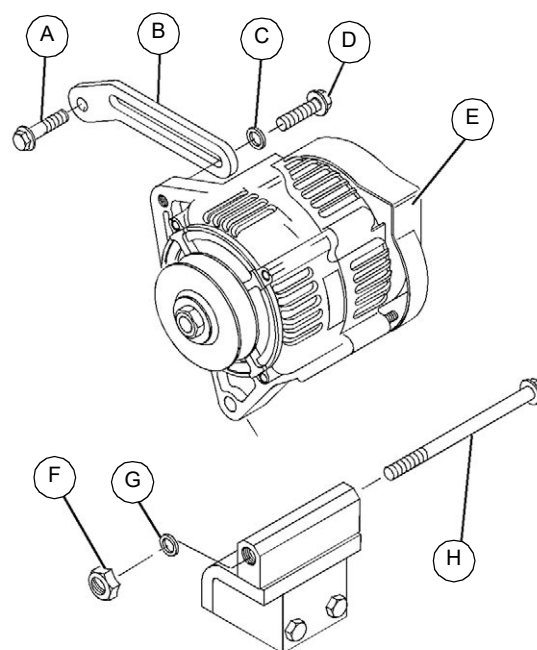


Figure 4-184. Remove Alternator

Installation Procedure

CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

1. See [Figure 4-185](#). If necessary, install adjusting link (B) and install bolt (A) finger tight.
2. Position alternator (E) on the mounting bracket.
3. Install bolt (H), washer (G) and nut (F) finger tight.
4. Install bolt (D) and washer (C) finger tight.
5. Install the V-Belt. See [V-Belts](#).

NOTE: Verify the alternator pulley is in alignment with the crankshaft pulley. Verify all fasteners are tightened.

6. Connect the harness assembly (not shown) to the alternator.

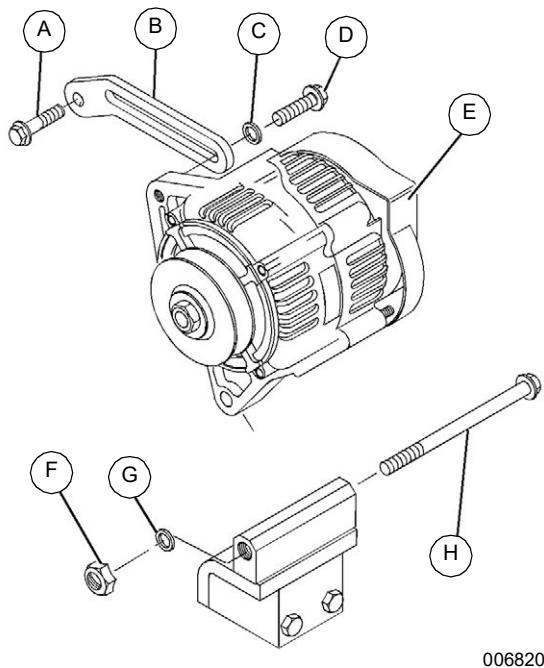


Figure 4-185. Install Alternator

7. Turn the battery disconnect switch ON.

Electric Starting Motor

Removal Procedure

WARNING

Accidental Start-up. Disconnect the negative battery cable, then the positive battery cable when working on unit. Failure to do so could result in death or serious injury.

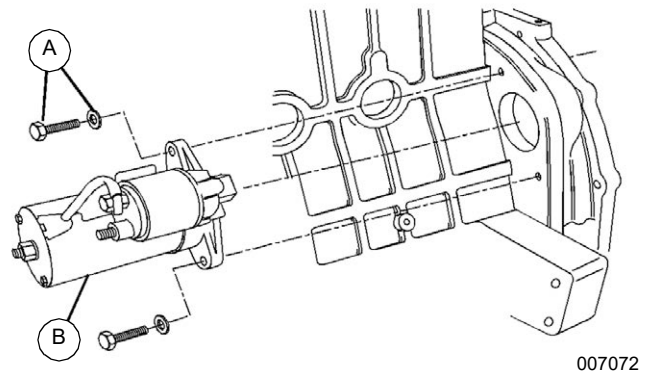
(000130)

IMPORTANT NOTE: Place a Do Not Operate tag at the Start/Stop switch location to inform personnel that the equipment is being worked on.

1. Disconnect battery.
2. Make temporary identification marks on the harness assemblies connected to the electric starting motor.
3. See [Figure 4-186](#). Disconnect the harness assemblies (not shown) from electric starting motor (B).
4. Remove fasteners (A) that secure electric starting motor (B).

NOTE: Support the weight of the electric starting motor as the fasteners are removed.

5. Remove electric starting motor (B) from the engine.

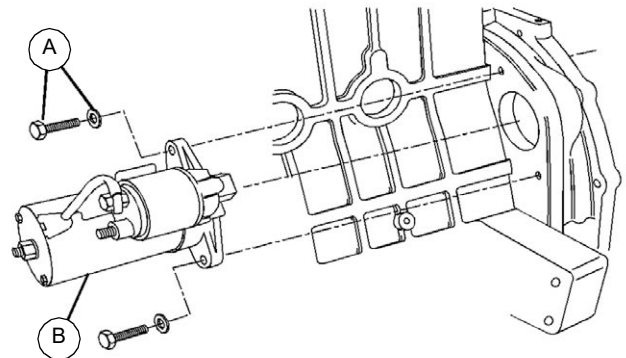


007072

Figure 4-186. Remove Electric Starter Motor

Installation Procedure

1. See [Figure 4-187](#). Install electric starting motor (B) to the engine.
2. Install fasteners (A) and tighten to 37 ft-lbs (50 Nm).
3. Connect the harness assemblies (not shown) to electric starting motor (B).



007072

Figure 4-187. Install Electric Starter Motor

4. Connect the battery.

Section 5: Operation

Lifting and Storage

Engine Lifting



⚠ WARNING

Personal injury. Failure to properly connect lifting cables, chains, or straps could result in death, serious injury, or property damage.

(000346)

⚠ WARNING

Personal Injury. Do not use lifting eye if there are signs of damage or corrosion. Doing so could result in death, serious injury, or property damage.

(000433)

⚠ WARNING

Personal Injury. Do not use lifting eye other than as directed. Doing so could result in death, serious injury, or property damage.

(000434)

⚠ WARNING

Personal Injury. Verify all fasteners are properly tightened prior to lifting unit. Failure to do so could result in death, serious injury, or property damage.

(000351)

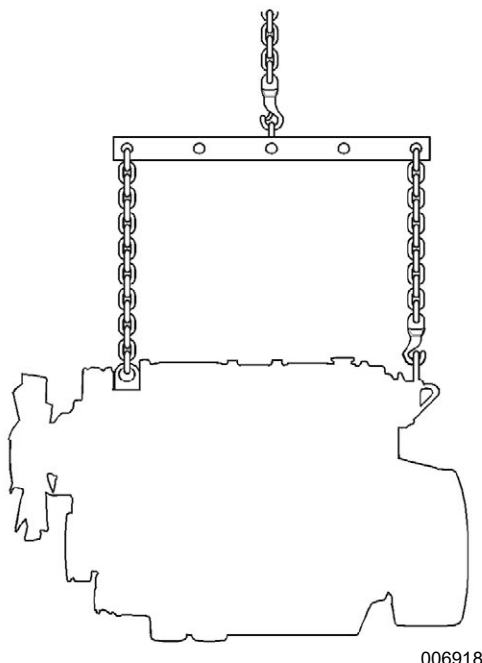


Figure 5-1. Engine Lifting

IMPORTANT NOTE: Never bend the eyeholes or brackets. Only load the eyeholes and the brackets under tension. Remember the capacity of an eyehole is less as the angle between the supporting members and the object becomes less than 90°. When it is necessary to remove a component at an angle, only use a link bracket properly rated for the weight.

Use a hoist to remove heavy components. Use an adjustable lifting beam to lift the engine. All supporting chains and cables should be parallel to each other. The chains and cables should be perpendicular to the top of the object being lifted.

Some removals require lifting the fixtures to obtain correct balance and safety.

To remove the engine **ONLY**, use the lifting eyes on the engine.

Lifting eyes are designed and installed for specific engine arrangements. Alterations to the lifting eyes or engine make the lifting eyes and the lifting fixtures obsolete. If alterations are made, verify correct lifting devices are provided. Contact an IASD for information regarding fixtures for correct engine lifting.

When lifting or moving the engine, use the following procedures to prevent engine damage:

- Do not tilt the engine to an extreme angle unless the lubricating oil is first drained from the oil pan
- Do not turn the engine onto a side or an end surface unless the lubricating oil is first drained from the oil pan
- The engine oil should be refilled to the correct level before the engine is started

Engine Storage

If the engine will not be started for several weeks, the lubricating oil will drain from the cylinder walls and from the piston rings. Rust can form on the cylinder walls which will cause increased engine wear and a reduction in engine service life.

Lubrication Storage

To help prevent excessive engine wear, use the following guidelines.

Complete all of the lubrication recommendations listed in [Maintenance Interval Schedule](#).

If engine is out of operation and use of the engine is not planned, special precautions should be made. If the engine will be stored for more than one month, a complete protection procedure is recommended.

Use the following guidelines:

1. Completely clean the outside of the engine.
2. Drain the fuel system completely and refill the system with preservative fuel. 1 772204 POWERPART Lay-Up 1 can be mixed with the

normal fuel to change the fuel into preservative fuel.

NOTE: If preservative fuel is not available, the fuel system can be filled with normal fuel. This fuel must be discarded at the end of the storage period together with the fuel filter elements.

3. Operate the engine until it reaches normal operating temperature. Stop any leaks from fuel, lubricating oil or air systems. Stop the engine and drain the lubricating oil from the oil pan.
4. Renew the canister(s) of the lubricating oil filter.
5. Fill the oil pan to the FULL mark on the engine oil level gauge with new, clean lubricating oil. Add POWERPART Lay-Up 2 to the oil to protect the engine against corrosion. If POWERPART Lay-Up 2 is not available, use a preservative of the correct specification instead of the lubricating oil. If a preservative is used, this must be drained completely at the end of the storage period and the oil pan must be refilled to the correct level with normal lubricating oil.

Cooling System



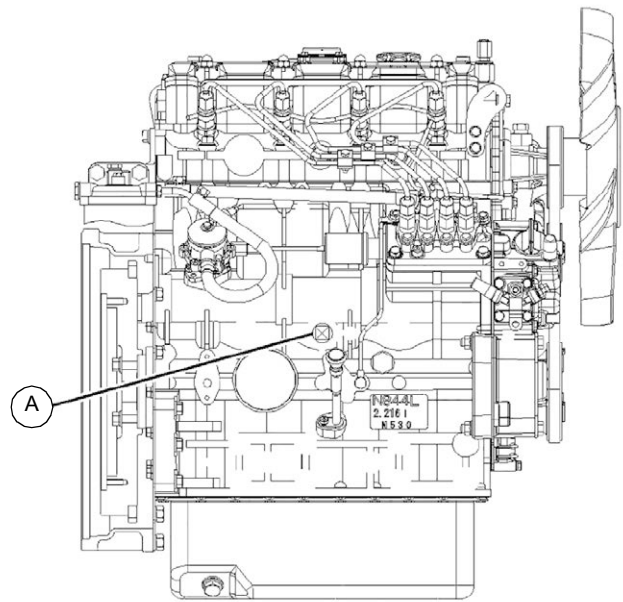
WARNING

Risk of burns. Contents under pressure. Do not remove the radiator pressure cap while engine is hot. Doing so could result in death or serious injury. (000322a)

NOTE: If freezing temperatures are expected, check the cooling system for adequate protection against freezing. See [Fluid Recommendations](#).

NOTE: To prevent frost damage, verify all coolant is removed from the engine. This is important if the system is drained after it has been flushed with water, or if an antifreeze solution too weak to protect the system from frost has been used.

1. Verify the engine is on level ground.
2. Remove the filler cap of the cooling system.
3. See [Figure 5-2](#). Remove the drain plug (A) from the side of the cylinder block to drain the engine. Verify the drain hole is not restricted.
4. Open the tap or remove the drain plug at the bottom of the radiator to drain the radiator. If the radiator does not have a tap or a drain plug, disconnect the hose at the bottom of the radiator.
5. Flush the cooling system with clean water.
6. Fit the drain plugs and the filler cap. Close the tap or connect the radiator hose.
7. Fill the cooling system with an approved antifreeze mixture to protect against corrosion.



006845

Figure 5-2. Drain Plug

NOTE: Certain corrosion inhibitors could cause damage to some engine components. Contact an IASD for advice.

8. Operate the engine for a short period to circulate the lubricating oil and the coolant in the engine.
9. Disconnect the battery. Put the battery into safe storage in a fully charged condition. Before the battery is put into storage, protect the terminals against corrosion. POWERPART Lay-Up 3 can be used on the terminals.
10. Clean the crankcase breather, if equipped. Seal the end of the pipe.
11. Remove the fuel injectors and spray POWERPART Lay-Up 2 for one or two seconds into each cylinder bore with the piston at BDC.
12. Slowly rotate the crankshaft for one complete revolution and then replace the fuel injectors.

Induction System

Remove the air filter assembly. If necessary, remove the pipes installed between the air filter assembly and turbocharger. Spray POWERPART Lay-Up 2 into the turbocharger. The duration of the spray is printed on the container. Seal the turbocharger with waterproof tape.

Exhaust System

Remove the exhaust pipe. Spray POWERPART Lay-Up 2 into the turbocharger. The duration of the spray is printed on the container. Seal the turbocharger with waterproof tape.

General Items

- If the lubricating oil filler is installed on the valve mechanism cover, remove the filler cap. If the lubricating oil filler cap is not installed on the valve mechanism cover, remove the valve mechanism cover. Spray POWERPART Lay-Up 2 around the rocker shaft assembly. Replace the filler cap or the valve mechanism cover.
- Seal the vent of the fuel tank or the fuel filler cap with waterproof tape.
- Remove the alternator drive belts and put the drive belts into storage.
- To prevent corrosion, spray the engine with 1 POWERPART Lay-Up 3. Do not spray the area inside the alternator.

When the engine protection has been completed in accordance with these instructions, no corrosion will occur. Generac is not responsible for damage which may occur when an engine is in storage after a period in service.

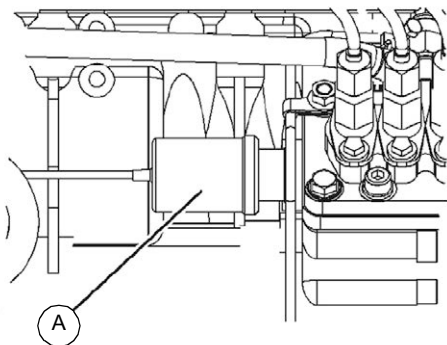
An IASD can assist in preparing the engine for extended storage periods.

Features and Controls

Fuel Shutoff

See [Figure 5-3](#). The fuel shutoff solenoid (A) is located on the fuel injection pump. When the fuel shutoff solenoid is activated, the solenoid moves the fuel rack to the OFF position.

If an electronically controlled governor has been installed the governor operates the fuel rack to stop the engine.



009809

Figure 5-3. Fuel Shutoff Solenoid

Engine Starting

Before Starting Engine



⚠ DANGER

Asphyxiation. Running engines produce carbon monoxide, a colorless, odorless, poisonous gas. Carbon monoxide, if not avoided, will result in death or serious injury.

(000103)

Before the engine is started, perform the required daily maintenance and any other periodic maintenance that is due. See [Maintenance Interval Schedule](#).

For the maximum service life of the engine, make a thorough inspection within the engine compartment before the engine is started.

- Look for the following items: oil leaks, coolant leaks, loose bolts and excessive dirt and grease
- Remove any excess dirt and grease buildup
- Repair any faults identified during the inspection
- Inspect the cooling system hoses for cracks and loose clamps
- Inspect the alternator and accessory drive belts for cracks, breaks and other damage
- Inspect the wiring for loose connections and worn or frayed wires
- Check the fuel supply
- Drain water from the water separator (if equipped)
- Open the fuel supply valve (if equipped)

IMPORTANT NOTE: All valves in the fuel return line must be open before and during engine operation to help prevent high fuel pressure.

If the engine has not been started for several weeks, fuel may have drained from the fuel system. Air may have entered the filter housing. Also, when fuel filters have been changed, some air pockets will be trapped in the engine. In these instances, prime the fuel system. See [Prime Fuel System](#).

- Do not start the engine or move any of the controls if there is a DO NOT OPERATE warning tag or similar warning tag attached to the start switch or to the controls
- Verify areas around rotating parts are clear
- Repair or replace any damaged or missing guards
- Disconnect any battery chargers not protected against the high current drain created when the electric starting motor is engaged
- Check electrical cables and check the battery for poor connections and corrosion
- Reset all of the shut-offs or alarm components (if equipped)

- Verify the oil level is between the ADD mark and FULL mark on the engine oil level gauge
- Verify the coolant level is filled to the FULL mark on the header tank

NOTE: If the engine is not equipped with a header tank maintain the coolant level within 0.5 in (13 mm) of the bottom of the filler pipe. If the engine is equipped with a sight glass, maintain the coolant level in the sight glass.

- Observe the air cleaner service indicator (if equipped) and service when the yellow diaphragm enters the red zone, or when the red piston locks in the visible position
- Verify any equipment driven by the engine has been disengaged from the engine
- Minimize electrical loads or remove any electrical loads

Perform the required daily maintenance and other periodic maintenance before the engine is started. Inspect the engine compartment. This inspection can help prevent major repairs at a later date. See [Maintenance Interval Schedule](#).

- Verify the engine has an adequate fuel supply
- Open the fuel supply valve (if equipped)

Starting the Engine



⚠ DANGER

Explosion. Do not use aerosol starting aids such as ether. Doing so could cause an explosion, which will result in death or serious injury.

(000566)

Refer to the OEM manual for your type of controls. Use the following procedure to start the engine.

1. Move the throttle lever to the low idle position before starting the engine.
2. Turn the engine start switch to the HEAT position.
3. Hold the engine start switch in the HEAT position for time shown in [Figure 5-4](#). This action will activate the glow plugs and aid in the starting of the engine.

IMPORTANT NOTE: Do not operate the glow plug for more than 30 seconds at a time.

4. When the glow plug indicator light is illuminated, turn the engine start switch to the START position and crank the engine.
5. When the engine starts, release the engine start switch.
6. Slowly move the throttle lever to the low idle position and allow the engine to idle.

NOTE: If the glow plug indicator light flashes rapidly for two to three seconds or fails to illuminate, a malfunction exists in the cold start system. Do not use ether or other starting fluids to start the engine.

7. If the engine does not start, release the engine start switch and allow the electric starting motor to cool. Then, repeat Step 2 through Step 6.
8. Turn the engine start switch OFF to stop the engine.

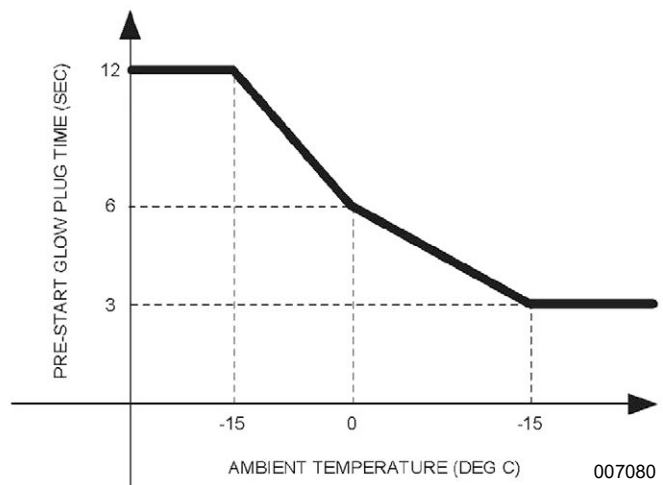


Figure 5-4. Starting the Engine

After Starting Engine

NOTE: In temperatures from 32 to 140 °F (0 to 60 °C), the warm-up time is approximately three minutes. In temperatures below 32 °F (0 °C), additional warm-up time may be required.

When the engine idles during warm-up, observe the following conditions:

- Check for any fluid or air leaks at idle RPM and at one-half full RPM (no load on the engine) before operating the engine under load
- Operate the engine at low idle until all systems achieve operating temperatures, and check all gauges during the warm-up period.

NOTE: Gauge readings should be observed and the data frequently recorded while the engine is operating. Comparing the data over time will help to determine normal readings for each gauge. Comparing data over time will also help detect abnormal operating developments. Significant changes in the readings should be investigated.

Engine Operation

Proper operation and maintenance are key factors in obtaining the maximum life and economy of the engine. If the directions in [Operation](#) and [Maintenance](#) are followed, costs can be minimized and engine service life can be maximized.

The engine can be operated at the rated RPM after the engine reaches operating temperature. The engine will reach normal operating temperature sooner during a low engine speed (RPM) and during a low-power demand. This procedure is more effective than idling the engine at no load. The engine should reach operating temperature in a few minutes.

Fuel Conservation Practices

The efficiency of the engine can affect fuel economy. Follow the recommended procedures to attain optimum performance for the life of the engine.

- Avoid spilling fuel

Fuel expands when the fuel is warmed up. The fuel may overflow from the fuel tank. Inspect fuel lines for leaks. Repair fuel lines as needed.

- Be aware of the properties of the different fuels and use only recommended fuels
- Avoid unnecessary idling

Shut off the engine rather than idle for long periods of time.

- Observe the air cleaner service indicator frequently and keep the air cleaner elements clean
- Maintain electrical systems

One damaged battery cell will overwork the alternator. This will consume excess power and excess fuel.

- Verify drive belts are correctly adjusted
- Verify all hose connections are tight
- Verify the driven equipment is in good working order
- Utilize heat from the jacket water system and the exhaust system when possible.
- Keep cooling system components clean and in good repair.
- Never operate the engine without water temperature regulators

Engine Stopping

Stopping the Engine

IMPORTANT NOTE: Stopping the engine immediately after it has been working under load can result in overheating and accelerated wear of the engine components. Avoiding hot engine shutdowns will maximize turbocharger shaft and bearing life.

NOTE: Individual applications will have different control systems. Verify the shut-off procedures are understood. Use the following general guidelines to stop the engine.

1. Remove the load from the engine. Reduce the engine speed (RPM) to low idle. Allow the engine to idle for five minutes to cool.
2. Stop the engine after the cool down period according to the shut-off system on the engine and turn the ignition key switch OFF. If necessary, refer to the instructions provided by the OEM.

Emergency Stopping



Equipment Damage. The emergency stop switch is not to be used to power down the unit under normal operating circumstances. Doing so could result in equipment damage. (000246b)

The OEM may have equipped the application with an emergency stop button. For more information about the emergency stop button, refer to the OEM information.

Verify any components for the external system that support the engine operation are secured after the engine is stopped.

After Stopping the Engine

NOTE: Before checking the engine oil, do not operate the engine for at least ten minutes to allow the engine oil to return to the oil pan.

- Verify the crankcase oil level is between the MIN and MAX mark on the engine oil level gauge
- If necessary, perform minor adjustments
- Repair any leaks and tighten any loose bolts
- If the engine is equipped with a service hour meter, note the reading and perform the maintenance in the [Maintenance Interval Schedule](#)
- Fill the fuel tank to help prevent accumulation of moisture in the fuel. Do not overfill the fuel tank

IMPORTANT NOTE: Only use antifreeze and coolant mixtures recommended in [Coolant Recommendations](#). Failure to do so can cause engine damage.

- Allow the engine to cool and check the coolant level
- If freezing temperatures are expected, check the coolant for correct antifreeze protection. The cooling system must be protected against freezing to the lowest expected outside temperature. Add the correct coolant and water mixture, if necessary.
- Perform all required periodic maintenance on all driven equipment, outlined in the instructions from the OEM.

Cold Weather Operation

During cold weather, the starting and operation of the diesel engine is dependent on the following items:

- Type of fuel used
- Viscosity of engine oil
- Operation of glow plugs
- Optional cold starting aid
- Battery condition
- Ambient air temperature and altitude

- Parasitic load of application
- Application hydraulic and transmission oil viscosities

This section will cover the following information:

- Potential problems caused by cold weather operation
- Suggest steps which can be taken to minimize starting problems and operating problems when the ambient air temperature is between 32 to 40 °F (0 to -40 °C).

The operation and maintenance of an engine in freezing temperatures is complex. This complexity is because of the following conditions:

- Weather conditions
- Engine applications

Recommendations are based on past proven practices. The information contained in this section provides guidelines for cold weather operation.

Tips for Cold Weather Operation

- If the engine will start, operate the engine until a minimum operating temperature of 177.8 °F (81 °C) is achieved to help prevent the intake and exhaust valves from sticking
- The cooling system and lubrication system for the engine do not lose heat immediately upon shutdown, so an engine can be shut down for a period of time and still be able to start readily
- Install the correct specification of engine lubricant before the beginning of cold weather
- Check all rubber parts (hoses, fan drive belts) weekly
- Check all electrical wiring and connections for any fraying or damaged insulation
- Keep all batteries fully charged and warm
- Fill the fuel tank at the end of each shift
- Check the air cleaners and the air intake daily, and check more often when operating in snow
- Verify the glow plugs are in working order. See [Test Glow Plugs](#).



⚠ DANGER

Explosion. Do not use aerosol starting aids such as ether. Doing so could cause an explosion, which will result in death or serious injury.

(000566)



⚠ WARNING

Explosion. Starting fluids are highly explosive. Always store away from sparks and heat. Failure to do so could result in death or serious injury.

(000568)

Viscosity of Engine Lubrication Oil

Correct engine oil viscosity is essential. Oil viscosity affects the amount of torque needed to crank the engine. See [Fluid Recommendations](#) for the recommended viscosity of oil.

Coolant Recommendations

Provide cooling system protection for the lowest expected outside temperature. See [Coolant Recommendations](#) for the recommended coolant mixture.

In cold weather, check the coolant often for the correct glycol concentration to ensure adequate freeze protection.

Engine Block Heaters (if equipped)

Engine block heaters heat the engine jacket water that surrounds the combustion chambers. This heat provides the following functions:

- Startability is improved
- Warm-up time is reduced

An electric block heater can be activated once the engine is stopped. An effective block heater is typically a 1,250/1,500 W unit. Contact an IASD for more information.

Idling the Engine

When idling after the engine is started in cold weather, increase the engine RPM from 1,000 to 1,200 RPM. This increase in RPM will warm up the engine more quickly. Maintaining an elevated low idle speed for extended periods will be easier with the installation of a hand throttle. The engine should not be "raced" to speed up the warm-up process.

While the engine is idling, the application of a light load (parasitic load) will assist in achieving the minimum operating temperature. The minimum operating temperature is 179.6 °F (82 °C).

Recommendations for Coolant Warm-Up

Warm up an engine that has cooled below normal operating temperatures due to inactivity. The warm-up should be performed before the engine is returned to full operation. During operation in very cold temperature conditions, damage to engine valve mechanisms can result from engine operation for short intervals. This can happen if the engine is started and stopped many times without being operated to warm up completely.

When the engine is operated below normal operating temperatures, fuel and oil are not completely burned in the combustion chamber. This fuel and oil causes soft carbon deposits to form on the valve stems. Generally, the deposits do not cause problems and the deposits are burned off during operation at normal engine operating temperatures.

When the engine is started and the engine is stopped many times without being operated to warm up completely, the carbon deposits become thicker. This can cause the following problems:

- Free operation of the valves is prevented
- Valves become stuck
- Pushrods may become bent
- Other damage to valve train components can result

For this reason, when the engine is started, it must be operated until the coolant temperature is 160 °F (71 °C) minimum. Carbon deposits on the valve stems will be kept at a minimum. The free operation of the valves and the valve components will be maintained.

In addition, the engine must be thoroughly warmed to keep other engine parts in better condition and the service life of the engine will be extended. Lubrication will be improved. There will be less acid and less sludge in the oil. This lubrication will provide longer service life for the engine bearings, the piston rings and other parts. However, limit unnecessary idle time to ten minutes to reduce wear and fuel consumption.

Water Temperature Regulator and Insulated Heater Lines

The engine is equipped with a water temperature regulator. When the engine coolant is below the correct operating temperature, jacket water circulates through the engine cylinder block and into the engine cylinder head. The coolant then returns to the cylinder block via an internal passage that bypasses the valve of the coolant temperature regulator. This system ensures coolant flows around the engine under cold operating conditions. The water temperature regulator begins to open when the engine jacket water has reached the correct minimum operating temperature. As the jacket water coolant temperature rises above the minimum operating temperature the water temperature regulator opens further allowing more coolant through the radiator to dissipate excess heat.

The progressive opening of the water temperature regulator operates the progressive closing of the bypass passage between the cylinder block and head. This system ensures maximum coolant flow to the radiator to achieve maximum heat dissipation.

NOTE: Air flow restriction devices, such as radiator shutters, are discouraged. Restriction of the air flow can result in high exhaust temperatures, power loss, excessive fan usage and reduction in fuel economy.

A cab heater is beneficial in very cold weather. The feed from the engine and the return lines from the cab should be insulated to reduce heat loss to the outside air.

Insulating the Air Inlet and Engine Compartment

When temperatures below -0.4 °F (-18 °C) will be frequently encountered, an air cleaner inlet located in the engine compartment may be specified. An air cleaner located in the engine compartment may also minimize the entry of snow into the air cleaner. Also, heat rejected by the engine helps to warm the intake air.

Additional heat can be retained around the engine by insulating the engine compartment.

Fuel and the Effect from Cold Weather

NOTE: Only use recommended grades of fuel. See [Fluid Recommendations](#).

Properties of diesel fuel can have a significant effect on engine cold start capability. Critical to the low temperature properties of diesel fuel is the acceptability for the minimum ambient temperature the engine is expected to see in operation. The following properties are used to define fuel low temperature capability:

- Cloud point
- Pour point
- Cold Filter Plugging Point (CFPP)

The cloud point of the fuel is the temperature at which waxes naturally found in diesel fuel begin to form crystals. The cloud point of the fuel must be below the lowest ambient temperature to prevent filters from plugging.

CFPP is a temperature at which a particular fuel will pass through a standardized filtration device. The CFPP gives an estimate of the lower operability temperature of fuel.

Pour point is the last temperature before the fuel flow stops and waxing of the fuel will start.

Be aware of these properties when diesel fuel is purchased. Consider the average ambient air temperature for the engine application. Engines fueled in one climate may not operate well if shipped to a colder climate. Problems can result due to changes in temperature.

Before troubleshooting for low power or for poor performance in the winter, check the fuel for waxing.

The following components can provide a means of minimizing fuel waxing problems in cold weather:

- Fuel heaters, which may be an OEM option
- Fuel line insulation, which may be an OEM option

Winter and arctic grades of diesel fuel are available in the countries and territories with severe winters. For more information see [Cold Weather Operation](#).

Another important fuel property which can affect cold start and operation of diesel engine is cetane number. For more information see [Fluid Recommendations](#).

Fuel Tanks

Condensation can form in partially filled fuel tanks. Top off the fuel tanks after operating the engine.

Fuel tanks should contain some provision for draining water and sediment from the bottom of the tanks. Some fuel tanks use supply pipes that allow water and sediment to settle below the end of the fuel supply pipe.

Some fuel tanks use supply lines that take fuel directly from the bottom of the tank. If the engine is equipped with this system, regular maintenance of the fuel system filter is important.

Drain the water and sediment from any fuel storage tank at the following intervals: weekly oil changes and refueling of the fuel tank. This will help prevent water and sediment from being pumped from the fuel storage tank and into the engine fuel tank.

Fuel Filters

It is possible a primary fuel filter is installed between the fuel tank and the engine fuel inlet. After changing the fuel filter, always prime the fuel system to remove air bubbles from the fuel system. See [Prime Fuel System](#).

The micron rating and the location of a primary fuel filter is important in cold weather operation. The primary fuel filter and the fuel supply line are the most common components affected by cold fuel.

Fuel Heaters (if equipped)

The OEM may equip the application with fuel heaters. If this is the case, disconnect an electric type of fuel heater in warm weather to prevent overheating of the fuel. If the type of fuel heater is a heat exchanger, the OEM should have included a bypass for warm weather. Verify the bypass is operational during warm weather to prevent overheating of the fuel.

For more information about fuel heaters, refer to the OEM information.

Refill Capacities

Lubrication System

The refill capacities for the engine crankcase reflect the approximate capacity of the crankcase or sump plus standard oil filters. Auxiliary oil filter systems will require additional oil. Refer to the OEM specifications for the capacity of the auxiliary oil filter.

Table 5-8. Refill Capacities (D2.2L Engine)

Compartment or System	Minimum	Maximum
Crankcase oil sump*	9.4 qt (8.9 L)	11.2 qt (10.6 L)
Total lubrication system**		

* More than one style of sump may be used on these engines. Use these values to estimate the refill capacity. Use the engine oil level gauge to fill the engine to the correct oil level. Record the result in this table. These values are the approximate capacities for the crankcase oil sump which includes the standard factory installed oil filters. Engines with auxiliary oil filters will require additional oil. Refer to the OEM specifications for the capacity of the auxiliary oil filter.

** The total lubrication system includes the capacity for the crankcase oil sump plus the capacity of factory installed oil filters and other filters added to the lubrication system. Enter the value for the capacity of the total lubrication system in this row.

Table 5-9. Refill Capacities (D1.5L Engine)

Compartment or System	Minimum	Maximum
Crankcase oil sump*	4.8 qt (4.5 L)	6.3 qt (6.0 L)
Total lubrication system**		

* These values are the approximate capacities for the crankcase oil sump which includes the standard factory installed oil filters. Engines with auxiliary oil filters will require additional oil. Refer to the OEM specifications for auxiliary oil filter capacity.

** The total lubrication system includes the capacity for the crankcase oil sump plus the capacity of factory installed oil filters and other filters added to the lubrication system. Enter the value for the capacity of the total lubrication system in this row.

Cooling System

To maintain the cooling system, the Total Cooling System capacity must be known. The approximate capacity is for the engine cooling system. External system capacities will vary among applications. See the OEM specifications for the external system capacity. This capacity information will be needed to determine the amount of coolant required for the Total Cooling System.

Table 5-10. Refill Capacities (D2.2L Engines)

Compartment or System	Capacity
Engine only	3.8 qt (3.6 L)
External system per OEM*	
Total cooling system**	

* The external system includes a radiator or an expansion tank with the following components: heat exchanger and piping. Refer to the OEM specifications. Enter the value for the capacity of the external system in this row.

** The total cooling system capacity includes the capacity for the engine plus the external system. Enter the value for the capacity of the total cooling system in this row.

Table 5-11. Refill Capacities (D1.5L Engines)

Compartment or System	Capacity
Engine only	2.7 qt (2.6 L)
External system per OEM*	
Total cooling system**	

* The external system includes a radiator or an expansion tank with the following components: heat exchanger and piping. Refer to the OEM specifications. Enter the value for the capacity of the external system in this row.

** The total cooling system capacity includes the capacity for the engine plus the external system. Enter the value for the capacity of the total cooling system in this row.

Fuel Recommendations

Table 5-12. Glossary

Abbreviation	Definition
ISO	International Standards Organization
ASTM	American Society for Testing and Materials
HFRR	High Frequency Reciprocating Rig for Lubricity testing of diesel fuels
FAME	Fatty Acid Methyl Esters
CFR	Co-ordinating Fuel Research
LSD	Low Sulfur Diesel
ULSD	Ultra Low Sulfur Diesel
RME	Rape Methyl Ester
SME	Soy Methyl Ester
EPA	Environmental Protection Agency of the United States

Diesel Fuel Requirements

Satisfactory engine performance is dependent on the use of a good quality fuel. The use of a good quality fuel will result in long engine life and acceptable exhaust emissions levels. The fuel must meet the minimum requirements stated in [Table 5-13](#), [Table 5-14](#) and [Table 5-15](#).

IMPORTANT NOTE: Read all footnotes in [Table 5-13](#).



Engine Damage. Do not operate with incorrect fuel. Doing so will result in reduced service life of the engine.

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Table 5-13. Specification for Distillate Diesel Fuel

Property	Units	Requirements	ASTM Test	ISO Test
Aromatics	% volume	35% maximum	D1319	ISO 3837
Ash	% weight	0.02% maximum	D482	ISO 6245
Carbon residue on 10% bottoms	% weight	0.35% maximum	D524	ISO 4262
Cetane number ²	—	40 minimum	D613/D6890	ISO 5165
Cloud point	°C	The cloud point must not exceed the lowest expected ambient temperature.	D2500	ISO 3015
Copper strip corrosion	—	No. 3 maximum	D130	ISO 2160
Density at 59 °F (15 °C) ³	Kg/m ³	801 minimum and 876 maximum	No equivalent test	ISO 3675, ISO 12185
Distillation	°C	10% at 539.6 °F (282 °C) maximum 90% at 680 °F (360 °C) maximum	D86	ISO 3405
Flash point	°C	Legal limit	D93	ISO 2719
Thermal stability	—	Minimum of 80% reflectance after aging for 180 minutes at 302 °F (150 °C)	D6468	No equivalent test
Pour point	°C	42.8 °F (6 °C) minimum below ambient temperature	D97	ISO 3016
Sulfur ^{1,4}	% mass	The sulfur level in the fuel is controlled by emissions regulations. See Table 5-14 and Table 5-15 .	D5453/D26222	ISO 20846, ISO 20884
Kinematic viscosity ⁵	"MM" "/S (cSt)"	Viscosity of fuel that is delivered to the fuel injection pump. "1.4 minimum/4.5 maximum"	D445	ISO 3405
Water and sediment	% weight	0.1% maximum	D1796	ISO 3734
Water	% weight	0.1% maximum	D1744	No equivalent test
Sediment	% weight	0.05% maximum	D473	ISO 3735
Gums and resins ⁶	mg/100 mL	10 mg per 100 mL maximum	D381	ISO 6246
Lubricity corrected wear scar diameter at 140 °F (60 °C) ⁷	mm	0.46 maximum	D6079	ISO 12156-1

1 This specification includes the requirements for Ultra Low Sulfur Diesel (ULSD). ULSD fuel will have ≤ 15 ppm (0.0015%) sulfur. Refer to ASTM D5453, ASTM D2622, or ISO 20846, ISO 20884 test methods. This specification includes the requirements for Low Sulfur Diesel (LSD). LSD fuel will have ≥ 500 ppm (0.05%) sulfur. Refer to following: ASTM 5453, ASTM 02622, ISO 20846 and ISO 20884 test methods.

2 A fuel with a higher cetane number is recommended to operate at a higher altitude or in cold weather.

3 Via standards tables, the equivalent API gravity for the minimum density of 801 kg/m³ (kilograms per cubic meter) is 45 and for the maximum density of 876 kg/m³ is 30.

4 Regional regulations, national regulations or international regulations can require a fuel with a specific sulfur limit. See all applicable regulations before selecting a fuel for a given engine application. Perkins fuel systems and engine components can operate on high sulfur fuels in territories that are non-emissions regulated. Fuel sulfur levels affect exhaust emissions. High sulfur fuels also increase the potential for corrosion of internal components. Fuel sulfur levels above 0.5% may significantly shorten the oil change interval.

5 The values of the fuel viscosity are the values as the fuel is delivered to the fuel injection pumps. Fuel should also meet the minimum viscosity requirement and the fuel should meet the maximum viscosity requirements at 104 °F (40 °C) of either the ASTM D445 test method or the ISO 3104 test method. If a fuel with a low viscosity is used, cooling of the fuel may be required to maintain 1.4 cSt or greater viscosity at the fuel injection pump. Fuels with a high viscosity might require fuel heaters to lower the viscosity to 4.5 cSt at the fuel injection pump.

6 Follow the test conditions and procedures for gasoline (motor).

7 The lubricity of a fuel is a concern with low sulfur and ultra low sulfur fuel. To determine the lubricity of the fuel, use the ISO 12156-1 or ASTM 06079 High Frequency Reciprocating Rig (HFRR) test. If the lubricity of a fuel does not meet the minimum requirements, contact your fuel supplier. Do not treat the fuel without contacting the fuel supplier. Some additives are not compatible. These additives can cause problems in the fuel system.

Diesel Fuel Characteristics

Cetane Number

Fuel with a high cetane number will give a shorter ignition delay. This will produce a better ignition quality. Cetane numbers are derived for fuels against proportions of cetane and heptamethylnonane in the standard CFR engine. Refer to ISO 5165 for the test method.

Cetane numbers in excess of 45 are normally expected from current diesel fuel. However, a cetane number of 40 may be experienced in some territories. The United States of America is one of the territories that can have a low cetane value. A minimum cetane value of 40 is required during average starting conditions. A higher cetane value may be required for operations at high altitudes or in cold weather operations.

Fuel with a low cetane number can be the root cause of problems during cold start.

Viscosity

Viscosity is the property of a liquid of offering resistance to shear or flow. Viscosity decreases with increasing temperature. This decrease in viscosity follows a logarithmic relationship for normal fossil fuel. The common reference is to kinematic viscosity. This is the quotient of the dynamic viscosity divided by the density. The determination of kinematic viscosity is normally by readings from gravity flow viscometers at standard temperatures. Refer to ISO 3104 for the test method.

The viscosity of the fuel is significant because fuel serves as a lubricant for the fuel system components. Fuel must have sufficient viscosity to lubricate the fuel system in both extremely cold temperatures and extremely hot temperatures. If the kinematic viscosity of the fuel is lower than 1.4 cSt at the fuel injection pump damage to the fuel injection pump can occur. This damage can be excessive scuffing and seizure. Low viscosity may lead to difficult hot restarting, stalling and loss of performance. High viscosity may result in seizure of the pump.

Generac recommends kinematic viscosities of 1.4 and 4.5 mm²/sec delivered to the fuel injection pump.

Density

Density is the mass of the fuel per unit volume at a specific temperature. This parameter has a direct influence on engine performance and a direct influence on emissions. This determines the heat output from a given injected volume of fuel. This is generally quoted in the following kg/m at 59 °F (15 °C).

A value of density of 841 kg/m is recommended to obtain the correct power output. Lighter fuels are acceptable but will not produce the rated power.

Sulfur

The level of sulfur is governed by emissions legislations. Regional regulation, national regulations or international regulations can require a fuel with a specific sulfur limit. The sulfur content of the fuel and the fuel quality must comply with all existing local regulations for emissions.

Table 5-14 and **Table 5-15** list the guidelines for the correct sulfur level for specific territories. Consult all applicable regulations before selecting the fuel for a given engine application.

Table 5-14. Fuel Requirements

Territory	Fuel Requirements from 2007	
EPA	Low sulfur (500 ppm) maximum	
EC	Sulfur/Power	Sulfur (1,000 ppm) maximum for greater than 19 kW
Non-regulated territories	Sulfur limit of less than 4,000 ppm	

Table 5-15. Fuel Requirements

Territory	Fuel Requirements from 2010		
EPA	Ultra low sulfur (15 ppm) maximum		
EC	Sulfur/Power	Ultra Low sulphur (10 ppm) maximum for less than or equal to 37 kW	Low sulfur (300 ppm) maximum for greater than 37 kW
	Models	D1.5L engines	D2.2L engines
Non-regulated territories	Sulfur limit of less than 4,000 ppm		

By using the test methods ASTM D5453, ASTM D2622, or ISO 20846 ISO 20884, the content of sulfur in low sulfur diesel (LSD) fuel must be below 500 ppm 0.05%. By using the test methods ASTM D5453, ASTM D2622, or ISO 20846 ISO 20884 the content of sulfur in ultra low sulfur (ULSD) fuel must be below 15 ppm 0.0015%. The lubricity of these fuels must not exceed wear scar diameter of 0.0181 in (0.46 mm). The fuel lubricity test must be performed on a HFRR, operated at 140 °F (60 °C). Refer to ISO 12156-1.

In some parts of the world and for some applications, high sulfur fuels above 0.5% by mass might only be available. Fuel with very high sulfur content can cause engine wear. High sulfur fuel will have a negative impact on emissions of particulates. High sulfur fuel can be used provided the local emissions legislation will allow it. High sulfur fuel can be used in countries that do not regulate emissions.

When only high sulfur fuels are available, it will be necessary that high alkaline lubricating oil is used in the engine or the lubricating oil change interval is reduced. See [Fluid Recommendations](#).

Lubricity

This is the capability of the fuel to prevent pump wear. The fluid lubricity describes the ability of the fluid to reduce the friction between surfaces are under load. This ability reduces the damage caused by friction. Fuel injection systems rely on the lubricating properties of the fuel.

The lubricity has particular significance to the current low viscosity fuel, low sulfur fuel and low aromatic fossil fuel. These fuels are made to meet stringent exhaust emissions. A test method for measuring the lubricity of diesel fuels has been developed and the test is based on the HFRR method operated at 140 °F (60 °C). Refer to ISO 12156 part 1 and CEC document F06-A-96 for the test method.

Lubricity wear scar diameter of 0.0181 in (0.46 mm) MUST NOT be exceeded. The fuel lubricity test must be performed on a HFRR, operated at 140 °F (60 °C). Refer to ISO 12156-1.

Distillation

This is an indication of the mixture of different hydrocarbons in the fuel. A high ratio of light weight hydrocarbons can affect the characteristics of combustion.

Coolant Recommendations

General Coolant Information



⚠ WARNING

Risk of burns. Do not open coolant system until engine has completely cooled. Doing so could result in serious injury.

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IMPORTANT NOTE: If the engine is to be stored in, or shipped to, an area with below freezing temperatures, the cooling system must be either protected to the lowest outside temperature or drained completely to prevent damage.

IMPORTANT NOTE: Frequently check the specific gravity of the coolant for proper freeze protection or for anti-boil protection.

Clean the cooling system for the following reasons:

- Contamination of the cooling system
- Overheating of the engine
- Foaming of the coolant

IMPORTANT NOTE: Never operate an engine without water temperature regulators in the cooling system.

Many engine failures are related to the cooling system. The following problems are related to cooling system failures: overheating, leakage of the water pump and plugged radiators or heat exchangers.

These failures can be avoided with correct cooling system maintenance. Cooling system maintenance is as important as maintenance of the fuel system and the lubrication system. Coolant quality is as important as fuel and oil quality.

Coolant is normally composed of three elements: water, additives and glycol.

Water

Water is used in the cooling system to transfer heat.

NOTE: Distilled water or deionized water is recommended for use in engine cooling systems.

Do not use the following types of water in cooling systems: hard water, softened water conditioned with salt and sea water.

If distilled water or deionized water is not available use water with the properties listed in [Table 5-16](#).

Table 5-16. Acceptable Water

Property	Maximum Limit
Chloride (Cl)	40 mg/L
Sulfate (SO ₄)	100 mg/L
Total hardness	170 mg/L
Total solids	340 mg/L
Acidity	pH of 5.5 to 9.0

For a water analysis, contact one of the following sources:

- Local water utility company
- Agricultural agent
- Independent laboratory

Additives

Additives help protect the metal surfaces of the cooling system. A lack of coolant additives or insufficient amounts of additives enable the following conditions to occur:

- Corrosion
- Formation of mineral deposits
- Rust
- Scale
- Foaming of the coolant

Many additives are depleted during engine operation. These additives must be replaced periodically.

Additives must be added at the correct concentration. Over-concentration of additives can cause the inhibitors to drop out-of-solution. The deposits can enable the following problems to occur:

- Formation of gel compounds
- Reduction of heat transfer
- Leakage of the water pump seal
- Plugging of radiators, coolers and small passages

Glycol

Glycol in the coolant helps protect against the following conditions:

- Boiling
- Freezing

- Cavitation of the water pump (for optimum performance, a 1:1 mixture of a water and glycol solution is recommended).

NOTE: Use a mixture that will provide protection against the lowest ambient temperature.

NOTE: 100% pure glycol will freeze at a temperature of -9 °F (-23 °C).

Most conventional antifreezes use ethylene glycol. Propylene glycol may also be used. In a 1:1 mixture with water, ethylene and propylene glycol provides similar protection against freezing and boiling. See [Table 5-17](#) and [Table 5-18](#).

IMPORTANT NOTE: Do not use propylene glycol in concentrations that exceed 50% glycol because of propylene glycol's reduced heat transfer capability. Use ethylene glycol in conditions that require additional protection against boiling or freezing.

Table 5-17. Ethylene Glycol

Concentration	Freeze Protection
50%	-33 °F (-36 °C)
60%	-60 °F (-51 °C)

Table 5-18. Propylene Glycol

Concentration	Freeze Protection
50%	-20 °F (-29 °C)

NOTE: To check the concentration of glycol in the coolant, measure the specific gravity of the coolant.

Coolant Recommendations

The following two coolants are used in Perkins diesel engines:

- **Preferred** - Perkins ELC (Extended Life Coolant)
- **Acceptable** - A commercial heavy-duty antifreeze that meets ASTM (American Society for Testing and Materials) D4985 specifications

IMPORTANT NOTE: Do not use a commercial coolant or antifreeze that only meets the ASTM D3306 specification. This type of coolant or antifreeze is made for light automotive applications.

A 1:1 mixture of water and glycol is recommended. This mixture will provide optimum heavy-duty performance as an antifreeze. This ratio may be increased to 1:2 water to glycol if extra freezing protection is required.

NOTE: A commercial heavy-duty antifreeze that meets ASTM D4985 specifications MAY require a treatment with an SCA at the initial fill. Read the label or the instructions provided by the OEM of the product.

In stationary engine applications that do not require anti-boil protection or freeze protection, a mixture of SCA and water is acceptable. A 6% to 8% concentration of SCA in those cooling systems. Distilled water or deionized water is preferred. Water which has the recommended properties may be used.

Table 5-19. Coolant Service Life

Coolant Type	Service Life
Perkins ELC	6,000 service hours or three years
Commercial heavy-duty anti-freeze that meets ASTM D4985	3,000 service hours or two years
Perkins POWERPART SCA	3,000 service hours or two years
Commercial SCA and water	3,000 service hours or two years

ELC (Extended Life Coolant)

ELC is provided for use in the following applications:

- Heavy-duty spark ignited gas engines
- Heavy-duty diesel engines

The anti-corrosion package for ELC is different from the anti-corrosion package for other coolants. ELC is an ethylene glycol base coolant. However, ELC contains organic corrosion inhibitors and anti-foam agents with low amounts of nitrite.

ELC is available in a 1:1 pre-mixed cooling solution with distilled water. The pre-mixed ELC provides freeze protection to -33 °F (-36 °C). The pre-mixed ELC is recommended for the initial fill of the cooling system. The pre-mixed ELC is also recommended for topping off the cooling system. Containers of several sizes are available. Contact an IASD for part numbers.

ELC Cooling System Maintenance

Correct Additions to the Extended Life Coolant

IMPORTANT NOTE: Use only Perkins products for pre-mixed or concentrated coolants. Do not mix coolant with other products.

IMPORTANT NOTE: To maintain the correct balance between the antifreeze and additives, maintain the recommended concentration of ELC. Lowering the proportion of antifreeze lowers the proportion of additive. This will lower the ability of the coolant to protect the system from pitting, cavitation, erosion and deposits.

NOTE: Do not use Standard Supplemental Coolant Additive (SCA).

NOTE: When using Perkins ELC, do not use standard SCA's or SCA filters.

ELC Cooling System Cleaning

NOTE: If the cooling system is already using ELC, cleaning agents are not required to be used at the specified coolant change interval. Cleaning agents are only required if the system has been contaminated by the addition of some other type of coolant or by cooling system damage.

Clean water is the only cleaning agent required when ELC is drained from the cooling system.

Before the cooling system is filled, the heater control (if equipped) must be set to HOT. Refer to the OEM to set the heater control. After the cooling system is drained and the cooling system is refilled, operate the engine until the coolant level reaches the normal operating temperature and until the coolant level stabilizes. Add the coolant mixture as needed to fill the system to the specified level.

Changing to Perkins ELC

To change from heavy-duty antifreeze to the Perkins ELC, perform the following steps:

IMPORTANT NOTE: Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids. Dispose of all fluids according to local regulations and mandates.

1. Drain the coolant into a suitable container.
2. Dispose of the coolant according to local regulations.
3. Flush the system with clean water to remove any debris.
4. Use Perkins cleaner to clean the system. Follow the instruction on the label.
5. Drain the cleaner into a suitable container. Flush the cooling system with clean water.
6. Fill the cooling system with clean water and operate the engine until the engine is warmed to 120 to 150 °F (49 to 66 °C).

IMPORTANT NOTE: Completely flush the cooling system with clear water until all signs of the cleaning agent are gone.

7. Drain the cooling system into a suitable container and flush the cooling system with clean water.

NOTE: The cooling system cleaner must be thoroughly flushed from the cooling system. Cooling system cleaner left in the system will contaminate the coolant. The cleaner may also corrode the cooling system.

8. Repeat Steps 6 and 7 until the system is completely clean.
9. Fill the cooling system with the Perkins Premixed ELC.

ELC Cooling System Contamination

IMPORTANT NOTE: Use only Perkins products for pre-mixed or concentrated coolants. Do not mix coolant with other products.

ELC cooling systems can withstand contamination to a maximum of 10% of conventional heavy duty antifreeze or SCA. If the contamination exceeds 10% of the total system capacity, perform ONE of the following procedures:

- Drain the cooling system into a suitable container. Dispose of the coolant according to local regulations. Flush the system with clean water. Fill the system with the Perkins ELC.
- Drain a portion of the cooling system into a suitable container according to local regulations. Then, fill the cooling system with premixed ELC. This should lower the contamination to less than 10%.
- Maintain the system as a conventional heavy duty coolant. Treat the system with an SCA. Change the coolant at the interval recommended for the conventional heavy duty coolant.

Commercial Heavy Duty Antifreeze and SCA (Supplement Coolant Additive)

IMPORTANT NOTE: Commercial heavy duty coolant which contains Amine as part of the corrosion protection system must not be used.

IMPORTANT NOTE: Never operate an engine without water temperature regulators in the cooling system.

Check the antifreeze (glycol concentration) to verify adequate protection against boiling or freezing. A refractometer is recommended for checking the glycol concentration.

The cooling system should be tested at 500 hour intervals for the concentration of SCA.

Additions of SCA are based on results of the test. A liquid SCA may be needed at 500 hour intervals.

See [Table 5-20](#) for part numbers and quantities of SCA.

Table 5-20. Perkins Liquid SCA

Part Number	Quantity
21825735	10

Adding SCA to Heavy Duty Coolant at the Initial Fill

Commercial heavy-duty antifreeze that meets ASTM D4985 specifications MAY require an addition of SCA at the initial fill. Read the label or the instructions provided by the OEM of the product.

Use the following equation to determine the amount of SCA required when the cooling system is initially filled:

$$V \times 0.045 = X$$

V is the total volume of the cooling system.

X is the amount of SCA that is required.

Table 5-21 is an example for using the equation above.

Table 5-21. Equation Example

Total Volume of Cooling System (V)	Multiplication Factor	Amount of SCA Required (X)
4 gal (15 L)	x 0.045	24 oz (0.7 L)

Adding SCA to Heavy Duty Coolant for Maintenance

Heavy duty antifreeze of all types REQUIRE periodic additions of an SCA.

Test the antifreeze periodically for the concentration of SCA. For the interval, see [Maintenance Interval Schedule](#). Test the concentration of SCA.

Additions of SCA are based on results of the test. The size of the cooling system determines the amount of SCA needed.

Use the equation below to determine the amount of SCA required, if necessary:

$$V \times 0.014 = X$$

V is the total volume of the cooling system.

X is the amount of SCA required.

Table 5-22 is an example for using the equation above.

Table 5-22. Equation Example

Total Volume of Cooling System (V)	Multiplication Factor	Amount of SCA Required (X)
4 gal (15 L)	x 0.014	7 oz (0.2 L)

Cleaning the System of Heavy Duty Anti-freeze

- Clean the cooling system after used coolant is drained or before the cooling system is filled with new coolant.
- Clean the cooling system whenever the coolant is contaminated or whenever the coolant is foaming.

Fluid Recommendations

General Lubricant Information

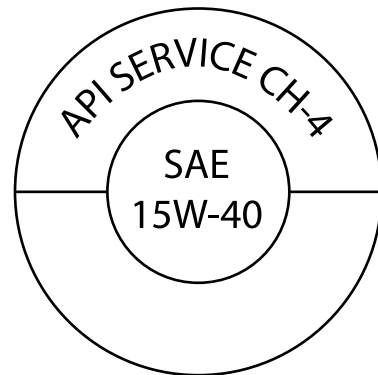
Because of government regulations regarding the certification of exhaust emissions from the engine, the lubricant recommendations must be followed.

Engine Manufacturers Association (EMA) Oils

The Engine Manufacturers Association Recommended Guideline on Diesel Engine Oil is recognized by Generac. For detailed information about this guideline, see the latest edition of EMA publication: EMA DHD -1.

API Oils

The Engine Oil Licensing and Certification System by the American Petroleum Institute (API) is recognized by Generac. For detailed information about this system, see the latest edition of the API publication No. 1509. Engine oils that bear the API symbol are authorized by API. See [Figure 5-5](#) for a typical API symbol example.



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Figure 5-5. Typical API Symbol

Diesel engine oils CC, CD, CD-2, and CE have not been API authorized classifications since January 1, 1996. [Table 5-23](#) summarizes the status of the classifications.

Table 5-23. API Classifications

Current	Obsolete
CF-4, CG-4, CH-4	CE
CF	CC, CD
CF-2*	CD-2*

* The classifications CD-2 and American Petroleum Institute CF-2 are for two-cycle diesel engines.

Terminology

Certain abbreviations follow the nomenclature of SAE J754. Some classifications follow SAE J183 abbreviations, and some classifications follow the EMA Recommended Guideline on Diesel Engine Oil. In addition to these definitions, there are other definitions that will be of assistance in purchasing lubricants. Recommended oil viscosities can be found in [Lubricant Viscosity Recommendations](#).

Engine Oil

Commercial Oils



Engine Damage. Failure to follow oil recommendations can result in shortened engine life.

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The performance of commercial diesel engine oils is based on API (American Petroleum Institute) classifications. These API classifications are developed to provide commercial lubricants for a broad range of diesel engines that operate at various conditions.

Only use commercial oils that meet the following classifications:

- EMA DHD-1 multi-grade oil (preferred oil)
- API CH-4 multi-grade oil (preferred oil)
- ACEAE5

To make the correct choice of a commercial oil, refer to the following explanations:

EMA DHD-1 – The Engine Manufacturers Association (EMA) has developed lubricant recommendations as an alternative to the API oil classification system. DHD-1 is a Recommended Guideline that defines a level of oil performance for these types of diesel engines: high speed, four stroke cycle, heavy duty and light duty. DHD-1 oils may be used when the following oils are recommended: API CH-4, API CG-4 and API CF-4.

DHD-1 oils are intended to provide superior performance in comparison to API CG-4 and API CF-4.

DHD-1 oils will meet the needs of high performance diesel engines operating in many applications. The tests and test limits used to define DHD-1 are similar to the new API CH-4 classification. Therefore, these oils will also meet the requirements for diesel engines that require low emissions. DHD-1 oils are designed to control the harmful effects of soot with improved wear resistance and improved resistance to plugging of the oil filter. These oils will also provide superior control of piston deposits for engines with either two-piece steel pistons or aluminum pistons.

All DHD-1 oils must complete a full test program with the base stock and with the viscosity grade of the finished commercial oil. The use of API Base Oil Interchange Guidelines are not appropriate for DHD-1 oils. This feature reduces the variation in performance that can occur when base stocks are changed in commercial oil formulations.

DHD-1 oils are recommended for use in extended oil change interval programs that optimize the life of the oil. These oil change interval programs are based on oil analysis. DHD-1 oils are recommended for conditions that demand a premium oil. Contact an IASD for specific guidelines for optimizing oil change intervals.

API CH-4 – API CH-4 oils were developed to meet the requirements of the new high performance diesel engines. Also, the oil was designed to meet the requirements of the low emissions diesel engines. API CH-4 oils are also acceptable for use in older diesel engines and in diesel engines that use high sulfur diesel fuel. API CH-4 oils may be used in engines that use API

CG-4 and API CF-4 oils. API CH-4 oils will generally exceed the performance of API CG-4 oils in the following criteria: deposits on pistons, control of oil consumption, wear of piston rings, valve train wear, viscosity control and corrosion.

Three new engine tests were developed for the API CH-4 oil. The first test specifically evaluates deposits on pistons for engines with the two-piece steel piston. This test (piston deposit) also measures the control of oil consumption. A second test is conducted with moderate oil soot. The second test measures the following criteria: wear of piston rings, wear of cylinder liners and resistance to corrosion. A third new test measures the following characteristics with high levels of soot in the oil: wear of the valve train, resistance of the oil in plugging the oil filter and control of sludge.

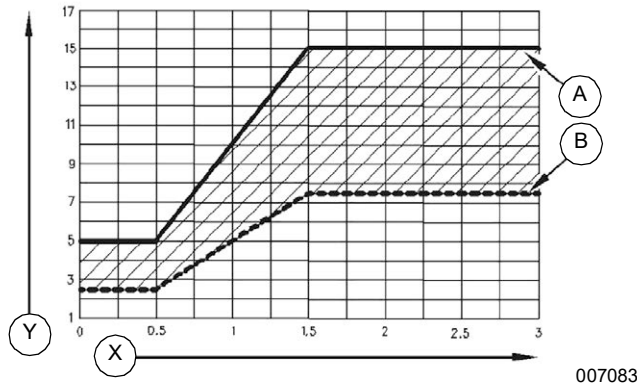
In addition to the new tests, API CH-4 oils have tougher limits for viscosity control in applications that generate high soot. The oils also have improved oxidation resistance. API CH-4 oils must pass an additional test (piston deposit) for engines that use aluminum pistons (single piece). Oil performance is also established for engines that operate in areas with high sulfur diesel fuel.

All of these improvements allow the API CH-4 oil to achieve optimum oil change intervals. API CH-4 oils are recommended for use in extended oil change intervals. API CH-4 oils are recommended for conditions that demand a premium oil. Contact an IASD for specific guidelines on optimizing oil change intervals.

Some commercial oils that meet the API classifications may require reduced oil change intervals. To determine the oil change interval, closely monitor the condition of the oil and perform a wear metal analysis.

Total Base Number (TBN) and Fuel Sulfur Levels for Diesel Engines

The TBN for an oil depends on the fuel sulfur level. For engines that use distillate fuel the minimum TBN of the new oil must be 10 times the fuel sulfur level. The TBN is defined by ASTM 02896. The minimum TBN of the oil is 5 regardless of fuel sulfur level. **Figure 5-6** demonstrates the TBN.



A	TBN of new oil
B	Change the oil when TBN deteriorates to 50% of the original TBN
X	Percentage of fuel sulfur by weight
Y	TBN by ASTM D2896

Figure 5-6. TBN for an Oil

Use the following guidelines for fuel sulfur levels that exceed 1.5%:

- Choose an oil with the highest TBN that meets one of these classifications: EMA DHD-1 and API CH-4.
- Reduce the oil change interval. Base the oil change interval on the oil analysis. Verify the oil analysis includes the condition of the oil and a wear metal analysis.

Excessive piston deposits can be produced by an oil with a high TBN. These deposits can lead to a loss of control of the oil consumption and to the polishing of the cylinder bore.

NOTE: Operating diesel engines with fuel sulphur levels over 0.5% will require shortened oil change intervals to help maintain adequate wear protection.

Table 5-24. Oil Change Interval per Sulfur Percentage

Sulfur Percentage in Fuel	Oil Change Interval
Lower than 0.5	Normal
0.5 to 1.0	0.75 of normal
Greater than 1.0	0.50 of normal

Lubricant Viscosity Recommendations

The correct SAE viscosity grade of oil is determined by the minimum ambient temperature during cold engine startup, and the maximum ambient temperature during engine operation.

See [Table 5-25](#) (minimum temperature) to determine the required oil viscosity for starting a cold engine.

See [Table 5-25](#) (maximum temperature) to select the oil viscosity for engine operation at the highest ambient temperature anticipated.

Generally, use the highest oil viscosity available to meet the requirement for the temperature at startup.

Table 5-25. Engine Oil Viscosity

EMA LRG-1 API CH-4 Viscosity Grade	Ambient Temperature	
	Minimum	Maximum
SAE 0W20	-40 °F (-40 °C)	50 °F (10 °C)
SAE 0W30	-40 °F (-40 °C)	86 °F (30 °C)
SAE 0W40	-40 °F (-40 °C)	104 °F (40 °C)
SAE 0W30	-22 °F (-30 °C)	86 °F (30 °C)
SAE 0W40	-22 °F (-30 °C)	104 °F (40 °C)
SAE 0W30	-4 °F (-20 °C)	104 °F (40 °C)
SAE 0W40	14 °F (-10 °C)	122 °F (50 °C)

Synthetic Base Stock Oils

Synthetic base oils are acceptable for use if they meet the performance requirements specified for the engine.

Synthetic base oils generally perform better than conventional oils in the following two areas:

- Synthetic base oils have improved flow at low temperatures especially in arctic conditions.
- Synthetic base oils have improved oxidation stability especially at high operating temperatures.

Some synthetic base oils have performance characteristics that enhance the service life of the oil. Generac does not recommend the automatic extending of the oil change intervals for any type of oil.

Re-refined Base Stock Oils

Re-refined base stock oils are acceptable for use if they meet the performance requirements specified by Generac. Re-refined base stock oils can be used exclusively in finished oil or in a combination with new base stock oils. The United States military specifications and the specifications of other heavy equipment manufacturers also allow the use of re-refined base stock oils that meet the same criteria.

The process used to make re-refined base stock oil should adequately remove all wear metals in the used oil and all the additives in the used oil. The process used to make re-refined base stock oil generally involves the process of vacuum distillation and hydro-treating the used oil. Filtering is adequate for the production of high quality, re-refined base stock oil.

Lubricants for Cold Weather

When an engine is started and operated in ambient temperatures below -4 °F (-20 °C), use multi-grade oils capable of flowing in low temperatures.

These oils have lubricant viscosity grades of SAE 0W or SAE 5W.

When an engine is started and operated in ambient temperatures below -22 °F (-30 °C), use a synthetic base stock multi-grade oil with an 0W viscosity grade or with a 5W viscosity grade. Use an oil with a pour point lower than -58 °F (-50 °C).

The number of acceptable lubricants is limited in cold weather conditions. Generac recommends the following lubricants for use in cold weather conditions:

First Choice – Use oil with an EMA DHD-1 Recommended Guideline. Use a CH-4 oil that has an API license. The oil should be SAE 0W20, SAE 0W30, SAE 0W40, SAE 5W30 or SAE 5W40 lubricant viscosity grade.

Second Choice – Use an oil with a CH-4 additive package. Although the oil has not been tested for the requirements of the API license, the oil must be SAE 0W20, SAE 0W30, SAE 0W40, SAE 5W30 or SAE 5W40.

IMPORTANT NOTE: Shortened engine life can result if second choice oils are used.

Aftermarket Oil Additives

The use of aftermarket additives in oil is not recommended. It is not necessary to use aftermarket additives to achieve the engine's maximum service life or rated performance. Fully formulated, finished oils consist of base oils and of commercial additive packages. These additive packages are blended into the base oils at precise percentages to help provide finished oils with performance characteristics that meet industry standards.

There are no industry standard tests that evaluate the performance or the compatibility of aftermarket additives in finished oil. Aftermarket additives may not be compatible with the finished oil's additive package, which could lower the performance of the finished oil. The aftermarket additive could fail to mix with the finished oil. This could produce sludge in the crankcase. The use of aftermarket additives in finished oils is discouraged.

To achieve the best performance, conform to the following guidelines:

- Select the correct oil, or a commercial oil that meets the EMA Recommended Guideline on Diesel Engine Oil or the recommended API classification.
- See [Table 5-25](#) to find the correct oil viscosity grade.
- At the specified interval, service the engine. Use new oil and install a new oil filter.
- Perform maintenance at the intervals specified in [Maintenance Interval Schedule](#).

Oil Analysis

Some engines may be equipped with an oil sampling valve. If oil analysis is required the oil sampling valve is used to obtain samples of the engine oil. The oil analysis will complement the preventive maintenance program.

The oil analysis is a diagnostic tool used to determine oil performance and component wear rates. Contamination can be identified and measured through the use of the oil analysis. The oil analysis includes the following tests:

- The Wear Rate Analysis monitors the wear of the engine metals. The amount of wear metal and type of wear metal in the oil is analyzed. The increase in the rate of engine wear metal in the oil is as important as the quantity of engine wear metal in the oil.
- Tests are conducted to detect contamination of the oil by water, glycol or fuel.
- The Oil Condition Analysis determines the loss of the oil lubricating properties. An infrared analysis is used to compare the properties of new oil to the properties of the used oil sample. This analysis allows technicians to determine the amount of deterioration of the oil during use. This analysis also allows technicians to verify the performance of the oil according to the specification during the entire oil change interval.

Section 6: Maintenance

System Pressure Release

Coolant System



WARNING

Risk of burns. Contents under pressure. Do not remove the radiator pressure cap while engine is hot. Doing so could result in death or serious injury. (000322a)

To relieve pressure from the coolant system, turn the engine OFF. Allow the cooling system pressure cap to cool. Remove the cooling system pressure cap slowly to relieve pressure.

Fuel System

To relieve the pressure from the fuel system, turn the engine OFF.

High Pressure Fuel Lines (if equipped)



DANGER

Hydraulic Fluid Injection. High-pressure, high-temperature hydraulic fluid can pierce skin and cause severe burns. Do not check for leaks with hands. Seek immediate medical attention in case of accident. Failure to protect body accordingly will result in death or serious injury. (000239)

The high pressure fuel lines are the fuel lines between the high pressure fuel pump and the high pressure fuel manifold and the fuel lines between the fuel manifold and cylinder head. These fuel lines are different from fuel lines on other fuel systems for the following reasons:

- The high pressure fuel lines are constantly charged with high pressure
- The internal pressures of the high pressure fuel lines are higher than other types of fuel system

Before any service or repair is performed on the engine fuel lines, perform the following tasks:

1. Stop the engine
2. Wait ten minutes

IMPORTANT NOTE: Do not loosen the high pressure fuel lines to remove air pressure from the fuel system.

Engine Oil

To relieve pressure from the lubricating system, turn the engine OFF.

Maintenance Interval Schedule

Daily

- *Check Cooling System Coolant Level*
- *Check Driven Equipment*
- *Inspect Engine Air Cleaner Service Indicator*
- *Clean Engine Air Pre-cleaner*
- *Check Engine Oil Level*
- *Drain Fuel System Water Separator*
- *Walkaround Inspection*

Every 50 Service Hours or Weekly

- *Drain Fuel Tank Water and Sediment*

Every 250 Service Hours or 6 Months

- *Inspect and Adjust Alternator Fan Belts*

Every 500 Service Hours

- *Replace Fuel System Filter*

Every 500 Service Hours or 1 Year

- *Check Battery Electrolyte Level*
- *Test or Add Cooling System Supplemental Coolant Additive (SCA)*
- *Clean or Replace Air Cleaner Element*
- *Inspect or Replace Engine Air Cleaner Element*
- *Change Engine Oil and Filter*
- *Inspect or Replace Hoses and Clamps*
- *Clean Radiator*

Every 1,000 Service Hours

- *Replace Alternator and Fan Belts*
- *Inspect or Adjust Engine Valve Lash*
- *Inspect Turbocharger (if equipped)*

Every 2,000 Service Hours

- *Inspect Aftercooler Core*
- *Inspect Alternator*
- *Replace Engine Crankcase Breather*
- *Inspect Engine Mounts*
- *Inspect Starting Motor*

Every 3,000 Service Hours

- *Replace Cooling System Water Temperature Regulator*
- *Test or Change Fuel Injector*
- *Inspect Water Pump*

Every 4,000 Service Hours

- *Clean or Test Aftercooler Core (Air-to-Air Aftercooler)*

Every 6,000 Service Hours or 3 Years

- *Change Coolant*

Every 12,000 Service Hours or 6 Years

- *Check Cooling System Coolant Level*

When Required

- *Replace Battery*
- *Disconnect Battery or Battery Cable*
- *Clean Engine*
- *Clean or Replace Air Cleaner Element*
- *Inspect or Replace Engine Air Cleaner Element*
- *Prime Fuel System*
- *Check Severe Service Application*

Clean or Test Aftercooler Core (Air-to-Air Aftercooler)

1. Remove the core. Refer to the OEM information for the correct procedure.
2. Turn the aftercooler core upside down to remove debris.

IMPORTANT NOTE: Maximum air pressure at the nozzle must be less than 30 psi (205 kPa) for cleaning purposes.

3. Pressurized air is the preferred method for removing loose debris. Direct the air in the opposite direction of the fan's air flow. Hold the nozzle approximately 0.25 in (6 mm) away from the fins. Slowly move the air nozzle in a direction that is parallel with the tubes. This will remove debris between the tubes.
4. Pressurized water may also be used for cleaning. The maximum water pressure for cleaning purposes must be less than 40 psi (275 kPa). Use pressurized water to soften mud. Clean the core from both sides.

IMPORTANT NOTE: A high concentration of caustic cleaner can cause leakage. Only use the recommended concentration of cleaner.

5. Back flush the core with a suitable cleaner.
6. Steam clean the core to remove any residue. Flush the fins of the aftercooler core. Remove any other trapped debris.
7. Wash the core with hot, soapy water. Rinse the core thoroughly with clean water.
8. Dry the core with compressed air. Direct the air in the reverse direction of the normal flow.
9. Inspect the core for cleanliness. Pressure test the core and repair if necessary.
10. Install the core. Refer to the OEM information for the correct procedure.
11. After cleaning, start the engine and accelerate the engine to high idle RPM. This will help in the removal of debris and drying of the core. Stop the engine. Use a light bulb behind the core to inspect for cleanliness and clean again if necessary.

Inspect Aftercooler Core

NOTE: Adjust the frequency of cleaning according to the effects of the operating environment.

Inspect the aftercooler for damaged fins, corrosion, dirt, grease, insects, leaves, oil and other debris. Clean the aftercooler, if necessary.

For air-to-air aftercoolers, use the same methods used for cleaning radiators.

IMPORTANT NOTE: Maximum air pressure at the nozzle must be less than 30 psi (205 kPa) for cleaning purposes.

After cleaning, start the engine and accelerate the engine to high idle RPM. This will help in the removal of debris and drying of the core. Stop the engine. Use a light bulb behind the core to inspect the core for cleanliness. Repeat the cleaning, if necessary.

Inspect the fins for damage. Bent fins may be opened with a "comb".

NOTE: If parts of the aftercooler system are repaired or replaced, a leak test is highly recommended.

Inspect condition of welds, mounting brackets, air lines, connections, clamps and seals. Make repairs if necessary.

Inspect Alternator

A scheduled inspection of the alternator is recommended. Inspect the alternator for loose connections and correct battery charging. Check the ammeter (if equipped) during engine operation to ensure correct battery and electrical system performance. Make repairs as necessary.

Check the alternator and the battery charger for correct operation. If the batteries are correctly charged, the ammeter reading should be very near zero. All batteries should be kept charged and warm. If the battery is too cold, the battery will not crank the engine. When the engine is not run for long periods of time or if the engine

is run for short periods, the batteries may not fully charge. A battery with a low charge will freeze more easily than a battery with a full charge.

Inspect and Adjust Alternator Fan Belts

Inspection

To maximize engine performance, inspect the belts for wear and cracking. Replace worn or damaged belts.

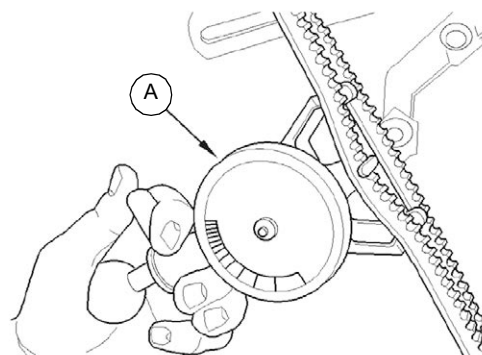
For applications that require multiple drive belts, replace the belts in matched sets. Replacing only one belt of a matched set will cause the new belt to carry more load because the older belt is stretched. The additional load on the new belt could cause the new belt to break.

If the belts are too loose, vibration causes unnecessary wear on the belts and pulleys. Loose belts may slip enough to cause overheating.

To accurately check the belt tension, a suitable gauge should be used.

See [Figure 6-1](#). Install gauge (A) at the center of the belt between the alternator and the crankshaft pulley and check the belt tension. The correct tension for a new belt is 90 lb (400 N) to 110 lb (489 N). The correct tension for a used belt that has been in operation for 30 minutes or more at the rated speed is 60 lb (267 N) to 80 lb (356 N).

If twin belts are installed, check and adjust the tension on both belts.

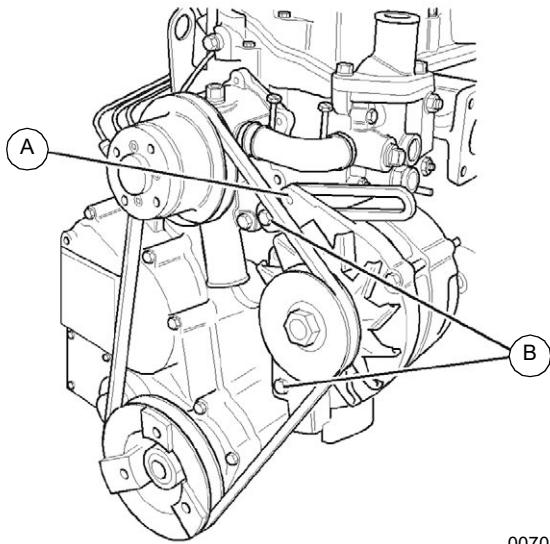


007000

Figure 6-1. Install Gauge

Adjustment

1. See [Figure 6-2](#). Loosen mounting bolts (B) and adjusting bolt (A).
2. Move the alternator to increase or decrease the belt tension.
3. Tighten adjusting bolt (A). Tighten mounting bolts (B). See [V-Belts](#) for the correct torque settings.



007010

Figure 6-2. Engine Mounting and Adjusting Bolts

Replace Alternator and Fan Belts

For applications that require multiple drive belts, replace the belts in matched sets. Replacing only one belt of a matched set will cause the new belt to carry more load because the older belt is stretched. The additional load on the new belt could cause the new belt to break.

NOTE: When new belts are installed, check the belt tension again after 20 hours of engine operation.

See [V-Belts](#) for the installation and removal procedure for the belt.

Replace Battery



⚠ WARNING

Explosion. Batteries emit explosive gases while charging. Keep fire and spark away. Wear protective gear when working with batteries. Failure to do so could result in death or serious injury.

(000137a)



⚠ WARNING

Battery Explosion. Remove battery cover before removing battery or battery cables. Failure to do so could cause an explosion, which will result in death or serious injury.

(000558)

1. Turn the engine OFF. Remove all electrical loads.
2. Turn off any battery chargers. Disconnect any battery chargers.
3. The NEGATIVE (-) cable connects the NEGATIVE (-) battery terminal to the NEGATIVE (-) terminal on the starting motor. Disconnect the cable from the NEGATIVE (-) battery terminal.
4. The POSITIVE (+) cable connects the POSITIVE (+) battery terminal to the POSITIVE (+) terminal

on the starting motor. Disconnect the cable from the POSITIVE (+) battery terminal.

⚠ WARNING

Environmental Hazard. Always recycle batteries at an official recycling center in accordance with all local laws and regulations. Failure to do so could result in environmental damage, death, or serious injury. (000228)

Always recycle batteries in accordance with local laws and regulations. Contact your local solid waste collection site or recycling facility to obtain information on local recycling processes. For more information on battery recycling, visit the Battery Council International website at: <http://batterycouncil.org>.

5. Remove the used battery.
6. Install the new battery.

NOTE: Before the cables are connected, verify the engine start switch is OFF.

7. Connect the cable from the starting motor to the POSITIVE (+) battery terminal.
8. Connect the NEGATIVE (-) cable to the NEGATIVE (-) battery terminal.

Check Battery Electrolyte Level



⚠ WARNING

Risk of burns. Batteries contain sulfuric acid and can cause severe chemical burns. Wear protective gear when working with batteries. Failure to do so could result in death or serious injury.

(000138a)

When the engine is not run for long periods of time or when the engine is run for short periods, the batteries may not fully recharge. Verify the battery is fully charged to help prevent the battery from freezing. If batteries are correctly charged, the ammeter reading should be very near zero when the engine is in operation.

1. Remove the filler caps. Maintain the electrolyte level to the FULL mark on the battery.

If the addition of water is necessary, use distilled water. If distilled water is not available use clean water that is low in minerals. Do not use artificially softened water.

2. Check the condition of the electrolyte with a suitable battery tester.
3. Install the caps.
4. Keep the batteries clean.
5. Clean the battery case with one of the following cleaning solutions:
 - Use a solution of 0.2 lb (0.1 kg) baking soda and 1 qt (1 L) of clean water.
 - Use a solution of ammonium hydroxide. Thoroughly rinse the battery case with clean water.

Disconnect Battery or Battery Cable



⚠ WARNING

Battery Explosion. Remove battery cover before removing battery or battery cables. Failure to do so could cause an explosion, which will result in death or serious injury.

(000558)

1. Turn the start switch OFF. Turn the ignition switch (if equipped) OFF and remove the key and all electrical loads.
2. Disconnect the negative battery terminal. Verify the cable cannot contact the terminal. When four 12 volt batteries are involved, two negative connections must be disconnected.
3. Remove the positive connection.
4. Clean all disconnected connection and battery terminals.
5. Use a fine grade of sandpaper to clean the terminals and the cable clamps. Clean the items until the surfaces are bright or shiny. DO NOT remove material excessively. Excessive removal of material can cause the clamps to not fit correctly. Coat the clamps and the terminals with a suitable silicone lubricant or petroleum jelly.
6. Tape the cable connections to help prevent accidental starting.
7. Proceed with necessary system repairs.
8. To connect the battery, connect the positive connection before the negative connector.

Change Coolant

⚠ CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

IMPORTANT NOTE: Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids. Dispose of all fluids according to local regulations and mandates.

Clean and flush the cooling system the recommended maintenance interval if the following conditions exist:

- The engine overheats frequently
- Foaming is observed
- Oil has entered the cooling system and the coolant is contaminated
- Fuel has entered the cooling system and the coolant is contaminated

NOTE: When the cooling system is cleaned, only clean water is needed.

NOTE: Inspect the water pump and the water temperature regulator after the cooling system has been drained. This is a good opportunity to replace the water pump, the water temperature regulator and the hoses, if necessary.

Drain Cooling System



⚠ WARNING

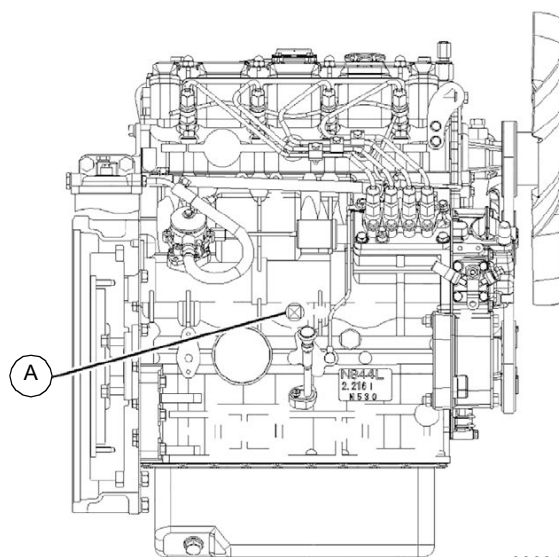
Risk of burns. Contents under pressure. Do not remove the radiator pressure cap while engine is hot. Doing so could result in death or serious injury.

(000322a)

1. Stop the engine and allow to cool. Loosen the cooling system filler cap slowly to relieve pressure. Remove the cooling system filler cap.
2. See [Figure 6-3](#). Open the drain cock or remove the drain plug (A) on the engine. Open the drain cock or remove the drain plug on the radiator. Allow the coolant to drain.

NOTE: The drain plug (A) and its opening on the engine block is 3/4 in BSPT (British Standard Pipe Taper) and is not directly compatible with standard imperial or metric plumbing and hardware fittings.

NOTE: The threads on the drain plug (A) are 1/4 in BSPP (British Standard Pipe Parallel) and are not directly compatible with standard imperial or metric plumbing and hardware fittings.



IMPORTANT NOTE: Dispose of used engine coolant or recycle. The full distillation procedure is the only acceptable method to reclaim the coolant.

For information regarding the disposal and the recycling of used coolant, contact an IASD.

Flush Cooling System

1. Flush the cooling system with clean water to remove any debris.
2. Close the drain cock or install the drain plug in the engine. Close the drain cock or install the drain plug on the radiator.

NOTE: Do not fill the cooling system faster than 1.3 gal (5 L) per minute to avoid air locks.

3. Fill the cooling system with clean water. Install the cooling system filler cap.
4. Start and run the engine at low idle until the temperature reaches 120 to 150 °F (49 to 66 °C).
5. Stop the engine and allow to cool. Loosen the cooling system filler cap slowly to relieve pressure. Remove the cooling system filler cap. Open the drain cock or remove the drain plug on the engine. Open the drain cock or remove the drain plug on the radiator. Allow the water to drain. Flush the cooling system with clean water.

Fill Cooling System

1. Close the drain cock or install the drain plug on the engine. Close the drain cock or install the drain plug on the radiator.

NOTE: Do not fill the cooling system faster than 1.3 gpm (5 Lpm) to avoid air locks.

2. Fill the cooling system with Commercial Heavy Duty Coolant. Add Supplemental Coolant Additive to the coolant. For the correct amount, See [Fuel Recommendations](#) for more information on cooling system specifications. Do not install the cooling system filler cap.
3. Start and run the engine at low idle. Increase the engine RPM to high idle. Run the engine at high idle for one minute to purge the air from the cavities of the engine block. Stop the engine.
4. Check the coolant level. Maintain the coolant level within 0.5 in (13 mm) below the bottom of the pipe for filling. Maintain the coolant level in the expansion bottle (if equipped) at the correct level.
5. Clean the cooling system filler cap. Inspect the gasket on the cooling system filler cap. If damaged, discard and install a new cap. If the gasket is not damaged, use a suitable pressurizing pump to pressure test the cooling system filler cap. The correct pressure for the cooling system filler cap is

stamped on the face of the cooling system filler cap. If the cooling system filler cap does not retain the correct pressure, install a new cooling system filler cap.

6. Start the engine. Inspect the cooling system for leaks and for correct operating temperature.

Check Cooling System Coolant Level

Engines with a Coolant Recovery Tank



WARNING

Risk of burns. Contents under pressure. Do not remove the radiator pressure cap while engine is hot. Doing so could result in death or serious injury.

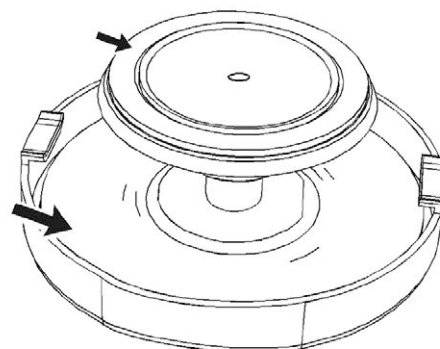
(000322a)

NOTE: The procedure that follows is for typical cooling systems. Refer to the OEM information for the correct procedures.

Check the coolant level when the engine is stopped and cool.

1. Observe the coolant level in the coolant recovery tank. Maintain the coolant level to COLD FULL mark on the coolant recovery tank.
2. Loosen filler cap slowly to relieve pressure. Remove the filler cap.
3. Pour the correct coolant mixture into the tank. See [Refill Capacities](#) for information on the correct mixture and type of coolant and cooling system capacity. Do not fill the coolant recovery tank above COLD FULL mark.
4. Clean filler cap and the receptacle. Install filler cap and inspect cooling system for leaks.

NOTE: The coolant will expand as the coolant heats up during normal engine operation. The additional volume will be forced into the coolant recovery tank during engine operation. When the engine is stopped and cool, the coolant will return to the engine.



006938

Figure 6-4. Filler Cap

Engines Without a Coolant Recovery Tank

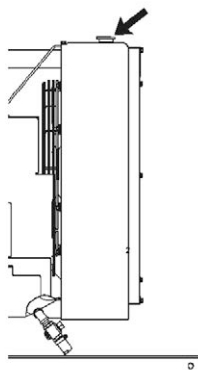


⚠ WARNING

Risk of burns. Contents under pressure. Do not remove the radiator pressure cap while engine is hot. Doing so could result in death or serious injury. (000322a)

Check the coolant level when the engine is stopped and cool.

1. Remove the cooling system filler cap slowly to relieve pressure.
2. Maintain the coolant level at the maximum mark that is correct for your application. If the engine is equipped with a sight glass, maintain the coolant level to the correct level in the sight glass.
3. Clean the cooling system filler cap and inspect the gasket. If the gasket is damaged, discard the old filler cap and install a new filler cap. If the gasket is not damaged, use a suitable pressurizing pump to pressure test the filler cap. The correct pressure is stamped on the face of the filler cap. If the filler cap does not retain the correct pressure, install a new filler cap.
4. Inspect the cooling system for leaks.



006848

Figure 6-5. Filler Cap Location

Test or Add Cooling System Supplemental Coolant Additive (SCA)

⚠ DANGER

Risk of Poisoning. Do not allow coolant to contact skin or eyes. If coolant is consumed, seek medical attention immediately. Failure to do so will result in death or serious injury. (000559)



⚠ WARNING

Risk of burns. Contents under pressure. Do not remove the radiator pressure cap while engine is hot. Doing so could result in death or serious injury. (000322a)

Test for SCA Concentration

IMPORTANT NOTE: Do not exceed the recommended six percent supplemental coolant additive concentration. Use a Coolant Conditioner Test Kit to check the concentration of the SCA.

NOTE: Excessive supplemental coolant additive concentration can form deposits on the higher temperature surfaces of the cooling system, reducing the engine's heat transfer characteristics. Reduced heat transfer could cause cracking of the cylinder head and other high temperature components. Excessive supplemental coolant additive concentration could also result in radiator tube blockage, overheating, and accelerated water pump seal wear. Never use both liquid supplemental coolant additive and the spin-on element (if equipped) at the same time. The use of those additives together could result in supplemental coolant additive concentration exceeding the recommended maximum.

Add SCA

1. Slowly loosen the cooling system filler cap to relieve pressure. Remove the cooling system filler cap.

NOTE: Always discard drained fluids according to local regulations.

2. If necessary, drain some coolant from the cooling system into a suitable container to allow space for the extra SCA.
3. Add the correct amount of SCA. See [Refill Capacities](#) for more information on SCA requirements.
4. Clean the cooling system filler cap and inspect the gasket. If the gasket is damaged, discard the old filler cap and install a new filler cap. If the gasket is not damaged, use a suitable pressurizing pump to pressure test the filler cap. The correct pressure is stamped on the face of the filler cap. If the filler cap does not retain the correct pressure, install a new filler cap.

Replace Cooling System Water Temperature Regulator

Replace the water temperature regulator before the water temperature regulator fails. This is a recommended preventive maintenance practice. Replacing the water temperature regulator reduces the chances for unscheduled downtime.

A water temperature regulator that fails in a partially opened position can cause overheating or over-cooling of the engine.

A water temperature regulator that fails in the closed position can cause excessive overheating. Excessive overheating could result in cracking of the cylinder head or piston seizure problems.

IMPORTANT NOTE: A water temperature regulator that fails in the open position will cause the engine operating temperature to be too low during partial load operation. Low engine operating temperatures during partial loads could cause an excessive carbon buildup inside the cylinders. This excessive carbon buildup could result in an accelerated wear of the piston rings and wear of the cylinder liner.

Refer to the [Water Temperature Regulator](#) for the replacement procedure of the water temperature regulator, or contact an IASD.

NOTE: If only the water temperature regulators are replaced, drain the coolant from the cooling system to a level below the water temperature regulator housing.

Check Driven Equipment

Refer to the OEM specifications for more information on the following maintenance recommendations for the driven equipment:

- Inspection
- Adjustment
- Lubrication
- Other maintenance recommendations

Perform any maintenance for the driven equipment which is recommended by the OEM.

Clean Engine



⚠ DANGER

Risk of fire. Allow fuel spills to completely dry before starting engine. Failure to do so will result in death or serious injury.

(000174)



⚠ WARNING

Electric Shock. Verify the electrical system is OFF before performing this procedure. Failure to do so could result in death or serious injury.

(000571)

Periodic cleaning of the engine is recommended. Steam cleaning the engine will remove accumulated oil and grease. A clean engine provides the following benefits:

- Easy detection of fluid leaks
- Maximum heat transfer characteristics
- Ease of maintenance

NOTE: Caution must be used to prevent electrical components from being damaged by excessive water when the engine is cleaned. Pressure washers and steam cleaners should not be directed at any electrical connectors or the junction of cables into the rear of the connectors. Avoid electrical components such as the alternator, and the starter. Protect the fuel injection pump from fluids to wash the engine.

Clean or Replace Air Cleaner Element



⚠ WARNING

Risk of fire. Never operate engine without the air cleaner installed. Operating engine without the air cleaner could result in death or serious injury.

(000249)

Service Air Cleaner Elements

If the air cleaner element becomes plugged, the air can split the material of the air cleaner element. Unfiltered air will drastically accelerate internal engine wear. Contact an IASD for the correct air cleaner element.

- Check the pre-cleaner (if equipped) daily for accumulation of dirt and debris. Remove any dirt and debris, as needed.
- Operating conditions (dust, dirt and debris) may require more frequent service of the air cleaner element.
- The air cleaner element may be cleaned up to six times if the element is properly cleaned and inspected.
- The air cleaner element should be replaced at least one time per year. This replacement should be performed regardless of the number of cleanings.

Replace the dirty paper air cleaner elements with clean air cleaner elements. Before installation, the air cleaner elements should be thoroughly checked for tears and holes in the filter material. Inspect the gasket or the seal of the air cleaner element for damage. Maintain a supply of suitable air cleaner elements for replacement purposes.

Clean the Air Cleaner Elements

NOTE: Certified air filter cleaning services are recommended for cleaning the air filter elements.

Observe the following guidelines for cleaning the filter element:

- Do not tap or strike the filter element to remove dust.
- Do not wash the filter element.
- Use low pressure compressed air to remove the dust from the filter element. Air pressure must not exceed 30 psi (207 kPa). Direct the air flow up the pleats and down the pleats from the inside of the filter element. Take extreme care to avoid damage to the pleats.
- Do not use air filters with damaged pleats, gaskets, or seals. Dirt entering the engine will cause damage to engine components.

The air cleaner element can be used up to six times if the element is properly cleaned and inspected. When the air cleaner element is cleaned, check for rips or tears in the filter material. The air cleaner element should be replaced at least once per year. This replacement should be performed regardless of the number of cleanings.

Use clean air cleaner elements while dirty elements are being cleaned.

IMPORTANT NOTE: Do not clean the air cleaner elements by bumping or tapping. This could damage the seals. Do not use elements with damaged pleats, gaskets or seals. Damaged elements will allow dirt to pass through.

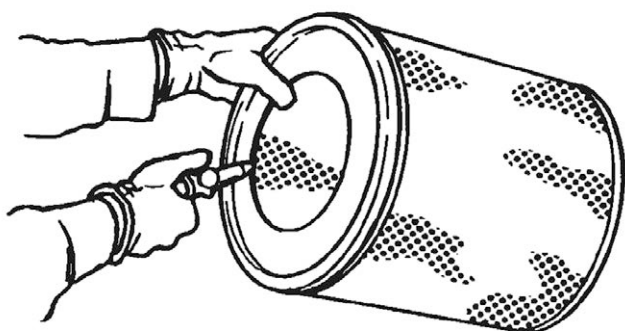
Visually inspect the air cleaner elements before cleaning. Inspect the air cleaner elements for damage to the seal, the gaskets, and the outer cover. Discard any damaged air cleaner elements.

There are two common methods used to clean air cleaner elements:

- Pressurized air
- Vacuum cleaning

Pressurized Air

Pressurized air can be used to clean air cleaner elements that have not been cleaned more than two times. Pressurized air will not remove deposits of carbon and oil. Use filtered, dry air with a maximum pressure of 30 psi (207 kPa).



006840

Figure 6-6. Clean Air Cleaner Elements

NOTE: When the air cleaner elements are cleaned, always begin with the clean side (inside) to force dirt particles toward the dirty side (outside).

Aim the hose so the air flows inside the element along the length of the filter to help prevent damage to the paper pleats. Do not aim the stream of air directly at the air cleaner element. Dirt could be forced further into the pleats.

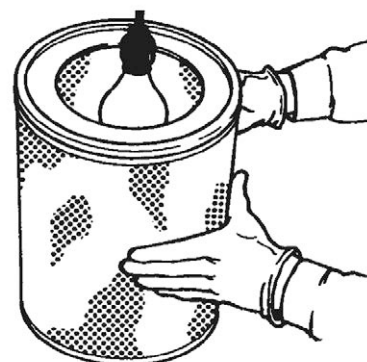
NOTE: See [Inspect Air Cleaner Elements](#).

Vacuum Cleaning

Vacuum cleaning is a good method for cleaning air cleaner elements which require daily cleaning because of a dry, dusty environment. Cleaning with pressurized air is recommended prior to vacuum cleaning. Vacuum cleaning will not remove deposits of carbon and oil.

NOTE: See [Inspect Air Cleaner Elements](#).

Inspect Air Cleaner Elements



006839

Figure 6-7. Inspect Air Cleaner Elements

Inspect the clean, dry air cleaner element. Use a 60 watt blue light in a dark room or in a similar facility. Place the blue light in the air cleaner element. Rotate and inspect for tears and holes. Inspect for light that may show through the filter material. If necessary, compare the air cleaner element to a new air cleaner element with the same part number.

Do not use an air cleaner element with tears or holes in the filter material. Do not use an air cleaner element with damaged pleats, gaskets or seals. Discard damaged air cleaner elements.

Storing Air Cleaner Elements

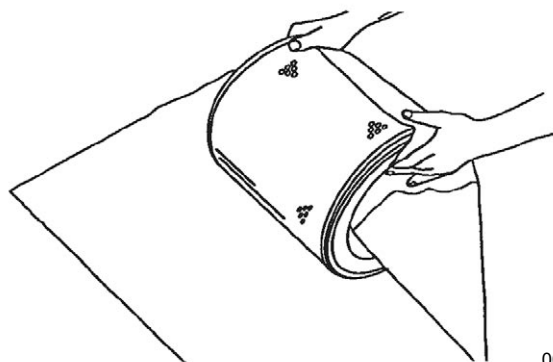
If an air cleaner element that passes inspection will not be used, the air cleaner element can be stored for future use.

Do not use paint, a waterproof cover, or plastic as a protective covering for storage. An air flow restriction may result. To protect against dirt and damage, wrap the air cleaner elements in Volatile Corrosion Inhibited (VCI) paper.

Place the air cleaner element into a box for storage. For identification, mark the outside of the box and mark the air cleaner element. Include the following information:

- Date of cleaning
- Number of cleanings

Store the box in a dry location.



007081

Figure 6-8. Store Air Cleaner Elements

Inspect or Replace Engine Air Cleaner Element



⚠ WARNING

Risk of fire. Never operate engine without the air cleaner installed. Operating engine without the air cleaner could result in death or serious injury.

(000249)

See **Inspect Engine Air Cleaner Service Indicator**.

A wide variety of air cleaners may be installed for use with this engine. See the OEM information for the correct procedure to replace the air cleaner.

Inspect Engine Air Cleaner Service Indicator

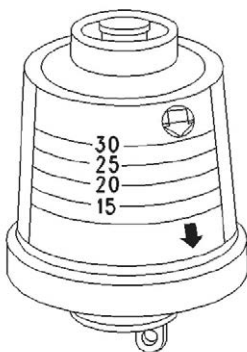
Some engines may be equipped with a different service indicator.

Some engines are equipped with a differential gauge for inlet air pressure. The differential gauge for inlet air pressure displays the difference in the pressure measured before the air cleaner element and the pressure measured after the air cleaner element. As the air cleaner element becomes dirty, the pressure differential rises. If the engine is equipped with a different type of service indicator, follow the OEM recommendations to service the air cleaner service indicator.

The service indicator may be mounted on the air cleaner element or in a remote location.

Observe the service indicator. The air cleaner element should be cleaned or replaced when one of the following conditions occur:

- The yellow diaphragm enters the red zone.
- The red piston locks in the visible position.



007065

Figure 6-9. Typical Service Indicator

Test the Service Indicator

Service indicators are important instruments.

- Check for ease of resetting. The service indicator should reset in less than three pushes.
- Check the movement of the yellow core when the engine is accelerated to the engine rated speed.

The yellow core should latch at the greatest vacuum attained.

If the service indicator does not reset easily, or if the yellow core does not latch at the greatest vacuum, the service indicator should be replaced. If the new service indicator will not reset, the hole for the service indicator may be restricted.

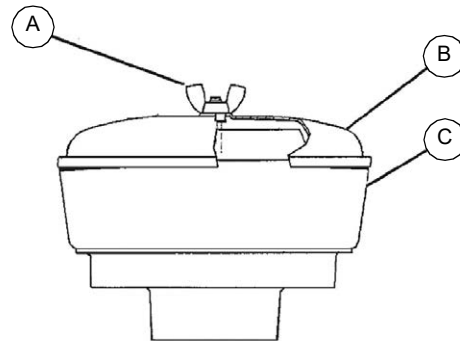
The service indicator may need to be replaced frequently in severely dusty environments.

Clean Engine Air Pre-cleaner

See **Figure 6-10**. Remove wing nut (A) and cover (B). Check for an accumulation of dirt and debris in body (C). Clean the body, if necessary.

After cleaning the pre-cleaner, install cover (B) and wing nut (A).

NOTE: When the engine is operated in dusty applications, more frequent cleaning is required.



006917

Figure 6-10. Typical Engine Air Pre-cleaner

Replace Engine Crankcase Breather

⚠ CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

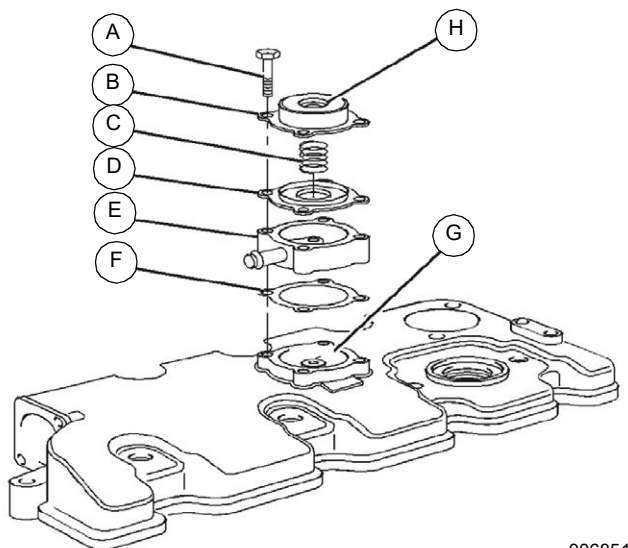
(000357)

1. See **Figure 6-11**. Loosen the screws (A) and remove the breather cover (B) from the valve mechanism cover.
2. Remove the spring (C). Remove the diaphragm and plate (D).
3. For turbocharged engines, remove the spacer (E) and the joint (F).
4. Clean the vent hole (H) and the cavity (G) in the valve mechanism cover.

IMPORTANT NOTE: Verify components of the breather assembly are installed correctly.

5. For turbocharged engines, install a new joint (F) and the spacer (E).

6. Install a new diaphragm and plate (D) for the breather assembly into the cavity (G) of the valve mechanism cover or the spacer (E) for turbocharged engines.
7. Install a new spring (C).
8. Install the breather cover (B) and the four screws (A). Tighten the screws.



006851

Figure 6-11. Engine Crankcase Breather

Inspect Engine Mounts

Inspect the engine mounts for deterioration and for correct bolt torque. Engine vibration can be caused by the following conditions:

- Incorrect mounting of the engine
- Deterioration of the engine mounts
- Loose engine mounts

Any engine mount that shows deterioration should be replaced. See the OEM information for the recommended torques.

NOTE: If applicable, see the OEM information for further information on the engine mounts and the correct bolt torque.

Check Engine Oil Level

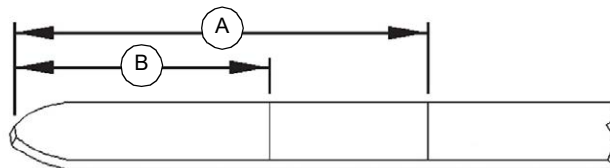
**WARNING**

Risk of burns. Allow engine to cool before draining oil or coolant. Failure to do so could result in death or serious injury.

(000139)

IMPORTANT NOTE: Perform this procedure with the engine OFF.

1. See **Figure 6-12**. Maintain the oil level between ADD mark (B) and FULL mark (B) on oil level gauge. Do not fill the crankcase above FULL mark (A).
2. Remove the oil filler cap and add oil, if necessary. Clean the oil filler cap. Install the oil filler cap.



007011

Figure 6-12. Engine Oil Dipstick

Change Engine Oil and Filter

**WARNING**

Risk of burns. Allow engine to cool before draining oil or coolant. Failure to do so could result in death or serious injury.

(000139)

CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

IMPORTANT NOTE: Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids. Dispose of all fluids according to local regulations and mandates.

Do not drain the oil when the engine is cold. As the oil cools, suspended waste particles settle on the bottom of the oil pan. The waste particles are not removed with the draining cold oil. Drain the crankcase with the engine stopped and the oil warm. This draining method allows the waste particles suspended in the oil to be drained correctly.

Failure to follow this recommended procedure will cause the waste particles to be recirculated through the engine lubrication system with the new oil.

Drain Engine Oil

After the engine has been run at the normal operating temperature, stop the engine. Use one of the following methods to drain the engine crankcase oil:

- If the engine is equipped with a drain valve, turn the drain valve knob counter-clockwise to drain the oil. After the oil has drained, turn the drain valve knob clockwise to close the drain valve.
- If the engine is not equipped with a drain valve, remove the oil drain plug to allow the oil to drain. After the oil has drained, the oil drain plug should be cleaned and installed.

Replace Oil Filter

1. Use a suitable tool to remove the oil filter.

NOTE: The following actions can be carried out as part of the preventive maintenance program.

2. Cut the oil filter open with a suitable tool. Break apart the pleats and inspect the oil filter for metal debris. An excessive amount of metal debris in the oil filter may indicate early wear or a pending failure.

Use a magnet to differentiate between the ferrous and non-ferrous metals found in the oil filter element. Ferrous metals may indicate wear on the steel and cast iron parts of the engine.

Non-ferrous metals may indicate wear on the aluminum parts, brass parts or bronze parts of the engine. Parts affected may include main bearings, rod bearings, turbocharger bearings and cylinder heads.

Due to normal wear and friction, it is not uncommon to find small amounts of debris in the oil filter. Contact an IASD if an excessive amount of debris is found in the oil filter.

NOTE: The oil cooler and the adapter are installed on engines with a turbocharger.

3. Clean the sealing surface of the cylinder block or oil cooler.
4. Apply clean engine oil to the new oil filter seal.

IMPORTANT NOTE: Do not fill oil filters with oil before installation.

5. Install the oil filter and tighten by hand. Do not over-tighten the oil filter.

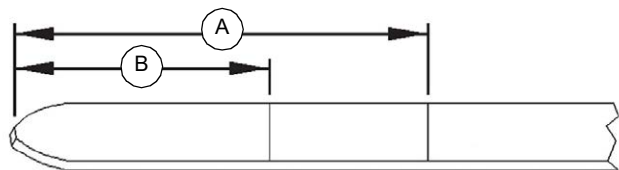
Fill Engine Crankcase

1. Remove the oil filler cap. See [Fluid Recommendations](#) for more information on lubricant specifications. Fill the crankcase with the correct amount of oil. See [Refill Capacities](#).

NOTE: If equipped with an auxiliary oil filter system or a remote oil filter system, follow the OEM or filter manufacturer's recommendations. Under-filling or over-filling the crankcase with oil can cause engine damage.

NOTE: Crank the engine with the fuel OFF. This will fill the oil filters before starting the engine. Do not crank the engine for more than 30 seconds.

2. Start the engine and run at LOW IDLE for two minutes. Perform this procedure to verify the lubrication system has oil and the oil filters are filled. Inspect the oil filter for oil leaks.
3. Stop the engine and allow the oil to drain back to the sump for a minimum of ten minutes.
4. See [Figure 6-13](#). Remove the oil level gauge to check the oil level. Maintain the oil level between the MIN (B) and MAX (A) marks on the oil level gauge.



007011

Figure 6-13. Engine Oil Dipstick

Inspect or Adjust Engine Valve Lash

⚠ WARNING

Accidental Start-up. Disconnect the negative battery cable, then the positive battery cable when working on unit. Failure to do so could result in death or serious injury. (000130)

⚠ WARNING

Equipment damage. Only qualified service personnel may install, operate, and maintain this equipment. Failure to follow proper installation requirements could result in death, serious injury, and equipment or property damage. (000182a)



⚠ WARNING

Risk of Burn. Allow the engine to cool before performing the following procedure. Failure to do so could result in serious injury. (000560)

Verify the engine is stopped before measuring the valve lash. The engine valve lash can be inspected and adjusted when the temperature of the engine is hot or cold. See [Inspect and Adjust Engine Valve Lash](#).

Test or Change Fuel Injector



⚠ DANGER

Risk of Fire. Fuel spilled or leaked onto hot surfaces or electrical components can cause a fire. Verify the battery is disconnected before performing this procedure. Failure to do so could result in death or serious injury. (000561)



⚠ WARNING

Vision loss. Eye protection is required when servicing unit. Failure to do so could result in vision loss or serious injury.

(000377)

⚠ CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

Generac recommends regular maintenance of the fuel injectors. The fuel injectors must be removed and tested by an IASD. The fuel injectors should not be cleaned as cleaning with incorrect tools can damage the nozzle. The fuel injectors should be renewed only if a fault with the fuel injectors occurs. Some problems that indicate new fuel injectors may be needed are listed below:

- The engine will not start or is difficult to start
- Not enough power
- The engine misfires or runs erratically
- High fuel consumption
- Black exhaust smoke
- The engine knocks or there is vibration in the engine
- Excessive engine temperature

For more information on the removal and installation of fuel injectors, see [Fuel Injector](#).

For more information on the testing of fuel injectors, see [Test Fuel Injector](#).

Identification of a Suspect Fuel Injector



⚠ DANGER

Hydraulic Fluid Injection. High-pressure, high-temperature hydraulic fluid can pierce skin and cause severe burns. Do not check for leaks with hands. Seek immediate medical attention in case of accident. Failure to protect body accordingly will result in death or serious injury. (000239)



⚠ WARNING

Moving Parts. Keep clothing, hair, and appendages away from moving parts. Failure to do so could result in death or serious injury.

(000111)

Operate the engine at a fast idle speed to identify the faulty fuel injector. Individually loosen and tighten the union nut for the high pressure pipe to each fuel injector. Do not loosen the union nut more than half a turn. There will be little effect on the engine speed when the union nut to the faulty fuel injector is loosened. Contact an IASD for further assistance.

Prime Fuel System

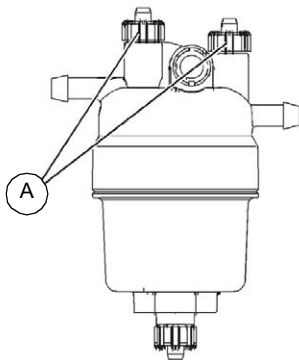
IMPORTANT NOTE: Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids. Dispose of all fluids according to local regulations and mandates.

If air enters the fuel system, it must be purged before the engine can be started. Air can enter the fuel system when the following events occur:

- The fuel tank is empty or has been partially drained
- The low pressure fuel lines are disconnected
- A leak exists in the low pressure fuel system
- The fuel filter is replaced

Primary Filter

See [Figure 6-14](#). Use the vent screws (A) to remove air from the primary filter before priming the fuel filters.



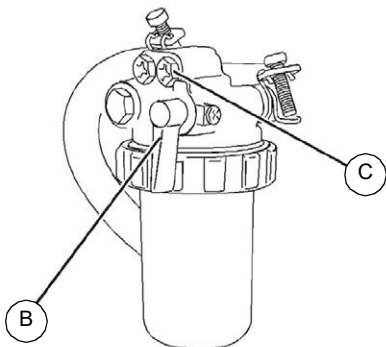
007032

Figure 6-14. Primary Fuel Filter

Fuel Filters

There are three types of fuel filters that may be installed on the engine:

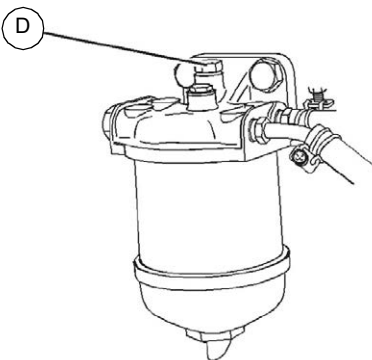
- Element ([Figure 6-15](#))
- Canister ([Figure 6-16](#))
- Spin-on filter with fuel priming pump ([Figure 6-17](#))



006916

B	Fuel valve
C	Vent screw

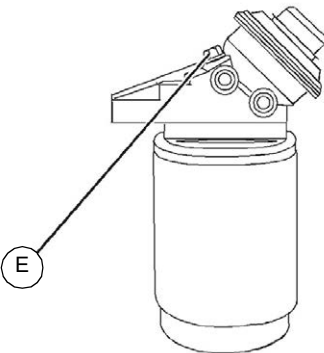
Figure 6-15. Element



006835

D	Vent screw
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Figure 6-16. Canister



007069

E	Vent screw
---	------------

Figure 6-17. Spin-on Filter with Fuel Priming Pump

Priming the System

See [Figure 6-14](#). Use the vent screws (A) to remove air from the primary filter. Operate the priming pump. When fuel free from air flows from the vent screw tighten the vent screw.

NOTE: Some fuel systems use gravity to prime the primary fuel filter. If gravity is used, verify the fuel tank is full and all stop valves in the fuel line are open.

There are three different types of systems that can be installed on the engine to prime the fuel system. See [Figure 6-18](#).

- Hand priming pump (F)
- In-line hand priming pump (G)
- Transfer pump (H)

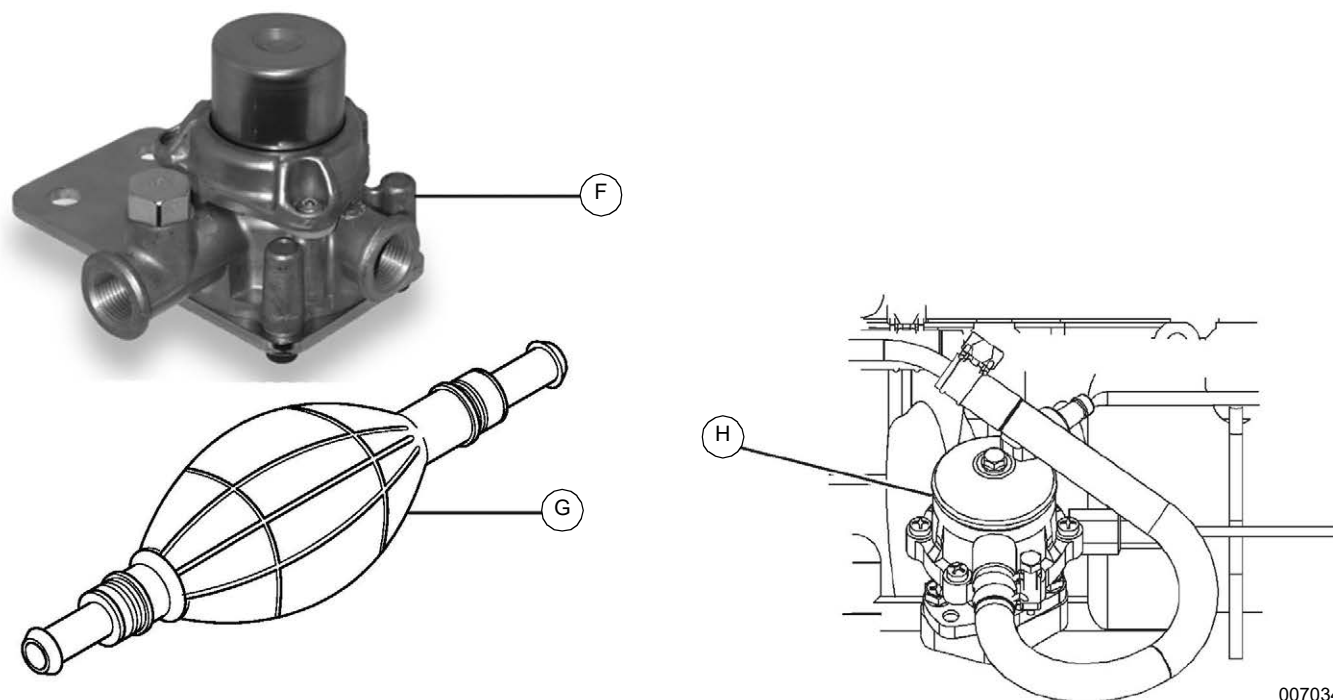


Figure 6-18. Pump Types

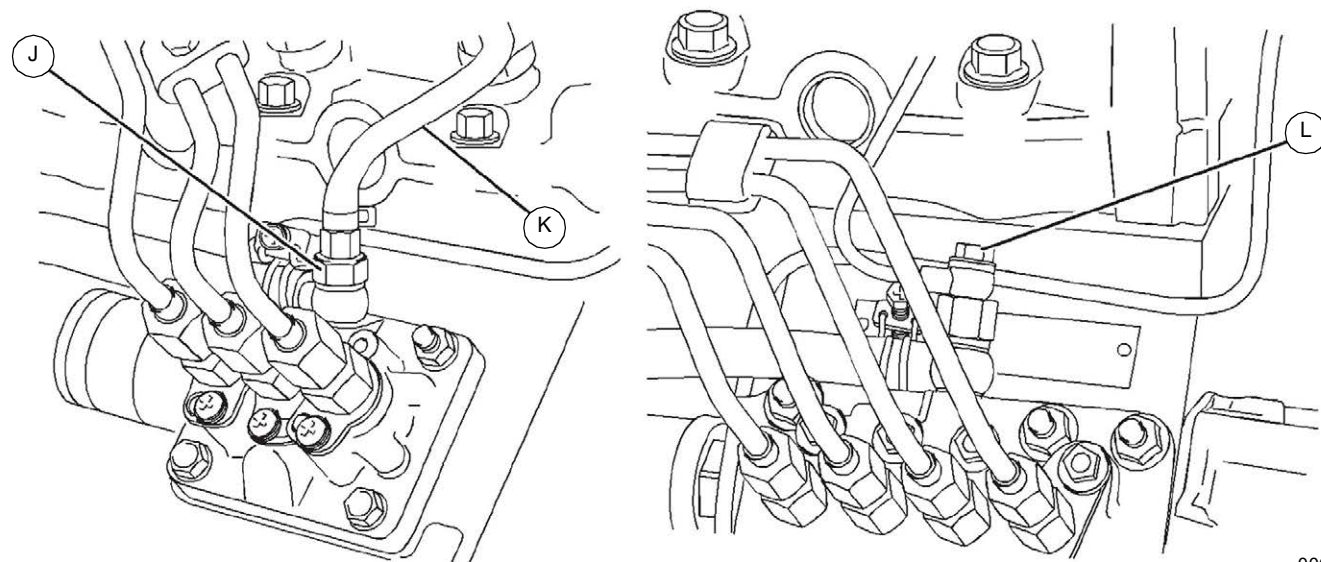


Figure 6-19. Fuel Return Line and Connector Bolts

Hand Priming Pump

IMPORTANT NOTE: Refer to [Figure 6-14](#) through [Figure 6-19](#) for the following procedure.

NOTE: See [Figure 6-18](#), item F to identify the hand priming pump.

1. Verify the fuel valve (B) for the fuel filter that has an element is in the ON position.
2. Loosen vent screw (C, D or E) on the fuel filter.

3. Operate hand priming pump (F). When fuel free from air flows from the vent screw tighten the vent screw.
4. Loosen connection (J or L) at the fuel injection pump.

NOTE: Fuel return line (K) may need to be removed to prime the fuel system.

5. Operate the hand priming pump. When fuel free from air flows from the connections, tighten the connecting bolt.
6. The engine should now be able to start. Operate the starting motor to start the engine.

NOTE: Do not operate the starting motor for more than 15 seconds. If the engine does not start after 15 seconds, stop and wait 30 seconds before trying again.

In-line Priming Pump

IMPORTANT NOTE: Refer to [Figure 6-14](#) through [Figure 6-19](#) for the following procedure.

NOTE: See [Figure 6-18](#), item G to identify the hand priming pump.

1. Verify the fuel valve (B) for the fuel filter with an element is in the ON position.
2. Loosen vent screw (C, D or E) on the fuel filter.
3. Operate in-line priming pump (G). When fuel free from air flows from the vent screw tighten the vent screw.
4. Loosen connection (J or L) at the fuel injection pump.

NOTE: Fuel return line (K) may need to be removed to prime the fuel system.

5. Operate the in-line priming pump. When fuel free from air flows from the connections tighten the connecting bolt.
6. The engine should now be able to start. Operate the starting motor to start the engine.

NOTE: Do not operate the starting motor for more than 15 seconds. If the engine does not start after 15 seconds, stop and wait for 30 seconds before trying again.

Electrical Priming Pump

IMPORTANT NOTE: Refer to [Figure 6-14](#) through [Figure 6-19](#) for the following procedure.

NOTE: See [Figure 6-18](#), item H to identify the hand priming pump.

1. Verify the fuel valve (B) for the fuel filter that has an element is ON.
2. Loosen vent screw (C, D or E) on the fuel filter.
3. Operate electrical priming pump (H). When fuel free from air flows from the vent screw tighten the vent screw. Switch off the electrical priming pump.
4. Loosen connection (J or L) at the fuel injection pump.

NOTE: Fuel return line (K) may need to be removed to prime the fuel system.

5. Operate the in-line priming pump. When fuel free from air flows from the connections tighten the connecting bolt.
6. The engine should now be able to start. Operate the starting motor to start the engine.

NOTE: Do not operate the starting motor for more than 15 seconds. If the engine does not start after 15 seconds, stop and wait for 30 seconds before trying again.

Fuel Transfer Pump

IMPORTANT NOTE: Refer to [Figure 6-14](#) through [Figure 6-19](#) for the following procedure.

NOTE: See [Figure 6-18](#), item I to identify the hand priming pump.

NOTE: To use the transfer pump, operate the starting motor. Do not operate the starting motor for more than 15 seconds. After 15 seconds, stop and wait for 30 seconds before operating the starting motor.

1. Verify the fuel valve (B) for the fuel filter with an element is in the ON position.
2. Loosen vent screw (C, D or E) on the fuel filter.
3. Operate fuel transfer pump (I). When fuel free from air flows from the vent screw tighten the vent screw. Stop the fuel transfer pump.
4. Loosen connection (J or L) at the fuel injection pump.

NOTE: Fuel return line (K) may need to be removed to prime the fuel system.

5. Operate the fuel transfer pump. When fuel free from air flows from the connections tighten the connecting bolt. Stop the transfer pump.
6. The engine should now be able to start. Operate the starting motor to start the engine.

NOTE: Do not operate the starting motor for more than 15 seconds. If the engine does not start after 15 seconds, stop and wait 30 seconds before trying again.

Replace Fuel System Filter



⚠ DANGER

Risk of Fire. Fuel spilled or leaked onto hot surfaces or electrical components can cause a fire. Verify the battery is disconnected before performing this procedure. Failure to do so could result in death or serious injury. (000561)

⚠ CAUTION

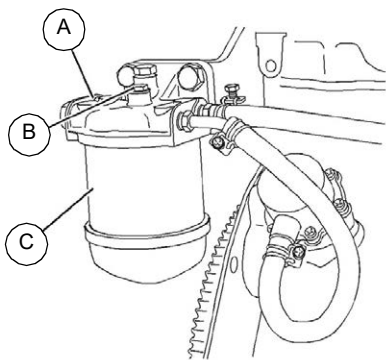
Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage.

(000357)

IMPORTANT NOTE: Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids. Dispose of all fluids according to local regulations and mandates.

Fuel Filter with Canister

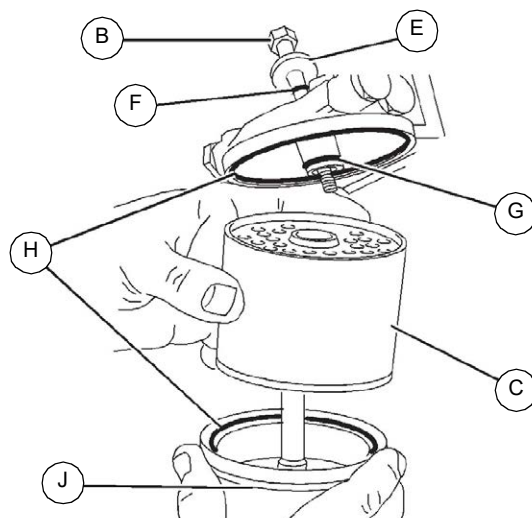
1. Close the fuel supply valve.
2. See [Figure 6-20](#). Clean the outside of the fuel filter assembly (A).
3. Remove setscrew (B).
4. Remove the canister (C). Verify any fluid is drained into a suitable container.



006963

Figure 6-20. Fuel Filter Canister

5. See [Figure 6-21](#). Assemble the following items: seals (H), seal (G), canister (C) and bowl (J). Place washer (E) and seal (F) on setscrew (B).
6. Fasten the assembly to the fuel filter base with setscrew (B).



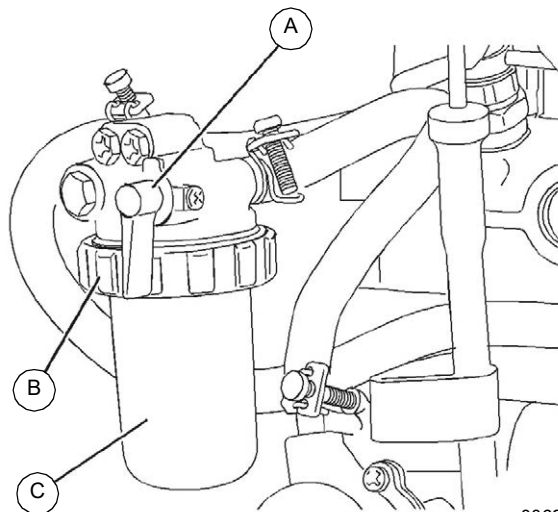
006947

Figure 6-21. Fuel Filter Canister

NOTE: The fuel system must be primed after the new filter is installed. See [Prime Fuel System](#).

Fuel Filter with Element

1. See [Figure 6-22](#). Close the fuel supply valve (A).
2. Clean the outside of the fuel filter assembly.
3. Loosen the locking ring (B).
4. Remove the casing (C) for the filter and the element. Drain any fluids into a suitable container.



006949

Figure 6-22. Fuel Filter with Element

NOTE: Do not fill the fuel filter with fuel. The fuel will not be filtered and the fuel could be contaminated. Contaminated fuel can damage the fuel system.

5. See [Figure 6-23](#). Assemble the following items: seal (E), filter element (F) and casing (C).
6. Install the assembled items to the filter base (D).

7. Install the locking ring (B) to the filter head. Rotate the locking ring to lock the assembly. The fuel system must be primed after the new filter is installed. See [Prime Fuel System](#).

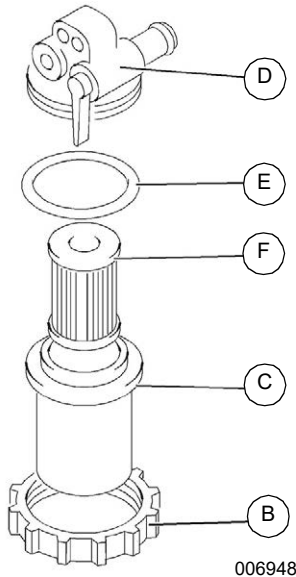


Figure 6-23. Fuel Filter with Element

Fuel Filter with Priming Pump

1. Close the fuel supply valve.
2. Clean the outside of the fuel filter assembly.
3. See [Figure 6-24](#). Use a suitable tool to remove the spin-on filter (A). Verify any fluid is drained into a suitable container.

NOTE: Do not fill the fuel filter with fuel. The fuel will not be filtered and the fuel could be contaminated. Contaminated fuel can damage your fuel system.

4. Install the new spin-on filter. Tighten the spin-on filter by hand.

NOTE: The fuel system will need to be primed after the new filter is installed. See [Prime Fuel System](#).

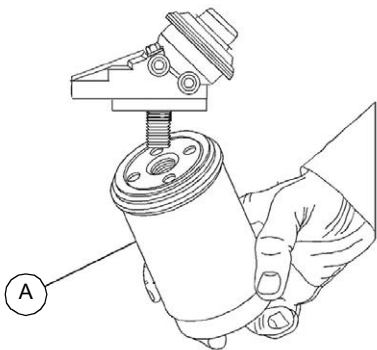


Figure 6-24. Spin-on Filter

Drain Fuel System Water Separator



⚠ DANGER

Risk of Fire. Fuel spilled or leaked onto hot surfaces or electrical components can cause a fire. Verify the battery is disconnected before performing this procedure. Failure to do so could result in death or serious injury. (000561)

IMPORTANT NOTE: Do not run engine with water separator more than half full.

1. See [Figure 6-25](#). Open drain (A). Catch the draining fluid in a suitable container. Dispose of the drained fluid correctly.
2. Close drain (A).

IMPORTANT NOTE: The water separator is under suction during normal engine operation. Verify the drain valve is tightened securely to help prevent air from entering the fuel system.

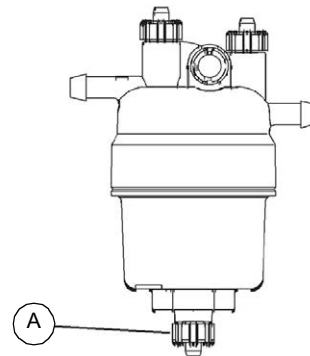


Figure 6-25. Water Separator

Drain Fuel Tank Water and Sediment

IMPORTANT NOTE: Care must be taken to ensure that fluids are contained during performance of inspection, maintenance, testing, adjusting and repair of the product. Be prepared to collect the fluid with suitable containers before opening any compartment or disassembling any component containing fluids. Dispose of all fluids according to local regulations and mandates.

Fuel Tank

Fuel quality is critical to the performance and to the service life of the engine. Water in the fuel can cause excessive wear to the fuel system.

Water can be introduced into the fuel tank when the fuel tank is being filled.

Condensation occurs during the heating and cooling of fuel. The condensation occurs as the fuel passes through the fuel system and the fuel returns to the fuel tank. This causes water to accumulate in fuel tanks. Draining the fuel tank regularly and obtaining fuel from reliable sources can help to eliminate water in the fuel.

Drain Water and Sediment

Fuel tanks should contain some provision for draining water and draining sediment from the bottom of the fuel tanks.

1. Open the drain valve on the bottom of the fuel tank to drain water and sediment.
2. Close the drain valve.

Check the fuel daily. Allow five minutes after the fuel tank has been filled before draining water and sediment from the fuel tank.

Fill the fuel tank after operating the engine to drive out moist air. This will help prevent condensation. Do not fill the tank to the top. The fuel expands as the fuel gets warm. The tank may overflow.

Some fuel tanks use supply pipes that allow water and sediment to settle below the end of the fuel supply pipe. Some fuel tanks use supply lines that take fuel directly from the bottom of the tank. If the engine is equipped with this system, regular maintenance of the fuel system filter is important.

Fuel Storage Tanks

Drain the water and the sediment from the fuel storage tank at the following intervals:

- Weekly
- Service intervals
- Refill of the tank

This will help prevent water or sediment from being pumped from the storage tank into the engine fuel tank.

If a bulk storage tank has been refilled or moved recently, allow adequate time for the sediment to settle before filling the engine fuel tank. Internal baffles in the bulk storage tank will also help trap sediment. Filtering fuel pumped from the storage tank helps to ensure the quality of the fuel. When possible, water separators should be used.

Inspect or Replace Hoses and Clamps

Inspect all hoses for leaks caused by the following conditions:

- Cracking
- Softness
- Loose clamps

Replace cracked or soft hoses. Tighten any loose clamps.

IMPORTANT NOTE: Do not bend or strike high pressure lines. Do not install bent or damaged lines, tubes or hoses. Repair any loose or damaged fuel and oil lines, tubes and hoses. Leaks can cause fires. Inspect all lines, tubes and hoses carefully. Tighten all connections to the recommended torque. Do not clip any other item to the high pressure lines.

Check for the following conditions:

- Damaged or leaking end fittings
- Chafed or cut outer covering
- Exposed wire used for reinforcement
- Ballooning outer covering
- Flexible part of the hose is kinked or crushed
- Armoring embedded in the outer covering

A constant torque hose clamp can be used in place of any standard hose clamp. Verify the constant torque hose clamp is the same size as the standard clamp.

Due to extreme temperature changes, the hose will harden. Hardening of the hoses will cause hose clamps to loosen. This can result in leaks. A constant torque hose clamp will help to prevent loose hose clamps.

Each installation application can be different. The differences depend on the following factors:

- Type of hose
- Type of fitting material
- Anticipated expansion and contraction of the hose
- Anticipated expansion and contraction of the fittings

Replace Hoses and Clamps

See the OEM information for further information on removing and replacing fuel hoses (if equipped).

The following describes a typical method of replacing coolant hoses. See the OEM information for further information on the coolant system and the hoses for the coolant system.



WARNING

Risk of burns. Contents under pressure. Do not remove the radiator pressure cap while engine is hot. Doing so could result in death or serious injury. (000322a)

1. Stop the engine and allow to cool.
2. Loosen the cooling system filler cap slowly to relieve pressure. Remove the cooling system filler cap.

NOTE: Drain the coolant into a suitable, clean container. The coolant can be reused.

3. Drain the coolant from the cooling system to a level that is below the hose being replaced.
4. Remove hose clamps.
5. Disconnect old hose.
6. Replace old hose with a new hose.
7. Install hose clamps with a torque wrench.

NOTE: For the correct coolant, see [Fluid Recommendations](#).

8. Refill the cooling system. Refer to the OEM information for more information on refilling the cooling system.

9. Clean the cooling system filler cap. Inspect the cooling system filler cap seals and replace the cap if damaged. Install the cooling system filler cap.
10. Start the engine. Inspect the cooling system for leaks.

Clean Radiator

The following describes a typical cleaning procedure for the radiator. Refer to the OEM information for more information on cleaning the radiator.

NOTE: Adjust the frequency of cleaning according to the effects of the operating environment.

Inspect the radiator for these items: Damaged fins, corrosion, dirt, grease, insects, leaves, oil and other debris. Clean the radiator, if necessary.

IMPORTANT NOTE: Maximum air pressure at the nozzle must be less than 30 psi (205 kPa) for cleaning purposes.

Pressurized air is the preferred method for removing loose debris. Direct the air in the opposite direction to the fan air flow. Hold the nozzle approximately 0.25 in (6 mm) away from the radiator fins. Slowly move the air nozzle in a direction parallel with the radiator tube assembly. This will remove debris from between the tubes.

Pressurized water may also be used for cleaning. The maximum water pressure for cleaning purposes must be less than 40 psi (275 kPa). Use pressurized water to soften mud. Clean the core from both sides.

Use a degreaser and steam for removal of oil and grease. Clean both sides of the core. Wash the core with detergent and hot water. Thoroughly rinse the core with clean water.

If the radiator is blocked internally, see the OEM Manual for information regarding flushing the cooling system.

After cleaning the radiator, start the engine. Allow the engine to operate at low idle speed for three to five minutes. Accelerate the engine to high idle. This will help in the removal of debris and the drying of the core. Slowly reduce the engine speed to low idle and then stop the engine. Use a light bulb behind the core to inspect the core for cleanliness. Repeat the cleaning, if necessary.

Inspect the fins for damage. Bent fins may be opened with a "comb". Inspect these items for good condition: Welds, mounting brackets, air lines, connections, clamps and seals. Make repairs, if necessary.

Check Severe Service Application

Severe service is the application of an engine that exceeds the current published standards for that engine. Generac maintains standards for the following engine parameters:

- Performance such as power range, speed range, and fuel consumption
- Fuel quality
- Operational altitude
- Maintenance intervals
- Oil selection and maintenance
- Coolant type and maintenance

- Environmental qualities
- Installation
- Fluid temperature in the engine

Refer to the standards for the engine or contact an IASD to determine if the engine is operating within the defined parameters.

Severe service operation can accelerate component wear. Engines operating under severe conditions may need more frequent maintenance intervals to ensure maximum reliability and retention of full service life.

Due to individual applications, it is not possible to identify every factor which can contribute to severe service operation. Contact an IASD for the unique maintenance necessary for the engine.

The operating environment, incorrect operating procedures and incorrect maintenance procedures can be factors which contribute to a severe service application.

Environmental Factors

Ambient temperatures – The engine may be exposed to extended operation in extremely cold or hot environments. Valve components can be damaged by carbon buildup if the engine is frequently started and stopped in very cold temperatures. Extremely hot intake air reduces engine performance.

Quality of the air – The engine may be exposed to extended operation in a dirty or dusty environment, unless the equipment is cleaned regularly. Mud, dirt and dust can encase components making maintenance very difficult. The buildup can contain corrosive chemicals.

Buildup – Compounds, elements, corrosive chemicals and salt can damage some components.

Altitude – Problems can arise when the engine is operated at higher altitudes than the intended settings for that application. Necessary adjustments should be made.

Incorrect Operating Procedures

- Extended operation at low idle
- Frequent hot shutdowns
- Operating at excessive loads
- Operating at excessive speeds
- Operating outside the intended application

Incorrect Maintenance Procedures

- Extending the maintenance intervals
- Failure to use recommended fuel, lubricants, coolant and antifreeze

Inspect Starting Motor

A scheduled inspection of the starting motor is recommended. If the starting motor fails, the engine may not start in an emergency situation.

Check the starting motor for correct operation. Check the electrical connections and clean the electrical connections. **Test Electric Starting System** for more information on the checking procedure and for specifications, or contact an IASD for assistance.

Inspect Turbocharger (if equipped)

A regular visual inspection of the turbocharger is recommended. Any fumes from the crankcase are filtered through the air inlet system. Byproducts from oil and combustion can collect in the turbocharger compressor housing. Over time, this buildup can contribute to loss of engine power, increased black smoke and overall loss of engine efficiency.

If the turbocharger fails during engine operation, damage to the turbocharger compressor wheel and engine may occur. Damage to the turbocharger compressor wheel can cause additional damage to the pistons, valves and cylinder head.

IMPORTANT NOTE: Turbocharger bearing failures can cause large quantities of oil to enter the air intake and exhaust systems. Loss of engine lubricant can result in serious engine damage. Minor leakage of oil into a turbocharger under extended low idle operation should not cause problems as long as a turbocharger bearing failure has not occurred. When a turbocharger bearing failure is accompanied by a significant engine performance loss (exhaust smoke or engine rpm up at no load), do not continue engine operation until the turbocharger is renewed.

A visual inspection of the turbocharger can minimize unscheduled downtime. A visual inspection of the turbocharger can also reduce the chance for potential damage to other engine parts.

Removal and Installation

NOTE: The turbochargers supplied are non-serviceable.

For options regarding the removal, installation, and replacement, contact an IASD. See **Turbocharger** for more information.

Inspecting

IMPORTANT NOTE: The compressor housing for the turbocharger must not be removed from the turbocharger for cleaning. The actuator linkage is connected to the compressor housing. If the actuator linkage is moved or disturbed the engine may not comply with emissions legislation.

1. Remove the pipe from the turbocharger exhaust outlet and remove the air intake pipe to the turbocharger. Visually inspect the piping for the presence of oil. Clean the interior of the pipes to prevent dirt from entering during reassembly.
2. Check for the presence of oil. If oil is leaking from the back side of the compressor wheel, there is a possibility of a failed turbocharger oil seal.

NOTE: The presence of oil may be the result of extended engine operation at low idle. The presence of oil may also be the result of a restriction of the line for the intake air (clogged air filters), which causes the turbocharger to slobber.

3. Inspect the bore of the housing of the turbine outlet for corrosion.
4. Fasten the air intake pipe and the exhaust outlet pipe to the turbocharger housing.

Walkaround Inspection

Inspect Engine for Leaks and Loose Connections

A walkaround inspection should only take a few minutes. When the time is taken to perform these checks, costly repairs and accidents can be avoided.

For maximum engine service life, make a thorough inspection of the engine compartment before starting the engine. Look for items such as oil leaks or coolant leaks, loose bolts, worn belts, loose connections and trash buildup. Make repairs as needed:

- Repair or replace damaged guards
- Wipe all caps and plugs before the engine is serviced to reduce the chance of system contamination

For any type of leak (coolant, lube or fuel) clean up the fluid. If leaking is observed, find the source and correct the leak. If leaking is suspected, check the fluid levels more often than recommended until the leak is found or fixed, or until the suspicion of a leak is proved to be unwarranted.



⚠ DANGER

Risk of fire. Allow fuel spills to completely dry before starting engine. Failure to do so will result in death or serious injury.

(000174)

- Verify cooling system hoses are correctly clamped and tight. Check for leaks. Check the condition of all pipes.
- Inspect water pump for coolant leaks.

NOTE: The water pump seal is lubricated by the coolant in the cooling system. It is normal for a small amount of leakage to occur as the engine cools down and the parts contract.

Excessive coolant leakage may indicate the need to replace the water pump seal. For the removal of the water pump and the installation of water pump or seal, refer to the **Water Pump** for more information or contact an IASD.

- Inspect the lubrication system for leaks at the front crankshaft seal, the rear crankshaft seal, the oil pan, the oil filters and the rocker cover

- Inspect the fuel system for leaks, and look for loose fuel line clamps or tie-wraps
- Inspect the piping for the air intake system and the elbows for cracks and for loose clamps
- Verify hoses and tubes are not contacting other hoses, tubes, wiring harnesses, etc
- Inspect the alternator belts and any accessory drive belts for cracks, breaks or other damage

Belts for multiple groove pulleys must be replaced as matched sets. If only one belt is replaced, the belt will carry more load than the belts that are not replaced. The older belts are stretched. The additional load on the new belt could cause the belt to break.

- Drain the water and the sediment from the fuel tank on a daily basis to ensure only clean fuel enters the fuel system.
- Inspect the wiring and the wiring harnesses for loose connections and for worn wires or frayed wires.
- Inspect the ground strap for a good connection and for good condition.
- Disconnect any battery chargers not protected against the current drain of the starting motor. Check the condition and the electrolyte level of the batteries, unless the engine is equipped with a maintenance free battery.
- Check the condition of the gauges and replace any that are cracked or cannot be calibrated.

Inspect Water Pump

A failed water pump may cause severe engine overheating problems that could result in the following conditions:

- Cracks in the cylinder head
- A piston seizure
- Other potential damage to the engine

NOTE: The water pump seal is lubricated by the coolant in the cooling system. It is normal for a small amount of leakage to occur as the engine cools down and parts contract.

Visually inspect the water pump for leaks. Renew the water pump seal or the water pump if there is an excessive leakage of coolant. See [Water Pump](#) for the disassembly and assembly procedure.

Section 7 Systems Operation Testing and Adjusting

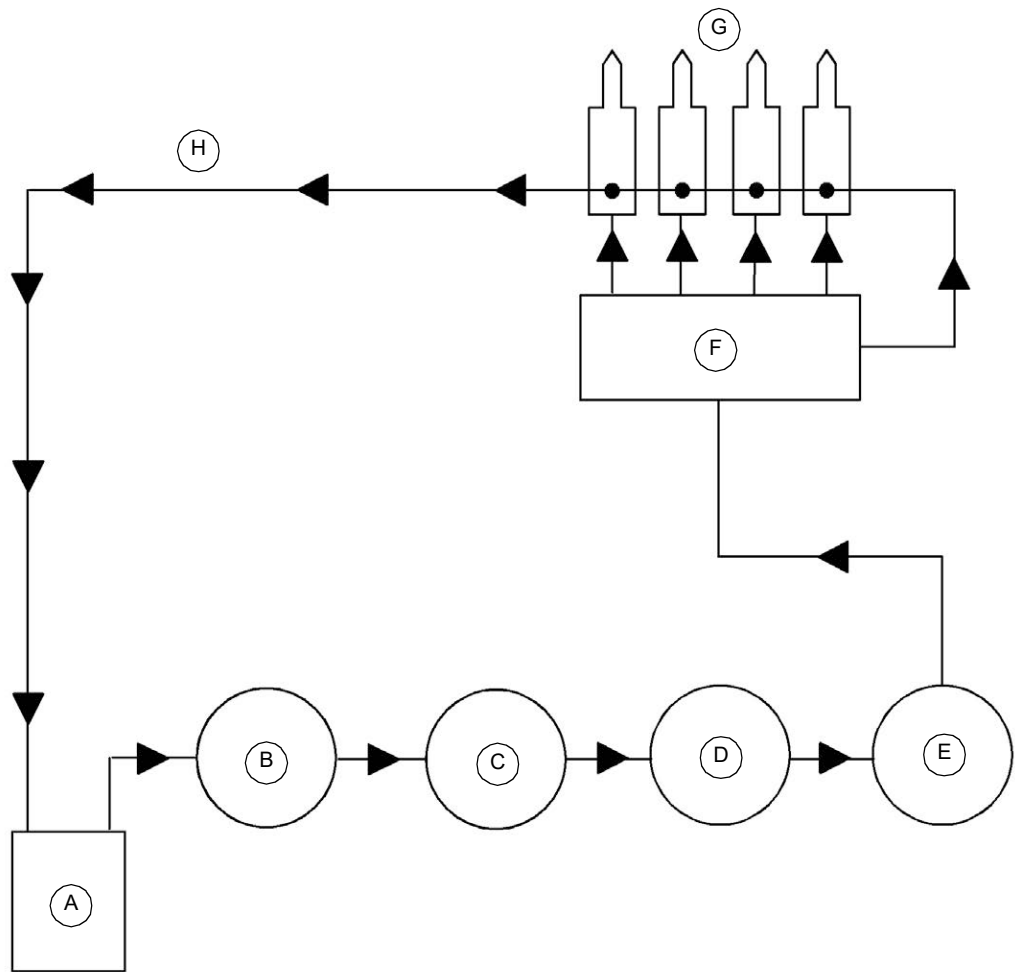
Fuel System

General Operation of the Fuel System

See **Figure 7-1**. When the engine is cranking, the fuel is pulled from the fuel tank (A) by the fuel priming pump (D). An optional sediment/water trap (C) may be installed between the fuel tank (A) and the fuel priming pump (D). The fuel priming pump forces the fuel through the fuel filter (E) to the fuel injection pump (F). The fuel injection pump sends fuel at high pressure to each fuel injector (G). The fuel injector sprays fuel into a pre-combustion chamber which slows the rate of combustion in the cylinder. The following items will result from reducing the rate of fuel combustion: prevention of engine knock, reduction of noise and reduction of emissions.

The fuel filter (E) can also function as a water separator. The fuel filter can be drained through a valve located at the bottom of the filter housing.

The fuel injection pump sends fuel at high pressure to each fuel injector (F). The fuel injector sprays fuel into a pre-combustion chamber which slows the rate of combustion in the cylinder. The following items will result from reducing the rate of fuel combustion: prevention of engine knock, reduction of noise and reduction of emissions.

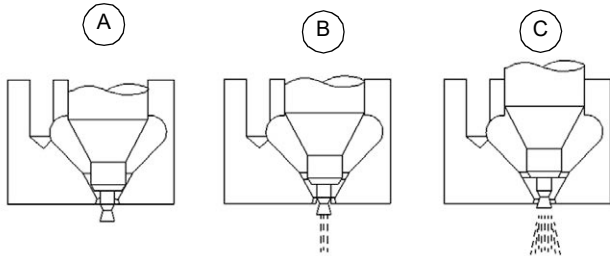


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A	Fuel tank	E	Fuel filter
B	Hand primer	F	Fuel injection pump
C	Sediment trap	G	Fuel injectors
D	Fuel priming pump	H	Fuel return line from fuel injection pump and fuel injectors to fuel tank

Figure 7-1. Fuel System

See **Figure 7-2**. The fuel injector injects fuel into the pre-combustion chamber at different angles during two phases. Most of the fuel is injected when the valve is fully open (C). This process is called indirect fuel injection. The results are more even combustion and complete combustion of the fuel at a reduced temperature. Improved fuel combustion will increase power output while reducing emissions and reducing fuel consumption.



006964

Figure 7-2. Fuel Injector Phases of Operation

See **Figure 7-1**. Excess fuel from the fuel injectors and the fuel injection pump flows through the fuel return line (G) and back to the fuel tank (A). The excess fuel aids the cooling of the fuel injectors. Also, the fuel return line removes any air that is trapped in the fuel injectors and the fuel injection pump.

The fuel injection pump needs fuel for lubrication. If the precision parts of the pump are not adequately lubricated, the components may be easily damaged. The engine must not be started until the fuel injection pump is full of fuel that is free of air.

The system must be primed when any part of the system is drained of fuel. The following list contains examples of both service and repairs when you must prime the system:

- The fuel filter is changed.
- The low pressure fuel line is removed.
- The fuel injection pump is removed.
- The fuel injectors are removed.
- The fuel tank is drained.
- A leak exists in the low pressure side of the fuel system.

to release air from the fuel injection pump and the fuel injectors, see **Prime Fuel System**.

Governor

The fuel rack is connected to the linkage, which controls the fuel injection pump. This linkage is located in the timing case (front housing).

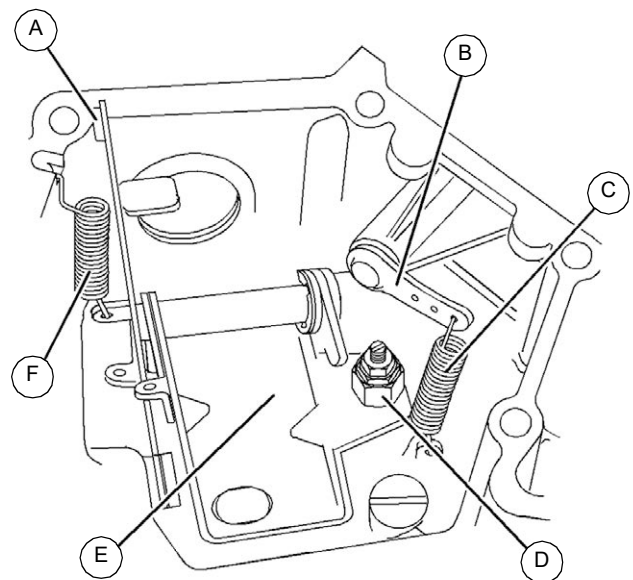
This engine has a mechanical governor to control engine speed. The governor operates for all engine RPM. The governor weight assembly is installed on the front of the gear of the camshaft. The other components of the governor are installed in the front housing.

See **Figure 7-3** and **Figure 7-4**. The movement of the governor weight assembly is transferred to the fuel rack on the fuel injection pump by the control lever (B), the governor lever (E) and the linkage to the fuel injection pump. The governor main spring (C) connects the governor lever to the control lever. The governor main spring controls the movement of the governor weight assembly on the camshaft. When the angle of the control lever changes, the tension on the governor main spring changes. This action controls the linkage to the fuel rack on the fuel injection pump, which controls the engine RPM.

The maximum fuel adjustment screw is mounted in the front housing. This adjustment regulates the fuel injection at high engine RPM. This adjustment should only be made by personnel with the correct training.

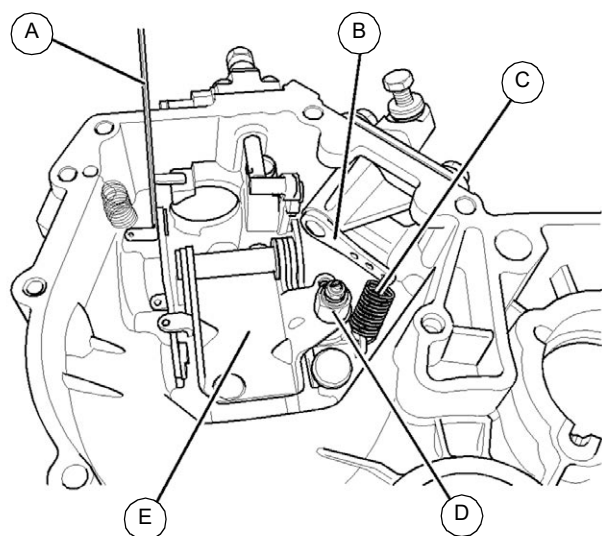
The fuel injection pump timing, the low idle, and the high idle are preset at the factory. Adjustments to the pump timing and idle RPM should only be made by personnel that have had the correct training. The timing for the fuel injection pump should only change if the camshaft or the cylinder block are replaced. The fuel injection pump timing should not change if the fuel injection pump is reinstalled with a shim of the same size.

The fuel rack automatically returns to the excess fuel position when the engine is stopped. The excess fuel position aids the cold starting of the engine. A spring connects the linkage to the fuel injection pump and mechanical stop control (B). When the engine is first started, the spring automatically increases the fuel flow to the cylinders.



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Figure 7-3. Governor Control Mechanism in the Front Housing without a BCD



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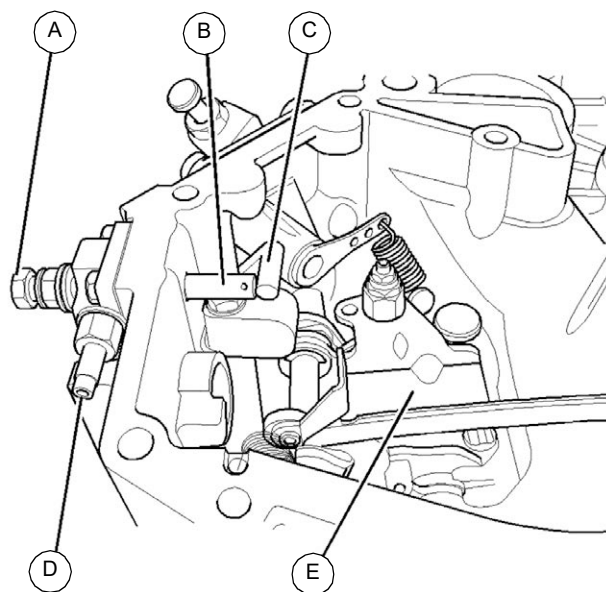
Figure 7-4. Governor Control Mechanism in the Front Housing with a BCD

Boost Compensation Device for Turbocharged Engines (if equipped)

If equipped, the Boost Compensation Device (BCD) can be installed on turbocharged engines. The BCD prevents over-fueling and the production of black smoke during acceleration from low idle.

See [Figure 7-5](#). When the engine is accelerated from a low RPM, the governor lever (E) on the fuel rack comes into contact with the BCD fuel stopper (C) to prevent excessive movement of the governor lever. This prevents over-fueling.

As the engine RPM increases, the increased intake manifold pressure is felt on the diaphragm in the BCD. As the BCD fuel stopper is moved by the diaphragm piston, the governor lever is allowed to operate over the full range.



006826

Figure 7-5. Boost Compensation Device (BCD)

Fuel Injection Pump

The electric fuel shutoff solenoid must be energized for fuel to flow through the system.

See [Figure 7-6](#). The fuel injection pump is a cassette type pump. The cassette type pump contains the following components: fuel delivery valves, fuel rack and pushrods. The fuel injection pump is installed directly into the cylinder block.

The part number and code letters for the fuel injection pump are stamped on the front of the pump. The fuel injection pump is a pressurized system that is totally enclosed. The pump sends the correct amount of fuel under high pressure at the correct time through the fuel injectors to the individual cylinders. The fuel injection occurs near the end of the compression stroke. The fuel injection pump regulates the amount of fuel delivered to the fuel injectors. This action controls the engine RPM by the governor setting or the position of the throttle control.

The camshaft is driven by the idler gear in the timing case. Lobes on the camshaft cause the pushrod for each cylinder to reciprocate. The reciprocating motion first draws the fuel. The reciprocating motion then pressurizes the fuel. A fuel delivery valve (B) for each cylinder acts as a check valve to prevent a loss of pressure to the fuel injector.

The correct operation of the fuel injection pump requires the pump to be completely full of fuel and empty of air. When the vent screw (E) is loosened, air can escape from the fuel injection pump.

The fuel injection pump will lubricate the components during operation.

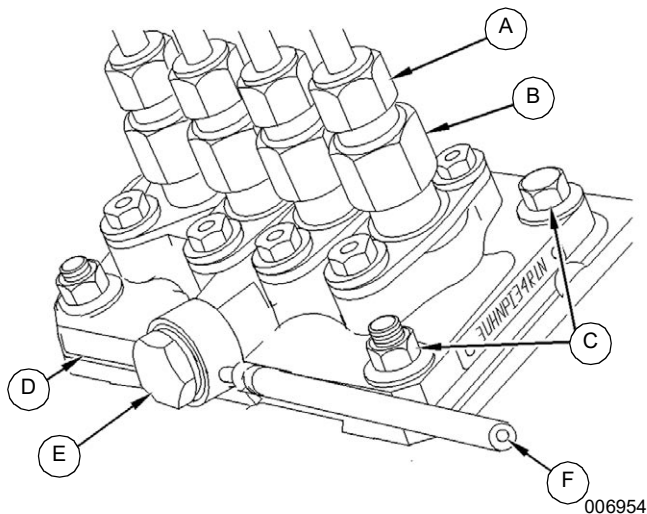


Figure 7-6. Fuel Injection Pump

Fuel Injectors

See **Figure 7-7**. The sealing washer (B) helps to prevent blowby. The sealing washer also sets the projection of the fuel injector (A) into the pre-combustion chamber. This projection affects the time required for combustion in the cylinder. If the projection is excessive, engine knock can occur at high RPM.

NOTE: When a fuel injector (A) is installed in the cylinder head, a new sealing washer (B) should be installed.

The operating pressure of the fuel injector is set and tested at the factory. See **Fuel Injector** for the pressure settings of the fuel injector. During operation, extra fuel is used as coolant and lubricant for components of the fuel injector.

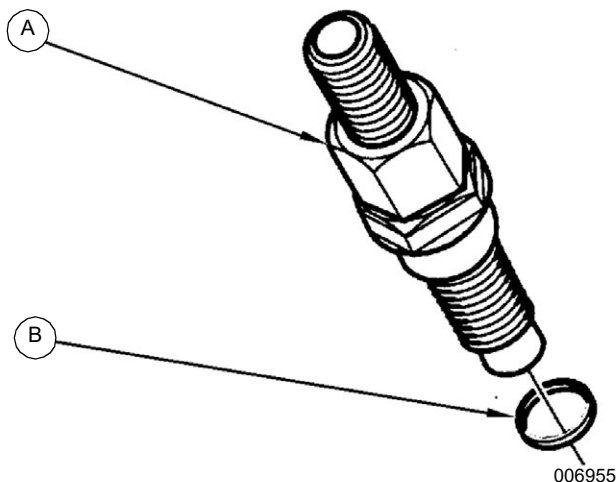


Figure 7-7. Fuel Injectors

Fuel Priming Pump

The fuel priming pump creates a vacuum to force fuel from the fuel tank. Pressure is created to pump the fuel to the fuel injection pump. The diaphragm prevents the leakage of the fuel into the cylinder block. The diaphragm

creates part of the pumping action of the fuel priming pump. The outlet valve and the inlet valve operate as check valves.

A lobe on the camshaft causes the arm to move up and down. When the arm moves down, the diaphragm assembly moves down. This down movement increases the size of the chamber above the diaphragm assembly. This suction opens the inlet valve which draws fuel into the chamber above the diaphragm assembly to the fuel priming pump.

When the arm moves up, the diaphragm assembly moves up. This up movement pressurizes the chamber above the diaphragm assembly. This pressure opens the outlet valve which allows fuel to flow out of the fuel priming pump toward the fuel injection pump.

Lift pump provides 0.5 gpm (1.8 Lpm) at approximately 2.9 psi (20 kPa).

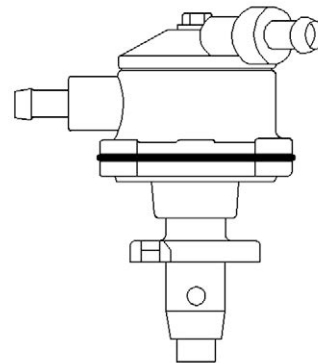


Figure 7-8. Fuel Priming Pump

Glow Plugs

Each cylinder has a glow plug to aid the cold starting of the engine. The glow plugs may be energized by two different methods:

- The ignition switch is moved to the auxiliary position (if equipped)
- The starting aid switch (if equipped) is turned ON

In cold ambient temperatures, energizing the glow plugs for six seconds will heat the cylinder sufficiently for easy starting of the engine.

Air Inlet and Exhaust System

Naturally aspirated engines pull outside air through an air cleaner directly into the inlet manifold. The air flows from the inlet manifold to the engine cylinders. The fuel is mixed with the air in the engine cylinders. After the fuel combustion occurs in the engine cylinder, the exhaust gases flow directly to the outside air through the exhaust manifold and the exhaust system.

Turbocharged engines pull outside air through an air cleaner into the air intake of the turbocharger. The suction is caused by the turbocharger compressor wheel. Then, the turbocharger compressor wheel compresses the air. The air flows through the inlet manifold which directs an even distribution of the air to each engine

cylinder. Air is pulled into the engine cylinder during the intake stroke of the piston. Then, the air is mixed with fuel from the fuel injectors. Each piston makes four strokes:

1. Intake

Air is drawn into the cylinder through the open inlet valve. Fuel is sprayed into the engine by the fuel injector.

2. Compression

The mixture of air and fuel is compressed in the cylinder to heat the mixture to the temperature of combustion.

3. Power

The mixture of air and fuel ignites at the top of the compression stroke. The expansion of gases from the combustion forces the piston downward. This force creates the power of the engine.

4. Exhaust

The piston moves upward to force the gases of combustion from the cylinder through the open exhaust valve.

The sequence of the strokes by all of the pistons in all of the engine cylinders provides constant air flow from the air inlet system during the engine operation. The valve mechanism cover contains a closed breather assembly. The gases in the valve cover, which are caused by blowby, pass from the crankcase to the inlet manifold. The breather is sealed from the outside air by a diaphragm. Above the diaphragm, the cover is vented to the outside air by a small vent hole so that pressure does not build up.

Turbocharger

A turbocharger increases the temperature and the density of the air sent to the engine cylinder. This condition causes a lower temperature of ignition to develop earlier in the compression stroke. The compression stroke is also timed in a more accurate way with the fuel injection. Surplus air lowers the temperature of combustion. This surplus air also provides internal cooling.

A turbocharger improves the following aspects of engine performance:

- Power output is increased.
- Engine torque is increased.
- Engine efficiency is increased.

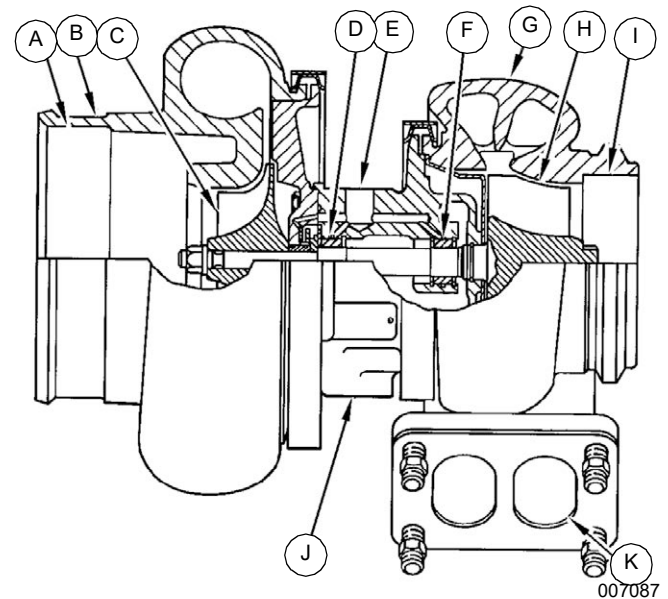
See **Figure 7-9**. A turbocharger is installed between the exhaust and intake manifolds. The turbocharger is driven by exhaust gases which flow through the exhaust inlet (K). The energy of the exhaust gas turns the turbine wheel (H). Then, the exhaust gas flows out of the turbine housing (G) through the exhaust outlet (I).

The turbine wheel (H) and the compressor wheel (C) are installed on the same shaft. Therefore, the turbine wheel (H) and the compressor wheel (C) rotate at the same RPM. The compressor wheel is enclosed by the compressor housing (B). The compressor wheel compresses the intake air. The intake air flows into the engine cylinders through the inlet valves of the cylinders.

The oil from the main gallery of the cylinder block flows through the oil inlet port (E) to lubricate the turbocharger bearings (D) and (F). The pressurized oil passes through the bearing housing of the turbocharger. The oil is returned through the oil outlet port (J) to the oil pan.

The turbocharger has a wastegate. The wastegate is controlled by the boost pressure. This allows some of the exhaust gases to bypass the turbine wheel at higher engine speeds. The wastegate is a type of flapper valve that automatically opens at a preset level of boost pressure to allow the exhaust gas to flow around the turbine. The wastegate allows the design of the turbocharger to be more effective at lower engine speeds.

Boost pressure is approximately 8.4 psi (58 kPa).



A	Air intake
B	Compressor housing
C	Compressor wheel
D	Bearing
E	Oil inlet port
F	Bearing
G	Turbine housing
H	Turbine wheel
I	Exhaust outlet
J	Oil outlet port
K	Exhaust inlet

Figure 7-9. Turbocharger Components

Cylinder Head and Valves

The valves and the rocker shaft assembly control the flow of air into the cylinders and out of the cylinders during engine operation. The cylinder head assembly has two valves for each cylinder. Each valve has one valve spring (C). The ports for the inlet valve (K) and the exhaust valve (J) are on the left side of the cylinder head.

The valve moves along a steel valve guide (E). The valve guides can be replaced.

The inlet valve and the exhaust valve are opened and closed by the rotation and movement of the following components:

- Crankshaft
- Idler gear
- Camshaft
- Valve lifters
- Pushrods
- Rocker arms
- Valve springs

The camshaft gear is driven by the idler gear. The camshaft gear, the idler gear and the crankshaft gear are timed together. When the camshaft turns, the valve lifters are moved up and down. The pushrods move the rocker arm levers. The rocker arm levers make the inlet valves and the exhaust valves open and close. This is in sequence with the firing order of the engine. The valve springs force the valves back to the closed position.

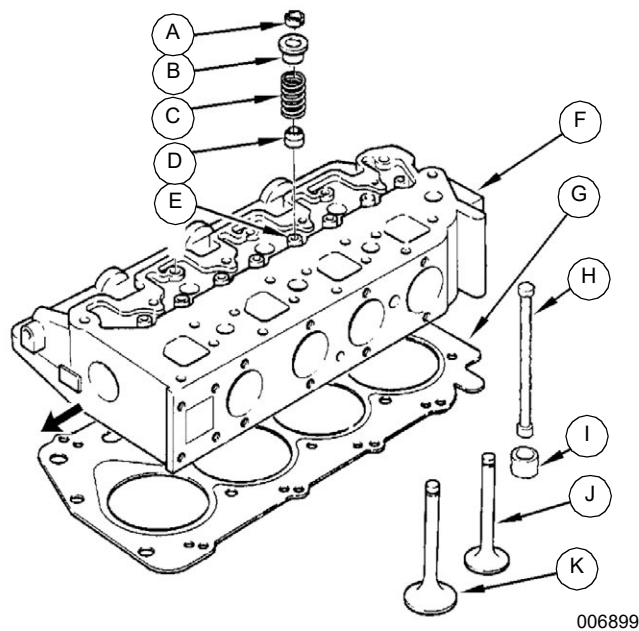


Figure 7-10. Cylinder Head and Valves

Lubrication System

The lubrication system contains the following components:

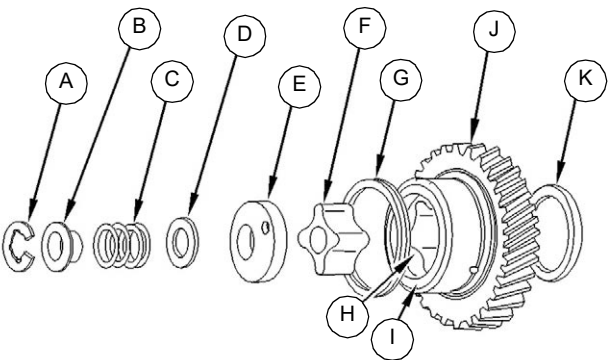
- Oil pump
- Engine oil relief valve
- Engine oil cooler (turbocharged engines)
- Oil filter
- Oil pan
- Oil strainer and suction pipe
- Oil level gauge
- Oil pressure switch
- An oil supply line to the cylinder head

The oil pump is contained within the idler gear. The engine oil relief valve is installed in the right side of the cylinder block. The oil filter is a disposable spin-on filter.

Pressure for the lubrication system is supplied by an engine oil pump which uses rotors. The oil pump is part of the idler gear (J). The idler gear is driven by the crankshaft gear.

The oil pump has an inner rotor (F) and an outer rotor (H). The axes of rotation of the rotors are off-center relative to each other. There is a pin that is inserted through a hole in the oil pump cover (E) into the outer rotor. The pin functions as a key to keep the outer rotor in a fixed position with the idler gear.

The outer rotor is pressed into the bush (I). The bush is pressed into the idler gear (J).



006928

A	Snap ring
B	Collar
C	Spring
D	Shim
E	Oil pump cover
F	Inner rotor
G	Spring
H	Outer rotor
I	Bush
J	Idler gear
K	Thrust washer

Figure 7-11. Idler Gear and Components

The inner rotor has four lobes which mesh with the five lobes of the outer rotor. When the outer lobe rotates, the distance increases between the lobes of the outer rotor and the lobes of the inner rotor to create suction. Then, the space between the lobes is filled with oil. When the distance decreases between the lobes, pressure is created. This pressure forces the oil into the chamber for the engine oil relief valve.

Lubricating oil from the oil pan flows through a strainer and a line to the suction side of the engine oil pump. The suction side is in the timing gear case. The lubricating oil flows from the outlet side of the pump to a relief valve. The relief valve is installed on the right side of the cylinder block. The lubricating oil, which flows around the relief valve, flows to the oil filter.

When the engine RPM increases, the flow rate of the oil pump increases. The increase in the flow rate from the oil pump causes the pressure to increase. The relief valve opens if the oil pressure is too high. When the oil pressure on the plunger of the relief valve is greater than the force of the spring in the relief valve, the relief valve opens. The lubricating oil which flows through the relief valve is returned to the oil pan.

The oil filter is installed on the right side of the cylinder block. Turbocharged engines have an engine oil cooler installed between the oil filter and the cylinder block. The oil flows through the oil filter into the main oil gallery. The main oil gallery is drilled through the total length of the right side of the cylinder block.

Oil flows from the main oil gallery through an externally mounted oil supply line to the cylinder head. An oil pressure switch measures the oil pressure at this location. This oil lubricates the rocker arm assembly. The oil passes through the rocker shaft to the bore of each rocker arm lever. Then, the oil flows from the rocker arm levers through holes located in the top of the rocker arm levers. The valve stems, the valve springs, and the tappets are lubricated by the splash and the mist of the oil.

The lubricating oil flows through drilled holes in the main oil gallery to passages in the main journals of the crankshaft. Then, the oil flows to the main bearings of the crankshaft. Also, the oil flows through passages in the crankshaft to the large end bearings of the connecting rods. The piston bearings, the pistons, and the cylinder bores are lubricated by the splash and the mist of the oil.

A hole is located in the bore of each main bearing. This hole allows oil to flow through passages that lubricate the journals of the camshaft for the valves. The bearing for the front journal receives oil from the front main journal of the crankshaft. The camshaft is lubricated by the splash of the oil.

The timing gears are lubricated by the splash of the oil. Lubricating oil from the timing case returns to the oil pan.

Cooling System

The coolant system contains the following components:

- Radiator
- Pressure cap for the radiator
- Fan for the radiator
- Drive pulley (if equipped) for the fan
- Water pump
- Drive pulley for the water pump
- Water temperature regulator
- Housing for the water temperature regulator
- Coolant temperature switch

The coolant flows from the bottom of the radiator to the centrifugal water pump. The water pump is installed on the cylinder block above the timing case. The water pump is driven by a pulley. The crankshaft pulley turns a belt which drives the pulley of the water pump.

The water pump forces the coolant to flow to the water temperature regulator. When the engine is cold, the water temperature regulator is closed. Then, the coolant flows directly into the cylinder head. When the engine warms, the water temperature regulator begins to open. Then, the regulator allows some of the coolant to flow to the top of the radiator.

The regulator opens fully when the engine reaches the normal operating temperature. When the regulator is fully open, the flow to the radiator is the maximum. However, the regulator does not close the flow of coolant into the cylinder head.

Coolant flows continuously through the cylinder head and the top of the cylinder block. This coolant flows into the back of the water pump from the cylinder block. This coolant then mixes with the coolant that is pumped from the radiator by the water pump.

The water temperature regulator maintains the correct engine temperature by adjusting the direct flow of coolant to the top of the radiator. The coolant is cooled by the radiator. Heat is removed from the coolant by cooler air which passes over the radiator fins. The fan causes a high volume of air to flow between the radiator fins to provide sufficient cooling. The coolant flows from the radiator through the bottom hose to the coolant pump.

The engine has a housing for the water temperature regulator. The housing is installed on the left side of the cylinder head.

The water temperature regulator housing for the engine is mounted vertically.

Basic Engine

Cylinder Head and Block

The cylinder head assembly has one inlet valve and one exhaust valve for each cylinder. Each valve has a single valve spring and a valve seal. The valve and the valve spring are held in position by a valve spring retainer and two collets. The valve seal fits over the top of the valve guide. The valve guides can be replaced.

The ports for the inlet and for the exhaust valves are on the left side of the cylinder head.

The cylinder block does not have cylinder liners. The cylinder walls are honed.

The valve mechanism cover is made from aluminum. The cover contains the following components:

- A closed breather assembly
- An oil filler cap
- A seal for the face toward the cylinder head
- Holes for four cap nuts
- Holes for two setscrews

The cap nuts are threaded onto studs. The steel studs are threaded into the cylinder head.

The setscrews are threaded onto the cover for the rocker shaft assembly.

The valve mechanism cover contains a crankcase breather. The gases in the valve cover, which are caused by blowby, pass from the crankcase to the inlet manifold.

Pistons and Connecting Rods

The pistons of the engine have a combustion chamber in the crown of the piston to provide an efficient mix of the fuel and the air.

The piston pin is off-center to reduce the noise level.

The pistons have two compression rings and an oil control ring.

The correct piston height is important to ensure the piston does not contact the cylinder head. The correct piston height also ensures the efficient combustion of fuel which is necessary to conform to requirements for emissions. The piston must not be machined to obtain the correct piston height.

A piston and connecting rod are matched to each cylinder.

One height of piston is available. The inside of the piston is marked "Shibaura". This reference mark faces the right side of the engine. See [Inspect Piston Height](#) for procedures to measure the piston height.

Identification marks on the connecting rod and the connecting rod cap must align. The identification marks face the right side of the engine.

The main bearing caps have a chamfered edge on one side. The chamfer faces the front of the engine.

Timing Gear Case and Gears

The timing gear case contains the following components:

- Front oil seal for the crankshaft pulley
- Bearing for the crankshaft gear
- Engine oil pump
- Adjustment screw for setting the maximum fuel position
- Adjustment screw for setting the maximum speed position
- Mechanical stop control
- Linkage for the fuel injection pump
- Fuel Control lever for the governor
- Throttle

The crankshaft for the D1.5L engine has four main journal bearings, and the crankshaft for the D2.2L engine has five main journal bearings. The crankshaft has a front bearing that is pressed in the cylinder block.

The thrust washers are hemispherical in shape. The D1.5L engine has two thrust washers on both sides of the bottom half of the holder for the rear main bearing. The D2.2L engine has a third thrust washer on the rear side of the upper half of the holder for the rear main bearing.

NOTE: The thrust washers on the D1.5L engine are machined into the aluminum holder for the rear main bearing.

The crankshaft has a front oil seal and a rear oil seal.

The timing case is made of aluminum. The timing gears are stamped with timing marks to ensure the correct assembly of the gears.

The number one piston is at the top center position on the compression stroke when the following timing marks are aligned:

- Idler gear, crankshaft gear, and camshaft gear
- Crankshaft pulley and timing case

The crankshaft gear turns the idler gear which then turns the camshaft gear.

The idler gear contains the oil pump. The idler gear turns the gear for the tachometer.

The crankshaft pulley drives the pulley on the water pump and the pulley which drives the alternator.

Electrical System

The electrical system has two separate circuits.

- Charging circuit
- Starting circuit

Some of the electrical system components are used in more than one circuit. The following items are common in each of the circuits:

- Battery
- Circuit breaker
- Cables
- Wires for the battery

The charging circuit is in operation when the engine is running. An alternator converts mechanical energy to electrical energy for the charging circuit. A voltage regulator in the circuit controls the electrical output to keep the battery at full charge.

If the engine has a disconnect switch, the starting circuit can operate only after the disconnect switch is put in the ON position.



Equipment Damage. The disconnect switch must be ON for the electrical system to function. Equipment damage will occur if disconnect switch is OFF.

(000572)

The starting switch is in operation only when the start switch is activated.

The charging circuit is connected through the ammeter. The starting circuit is not connected through the ammeter.

Automatic Shutdown Conditions

The engine will shut down if the following conditions continue for more than ten seconds while the engine is started or for two seconds while the engine is operated:
Coolant temperature exceeds 230 ± 5 °F (110 ± 3 °C)

Coolant temperature exceeds the following values:

Engine oil pressure switch located on the valve mechanism cover 4.27 psi (29 kPa)

Engine oil pressure switch located on the cylinder block 14.22 psi (98 kPa)

NOTE: There is no automatic shutdown for low coolant level.

NOTE: The capacity of the diode must be 3 amperes with a reverse voltage of 600 volts.

NOTE: The rated current draw of the engine oil pressure switch is 0.42 amperes (5 watts maximum bulb).

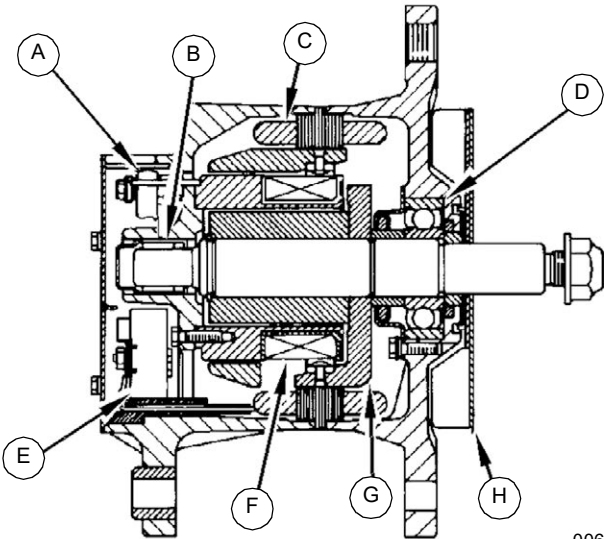
NOTE: If necessary, a delay fuse should be used for replacement.

Charging System Components



Equipment Damage. Never operate the alternator without the battery in the circuit. Making or breaking an alternator connection can result in damage to the regulator. (000573)

Alternator



006836

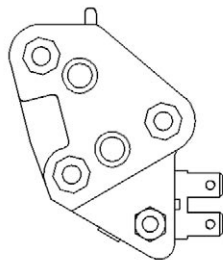
A	Regulator
B	Roller bearing
C	Stator winding
D	Ball bearing
E	Rectifier bridge
F	Field winding
G	Rotor assembly
H	Fan

Figure 7-12. Alternator

The alternator is driven by the crankshaft pulley through a belt. When the engine is running, the pulley rotates the shaft inside the alternator. The rotor assembly is attached to the shaft. The rotor assembly has many magnetic poles. The magnetic poles are similar to fingers. An air space exists between each of the opposite poles. The poles have residual magnetism that produces a small amount of magnet-like lines of force (magnetic field). This magnetic field is produced between the poles. As the rotor assembly begins to turn between the field winding and the stator windings, a small amount of alternating current (AC) is produced in the stator windings. The alternating current is produced from the small magnetic lines of force created by the residual magnetism of the poles. The AC is changed into direct current (DC) when

the current passes through the diodes of the rectifier bridge. Most of this current provides the battery charge and the supply for the low amperage circuit. The remainder of the current is sent to the field windings. The DC current flow through the field windings (wires around an iron core) increases the strength of the magnetic lines of force. These stronger magnetic lines of force increase the amount of AC produced in the stator windings. The increased speed of the rotor assembly also increases the current output of the alternator and the voltage output of the alternator.

Regulator



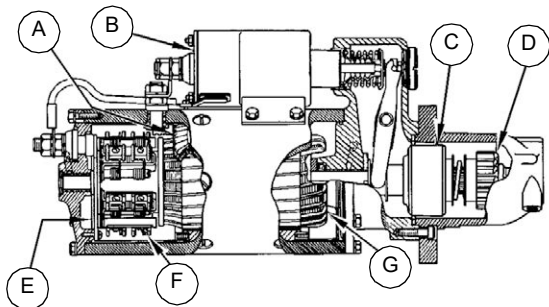
007037

Figure 7-13. Regulator Assembly

The voltage regulator is a solid-state electronic switch. The voltage regulator senses the voltage of the system. The regulator then uses switches to control the current to the field windings. This controls the voltage output to meet the electrical demand of the system.

Starting System Components

Electric Starting Motor



007071

A	Field
B	Solenoid
C	Clutch
D	Starter pinion
E	Commutator
F	Brush assembly
G	Armature

Figure 7-14. Starter Motor

The starting motor rotates the engine flywheel at a rate that is fast enough to start the engine.

See [Figure 7-14](#). The starting motor has a solenoid (B). When the start switch is activated, the solenoid (B) will move the starter pinion (D) to engage the starter pinion (D) to the ring gear on the engine flywheel. The starter pinion (D) and the ring gear will engage when the circuit between the battery and the starting motor is closed by the electric contacts in the solenoid (B). When the circuit between the battery and the starting motor is complete, the starter pinion (D) will rotate the engine flywheel. A clutch provides protection for the starting motor so the engine cannot turn the starting motor too fast. When the switch is released, the starter pinion (D) will move away from the ring gear.

Solenoid

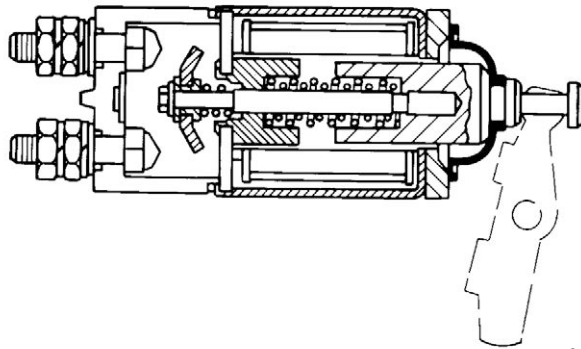


Figure 7-15. Solenoid

A solenoid is an electromagnetic switch that performs two basic functions:

- The solenoid closes the high current starting motor circuit with a low current start switch circuit.
- The solenoid engages the starting motor pinion with the ring gear.

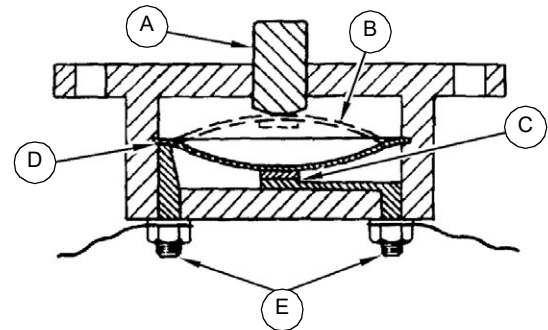
The solenoid has windings (one set or two sets) around a hollow cylinder. A plunger with a spring load device is inside of the cylinder. The plunger can move forward and backward. When the start switch is closed and electricity is sent through the windings, a magnetic field is created. The magnetic field pulls the plunger forward in the cylinder. This moves the shift lever for the pinion drive gear to engage with the ring gear. The front end of the plunger then makes contact across the battery and across the motor terminals of the solenoid. The starting motor then begins to turn the flywheel of the engine.

When the start switch is opened, current no longer flows through the windings. The spring now returns the plunger to the original position. At the same time, the spring moves the pinion gear away from the flywheel.

When two sets of windings in the solenoid are used, the windings are called the hold-in winding and the pull-in winding. Both of the windings wind around the cylinder an equal number of times. The pull-in winding uses a wire with a larger diameter to produce a stronger magnetic field. When the start switch is closed, part of the current flows from the battery through the hold-in winding. The remainder of the current flows through the pull-in windings, to the motor terminal, and then to the ground. When the solenoid is fully activated, the current is shut off through the pull-in windings. Only the smaller hold-in windings are in operation for the extended period of time necessary for the engine to be started. The solenoid will now take a smaller amount of current from the battery. Heat created by the solenoid will be kept at an acceptable level.

Other Components

Circuit Breaker



A	Reset button
B	Disc in open position
C	Contacts
D	Disc
E	Battery circuit terminals

Figure 7-16. Circuit Breaker

See [Figure 7-16](#). The circuit breaker is a switch that opens the battery circuit if the current in the electrical system is higher than the rating of the circuit breaker. The metal disc (B) is activated by heat. As the current in the electrical system increases, the temperature of the metal disc (B) will increase. The heat caused by the excessive current will cause a distortion in the metal disc (B). When a distortion occurs in the metal disc (B), the contacts (C) open. An open circuit breaker can be reset when the metal disc becomes cooler. Push the reset button (A) to close the contacts (C) and reset the circuit breaker.

Inspect Fuel System

A problem with the components that send fuel to the engine can cause low fuel pressure. This can decrease engine performance.

1. Check the fuel level in the fuel tank. Verify the vent in the fuel cap is not filled with dirt.
2. Check all fuel lines for fuel leakage. The fuel lines must be free from restrictions and faulty bends. Verify the fuel return line is not collapsed.
3. Inspect the fuel filter for excessive contamination. If necessary, install a new fuel filter. Determine the source of contamination. Make the necessary repairs.
4. Service the primary fuel filter (if equipped).
5. Operate the hand priming pump (if equipped). If excessive resistance is felt, inspect the fuel pressure regulating valve. If uneven resistance is felt, test for air in the fuel. See [Test Air in Fuel](#).

6. Remove any air that may be in the fuel system. See [Prime Fuel System](#).

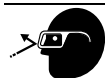
Test Air in Fuel



⚠ WARNING

Moving Parts. Keep clothing, hair, and appendages away from moving parts. Failure to do so could result in death or serious injury.

(000111)



⚠ WARNING

Vision Loss. Eye protection is required for the following procedure. Failure to wear proper eye protection could result in serious injury to the eyes.

(000248)



⚠ WARNING

Risk of Burn. Use protective gloves when handling hot parts. Failure to do so could result in serious injury.

(000386)

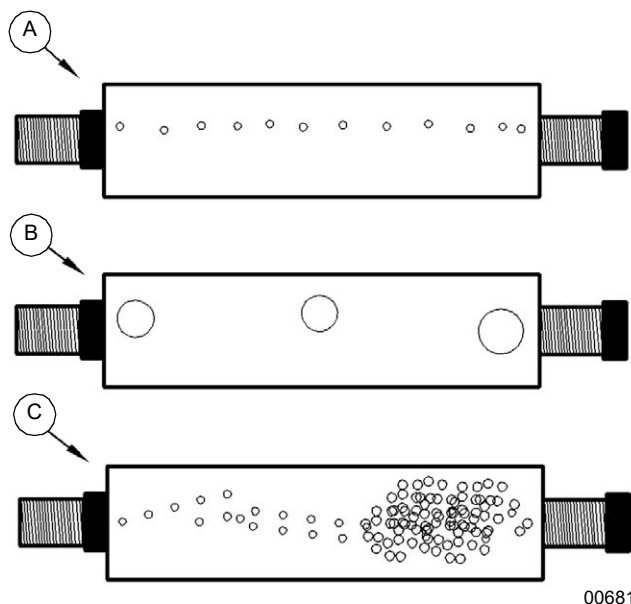
This procedure checks for air in the fuel system. This procedure also assists in finding the source of the air.

1. Examine the fuel system for leaks. Verify the fuel line fittings are properly tightened. Check the fuel level in the fuel tank. Air can enter the fuel system on the suction side between the fuel priming pump and the fuel tank.
2. Install a suitable fuel flow tube with a visual sight gauge in the fuel return line. When possible, install the sight gauge in a straight section of the fuel line that is at least 12 in (304.8 mm) long. Do not install the sight gauge near the following devices that create turbulence:
 - Elbows
 - Relief valves
 - Check valves

Observe the fuel flow during engine cranking. Look for air bubbles in the fuel. If there is no fuel that is present in the sight gauge, prime the fuel system. See [Prime Fuel System](#). If the engine starts, check for air in the fuel at varying engine speeds. When possible, operate the engine under the conditions which have been suspect.

- A steady stream of small bubbles with a diameter of approximately 0.063 in (1.60 mm) is an acceptable amount of air in the fuel ([Figure 7-17](#), item A).
- Bubbles with a diameter of approximately 0.250 in (6.35 mm) are also acceptable if there is two second to three second intervals between bubbles ([Figure 7-17](#), item B).

- Excessive air bubbles in the fuel are not acceptable ([Figure 7-17](#), item C).



006819

Figure 7-17. Air Bubbles in Fuel

If excessive air is seen in the sight gauge in the fuel return line, install a second sight gauge at the inlet to the fuel priming pump. If a second sight gauge is not available, move the sight gauge from the fuel return line and install the sight gauge at the inlet to the fuel priming pump. Observe the fuel flow during engine cranking. Look for air bubbles in the fuel. If the engine starts, check for air in the fuel at varying engine speeds.

If excessive air is not seen at the inlet to the fuel priming pump, the air is entering the system after the fuel priming pump. See [Prime Fuel System](#).

If excessive air is seen at the inlet to the fuel priming pump, air is entering through the suction side of the fuel system.

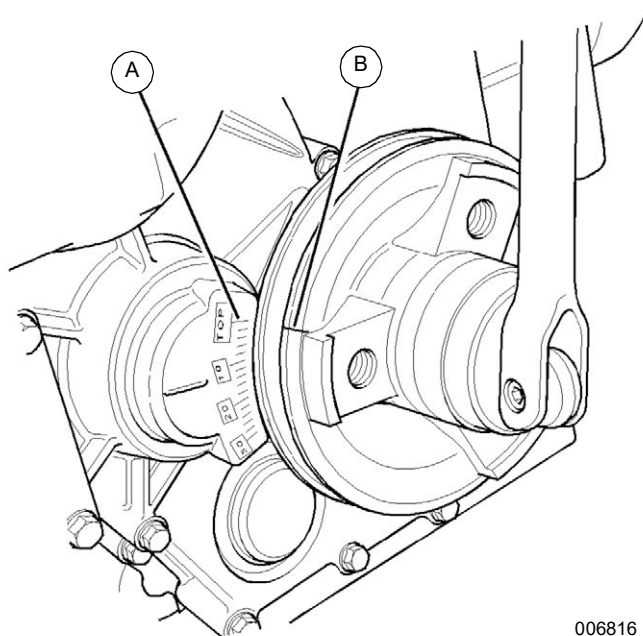
NOTE: Do not use more than 8 psi (55 kPa) to pressurize the fuel tank.

3. Pressurize the fuel tank to 5 psi (35 kPa). Do not use more than 8 psi (55 kPa) to avoid damage to the fuel tank. Check for leaks in the fuel lines between the fuel tank and the fuel priming pump. Repair any leaks. Check the fuel pressure to verify the fuel transfer pump is operating properly. For information about checking the fuel pressure, see [Inspect Fuel System](#).
4. If the source of the air is not found, disconnect the supply line from the fuel tank and connect an external fuel supply to the inlet of the fuel priming pump. If this corrects the problem, repair the fuel tank or the stand pipe in the fuel tank.

Finding Top Center Position for No. 1 Piston

This procedure sets the No. 1 piston at the Top Dead Center (TDC) position on the compression stroke.

1. Remove the valve mechanism cover. See [Valve Mechanism Cover](#).
2. Remove glow plugs. See [Glow Plugs](#).
3. See [Figure 7-18](#). Rotate the engine crankshaft until the exhaust valve for No. 1 cylinder is nearly closed and the inlet valve for No. 1 cylinder is starting to open. Rotate the crankshaft approximately 360 degrees and align the timing mark (B) on the crankshaft pulley with the TOP mark (A) on the timing case.
4. Install the glow plugs. See [Glow Plugs](#).
5. Install the valve mechanism cover. See [Valve Mechanism Cover](#).



006816

Figure 7-18. Set No. 1 Piston at TDC

Check Fuel Injection Timing

Fuel injection timing is set in the factory. Fuel injection timing cannot change during operation.

NOTE: If a fuel injection pump is removed, replace the old shims with new shims of the same thickness. Old shims should not be reused.

NOTE: If the shim thickness is 0.0197 in (0.5 mm) less, install beaded shims. If the shim thickness is above 0.0197 in (0.5 mm), install the non-beaded shim. Then install the beaded shim. Verify the beading on the beaded shim faces upward, toward the bottom of the fuel injection pump.

Select the correct combination of shims from the following table. Part numbers for the D2.2L engine can be found in 10000019509, and part numbers for the D1.5L engine can be found in A0000262802.

D2.2L Engines	
Shim Thickness - in (mm)	Beaded
0.008 (0.2)	Yes
0.012 (0.3)	Yes
0.016 (0.4)	Yes
0.02 (0.5)	Yes
0.02 (0.5)	No

D1.5L Engines	
Shim Thickness - in (mm)	Beaded
0.008 (0.2)	Yes
0.012 (0.3)	Yes
0.016 (0.4)	Yes
0.02 (0.5)	Yes
0.02 (0.5)	No

If the original shim thickness cannot be determined, contact an IASD. The user must provide the full engine number of the product that is serviced.

Test Fuel Injector

Perform the following procedures to determine if a fuel injector does not work correctly.

1. Run the engine at low idle.
2. See [Figure 7-19](#). Loosen the nut for the fuel supply line at each fuel injector (A). Listen for the low idle to decrease or become rough when the nuts are loosened at each cylinder.

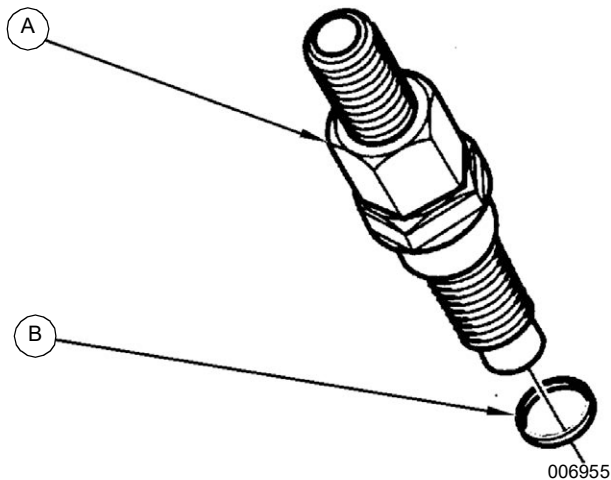


Figure 7-19. Fuel Injector

3. The fuel injector may be faulty if the following items occur during the test:
 - Engine RPM does not decrease
 - The engine continues to run properly
4. If the fuel injector is worn or damaged, remove for additional testing.

NOTE: If leakage at the nut for the fuel supply line occurs, verify the fuel supply line and the nut for the fuel supply line are correctly aligned with the inlet connection of the fuel injector. Do not tighten the nut for the fuel supply line on the high pressure fuel line more than the recommended torque. If the nut is tightened more, the fuel line may become restricted or the threads of the fuel injector and the nut may be damaged.

Inspection and Cleaning of the Fuel Injectors

Before a fuel injector is tested, remove any loose carbon from the tip of the nozzle. Do not use abrasive material or a wire brush to clean the nozzle.

NOTE: Extra care should be used in handling the fuel injector to prevent damage to the nozzle tip. A scratch or a burr could cause needle leakage or spray distortion. Dirt in the orifices of the nozzle tip can damage engine components. The dirt can also distort the spray pattern of the nozzle.

Leakage Test

This procedure tests for leakage around all components of the fuel injector.

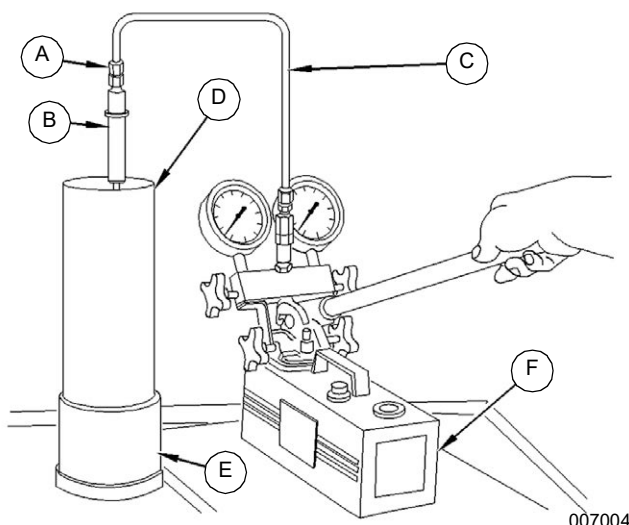
1. See [Figure 7-20](#). Connect the fuel injector (B) to suitable tooling. Position fuel injector (B) so the direction of the fuel spray is into the extension (D) and the fuel collector (E).
2. Pump the pressure to about 294 psi (2,030 kPa) below the opening pressure of the fuel injector (B).

NOTE: See [Fuel Injector](#) for the correct setting of the opening pressure.

3. Release the handle. When the pressure begins to decrease, note the time required for the pressure to decrease to approximately 0 psi (0 kPa). If the fuel injector (B) is not faulty, the time will be between 5 and 45 seconds.

NOTE: If the time required for the pressure to decrease to 0 psi (0 kPa) is less than five seconds, too much fuel is leaking around the valve needle. Replace fuel injector (B).

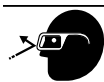
4. Look for any leakage in the connection from the adapter to the fuel injector (B). If leakage is visible, verify the components are tightened to the correct torque.
5. If leakage continues to occur, unscrew the body of the fuel injector (B) from the nozzle nut. Inspect the threaded surfaces for foreign particles and damage.
6. If the faces are damaged, replace fuel injector (B).
7. If the faces are not damaged, clean the faces. Repeat Steps 1 through 3. If leakage continues to occur, replace fuel injector.
8. Pump the pressure of the injector tester to approximately 294 psi (2,030 kPa) below the opening pressure of fuel injector (B). Fuel should not collect on the tip of the fuel injector (B) in a sufficient quantity to drip from the tip for at least ten seconds. A light dampness is acceptable. If the results of the tests are not acceptable, replace fuel injector (B).



A	Adapter
B	Fuel injector
C	Tube assembly
D	Extension
E	Fuel collector
F	Filter

Figure 7-20. Fuel Injector Leakage Test

Pressure Test



⚠ WARNING

Vision Loss. Eye protection is required for the following procedure. Failure to wear proper eye protection could result in serious injury to the eyes.

(000248)



⚠ WARNING

Fluid Injection. This machine produces high-pressure fluid streams that can pierce skin. Fluid injection could result in death or serious injury.

(000106b)

IMPORTANT NOTE: Do not use dirty test fluids to test fuel injectors. Verify the correct service tools are used.

This procedure is a test of the opening pressure of the fuel injector.

1. See [Figure 7-20](#). Connect the fuel injector (B) to suitable tooling.

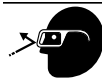
NOTE: The spray from the fuel injector (B) must be directed into the extension (D) and the fuel collector (E).

NOTE: Use clean calibration fluid when the fuel injectors are tested. The calibration fluid should be equivalent to SAE J-967 (ISO 4113) oil.

2. Close the gauge protector valve. Close the shutoff valve. Open the pump isolator valve. Flush the fuel injector (B) by operating the nozzle tester. Operate the nozzle tester for 10 to 15 strokes at a rate of approximately 60 strokes per minute.
3. Open the gauge protector valve. Slowly increase the pressure until the valve in the fuel injector (B) opens. Note the highest pressure indication on the dial indicator before the pointer moves to 0 psi (0 kPa). This highest pressure indication is the opening pressure of fuel injector (B). The opening pressure occurs when the valve needle is lifted from the nozzle seat.

NOTE: See [Fuel Injector](#) for the correct pressure settings. If the opening pressure is not within the range of the setting which is given in the table, replace fuel injector (B).

Test for Nozzle Spray Pattern



⚠ WARNING

Vision Loss. Eye protection is required for the following procedure. Failure to wear proper eye protection could result in serious injury to the eyes.

(000248)



⚠ WARNING

Fluid Injection. This machine produces high-pressure fluid streams that can pierce skin. Fluid injection could result in death or serious injury.

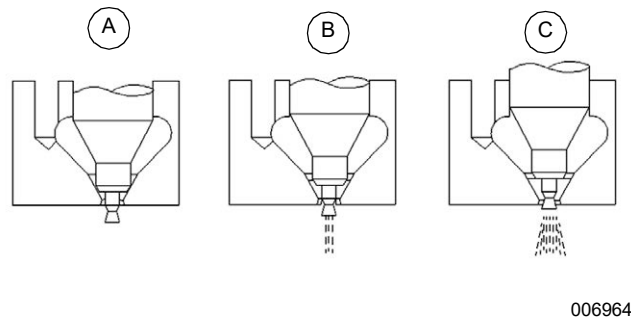
(000106b)

NOTE: The pressure adjustment must be correct before testing the spray pattern.

This procedure tests for the correct spray pattern for the fuel injectors.

1. See [Figure 7-20](#). Connect the fuel injector (B) to suitable tooling.
2. Position the fuel injector (B) so the direction of the fuel spray is into the extension (D) and the fuel collector (E).
3. Close the gauge protector valve and the shutoff valve. Open the pump isolator valve.

NOTE: See [Figure 7-21](#). The spray angle is 4 degrees from the vertical when the nozzle is fully open.



A	Closed nozzle
B	Open nozzle
C	Fully open nozzle

Figure 7-21. Fuel Injector Phases of Operation

4. See [Figure 7-20](#). Quickly pump the nozzle tester and look at the spray pattern when the fluid begins to flow through the orifices of fuel injector (B).

NOTE: The spray flows around the tip of the nozzle. A difference in the nozzle spray patterns indicates the fuel injector (B) is faulty. Refer to the correct spray pattern in [Figure 7-21](#).

Observe the following characteristics of the spray:

- Drops of fuel should not be in the spray.
- Fuel should be sprayed in the shape of a cone around the nozzle.
- Fuel should be sprayed evenly in a 360 degree pattern around the nozzle.

NOTE: See [Figure 7-20](#). Verify the gauge protector valve is closed before the fuel injector (B) is removed from the nozzle tester. Closing the valve will prevent damage to the pressure gauge.

Test Fuel Quality

Use the following procedure to test for problems regarding fuel quality:

1. Determine if water or contaminants are present in the fuel. Check the water separator (if equipped). If a water separator is not present, proceed to Step 2. Drain the water separator, if necessary. A full fuel tank minimizes the potential for overnight condensation.

NOTE: A water separator can appear to be full of fuel when the water separator is actually full of water.

2. Determine if contaminants are present in the fuel. Remove a sample of fuel from the bottom of the fuel tank. Visually inspect the fuel sample for

contaminants. The color of the fuel is not necessarily an indication of fuel quality. However, fuel that is black, brown or similar to sludge can be an indication of the growth of bacteria or oil contamination. In cold temperatures, cloudy fuel indicates the fuel may not be suitable for the operating conditions. See [Fuel Recommendations](#).

3. If fuel quality is still suspected as a possible cause of problems regarding engine performance, disconnect the fuel inlet line, and temporarily operate the engine from a separate source of fuel known to be good. This will determine if the problem is caused by fuel quality. If fuel quality is determined to be the problem, drain the fuel system and replace the fuel filters. Engine performance can be affected by the following characteristics:
 - Cetane number of the fuel
 - Air in the fuel
 - Other fuel characteristics

Prime Fuel System



⚠ DANGER

Risk of Fire. Fuel spilled or leaked onto hot surfaces or electrical components can cause a fire. Verify the battery is disconnected before performing this procedure. Failure to do so could result in death or serious injury. (000561)

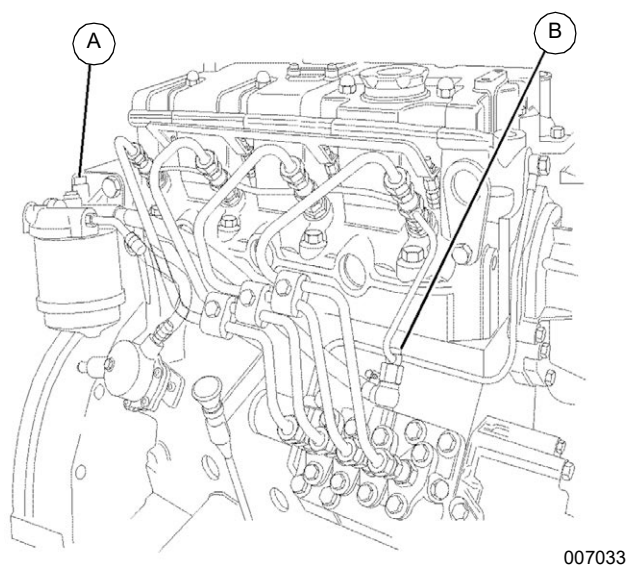
⚠ CAUTION

Engine Damage. Do not perform this procedure in a dirty or dusty environment. Doing so could result in engine damage. (000357)

IMPORTANT NOTE: Use a suitable container to catch any fuel that might spill. Clean up any spilled fuel immediately.

Use the following procedure to remove air from the fuel system.

1. See [Figure 7-22](#). Loosen the vent screw (A) on the fuel filter.
2. Operate the hand primer until fuel that is free from air flows from the vent screw. Tighten the vent screw (A).
3. Loosen the vent screw (B) on the fuel injection pump.
4. Operate the hand primer until fuel that is free from air flows from the vent screw. Tighten the vent screw (B).



007033

Figure 7-22. Fuel System

5. Attempt to start the engine with the electric starting motor for a maximum of 15 seconds. If the engine does not start, wait for 30 seconds before attempting to start the engine again. After the engine has started, check all the fuel lines and components for leaks.

Adjust Governor

Table 7-26. Required Tools

Part Description	Qty.
Dial Indicator graduated in 0.0004 in (0.01 mm)	1
0.787 in (20 mm) long extension	1
Dial holder	1
Adapter for cylinder block	1
Calibration spring	1
Allen head screw (M4 x 20 mm x 0.70)	1
Optical tachometer	1

NOTE: The setting of the front housing is only necessary if the front housing is damaged or if the front housing must be renewed.

Record Governor Settings

NOTE: The engine must be in running condition to carry out the governor adjustment procedure. If the engine cannot be returned to a running condition contact an IASD.

NOTE: If the engine is fitted with a Boost Compensation Device (BCD) contact an IASD.

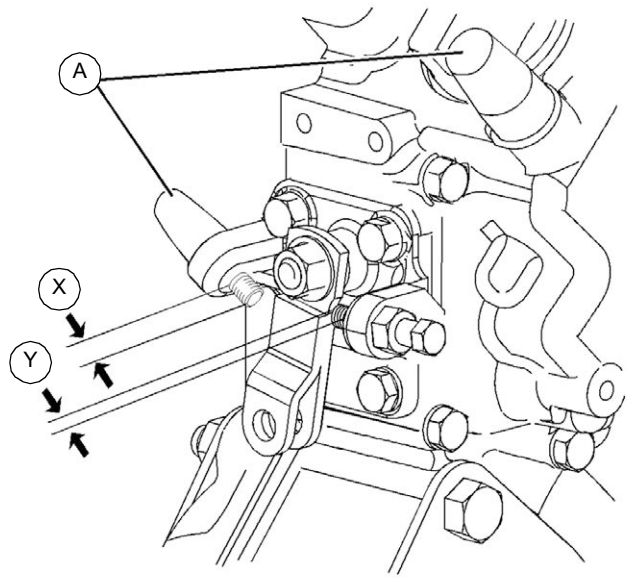
1. If the front housing has been removed, install the front housing. See [Housing \(Front\)](#).

NOTE: The setting of the low idle stop screw and the high idle stop screw must be recorded. The settings are recorded to verify governor operation is restored.

NOTE: Engine speed must be recorded from the crankshaft.

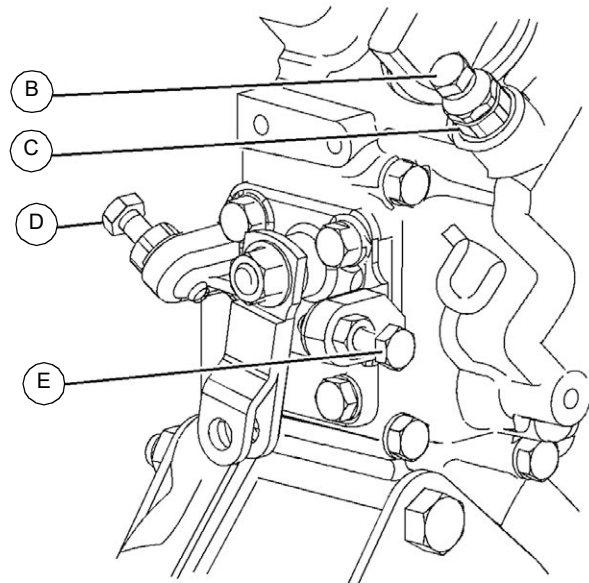
2. Operate the engine until the normal operating temperature is reached.
3. See [Figure 7-23](#) and [Figure 7-24](#). Use an optical tachometer to record the engine RPM at low idle.
4. Accelerate the engine to high idle. Use an optical tachometer and record the engine RPM as "Speed A".
5. Remove tamper proof caps (A).
6. Record Protrusion (X) of the high idle stop screw (D).
7. Record Protrusion (Y) of the low idle stop screw (E).
8. Release locknut (C). Turn high idle trimmer screw (B) for four turns counterclockwise and tighten locknut (C).
9. Accelerate the engine to high idle. Use an optical tachometer and record the engine RPM as "Speed B".

NOTE: The recorded "Speed B" will be lower than the recorded "Speed A".



006984

Figure 7-23. Record Governor Settings

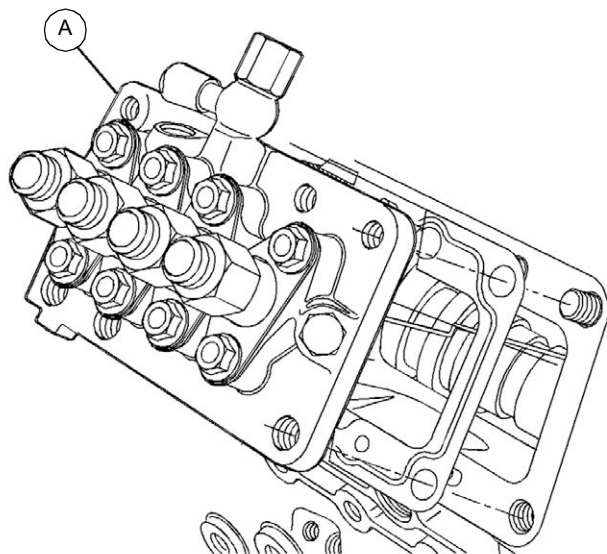


007035

Figure 7-24. Record Governor Settings

Remove the Old Front Housing

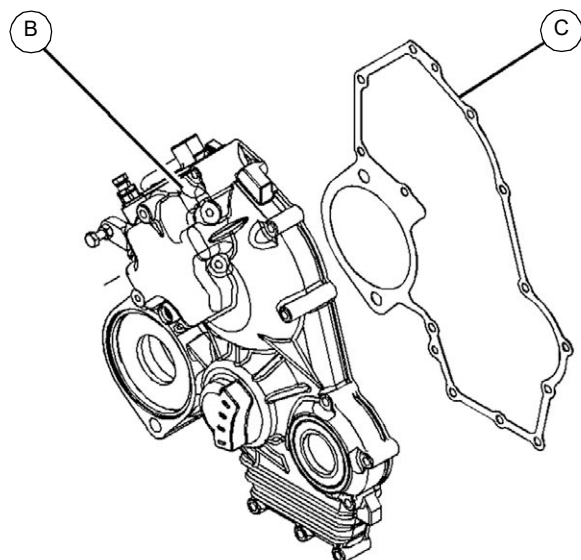
1. See [Figure 7-25](#). Remove fuel injection pump (A). See [Fuel Injection Pump](#).
2. Remove the crankshaft pulley. See [Crankshaft Pulley](#).



007050

Figure 7-25. Remove Old Front Housing

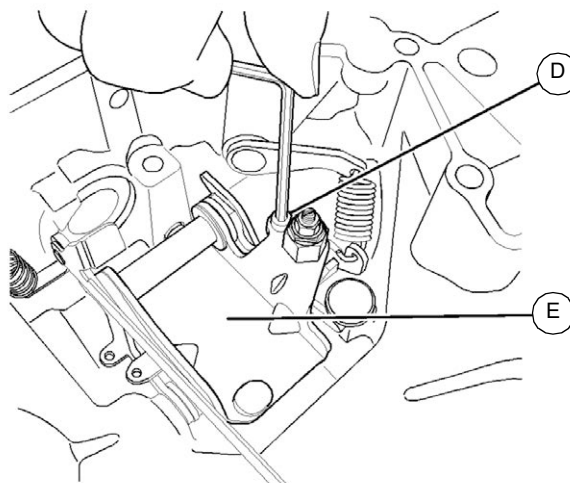
3. See [Figure 7-26](#). Remove front housing (B). See [Housing \(Front\)](#).
4. Remove the gasket (C). Verify the face of the front plate is clean.



007047

Figure 7-26. Remove Old Front Housing

5. See [Figure 7-27](#). Install Allen head screw (M4 x 20 mm x 0.70) (D) to lever assembly (E). Tighten Allen head screw (M4 x 20 mm x 0.70) (D) to **3.540 in-lbs** (0.4 Nm).



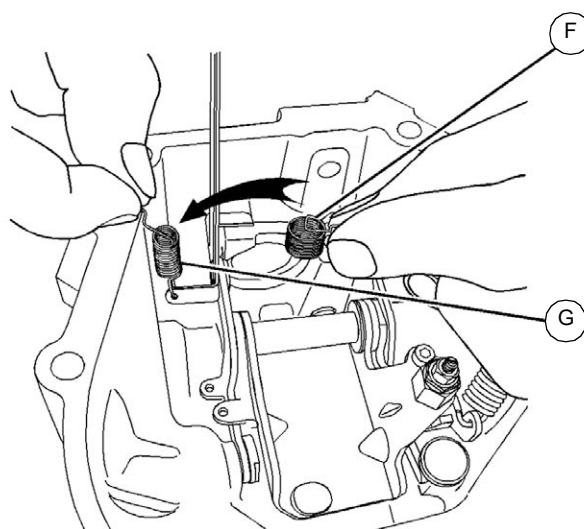
007048

Figure 7-27. Remove Old Front Housing

6. See [Figure 7-28](#). Remove start spring (G) and replace the start spring with calibration spring (F).
7. Install the front housing. See [Housing \(Front\)](#).

NOTE: Do not install a gasket.

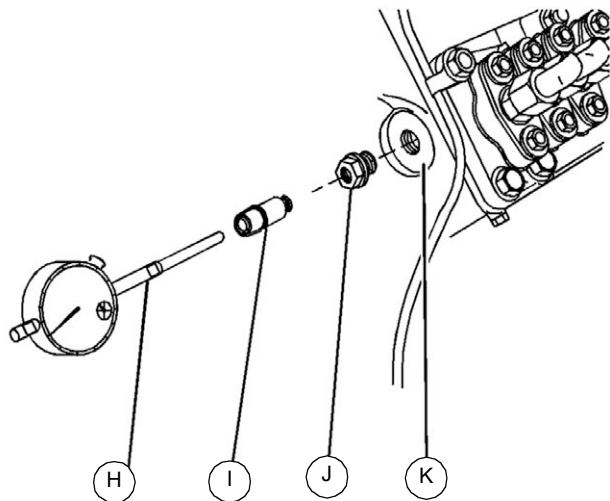
8. Install the fuel injection pump to the cylinder block. Install the original shim. See [Fuel Injection Pump](#).



007049

Figure 7-28. Remove Old Front Housing

9. See [Figure 7-29](#). Install adapter for cylinder block (J) into cylinder block (K).
10. Install dial holder (I) into adapter for cylinder block (J).
11. Assemble dial indicator and extension (H).
12. Install dial indicator and extension (H) into dial holder (I) until the extension on dial indicator (H) touches the rack of the fuel injection pump.

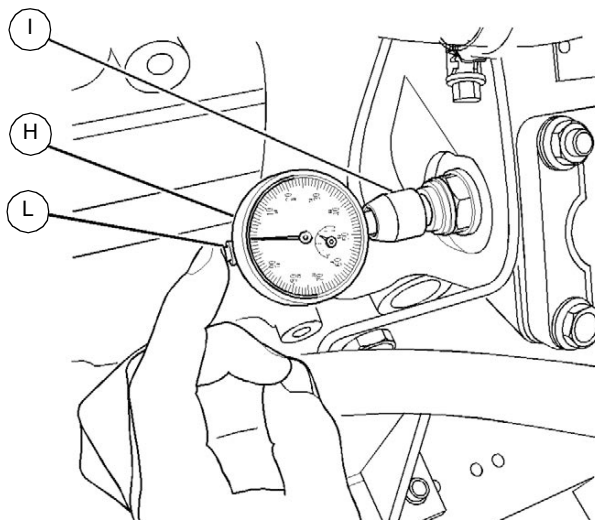


007051

Figure 7-29. Remove Old Front Housing

13. See [Figure 7-30](#). Push dial indicator and extension (H) into dial holder (I) a further 0.0787 in (2 mm). Lock dial holder (I) to retain dial indicator (H) in position. Set dial indicator (H) to read 0.0787 ± 0.0004 in (2.00 ± 0.01 mm).
14. Push plunger (L) of dial indicator (H) several times to seat the rack of the fuel injection pump. When the plunger is gently released, verify the dial indicator (H) reads 0.0787 ± 0.0004 in (2.00 ± 0.01 mm).

NOTE: If the rack of the fuel injection pump has heavy resistance to movement, remove dial indicator (H). Remove the fuel injection pump. Wash the fuel injection pump with a suitable cleaner. Install the fuel injection pump. Repeat steps 12 through 14.



007052

Figure 7-30. Remove Old Front Housing

Setting the New Front Housing

NOTE: See [Figure 7-30](#). Do not move the position of dial indicator (H), dial holder (I) or adapter (J) in the cylinder block.

1. Pull plunger (L) of dial indicator (H) until the plunger has reached the fully out position. Use a suitable clip to retain the plunger in this position.
2. Remove the fuel injection pump. See [Fuel Injection Pump](#).
3. Remove the original front housing. See [Housing \(Front\)](#).
4. Disassemble the original front housing. See [Housing \(Front\)](#).
5. Assemble the replacement front housing with the parts that were removed from the original front housing. Install a new fuel screw and a new locknut. See [Housing \(Front\)](#).
6. See [Figure 7-28](#). Remove adapter for cylinder block (F) from the original front housing. Install adapter for cylinder block (F) to the lever assembly of the replacement front housing. Tighten adapter for cylinder block (F) to 3.5 **in-lbs** (0.4 Nm).
7. Remove calibration spring (F) from the original front housing. Install calibration spring (F) into the replacement front housing.
8. Install the replacement front housing. See [Housing \(Front\)](#).

NOTE: Do not install a gasket.

9. Install the fuel injection pump to the cylinder block. Install the original shim. See [Fuel Injection Pump](#).
10. See [Figure 7-30](#). Remove the clip from plunger (L) of dial indicator (H).

Setting the Fuel Screw

1. See [Figure 7-31](#) and [Figure 7-32](#). Release the plunger on dial indicator (H). Do not move the position of dial indicator (H), dial holder (I), or adapter (J) in the cylinder block.
2. Use a ring spanner (box wrench) (N) to retain the locknut of the fuel screw in position.
3. Use an Allen wrench (M) to rotate the Allen head screw. Observe the reading on the dial indicator. Adjust the Allen head screw until the indicator reads 0.0787 ± 0.0004 in (2.00 ± 0.01 mm).
4. Tighten the locknut to 55 **in-lbs** (6 Nm) to retain the Allen head screw in position. The fuel screw is now set.
5. Verify the reading of dial indicator (H) is 0.0787 ± 0.0004 in (2.00 ± 0.01 mm).

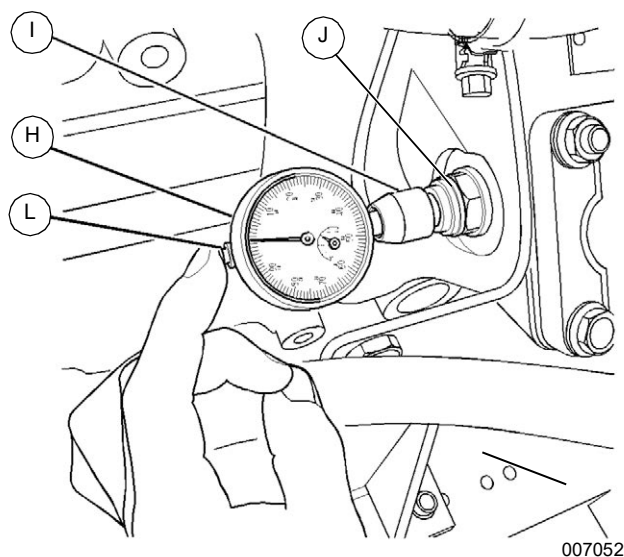


Figure 7-31. Set Fuel Screw

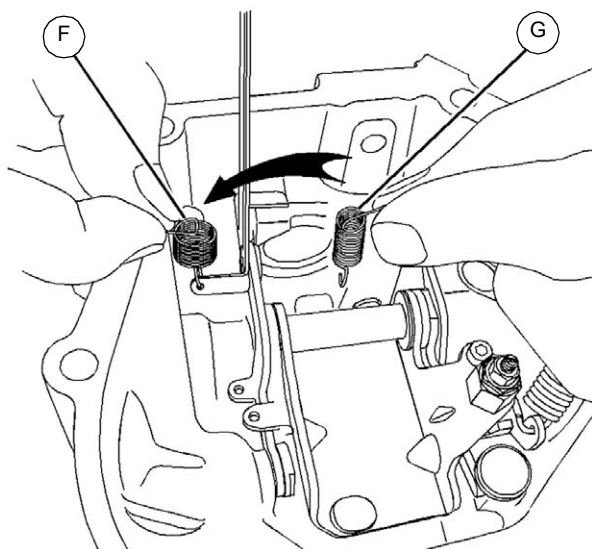


Figure 7-33. Final Installation

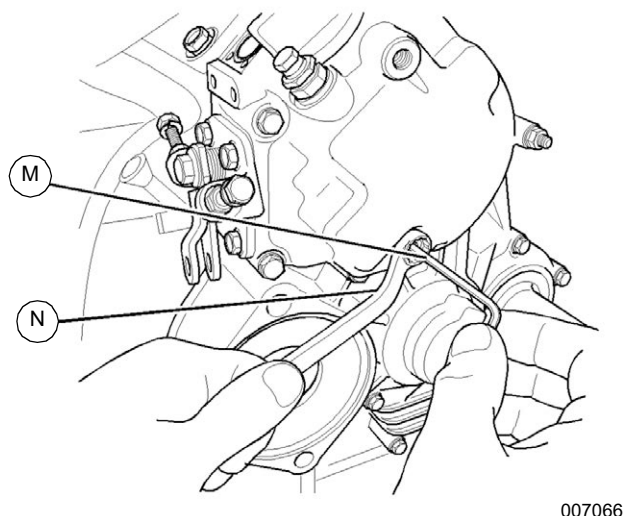


Figure 7-32. Set Fuel Screw

Final Installation of the New Front Housing

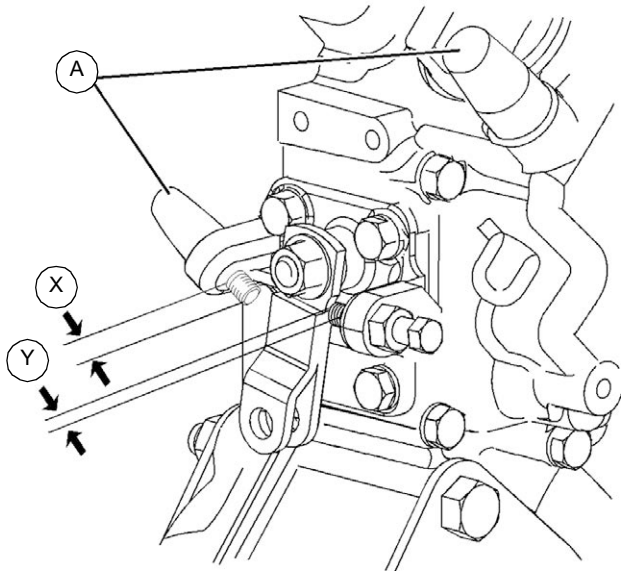
1. See [Figure 7-29](#). Remove dial indicator (H) and dial holder (I).
2. Remove adapter for cylinder block (J).
3. Remove the fuel injection pump. See [Fuel Injection Pump](#).
4. Remove the front housing. See [Housing \(Front\)](#).
5. See [Figure 7-33](#). Remove calibration spring (F). Install original start spring (G).
6. See [Figure 7-27](#). Remove Allen head screw (M4 x 20 mm x 0.70) (D) from lever assembly (E).
7. See [Figure 7-26](#). Install a new gasket (C). Verify the gasket is clean and free from damage.
8. Install the replacement front housing (B). See [Housing \(Front\)](#).
9. See [Figure 7-34](#), [Figure 7-35](#) and [Figure 7-36](#). Set the protrusion (X) of the high idle stop screw (D) to the same dimensions as the stop screws on the original front housing.
10. Set the protrusion (Y) of the low idle stop screw (E) to the same dimensions as the stop screws on the original front housing.
11. Release locknut (C). Set the high idle trimmer screw (B) to the fully out position. Turn the screw clockwise for two turns. Tighten locknut (C).
12. Install the fuel injection pump to the cylinder block. See [Fuel Injection Pump](#).
13. Start the engine and allow the engine to reach normal operating temperature.
14. Run the engine at low idle and adjust the low idle stop screw (E) to the speed that was previously recorded, prior to the original front end removal.
15. Run the engine at high idle and adjust the high idle stop screw (D) to the speed that was previously recorded as "Speed B".
16. Run the engine at high idle and release the locknut (C). Adjust the high idle trimmer screw (B) to the speed that was previously recorded as "Speed A". Tighten the locknut (C).

NOTE: The recorded "Speed A" will be higher than the recorded "Speed B".

17. Verify correct operation of the engine.

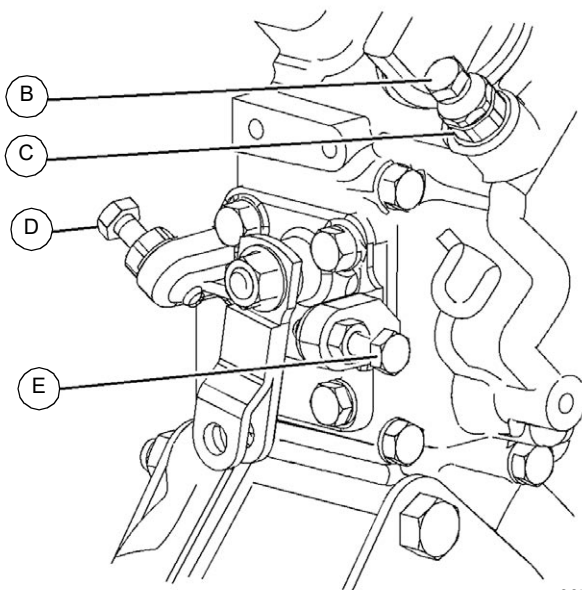
18. Install tamper proof caps (A) to the high idle trimmer screw (B) and the high idle screw (D). Install the tamper proof cap (not shown) to fuel screw (O).

NOTE: Care should be taken when the tamper proof cap is installed to the fuel screw.



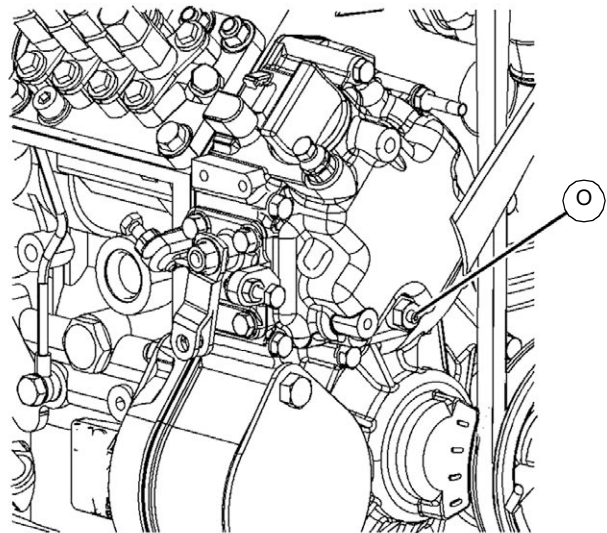
006984

Figure 7-34. Final Installation



007035

Figure 7-35. Final Installation



006939

Figure 7-36. Final Installation

Inspect Air Inlet and Exhaust System



WARNING

Hot Surfaces. When operating machine, do not touch hot surfaces. Keep machine away from combustibles during use. Hot surfaces could result in severe burns or fire. (000108)



WARNING

Moving Parts. Keep clothing, hair, and appendages away from moving parts. Failure to do so could result in death or serious injury. (000111)

A visual inspection should be made to the air inlet and exhaust system. Verify there are no signs of leaks.

There will be a reduction in the performance of the engine if there is a restriction in the air inlet system or the exhaust system.

NOTE: Maximum exhaust back pressure is 40.9 in H₂O (10.2 kPa) at 1,800 RPM.

1. Inspect the engine air cleaner inlet and ducting to ensure the passageway is not blocked or collapsed.
2. Inspect the engine air cleaner element. Replace a dirty engine air cleaner element with a clean engine air cleaner element.
3. Check for dirt tracks on the clean side of the engine air cleaner element. If dirt tracks are observed, contaminants are flowing past the engine air cleaner element and the seal for the engine air cleaner element.
4. For engines with plastic valve mechanism covers, if you experience excessive crankcase pressure,

remove the valve mechanism cover and check the end of the shroud for a skin of plastic. If the end of the shroud has a skin of plastic, remove the skin of plastic. Verify all of the debris is removed.

5. If the breather tube is made of plastic, use low pressure air to check for a blockage in the breather tube. If a blockage is inside of the connector, the cover assembly must be replaced. A broken valve mechanism cover will result if you try to remove the connection.

Test Wastegate



WARNING

Hot Surfaces. When operating machine, do not touch hot surfaces. Keep machine away from combustibles during use. Hot surfaces could result in severe burns or fire.

(000108)

NOTE: The turbocharger is a non-serviceable item. The pressure for the wastegate can be checked but not adjusted.

Before the wastegate is tested, verify the inlet air restriction is within the specifications for your engine. Verify the exhaust system restriction is within the specifications for your engine. See [Air Inlet and Exhaust System](#).

Inspect Wastegate

The wastegate controls the amount of exhaust gas allowed to bypass the turbine side of the turbocharger. This valve then controls the RPM of the turbocharger.

When the engine operates in conditions of low boost, a spring presses against a diaphragm in the canister. The actuating rod will move and the wastegate will close. Then, the turbocharger can operate at maximum performance.

When the boost pressure increases against the diaphragm in the canister, the wastegate will open. The RPM of the turbocharger becomes limited because a portion of the exhaust gases bypass the turbine wheel of the turbocharger. The following levels of boost pressure indicate a problem with the wastegate:

- Too high in full load conditions
- Too low in all low boost conditions

The boost pressure controls the maximum RPM of the turbocharger, because the boost pressure controls the position of the wastegate. The following factors also affect the maximum RPM of the turbocharger:

- The engine rating
- The horsepower demand on the engine
- The high idle RPM
- Inlet air restriction
- Exhaust system restriction

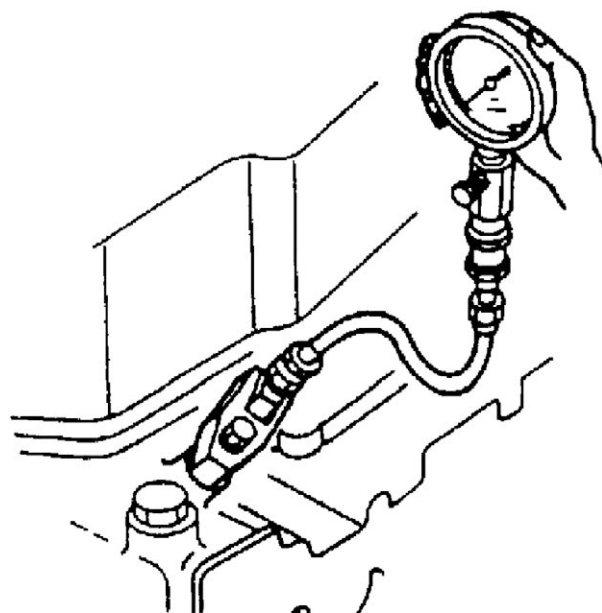
Test Wastegate for Proper Operation

1. Remove the heat shield from the turbocharger. Remove the guard for the wastegate.
2. Remove the boost line from the wastegate. Connect an air supply to the wastegate that can be adjusted accurately.
3. Fasten a dial gauge to the turbocharger so the end of the actuator rod is in contact with the plunger of the dial gauge. This will measure axial movement of the actuator rod.
4. Slowly apply air pressure to the wastegate so the actuator rod moves 0.039 in (1.0 mm). The air pressure should be within 7.5 psi (52 kPa). Verify the dial gauge returns to zero when the air pressure is released. Repeat the test several times to ensure an accurate reading.
5. Contact an IASD if wastegate operation is not correct.

Test Compression

The following conditions can affect the results of the cylinder compression test:

- The battery is in good condition
- The battery is fully charged
- The starter motor operates correctly
- The valve lash is set correctly
- The compression gauge is accurate



006837

Figure 7-37. Test Engine Compression

1. Remove the fuel injector from the cylinder to measure the compression for that cylinder.
2. Connect a suitable compression gauge to the cylinder.

3. Disconnect the fuel shutoff solenoid.
4. Operate the starter motor and record the pressure on the compression gauge.
5. Repeat for each cylinder.

NOTE: Compression tests should only be used to compare pressures between cylinders of an engine. If one or more cylinders vary more than 51 psi (350 kPa) then those cylinders may be damaged.

NOTE: The cylinder compression test should not be the only test for determining the condition of an engine.

Table 7-27. Compression Pressure*

Standard at Assembly	Repair Limit
426 psi (2,940 kPa)	355 psi (2,450 kPa)

* The compression pressure is taken at 250 RPM.

6. Repair procedures must be taken if the compression is lower than the repair limit.

IMPORTANT NOTE: Measure the compression on all of the cylinders. If all of the cylinders are not checked an improper diagnosis may result. The compression pressure will vary with the change in engine RPM. It is necessary to keep the engine RPM constant for all cylinders when you are taking a compression reading.

Inspect and Adjust Engine Valve Lash

DANGER

Accidental Starting. Turn ignition switch OFF and place a DO NOT OPERATE tag at the ignition switch location before performing this procedure. Failure to do so could result in death or serious injury. (000562)



WARNING

Risk of Burn. Allow the engine to cool before performing the following procedure. Failure to do so could result in serious injury. (000560)

IMPORTANT NOTE: Do not use the starter to turn the flywheel.

Set Valve Lash

NOTE: The valve lash setting is for a cold engine.

Table 7-28. Valve Lash Clearances

Inlet Valve	Exhaust Valve
0.008 ± 0.0020 in (0.2 ± 0.05 mm)	0.008 ± 0.0020 in (0.2 ± 0.05 mm)

Adjust Valve Lash

If the valve lash requires adjustment several times in a short period of time, excessive wear exists in a different part of the engine. Repair the problem to prevent more damage to the engine.

Not enough valve lash can be the cause of rapid wear of the camshaft and valve lifters. Not enough valve lash can indicate that the seats for the valves are worn.

Valves become worn due to the following causes:

- Incorrect operation of fuel injectors
- Excessive dirt and oil are present on the filters for the inlet air
- Incorrect fuel settings on the fuel injection pump
- The load capacity of the engine is frequently exceeded

Too much valve lash can cause broken valve stems, springs, and spring retainers. Too much valve lash can be an indication of the following problems:

- Worn camshaft and valve lifters
- Worn rocker arms
- Bent pushrods
- Broken socket on the upper end of a pushrod
- Loose adjustment screw for the valve lash

If the camshaft and valve lifters show rapid wear, look for fuel in the lubrication oil or dirty lubrication oil as a possible cause.

The valve lash is measured between the top of the valve stem and the rocker arm lever.

NOTE: No. 1 cylinder is at the front of the engine.

Remove the valve mechanism cover and perform the following procedure to adjust the valve lash:

NOTE: When these procedures are performed, the front housing must be installed.

Adjust Valve Lash for 1.5L Engines

1. Rotate the crankshaft in a clockwise direction that is viewed from the front of the engine. When the inlet valve of the No. 1 cylinder has started to open and the exhaust valve of the No. 1 cylinder has not completely closed, check the valve lash of the inlet valve of No. 2 cylinder and the exhaust valve of No. 3 cylinder. If necessary, make adjustment by following steps 2 through 4.

2. Loosen the valve adjustment screw locknut that is on adjustment screw (2).
3. Place the appropriate feeler gauge (1) between the rocker arm and the valve. Turn adjustment screw (2) while the valve adjustment screw locknut is being held from turning. Adjust the valve lash until the correct specification is achieved.
4. After each adjustment, tighten the valve adjustment screw locknut while adjustment screw (2) is being held from turning.
5. Rotate the crankshaft in a clockwise direction that is viewed from the front of the engine. When the inlet valve of the No. 2 cylinder has started to open and the exhaust valve of the No. 2 cylinder has not completely closed, check the valve lash of the inlet valve for No. 3 cylinder and the exhaust valve for No. 1 cylinder.

NOTE: If adjustment is necessary, refer to Steps 2 through 4 above.

6. Rotate the crankshaft in a clockwise direction that is viewed from the front of the engine. When the inlet valve of the No. 3 cylinder has started to open and the exhaust valve of the No. 3 cylinder has not completely closed, check the valve lash of the inlet valve for No. 1 cylinder and the exhaust valve for No. 2 cylinder.

NOTE: If adjustment is necessary, refer to Steps 2 through 4 above.

Adjust Valve Lash for 2.2L Engines

1. See [Figure 7-38](#), [Figure 7-39](#) and [Figure 7-40](#). Rotate the crankshaft in a clockwise direction when viewed from the front of the engine. When the inlet valve of the No. 4 cylinder has started to open and the exhaust valve of the No. 4 cylinder has not completely closed, verify the Top mark (C) on the timing case aligns with single dot (E) on the crankshaft pulley. Check the valve lash of the inlet valve and the exhaust valve of the No. 1 cylinder. If necessary, make an adjustment.

NOTE: The single dot (E) on the crankshaft pulley is the reference point for the top center position of No. 1 and No. 4 cylinders. The middle dot (F) on the crankshaft pulley is the top center reference point for No. 2 and No. 3 cylinders.

2. Loosen the valve adjustment screw locknut on the adjustment screw (B).
3. Place the appropriate feeler gauge (A) between the rocker arm and the valve. Turn adjustment screw (B) while the valve adjustment screw locknut is being held from turning. Adjust the valve lash until the correct specification is achieved.

4. After each adjustment, tighten the valve adjustment screw locknut while adjustment screw (B) is being held from turning.
5. Rotate the crankshaft clockwise 180°. When the inlet valve of the No. 2 cylinder has started to open and the exhaust valve of the No. 2 cylinder has not completely closed, verify the top mark (C) on the timing case aligns with middle dot (F) on the crankshaft pulley. Check the valve lash of the inlet valve and the exhaust valve of the No. 3 cylinder. If necessary, make an adjustment.

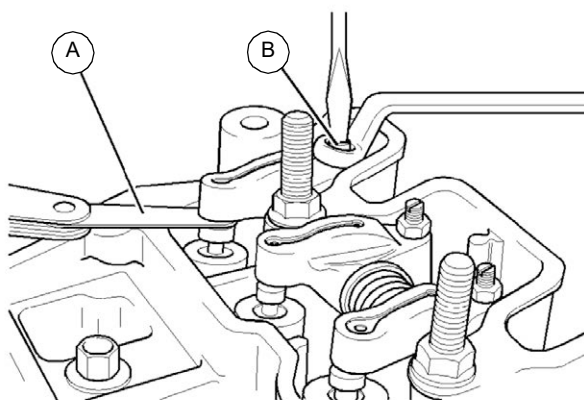
NOTE: If adjustment is necessary, refer to Steps 2 through 4 above.

6. Rotate the crankshaft in a clockwise direction 180°. When the inlet valve of the No. 1 cylinder has started to open and the exhaust valve of the No. 1 cylinder has not completely closed, verify the top mark (C) on the timing case aligns with single dot (E) on the crankshaft pulley. Check the valve lash of the inlet valve and the exhaust valve of the No. 4 cylinder. If necessary, make an adjustment.

NOTE: If adjustment is necessary, refer to Steps 2 through 4.

7. Rotate the crankshaft in a clockwise direction 180°. When the inlet valve of the No. 3 cylinder has started to open and the exhaust valve of the No. 3 cylinder has not completely closed, verify the top mark (C) on the timing case aligns with middle dot (F) on the crankshaft pulley. Check the valve lash of the inlet valve and the exhaust valve of the No. 2 cylinder. If necessary, make an adjustment.

NOTE: If adjustment is necessary, refer to Steps 2 through 4.



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Figure 7-38. Adjust Valve Lash

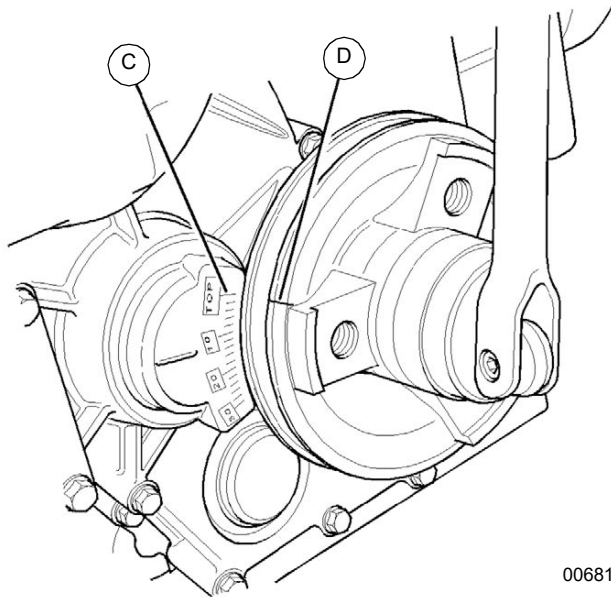


Figure 7-39. Adjust Valve Lash

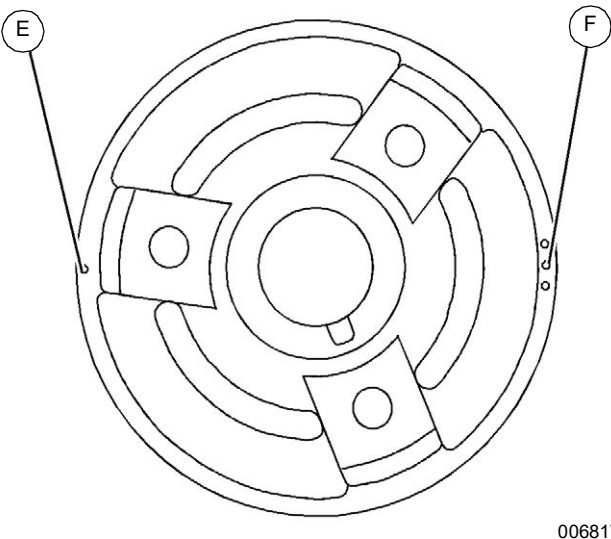


Figure 7-40. Adjust Valve Lash

Inspect Valve Depth

1. See [Figure 7-41](#). Use a dial gauge (A) with a dial gauge holder (B) to check the depths of the inlet valves and the exhaust valves below the face of the cylinder head. Use the cylinder head face to zero the dial gauge (A).
2. Position the dial gauge holder (B) and the dial gauge (A) to measure the valve depth. Measure the depth of the inlet valve and the exhaust valve before the valve springs are removed.
3. See [Cylinder Head Valves](#) for the minimum, the maximum, and the service wear limits for the valve depth below the cylinder head face.

4. If the valve depth below the cylinder head face exceeds the service limit, use a new valve to check the valve depth. If the valve depth still exceeds the service limit, renew the cylinder head or renew the valve seat inserts (if equipped). If the valve depth is within the service limit, renew the valves.
5. Inspect the valves for cracks and other damage. Check the valve stems for wear. Verify the valve springs are the correct length under the test force. See [Cylinder Head Valves](#) for the dimensions and tolerances of the valves and valve springs.

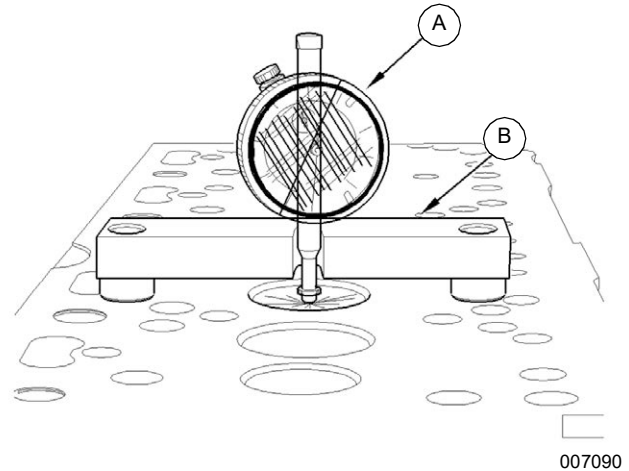
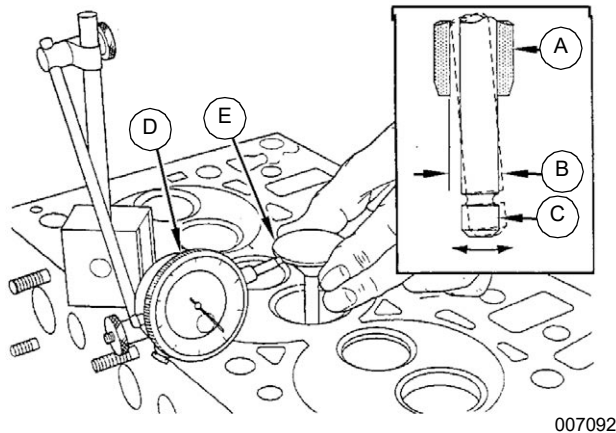


Figure 7-41. Inspect Valve Depth

Inspect Valve Guide

Perform this test to determine if a valve guide should be replaced.

1. Place a new valve in the valve guide.
2. Place the dial indicator with the magnetic base on the face of the cylinder head.
3. Lift the edge of the valve head to a distance of 0.60 in (15.0 mm).
4. Move the valve in a radial direction away from the dial indicator. Verify the valve moves away from the dial indicator as far as possible. Position the contact point of the dial indicator on the edge of the valve head. Set the position of the needle of the dial indicator to zero.
5. Move the valve in a radial direction toward the dial indicator as far as possible. Note the distance of movement which is indicated on the dial indicator. If the distance is greater than the maximum clearance of the valve in the valve guide, replace the valve guide. See [Cylinder Head Valves](#) for the maximum clearance of the valve in the valve guide.



A	Valve guide
B	Radial movement of valve in valve guide
C	Valve stem
D	Dial indicator
E	Valve head

Figure 7-42. Inspect Valve Guide

Test Engine Oil Pressure

An oil pressure gauge with a defect can indicate low oil pressure.

Use a suitable gauge that measures the oil pressure in the engine.

1. Verify the engine is filled to the correct oil level.
2. Connect the gauge to a pressure tap location for engine oil.
3. Operate the engine. Allow the engine to reach normal operating temperature.
4. Keep the oil temperature constant with the engine at the rated RPM. Read the pressure gauge.
5. See [Table 7-29](#) to determine if the engine oil pressure is in tolerance.

Table 7-29. Oil Pressure*

Oil pressure at high idle	28.4 to 64 psi (196 to 441 kPa)
Oil pressure at low idle	7.1 psi (49 kPa) or more

* Oil temperature must be 176 to 230 °F (80 to 110 °C)

Troubleshoot the cause of the problem and correct the problem if the results of the test do not fall within the pressure range in [Table 7-29](#). Engine failure or a reduction in engine life can be the result if engine operation is continued with oil pressure outside this range.

NOTE: A record of engine oil pressure at regular intervals can be used as an indication of possible problems of the engine. A record of engine oil pressure at regular intervals can also be used as an indication of possible damage to the engine. The engine should be inspected and the problem corrected if there is a sudden increase or a sudden decrease of 10 psi (70 kPa) in oil pressure.

Inspect Engine Oil Pump

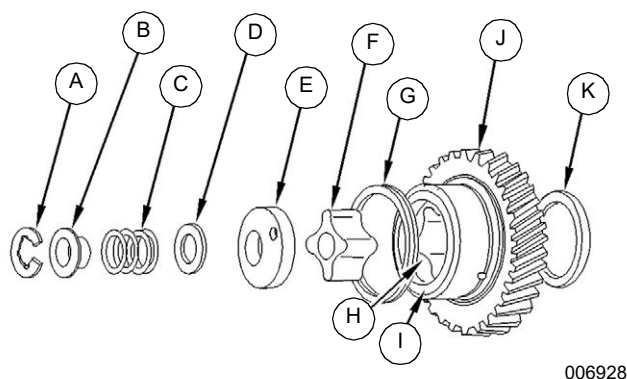
If any part of the engine oil pump is worn enough to affect performance of the engine oil pump, replace the engine oil pump.

Perform the following procedures to inspect the engine oil pump. See [Engine Oil Pump](#).

1. See [Figure 7-43](#). Remove the front housing. Remove C-clip (A).

NOTE: If the front housing is removed, do not turn the crankshaft.

2. Disassemble the engine oil pump. See [Engine Oil Pump](#). Do not remove bushing (I) from idler gear (J) unless damage is observed.
3. Clean and inspect all parts for cracks in the metal or other damage. Look for wear on the components. Verify bushing (I) is not loose in idler gear (J).

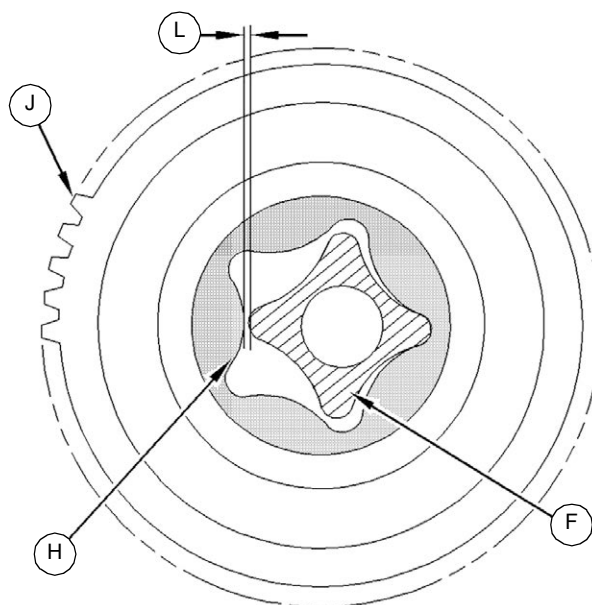


006928

A	C-clip
B	Collar
C	Spring
D	Shim
E	Oil pump cover
F	Inner rotor
G	Spring
H	Outer rotor
I	Bushing
J	Idler gear
K	Thrust washer

Figure 7-43. Idler Gear and Components of Engine Oil Pump

- See [Figure 7-44](#). Install the idler gear and the inner rotor on the shaft. Measure clearance (L) between inner rotor (F) and outer rotor (H).
- See [Engine Oil Pump](#).

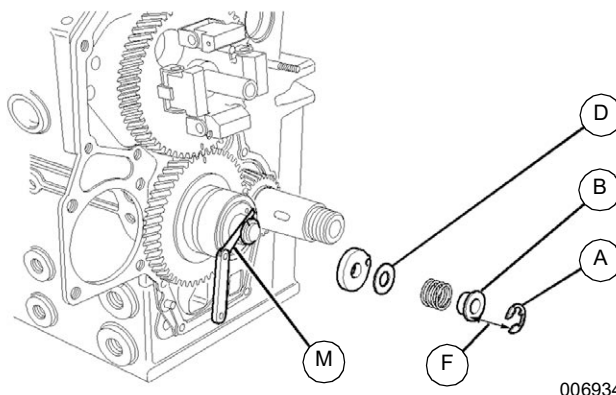


006841

Figure 7-44. Clearance Between Inner and Outer Rotor

- See [Figure 7-45](#). Use feeler gauge (M) to measure the distance between C-clip (A) and collar (B).
- When the components of the engine oil pump are installed on the front housing, measurement (F) between C-clip (A) and collar (B) must be between 0.004 to 0.006 in (0.10 to 0.15 mm) for new components. dimension B must be not more than 0.008 in (0.20 mm) for used parts.

The distance between the faces is adjusted with shims (D). The thicknesses of the shims are given in [Engine Oil Pump](#).



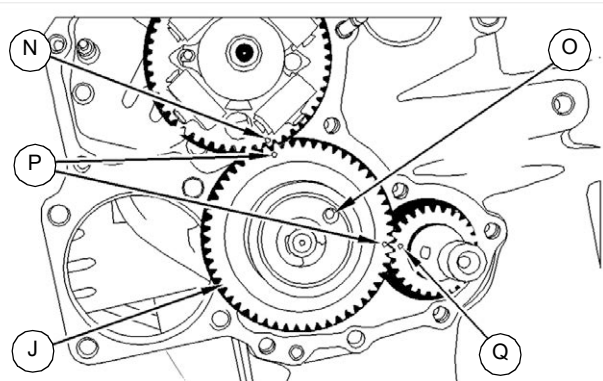
006934

A	C-clip
B	Collar
C	Shim
D	Feeler gauge

Figure 7-45. Face of Cover for Oil Pump Above Face of Idler Gear

8. Install the components of the engine oil pump and the front housing on the cylinder block. Verify the two dowels align with the bushing, the outer rotor, and the holes in the idler gear.
9. See [Figure 7-46](#). Align the timing mark (P) on idler gear (J) with the timing mark on crankshaft gear (Q) when installing the idler gear on the oil pump shaft. Also, align the other timing mark (P) on idler gear (J) with the timing mark on the camshaft gear (N). The hole (O) in oil pump cover should align with the locating pin in the front housing.

NOTE: When the front housing is not installed on the engine, do not rotate the crankshaft.



007016

Figure 7-46. Oil Pump and Front Housing Components

Inspect Excessive Bearing Wear

When some components of the engine show bearing wear in a short time, the cause can be a restriction in an oil passage.

An engine oil pressure indicator may show there is enough oil pressure, but a component is worn due to a lack of lubrication. In such a case, look at the passage for the oil supply to the component. A restriction in an oil supply passage will not allow enough lubrication to reach a component. This will result in early wear.

Inspect Excessive Engine Oil Consumption

Engine Oil Leaks on the Outside of Engine

Check for leakage at the seals at each end of the crankshaft. Look for leakage at the gasket for the engine oil pan and all lubrication system connections. Look for any engine oil that may be leaking from the crankcase breather. This can be caused by combustion gas leakage around the pistons. A dirty crankcase breather will cause high pressure in the crankcase. A dirty crankcase breather will cause the gaskets and the seals to leak.

Engine Oil Leaks into Combustion Area of Cylinders

Engine oil leaking into the combustion area of the cylinders can be the cause of blue smoke. There are several possible ways for engine oil to leak into the combustion area of the cylinders:

- Leaks between worn valve guides and valve stems
- Worn components or damaged components (pistons, piston rings, or dirty return holes for the engine oil)
- Incorrect installation of the compression ring or the intermediate ring
- Leaks past the seal rings in the turbocharger shaft
- Overfilling the crankcase
- Wrong oil level gauge or guide tube
- Sustained operation at light loads

Excessive consumption of engine oil can also result if engine oil with the wrong viscosity is used. Engine oil with a thin viscosity can be caused by increased engine temperature.

Check Cooling System (Overheating)

Above normal coolant temperatures can be caused by many conditions. Use the following procedure to determine the cause of above normal coolant temperatures:

1. Check the coolant level in the cooling system. If the coolant level is too low, air will get into the cooling system. Air in the cooling system will cause a reduction in coolant flow and bubbles in the coolant. Air bubbles will keep the coolant away from the engine parts, which will prevent the transfer of heat to the coolant. Low coolant level is caused by leaks or incorrectly filling the expansion tank.
2. If contamination of the coolant is suspected, see [Cooling System](#) for the correct specification of coolant.
3. Check for air in the cooling system. Air can enter the cooling system in different ways. The most common causes of air in the cooling system are not filling the cooling system correctly and combustion gas leakage into the cooling system. Combustion gas can get into the system through inside cracks, a damaged cylinder head, or a damaged cylinder head gasket. Air in the cooling system causes a reduction in coolant flow and bubbles in the coolant. Air bubbles keep the coolant away from the engine parts, which prevents the transfer of heat to the coolant.
4. Check the sending unit. In some conditions, the temperature sensor in the engine sends signals to a sending unit. The sending unit converts these signals to an electrical impulse which is used by a

mounted gauge. If the sending unit malfunctions, the gauge can show an incorrect reading. Also if the electric wire breaks or if the electric wire shorts out, the gauge can show an incorrect reading.

5. Check radiator.
 - Check radiator for a restriction to coolant flow. Check radiator for debris, dirt or deposits on the inside of the core. Debris, dirt or deposits will restrict the flow of coolant through the radiator.
 - Check for debris or damage between the fins of the radiator core. Debris between the fins of the radiator core restricts air flow through the radiator core. **Inspect Cooling System.**
6. Check the filler cap. A pressure drop in the cooling system can cause the boiling point to be lower. This can cause the cooling system to boil. See **Test Cooling System.**
7. Check the fan and fan shroud. The fan shroud must be the proper size and the fan shroud must be positioned correctly.
8. Check for loose drive belts.
 - A loose fan drive belt will cause a reduction in the air flow across the radiator. Check the fan drive belt for proper belt tension. Adjust the tension of the fan drive belt, if necessary. See **Table 3-7.**
 - A loose water pump drive belt will cause a reduction in coolant flow through the radiator. Check the water pump drive belt for proper belt tension. Adjust the water pump drive belt's tension, if necessary. See **Table 3-7.**
9. Check the cooling system hoses and clamps. Damaged hoses with leaks can normally be seen. Hoses that have no visual leaks can soften during operation. The soft areas of the hose can become kinked or crushed during operation. These areas of the hose can cause a restriction in the coolant flow. Hoses become soft or crack after a period of time. The inside of a hose can deteriorate, and the loose particles of the hose can cause a restriction of the coolant flow.
10. Check for a restriction in the air inlet system. A restriction of the air coming into the engine can cause high cylinder temperatures. High cylinder temperatures require higher than normal temperatures in the cooling system.
11. Check for a restriction in the exhaust system. A restriction of the air coming out of the engine can cause high cylinder temperatures.
 - Visually inspect the exhaust system.
 - Check for damage to exhaust piping. Check for damage to the exhaust elbow. If no damage is found, check the exhaust system for restrictions.
12. Check the water temperature regulator. A water temperature regulator that does not open, or a

water temperature regulator that only opens part of the way can cause overheating. See **Test Water Temperature Regulator.**

13. Check the water pump. A water pump with a damaged impeller does not pump enough coolant for correct engine cooling. Remove the water pump and check for damage to the impeller.
14. Consider high outside temperatures. When outside temperatures are too high for the rating of the cooling system, there is not enough of a temperature difference between the outside air and coolant temperatures. The maximum temperature of the ambient air that enters the engine should not exceed 120 °F (50 °C).
15. When a load applied to the engine is too large, the engine RPM does not increase with an increase of fuel. This lower engine RPM causes a reduction in coolant flow through the system. This combination of less air and less coolant flow during high input of fuel will cause above normal heating.
16. Timing of the engine which is incorrect may also cause overheating of the engine. Late timing creates more heat in the engine. Early timing creates less heat in the engine.

NOTE: If the timing of the engine is incorrect, the exhaust valves may be burned and damage to the exhaust manifold may occur.

Inspect Cooling System



⚠ DANGER

Hydraulic Fluid Injection. High-pressure, high-temperature hydraulic fluid can pierce skin and cause severe burns. Do not check for leaks with hands. Seek immediate medical attention in case of accident. Failure to protect body accordingly will result in death or serious injury. (000239)

IMPORTANT NOTE: If a pressure indication is shown on the indicator, push the release valve to relieve pressure before removing any hose from the radiator.

Cooling systems that are not regularly inspected are the cause for increased engine temperatures. Visually inspect the cooling system before any tests are performed.

1. Check the coolant level in the cooling system.
2. Look for leaks in the system.

NOTE: A small amount of coolant leakage across the surface of the water pump seals is normal. This leakage is required to provide lubrication for this type of seal. A hole is provided in the water pump housing to allow this lubricant to drain from the pump housing. Intermittent leakage of small amounts of coolant from this hole is not an indication of water pump seal failure.

3. Verify air flow through the radiator does not have a restriction. Look for bent core fins between the

folded cores of the radiator. Also, look for debris between the folded cores of the radiator.

4. Inspect the drive belts for the fan.
5. Check for damage to the fan blades.
6. Look for air or combustion gas in the cooling system.
7. Inspect the filler cap, and check the surface that seals the filler cap. This surface must be clean.
8. Look for large amounts of dirt in the radiator core. Look for large amounts of dirt on the engine. Remove the dirt from the radiator core and the engine.
9. Fan shrouds that are loose or missing cause poor air flow for cooling.

Test Cooling System



WARNING

Risk of burns. Contents under pressure. Do not remove the radiator pressure cap while engine is hot. Doing so could result in death or serious injury.

(000322a)

IMPORTANT NOTE: Remove filler cap slowly to relieve pressure only when engine is stopped and radiator cap is cool enough to touch with your bare hand.

NOTE: When a diagnosis is made of a cooling system problem, temperature and pressure must be checked. The cooling system pressure will have an effect on the cooling system temperature. For an example, refer to [Figure 7-47](#). This will show the effect of pressure on the boiling point (steam) of water. This will also show the effect of height above sea level.

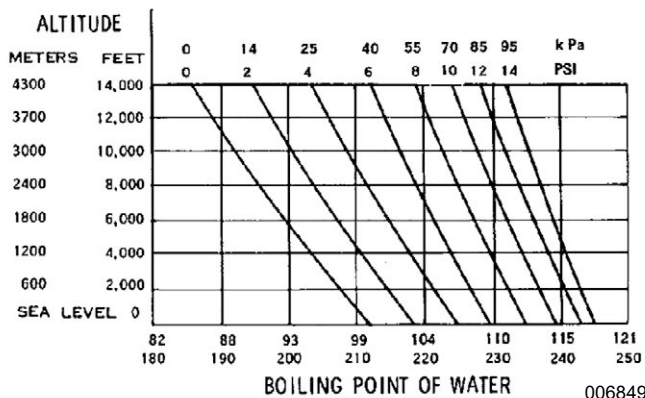


Figure 7-47. Cooling System Pressure at Specific Altitudes and Boiling Points of Water

The coolant level must be to the correct level to check the coolant system. The engine must be cold and the engine must not be running.

After the engine is cool, loosen the pressure cap to relieve the pressure out of the cooling system. Then remove the pressure cap.

The level of the coolant should not be more than 0.5 in (13 mm) from the bottom of the filler pipe. If the cooling system is equipped with a sight glass, the coolant should be to the correct level in the sight glass.

Check Filler Cap

See [Figure 7-48](#). One cause for a pressure loss in the cooling system can be a faulty seal on the radiator pressure cap.

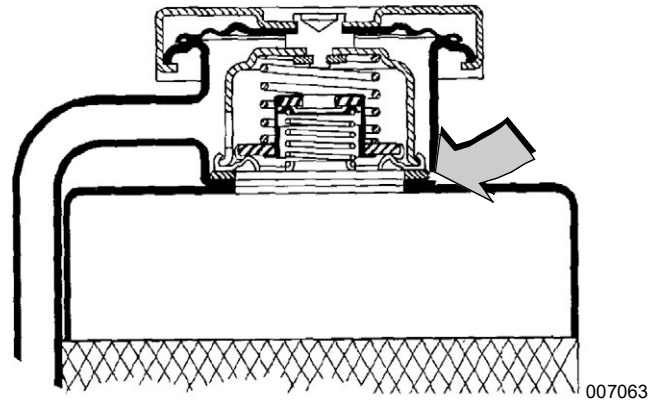


Figure 7-48. Cooling System Pressure at Specific Altitudes and Boiling Points of Water

To check for the amount of pressure that opens the filler cap, use the following procedure:



WARNING

Risk of burns. Contents under pressure. Do not remove the radiator pressure cap while engine is hot. Doing so could result in death or serious injury.

(000322a)

IMPORTANT NOTE: Remove filler cap slowly to relieve pressure only when engine is stopped and radiator cap is cool enough to touch with your bare hand.

1. After the engine cools, carefully loosen the filler cap. Slowly release the pressure from the cooling system. Then, remove the filler cap.
2. Inspect the pressure cap carefully. Look for damage to the seal. Look for damage to the surface that seals. Remove any debris on the cap, the seal, or the sealing surface.
3. Carefully inspect the filler cap. Look for any damage to the seals and to the sealing surface. Inspect the following components for any foreign substances:

- Filler cap
- Seal
- Surface for seal

4. Remove any deposits and material found on these items.
5. Install pressure cap onto a suitable pressurizing pump.
6. Observe the exact pressure that opens the filler cap.
7. Compare the pressure to the pressure rating found on top of the filler cap.
8. Replace the filler cap if damaged.

Test Radiator and Cooling System for Leaks



⚠ WARNING

Risk of burns. Contents under pressure. Do not remove the radiator pressure cap while engine is hot. Doing so could result in death or serious injury. (000322a)

IMPORTANT NOTE: Remove filler cap slowly to relieve pressure only when engine is stopped and radiator cap is cool enough to touch with your bare hand.

1. When the engine has cooled, loosen the filler cap to the first stop. Allow the pressure to release from the cooling system. Then remove the filler cap.
2. Verify the coolant covers the top of the radiator core.
3. Install a suitable pressurizing pump onto the radiator.
4. Use the pressurizing pump to increase the pressure to 3 psi (20 kPa) more than the operating pressure of the filler cap.
5. Check the radiator for leakage on the outside.
6. Check all connections and hoses of the cooling system for leaks.

The radiator and the cooling system do not have leakage if all of the following conditions exist:

- You do NOT observe any leakage after five minutes
- The dial indicator remains constant beyond five minutes

The inside of the cooling system has leakage only if the following conditions exist:

- The reading on the gauge goes down
- You do NOT observe any outside leakage

7. Make any repairs, as required.

Test Water Temperature Regulator



⚠ DANGER

Hydraulic Fluid Injection. High-pressure, high-temperature hydraulic fluid can pierce skin and cause severe burns. Do not check for leaks with hands. Seek immediate medical attention in case of accident. Failure to protect body accordingly will result in death or serious injury. (000239)

IMPORTANT NOTE: If a pressure indication is shown on the indicator, push the release valve to relieve pressure before removing any hose from the radiator.

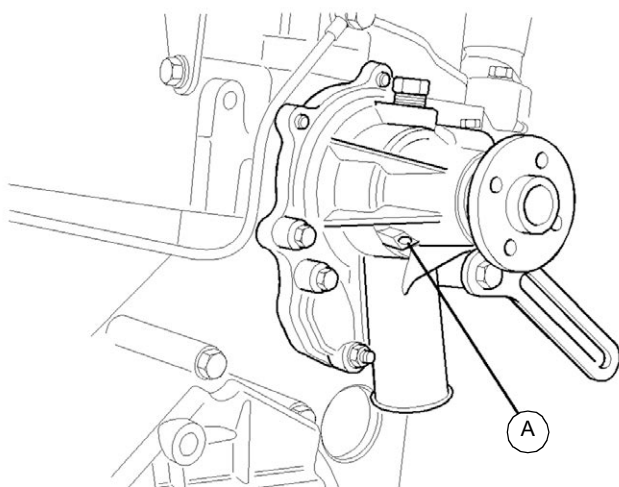
1. Remove the water temperature regulator from the engine.
2. Heat water in a pan until the temperature of the water is equal to the fully open temperature of the water temperature regulator. See [Water Temperature Regulator](#) for the fully open temperature of the water temperature regulator. Stir the water in the pan. This will distribute the temperature throughout the pan.
3. Hang the water temperature regulator in the pan of water. The water temperature regulator must be below the surface of the water. The water temperature regulator must be away from the sides and the bottom of the pan.
4. Keep the water at the correct temperature for ten minutes.
5. Remove the water temperature regulator and immediately measure the opening of the water temperature regulator. See [Water Temperature Regulator](#) for the minimum opening distance of the water temperature regulator at the fully open temperature. If the distance is less than the amount listed in the manual, replace the water temperature regulator.

Inspect Water Pump

1. See [Figure 7-49](#). Start the engine and inspect the water pump for excessive leaks at the vent hole (A).

NOTE: A small amount of coolant leakage across the surface of the seal for the jacket water pump is normal. This leakage is required to provide lubrication for this type of seal. A hole is provided in the water pump housing to allow this lubricant to drain from the water pump housing. Intermittent leakage of small amounts of coolant from this hole is not an indication of water pump seal failure.

2. Inspect the water pump shaft for unusual noise, excessive looseness and vibration of the bearings.



007104

Figure 7-49. Inspect Water Pump

Inspect Piston Ring Groove

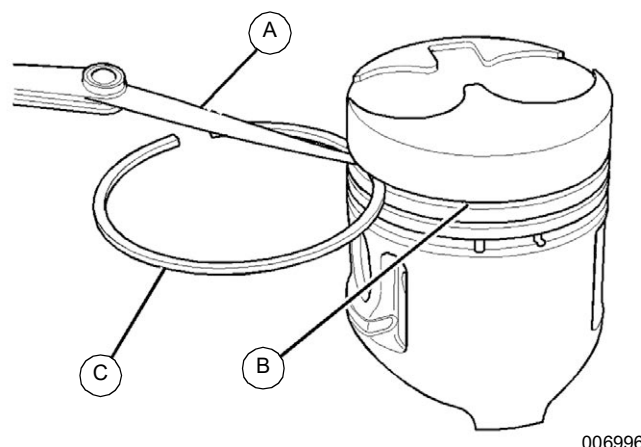
Inspect Piston and Piston Rings

1. Check the piston for wear and other damage.
2. Verify the piston rings are free to move in the grooves and that the rings are not broken.

Inspect Clearance of Piston Ring

1. Remove the piston rings and clean the grooves and the piston rings.
2. See [Figure 7-50](#). Fit new piston rings (C) in the piston grooves (B).
3. Check the clearance for the piston ring by placing a suitable feeler gauge (A) between piston groove (B) and the top of piston ring (C). See [Piston and Rings](#) for the dimensions.

NOTE: The pistons have a tapered top groove and the piston ring is wedged. The clearance for the top piston ring cannot be checked by the above method when this occurs.



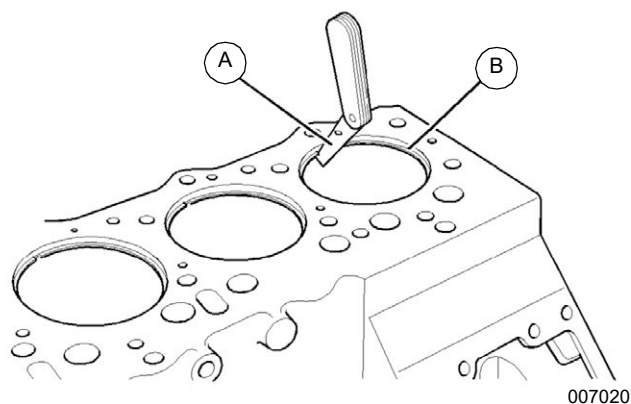
006996

Figure 7-50. Inspect Piston Ring Clearance

Inspect Piston Ring End Gap

1. Clean all carbon from the top of the cylinder bores.
2. See [Figure 7-51](#). Place each piston ring (B) in the cylinder bore.
3. Use a suitable feeler gauge (A) to measure piston ring end gap. See [Piston and Rings](#) for the dimensions.

NOTE: The coil spring must be removed from the oil control ring before the gap of the oil control ring is measured.



007020

Figure 7-51. Inspect Piston Ring End Gap

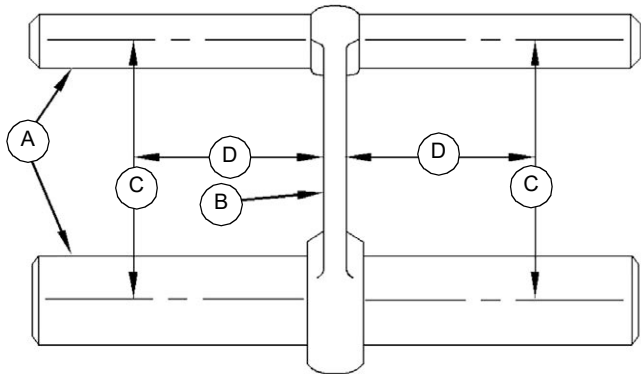
Inspect Connecting Rod

This procedure determines the following characteristics of the connecting rod:

- The distortion of the connecting rod
 - The parallel alignment of the bores of the connecting rod
1. See [Figure 7-52](#). Use the appropriate tools to measure the distances for the connecting rod (B):
 - Appropriate gauges for measuring distance
 - Measuring pins (A)

NOTE: The connecting rod bearings should be removed before taking measurements.

2. Measure connecting rod for distortion and parallel alignment between the bores. The measurements must be taken at distance (D). Distance (D) has a value of 3.94 in (100 mm) from both sides of the connecting rod.
3. Measure length (C).



006844

Figure 7-52. Inspect Connecting Rod

NOTE: The total difference in measurements of length (B) from each side should not vary more than ± 0.0031 in (± 0.08 mm).

4. Inspect the piston pin bearing and the piston pin for wear and damage.
5. Measure the clearance of the piston pin in the piston pin bearing. See [Connecting Rod](#) for clearance dimensions.

Inspect Connecting Rod Bearings

Check the connecting rod bearings and the connecting rod bearing journal for wear or other damage.

Connecting rod bearings are available with a smaller inside diameter than the original size bearings. These bearings are for crankshafts that have been ground.

Inspect Main Bearings

Check the main bearings for wear or damage. Replace both halves of the bearings and check condition of the other bearings if a main bearing is worn or damaged.

Main bearings are available with a smaller inside diameter than the original size bearings. These bearings are for main bearing journals that have been ground.

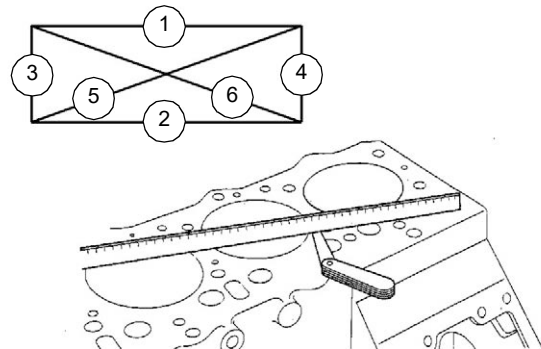
Inspect Cylinder Block

1. See [Figure 7-53](#). Use a straight edge and a feeler gauge to check the six positions for flatness.
2. Inspect the top of the cylinder block for cracks, damage, and warpage.

3. Inspect each cylinder bore. There should be no scoring, rust or corrosion. Use a suitable gauge to measure each cylinder bore. Measure the area of each cylinder bore that is in contact with the top, middle and bottom piston rings. Each cylinder bore should be measured at 90° to the crankshaft.

IMPORTANT NOTE: The flex-hone process must not be used on this engine.

If the cylinder bores exceed the service limit, the cylinder block must be renewed. See [Cylinder Block](#) for the bore diameters and tolerances.

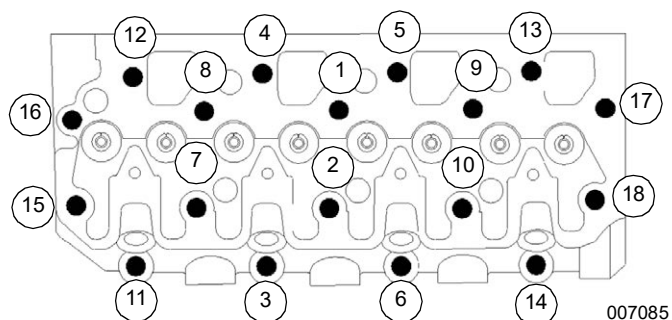


006871

Figure 7-53. Inspect Cylinder Block

Inspect Cylinder Head

1. Remove the bolts in the cylinder head in the reverse sequence shown in [Figure 7-54](#) and [Figure 7-55](#). Remove the cylinder head from the engine.
2. Clean the cylinder head thoroughly. Verify the contact surface of the cylinder head and the contact surface of the cylinder block are clean, smooth, and flat.
3. Inspect the bottom surface of the cylinder head for pitting, corrosion, and cracks. Inspect the area around the valve seats and the holes for the fuel injectors.



007085

Figure 7-54. Bolt Tightening Sequence for G2.2L Engines

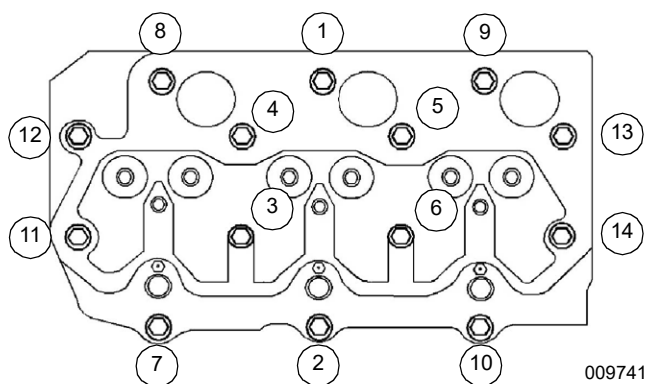


Figure 7-55. Bolt Tightening Sequence for G1.5L Engines

4. See [Figure 7-56](#). Put the cylinder head (A) on suitable supports (B).

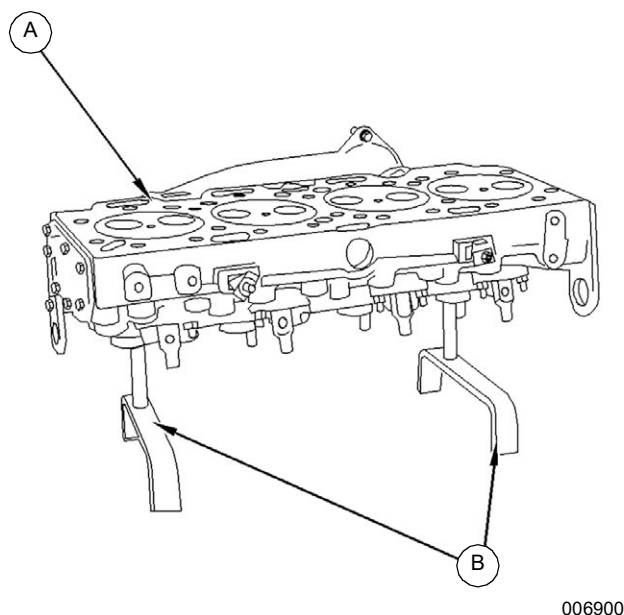


Figure 7-56. Support Cylinder Head

5. See [Figure 7-57](#). Use a straight edge and a feeler gauge to check the six positions of the cylinder head for distortion. See [Cylinder Head](#) for the requirements of flatness.

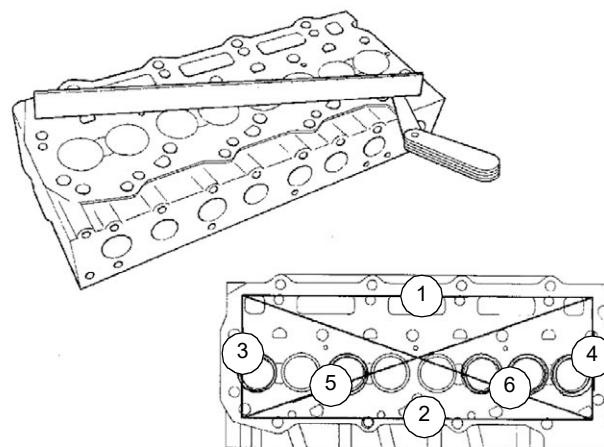


Figure 7-57. Check Cylinder Head for Distortion

Inspect Piston Height

If the height of the piston above the cylinder block is not within the tolerance given in the [Piston and Rings](#), the bearing for the piston pin must be checked. See [Inspect Connecting Rod](#). If any of the following components are replaced or re-machined, the piston height above the cylinder block must be measured:

- Crankshaft
- Cylinder head
- Connecting rod
- Bearing for the piston pin

The correct piston height must be maintained so the engine conforms to the standards for emissions.

NOTE: The top of the piston must not be machined. If the original piston is installed, verify the original piston is assembled to the correct connecting rod and installed in the original cylinder.

1. See [Figure 7-58](#). Use dial gauge (A) and dial gauge holder (B) to measure piston height above the cylinder block. Use the cylinder block face to zero the dial gauge (A).
2. Rotate the crankshaft until the piston is at the approximate top center.
3. Position the dial gauge holder (B) and the dial gauge (A) to measure the piston height above the cylinder block. Slowly rotate the crankshaft to determine when the piston is at the highest position. Record this dimension. Compare this dimension with the dimensions given in [Cylinder Head](#).

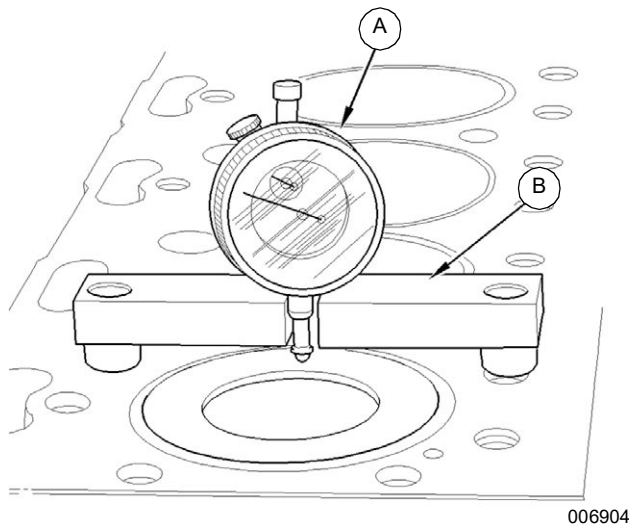


Figure 7-58. Dial Gauge and Holder

Inspect Flywheel

Align Flywheel Face

1. See [Figure 7-59](#). Install dial indicator.
2. Set the pointer of dial indicator to 0 in (0 mm).
3. Turn the flywheel. Read dial indicator for every 90°.

NOTE: During the check, keep the crankshaft pressed toward the front of the engine to remove any end clearance.

4. Calculate the difference between the lowest measurement and the highest measurement of the four locations. This difference must not be greater than 0.001 in (0.03 mm) for every 1.0 in (25 mm) of the radius of the flywheel. The radius of the flywheel is measured from the axis of the crankshaft to the contact point of the dial indicator.

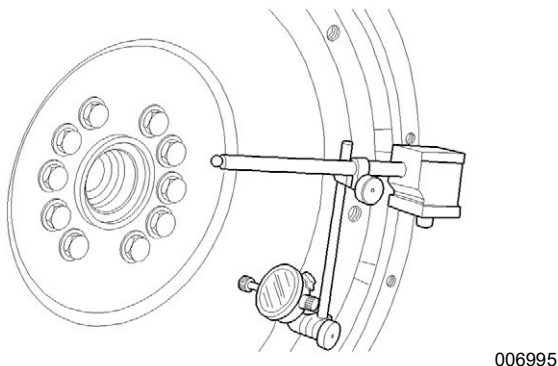


Figure 7-59. Dial Indicator

Flywheel Runout

1. See [Figure 7-60](#). Install dial indicator.

2. Set the pointer of dial indicator to 0 in (0 mm).
3. Turn the flywheel. Read the dial indicator for every 90°.
4. Calculate the difference between the lowest measurement and the highest measurement of the four locations. This difference must not be greater than 0.008 in (0.20 mm).

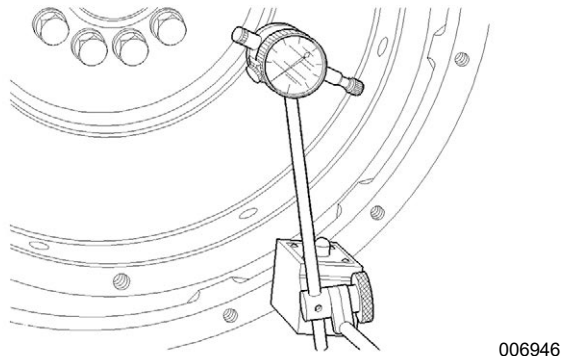


Figure 7-60. Dial Indicator

Inspect Flywheel Housing

Face Runout (Axial Eccentricity) of Flywheel Housing

1. See [Figure 7-61](#). Fasten a dial indicator to the flywheel so the anvil of the dial indicator will contact the face of the flywheel housing.
2. Put a force on the crankshaft toward the rear before the dial indicator is read at each point.

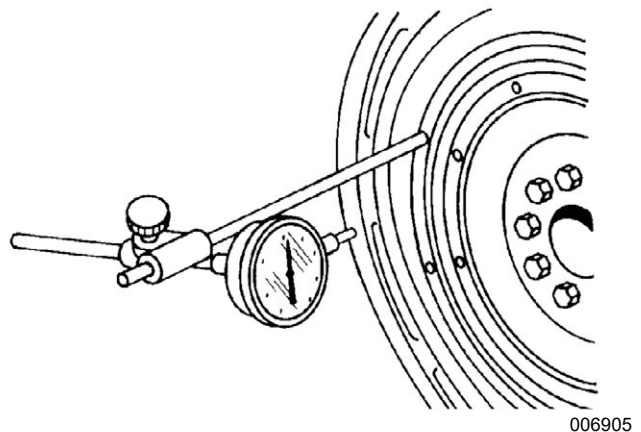
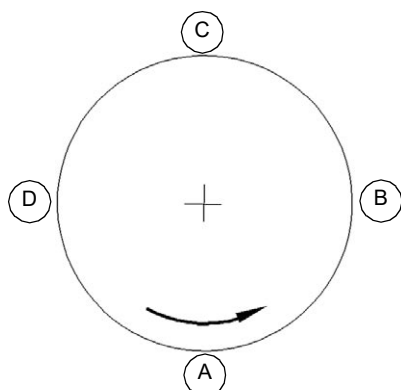


Figure 7-61. Dial Indicator Group

3. See [Figure 7-62](#). Turn the flywheel while the dial indicator is set at 0.00 in (0.0 mm) at location (A). Read dial indicator at locations (B), (C), and (D).
4. The difference between the lower measurements and the higher measurements performed at all four points must not be more than 0.008 in (0.20 mm)

for an SAE 5 flywheel housing. The difference between the lower measurements and the higher measurements performed at all four points must not be more than 0.009 in (0.23 mm) for an SAE 4 flywheel housing. These measurements are the maximum permissible face runout (axial eccentricity) of the flywheel housings.



006935

Figure 7-62. Face Runout of Flywheel Housing

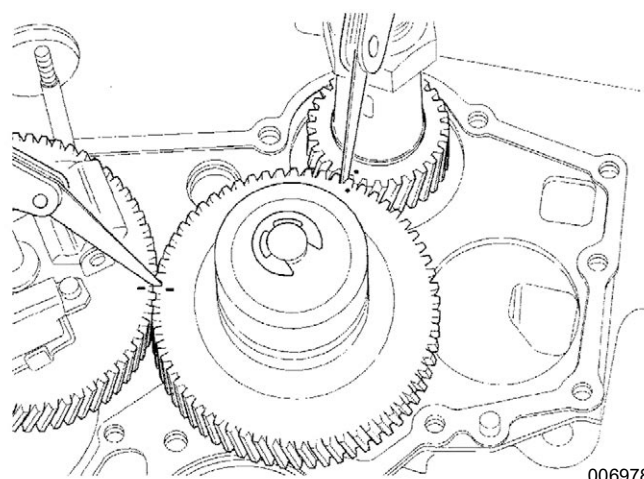
Bore Runout (Radial Eccentricity) of Flywheel Housing

1. See [Figure 7-61](#). Fasten a dial indicator to the flywheel so the anvil of the dial indicator will contact the bore of the flywheel housing.
2. See [Figure 7-62](#). Turn the flywheel while the dial indicator is set at 0.00 in (0.0 mm) at location (A). Read the dial indicator at locations (B), (C), and (D).
3. The difference between the lower measurements and the higher measurements performed at all four points must not be more than 0.008 in (0.20 mm) for an SAE 5 flywheel housing. The difference between the lower measurements and the higher measurements performed at all four points must not be more than 0.009 in (0.23 mm) for an SAE 4 flywheel housing. These measurements are the maximum permissible bore runout (radial eccentricity) of the flywheel housings.

Inspect Gear Group

1. Inspect the gears for wear or damage. If one or more of the gears are worn or damaged, use new parts for replacement.
2. See [Figure 7-63](#). Measure backlash of the gears with a feeler gauge. If backlash measurement is greater than the maximum, install new gears.

NOTE: See [Gear Group \(Front\)](#) for the minimum and maximum backlash measurements.



006978

Figure 7-63. Gear Group

Test Alternator

1. Put the positive lead (+) of a suitable multimeter on the BATT terminal of the alternator. Put the negative (-) lead on the GROUND terminal or on the frame of the alternator. Put a suitable ammeter around the positive output wire of the alternator.
2. Turn off all electrical accessories. Turn off the fuel to the engine. Operate the starter motor for 30 seconds. Wait for two minutes to cool the starter motor. If the electrical system appears to operate correctly, operate the starter motor again for 30 seconds.

NOTE: Operating the starter motor for 30 seconds partially discharges the batteries to do a charging test. If the battery has a low charge, do not perform this step. Jump start the engine or charge the battery before the engine is started.

3. Start the engine and run at high idle.
4. Check output current of the alternator. The initial charging current should be equal to the minimum full load current or greater than the minimum full load current. See [Alternator and Regulator](#) for the correct minimum full load current.
5. After approximately ten minutes of operating the engine at high idle, the output voltage of the alternator should be 14.0 ± 0.5 volts. See [Table 7-30](#).
6. After ten minutes of engine operation, the charging current should decrease to approximately 10 amperes. The actual length of time for the decrease to 10 amperes depends on the following conditions:
 - The battery charge
 - The ambient temperature
 - The rpm of the engine

See [Table 7-30](#).

Table 7-30. Fault Conditions and Possible Causes

Current at start-up	Voltage is below specifications after ten minutes	Voltage is within specifications after ten minutes	Voltage is above specifications after ten minutes
Less than the specifications	Replace alternator. Check circuit of the ignition switch.	Turn on all accessories. If voltage decreases below the specifications, replace alternator.	–
Decreases after matching specifications	Replace alternator.	The alternator and battery match the specifications. Turn on all accessories to verify voltage stays within specifications.	Replace alternator.
Voltage consistently exceeds specifications	Test battery. Test alternator again.	The alternator operates within the specifications. Test battery.	Replace alternator. Inspect battery for damage.

Test Battery



⚠ WARNING

Explosion. Do not disconnect battery cable while charging unit is operating. Doing so could result in death or serious injury.

(000574)

⚠ CAUTION

Equipment damage. Unit should not be powered. Battery charger must be OFF before removing battery cable. Cable removal on a powered unit can cause equipment damage.

(000314)

The correct procedures to test the battery can be found in the OEM manual.

Test Charging System

The condition of charge in the battery at each regular inspection will show if the charging system is operating correctly. A repair is necessary when the battery is constantly in a low condition of charge or a large amount of water is needed. There are no adjustments on maintenance free batteries. A large amount of water would be more than one ounce of water per cell per week or every 100 service hours.

Whenever possible, test the charging unit and voltage regulator on the engine. This test will use wiring and components that are a permanent part of the system. Off-engine testing or bench testing will test the charging unit and voltage regulator operation. This testing will give an indication of needed repair. After repairs are made, perform a test to prove the units have been repaired to the original condition of operation.

Alternator Regulator

The charging rate of the alternator should be checked when an alternator is charging the battery too much or not charging the battery enough.

Alternator output should be 14 ± 0.5 volts. No adjustment can be made to change the rate of charge on the alternator regulator. If the rate of charge is not correct, a replacement of the regulator is necessary. For individual alternator output, see [Alternator and Regulator](#).

Test Coolant Temperature Switch

This procedure tests the coolant temperature switch for correct operation. Perform this procedure when the engine is running.

1. Use a suitable digital multimeter to measure resistance of the coolant temperature switch. When coolant temperature is 122°F (50°C) or less, disconnect the wire connector from the coolant temperature switch.
2. Place one lead of the multimeter on the connector of the coolant temperature switch. Place the other lead of the multimeter on a ground for the coolant temperature switch.

Measure the resistance. The resistance should be 54 ± 2 Ohms.

3. Repeat the previous step when the temperature of the coolant is approximately 248°F (120°C). The resistance should not be less than 15 Ohms.
4. If the resistance through the coolant temperature switch is not within the given resistances, replace the coolant temperature switch.

Test Electric Starting System

Use a suitable multimeter in the DC volts range to find starting system components which do not function.

Move the start control switch to activate the starter solenoid. The operation of the starter solenoid can be heard as the pinion of the starter motor is engaged with the ring gear on the engine flywheel.

If the solenoid for the starter motor will not operate, it is possible the current from the battery did not reach the solenoid. Fasten one lead of the multimeter to the connection (terminal) for the battery cable on the solenoid. Connect the other lead to a good ground. A zero reading indicates there is a broken circuit from the battery. More testing is necessary when there is a voltage reading on the multimeter.

Solenoid operation also closes the electric circuit to the starter motor. Connect one lead of the multimeter to the solenoid connection (terminal) fastened to the starter motor. Connect the other lead to a good ground. Activate the starter solenoid and look at the multimeter. A voltage reading of the battery shows that the problem is in the starter motor. The starter motor must be removed for further testing. A zero reading on the multimeter shows the solenoid contacts do not close. This indicates the need for repair to the solenoid, or to the clearance for the starter motor pinion.

Perform a test. Fasten one multimeter lead to the connection (terminal) for the small wire at the solenoid and fasten the other lead to the ground. Look at the multimeter and activate the starting solenoid. A voltage reading shows that the problem is in the solenoid. A zero reading indicates the problem is in the start switch or the wires for the start switch.

Fasten one multimeter lead to the start switch at the connection (terminal) for the wire from the battery. Fasten the other lead to a good ground. A zero reading indicates a broken circuit from the battery. Check the circuit breaker and wiring. If there is a voltage reading, the problem is in the start switch or the wires for the start switch.

A starter motor that operates too slowly can have an overload due to excessive friction in the engine being started. Slow operation of the starter motor can also be caused by a short circuit, loose connections and dirt in the starter motor.

Test Engine Oil Pressure Switch

This procedure tests the engine oil pressure switch for correct operation.

1. Turn the ignition switch ON, but do not start the engine. The warning light for low engine oil pressure should turn on.

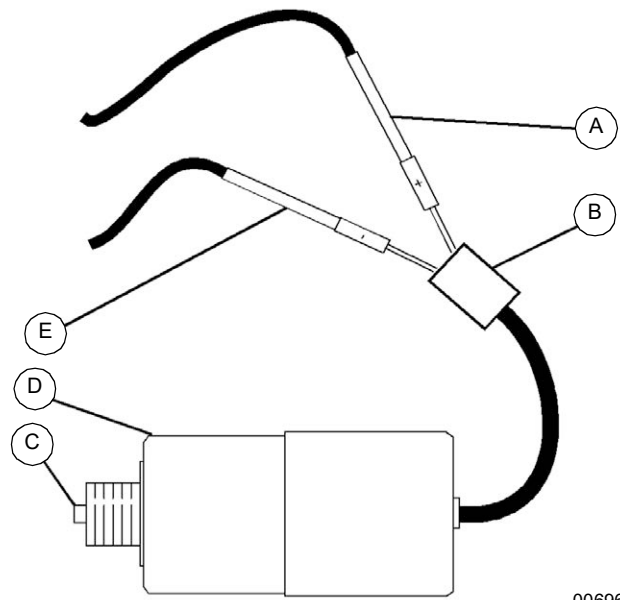
- If the warning light does not turn on, check the electrical circuit to the warning light and the engine oil pressure switch for shorts or an open circuit. Check the bulb for the warning light for damage.
- If the circuit is correct, replace the engine oil pressure switch. If the circuit is repaired or if the engine oil pressure switch or warning lamp are replaced, repeat this step to ensure the warning light turns on.

NOTE: The engine oil pressure switch may be mounted on either the cylinder block or the valve mechanism cover.

2. Remove the engine oil pressure switch from the cylinder block or from the valve mechanism cover. Disconnect the wire connector on the wiring harness from the engine oil pressure switch. Use a 12 volt battery as a power source. Connect the terminals of the battery, the engine oil pressure switch and the leads of a suitable digital multimeter in series. Measure the current through the engine oil pressure switch.
3. The current should be 0.30 to 0.40 amperes. If the current through the engine oil pressure switch is greater than 0.42 amperes, replace the engine oil pressure switch.

Test Fuel Shutoff Solenoid

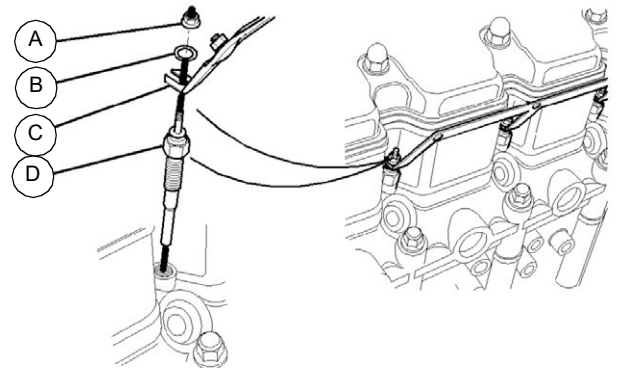
1. Remove the fuel shutoff solenoid from the engine. See [Fuel Shutoff Solenoid](#).
2. See [Figure 7-64](#). Use a 12 volt battery as a power source. Connect the lead from the battery positive terminal (A) to one of the pins in the electrical connector (B) on the fuel shutoff solenoid. Connect the lead from the battery negative terminal (E) to the remaining pin in the electrical connector (B) on the fuel shutoff solenoid.
 - The plunger (C) on the fuel shutoff solenoid (D) will extend the maximum distance when the leads are connected.
 - If the plunger does not move or if the plunger does not extend the maximum distance, replace the fuel shutoff solenoid.
3. Use a suitable digital multimeter to measure the current through the fuel shutoff solenoid. Connect the battery, the fuel shutoff solenoid, and the leads of the multimeter in series.
 - The current should be 1.0 ± 0.1 amperes. If the current is not within this tolerance, replace the fuel shutoff solenoid.
4. Install the fuel shutoff solenoid on the engine. See [Fuel Shutoff Solenoid](#).



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Figure 7-64. Fuel Shutoff Solenoid

Test Glow Plugs



006979

Figure 7-65. Glow Plug Components

Continuity Test

1. See [Figure 7-65](#). Connect a suitable test light between a ground and the terminal of the glow plug (D). Choose the glow plug farthest from the battery. Turn the starting aid switch ON.
 - If the circuit is correct, the test light should turn on.
 - If the test light does not turn on, inspect the circuit for loose connections, broken wires, and other damage.
2. Loosen the nut (A) on each glow plug and then disconnect the connectors (C) from the glow plugs (D).
3. Use a suitable digital multimeter to measure the resistance between the electrical terminal of the

glow plug (D) and ground. The point for the ground should be as close to the glow plug as possible.

Maximum resistance for the glow plugs: 0.0139 Ohm

4. Connect the test light to the positive terminal of the battery and to the electrical terminal of the glow plug (D). If the light turns on, the continuity in the glow plug is correct. If the light does not turn on, replace the glow plug. Repeat this step for each glow plug.
5. Install the connector (C) on each glow plug (D) and tighten the nut (A).

Glow Plug Circuit

1. See **Figure 7-65**. Disconnect the power supply cable from the connector (C). Loosen the nut (A) and remove the connector (C) from the electrical terminal of the glow plug (D).
2. Connect the leads of a suitable ammeter between the power supply cable and the terminal of a glow plug and then connect the leads of the digital multimeter between the terminal of the glow plug and a ground. Turn the starting aid switch ON. Observe the amperage reading of the ammeter and the voltage reading on the multimeter.

Maximum current through the glow plug 11 amperes

Current through the glow plug after

6 seconds 11.5 ± 1.5 amperes

NOTE: The voltage should not drop more than 0.5 volts.

3. Perform step 2 for each glow plug.
4. If the multimeter reading is below the specified amperes, the glow plug should be replaced. If there is no reading, replace the glow plug.
5. If the voltage reading of the multimeter is zero, inspect the starting aid switch and the power supply for faulty components and loose connections.
6. See **Figure 7-65**. When the tests are complete, install the connector (C) to the terminal of each of the glow plugs (D) and tighten the nut (A). Connect the cable for the power supply to the glow plugs.

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